

AFFINE GROUP FUNCTORS

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0. INTRODUCTION

List of topics. – We plan to discuss the following topics :

- (1) Yoneda’s lemma and Representable functors (MIL : A.31, A.32, A.33).
- (2) Hopf algebras and examples (MIL : 3.a, 3.b, 3.c).
- (3) Subgroups and homomorphisms of affine group functors (MIL : 3.d, 3.e)
- (4) Diagonalizable group functors (MIL : 12.a, 12.b, 12.c).
- (5) Finitely generated projective modules and Cartier duality (MIL : 11.c, WAT : 2.4).
- (6) Linear representations of affine algebraic group functors (MIL : 4.a, 4.b, 4.c, 4.d, 4.e).
- (7) Connectedness, Smoothness etc. (MIL : Mostly selected topics in Chapter 1 and miscellaneous).
- (8) Finite and Etale group schemes (MIL : Selected topics from Chapter 11)
- (9) Lie algebras associated to algebraic group functors (MIL : 10.a, 10.b, 10.c, 10.d, 10.e).
- (10) Tori (MIL : 12.e, 12.f, 12.g, 12.h, 12.i)
- (11) Unipotent groups and Jordan decomposition (MIL : 14.a, 14.b, 14.c, 9.b)
- (12) Nilpotent and solvable group functors (MIL : 6.e, 6.f, Chapter 16)
- (13) Quotients (MIL : 5.a, 5.b, 5.c, Appendix B)
- (14) Cohomology and descent theories (MIL : Chapter 15, A.j, A.o) (WAT : Chapter 17, 18)

We will first cover the topics 1 to 6 in the above list by-and-large using the language of representable functors. In these, we will be forced to define the notion of a “closed subfunctor” or “closed subgroup functor” in an ad-hoc fashion : the reason is that the defining conditions for Zariski topology on an affine scheme (Spec) do not have their exact analogues in the language of functors. However, this issue can be mitigated : each result we prove here is an “if and only if” statement yielding an equivalent Hopf algebraic fact for which the corresponding fact about affine group schemes exists. Then, before proceeding to topics 7-14, we will talk about the analogues of the main results encountered thus far in the setup of group schemes of finite type over a field; this would require considerable use of scheme theoretic machinery. This is in preparation for the topics 7-14 which require heavier doses of scheme theory.

Assumption. – *Throughout this note we will assume k is a fixed base ring which is commutative and unital. There will be no further implicit assumptions unless otherwise explicitly mentioned in a certain context. Many a times we will take k to be a field, and also sometimes we will assume k to be an algebraically closed field¹.*

1. REPRESENTABLE FUNCTORS AND HOPF ALGEBRAS

Let $\mathcal{C}$ be any category with its object *class* denoted by $\text{Ob}(\mathcal{C})$ and the *set* of arrows between two of its objects X, Y denoted by $\text{Hom}_{\mathcal{C}}(X, Y)$.

Definition 1.1. – For an object $X \in \text{Ob}(\mathcal{C})$ we define the functor $h^X : \mathcal{C} \rightarrow \mathbf{Set}$ which assigns to an object Y the set $h^X(Y) := \text{Hom}_{\mathcal{C}}(X, Y)$; and an arrow $f : Y \rightarrow Y'$ is assigned to $h^X(f) := [g \mapsto f \circ g] : \text{Hom}_{\mathcal{C}}(X, Y) \rightarrow \text{Hom}_{\mathcal{C}}(X, Y')$.

Definition 1.2. – Let $F : \mathcal{C} \rightarrow \mathbf{Set}$ be a functor. Consider two objects X, Y in $\mathcal{C}$ and an element $a \in FX$. We define the map

$$\omega_{a,Y} : \text{Hom}_{\mathcal{C}}(X, Y) \rightarrow FY$$

which is explicitly given as $g \mapsto (Fg)(a)$.

Lemma 1.3. – *Let $F : \mathcal{C} \rightarrow \mathbf{Set}$ be a functor. We have :*

¹But this will never be a global assumption which is typical of standard textbooks (e.g. Humphreys)

- (1) For every object X of $\mathcal{C}$ and every $a \in FX$, the family of maps $(\omega_{a,Y} \mid Y \in \text{Ob}(\mathcal{C}))$ is a natural transformation denoted $\omega_a : h^X \rightarrow F$.
- (2) (Yoneda's Lemma) – For every natural transformation $\eta : h^X \rightarrow F$, there exists a unique $a \in FX$ such that $\eta = \omega_a$ i.e we have a bijection of sets :

$$\boxed{\text{Nat}(h^X, F) \simeq FX}$$

Proof. – For (1) given two objects $Y, Y' \in \text{Ob}(\mathcal{C})$ and some $f \in \text{Hom}_{\mathcal{C}}(Y, Y')$ we have to show that $Ff \circ \omega_{a,Y} = \omega_{a,Y'} \circ h^X(f)$. Indeed, consider an arrow $g \in \text{Hom}_{\mathcal{C}}(X, Y)$ and note that

$$(Ff \circ \omega_{a,Y})(g) = Ff(Fg(a)) = (Ff \circ Fg)(a) = F(f \circ g)(a) = (\omega_{a,Y'} \circ h^X(f))(a).$$

Now for (2), suppose $\eta : h^X \rightarrow F$ is a natural transformation. Set $a := \eta_X(1_X)$. Now we show that $\omega_a = \eta$. So let Y be an object in $\mathcal{C}$ and $g \in h^X(Y) = \text{Hom}_{\mathcal{C}}(X, Y)$. Then we have

$$\eta_Y(g) = \eta_Y(g \circ 1_X) = \eta_Y(h^X(g)(1_X)) = (\eta_Y \circ h^X(g))(1_X) = (Fg \circ \eta_X)(1_X) = (Fg)(a) = \omega_{a,Y}(g)$$

whence the assertion.

To show uniqueness, suppose $a, b \in FX$ such that $\omega_a = \omega_b = \eta$. Then $\omega_{a,X}(1_X) = F(1_X)(a) = a$ and $\omega_{b,X}(1_X) = F(1_X)(b) = b$, and hence $a = b$. $\square$

Corollary 1.4 (Yoneda bijection 1). – Taking $F = h^Y$ we have a bijection of sets

$$\boxed{\text{Nat}(h^X, h^Y) \simeq \text{Hom}_{\mathcal{C}}(Y, X)}$$

In particular, a morphism $f : Y \rightarrow X$ corresponds to $\omega_f : h^X \rightarrow h^Y$.

Fact 1.5. – If $f : Y \rightarrow X$ and $g : Z \rightarrow Y$ then $\omega_{f \circ g} = \omega_g \circ \omega_f$.

Proof. – Let $T \in \text{Ob}(\mathcal{C})$ and $j \in h^X(T) = \text{Hom}(X, T)$. Then $\omega_{f,T}(j) = h^Y(j)(f) = j \circ f$ and $\omega_{g,T}(j \circ f) = (j \circ f) \circ g$. And we have $\omega_{f \circ g,T}(j) = j \circ (f \circ g)$. But by associativity of composition we know

$$(j \circ f) \circ g = j \circ (f \circ g).$$

$\square$

Fact 1.6. – In the notation of the above corollary, $f : Y \rightarrow X$ is an isomorphism if and only if $\omega_f : h^X \rightarrow h^Y$ is a natural isomorphism.

Proof. – Let $f : Y \rightarrow X$ be an isomorphism with inverse $g : X \rightarrow Y$. Then, by the previous exercise

$$\omega_f \circ \omega_g = \omega_{g \circ f} = \omega_{1_Y} = 1_{h^Y} \quad \text{and} \quad \omega_g \circ \omega_f = \omega_{f \circ g} = \omega_{1_X} = 1_{h^X}.$$

Conversely, suppose ω_f has inverse $\eta : h^Y \rightarrow h^X$, which by Yoneda, is of the form ω_g for some $g : X \rightarrow Y$. But then again by previous exercise we have

$$\omega_{g \circ f} = \omega_f \circ \omega_g = 1_{h^Y} = \omega_{1_Y} \quad \text{and} \quad \omega_{f \circ g} = \omega_g \circ \omega_f = 1_{h^X} = \omega_{1_X}.$$

Thus $g \circ f = 1_Y$ and $f \circ g = 1_X$ by evaluating at 1_Y and 1_X respectively. $\square$

Definition 1.7. – A functor $F : \mathcal{C} \rightarrow \mathbf{Set}$ is called *representable* if it is isomorphic to h^X for some $X \in \text{Ob}(\mathcal{C})$; then we say X *represents* F .

Remark 1.8. – By the previous exercise, if F is a representable functor represented by X and Y , then $X \simeq Y$.

Lemma 1.9 (Yoneda bijection 2). – A functor $F : \mathcal{C} \rightarrow \mathbf{Set}$ is representable $\iff$ there exists a pair (X, a) with $X \in \text{Ob}(\mathcal{C})$ and $a \in FX$ having the following property :

$$\boxed{\forall Y \in \text{Ob}(\mathcal{C}) \text{ and } \forall b \in FY, \exists! \varphi_b \in \text{Hom}_{\mathcal{C}}(X, Y) \text{ such that } b = F(\varphi_b)(a)}$$

Proof. – Suppose $F : \mathcal{C} \rightarrow \mathbf{Set}$ is a representable functor and $\omega_a : h^X \xrightarrow{\sim} F$ is an isomorphism for some $a \in FX$. We consider the pair (X, a) . Let $Y \in \text{Ob}(\mathcal{C})$ and $b \in FY$. Now we know that $\omega_{a,Y} : h^X(Y) \rightarrow FY$ is a bijection, so there exists a unique $\varphi_b \in \text{Hom}(X, Y)$ such that $\omega_{a,Y}(\varphi_b) = F(\varphi_b)(a) = b$ as required. Conversely, suppose (X, a) is a pair with the given property. But the property is precisely saying that $\omega_{a,Y} : h^X(Y) \rightarrow FY$ is a bijection for each object Y , and so it is a natural isomorphism. This completes the proof. $\square$

Definition 1.10. – Given a representable functor $F : \mathcal{C} \rightarrow \mathbf{Set}$ the pair (X, a) given by the proposition above is called a *universal pair* for F .

It is often useful to have a more intrinsic description of the Yoneda bijection of Corollary 1.4. More precisely, suppose E and F are representable functors represented by universal pairs (X, a) and (Y, b) with $a \in EX$ and $b \in FY$. Let $\eta : E \rightarrow F$ be a natural transformation. We consider the element $\eta(X)(a) \in FX$. Then the morphism $\varphi_{\eta(X)(a)} \in \text{Hom}(Y, X)$ is the unique one such that $F(\varphi_{\eta(X)(a)})(b) = \eta(X)(a)$. Thus we have

$$\eta \mapsto \varphi_{\eta(X)(a)} : \text{Nat}(E, F) \rightarrow \text{Hom}(Y, X).$$

On the other hand, given $f \in \text{Hom}(Y, X)$ we define a natural transformation $\eta_f : E \rightarrow F$ as follows : for each object Z and each $c \in EZ$, consider the unique map $\varphi_c : X \rightarrow Z$ such that $E(\varphi_c)(a) = c$, then we set

$$\eta_f(Z)(c) := F(\varphi_c \circ f)(b) \in FZ.$$

But $\varphi_{\eta_f(Z)(c)} \in \text{Hom}(Y, Z)$ is the unique morphism such that $F(\varphi_{\eta_f(Z)(c)})(b) = \eta_f(Z)(c)$. Thus we get

$$\varphi_{\eta_f(Z)(c)} = \varphi_c \circ f.$$

We check that $\eta_f : E \rightarrow F$ is indeed a natural transformation. Let $g : Z \rightarrow Z'$ be any morphism. We must show that

$$F(g) \circ \eta_f(Z) = \eta_f(Z') \circ E(g).$$

Fix any $c \in E(Z)$. Then we know

$$E(\varphi_c)(a) = c.$$

Applying $E(g)$ to both sides we get

$$E(g)(c) = E(g \circ \varphi_c)(a)$$

thus

$$\varphi_{E(g)(c)} = g \circ \varphi_c.$$

Now we compute both sides of naturality at c . The left hand side is

$$F(g)(\eta_f(Z)(c)) = F(g)(F(\varphi_c \circ f)(b)) = F(g \circ \varphi_c \circ f)(b).$$

And the right hand side is

$$\eta_f(Z')(E(g)(c)) = F(\varphi_{E(g)(c)} \circ f)(b) = F(g \circ \varphi_c \circ f)(b).$$

Thus $\eta_f : E \rightarrow F$ is natural.

Finally, we check that $\eta \mapsto \varphi_{\eta(X)(a)}$ and $f \mapsto \eta_f$ are inverses of each other. Indeed, we start with any $\eta : E \rightarrow F$ and set $f := \varphi_{\eta(X)(a)}$. We will show that $\eta_f = \eta$. Fix an object Z and an element $c \in E(Z)$. Now $\varphi_c : X \rightarrow Z$ is the unique morphism such that $E(\varphi_c)(a) = c$. We compute :

$$\eta_f(Z)(c) = F(\varphi_c \circ f)(b) = F(\varphi_c \circ \varphi_{\eta(X)(a)})(b).$$

Now $F(\varphi_{\eta(X)(a)})(b) = \eta(X)(a)$ thus $\eta_f(Z)(c) = F(\varphi_c)(\eta(X)(a))$. But by naturality of η we have

$$F(\varphi_c)(\eta(X)(a)) = \eta(Z)(E(\varphi_c)(a)) = \eta(Z)(c).$$

Thus we get $\eta_f(Z)(c) = \eta(Z)(c)$ for all objects Z and all $c \in EZ$ and hence $\eta_f = \eta$ as asserted.

Now we start with any $f : Y \rightarrow X$ and form $\eta_f : E \rightarrow F$. We wish to show that $f = \varphi_{\eta_f(X)(a)}$. Now observe that

$$\eta_f(X)(a) = F(\varphi_a \circ f)(b) = F(\text{id}_X \circ f)(b) = F(f)(b)$$

since $\varphi_a : X \rightarrow X$ is the unique map such that $E(\varphi_a)(a) = a$ and so $\varphi_a = \text{id}_X$. Hence we have

$$\varphi_{\eta_f(X)(a)} = \varphi_{F(f)(b)}.$$

But $\varphi_{F(f)(b)} = f$ since $F(\varphi_{F(f)(b)})(b) = F(f)(b)$ and by uniqueness we get the required equality. We can therefore reformulate Corollary 1.4 as follows :

Fact 1.11 (Yoneda bijection 3). – Let E and F be representable functors with universal pairs (X, a) and (Y, b) respectively. Then we have

$$\boxed{\text{Nat}(E, F) \simeq \text{Hom}(Y, X)}$$

where the bijection is given by

$$\eta \mapsto \varphi_{\eta(X)}(a)$$

with inverse

$$\eta_f \leftarrow f$$

where η_f is defined by the relation

$$\boxed{\varphi_{\eta_f(Z)}(c) = \varphi_c \circ f}$$

for every object Z and every $c \in EZ$.

Proposition 1.12. – Let $F : \mathcal{C} \rightarrow \text{Set}$ be a representable functor with a universal pair (X, a) . Suppose (X', a') is another pair with $X' \in \text{Ob}(\mathcal{C})$ and $a' \in FX'$. Then (X', a') is also a universal pair for F if and only if there exists an isomorphism $g : X \xrightarrow{\sim} X'$ such that $Fg(a) = a'$. Furthermore, in this case the isomorphism g is also unique.

Proof. – By the universal property of (X, a) in Proposition 1.9, we have a unique $g : X \rightarrow X'$ such that $Fg(a) = a'$. Similarly, we have a unique $g' : X' \rightarrow X$ such that $Fg'(a') = a$. It follows that

$$F(g' \circ g)(a) = F(g') \circ F(g)(a) = F(g')(a') = a.$$

But taking $Y = X$ and $b = a$ in Proposition 1.9, we know that the unique morphism $f : X \rightarrow X$ such that $Ff(a) = a$ must be 1_X . Thus $g' \circ g = 1_X$. Similarly we can show that $g \circ g' = 1_{X'}$.

Conversely, suppose there is an isomorphism $g : X \xrightarrow{\sim} X'$ such that $Fg(a) = a'$. Suppose Y is an object and $b \in FY$. Since (X, a) is a universal pair, there is a unique $f : X \rightarrow Y$ such that $Ff(a) = b$. Consider the morphism $h := f \circ g^{-1} : X' \rightarrow Y$. Observe that

$$Fh(a') = (Ff) \circ (Fg)^{-1}(a') = Ff(a) = b.$$

If $h' : X' \rightarrow Y$ is yet another morphism such that $Fh'(a') = b$, we consider the morphism $h' \circ g : X \rightarrow Y$, and observe that

$$F(h' \circ g)(a) = Fh'(Fg(a)) = Fh'(a') = b.$$

By the fact that (X, a) is a universal pair it is forced that $f = h' \circ g$, whence $h' = f \circ g^{-1} = h$. Thus we have shown that there is a unique morphism $h : X' \rightarrow Y$ such that $Fh(a') = b$, proving that (X', a') is also a universal pair. $\square$

From this point onwards we set $\mathcal{C} := \text{Alg}_k$ for a commutative unital ring k .

Definition 1.13. – Let k be a commutative unital ring. A functor $F : \text{Alg}_k \rightarrow \text{Set}$ is called a *k-functor* and if further F is representable then it is called *affine k-functor* or *affine functor* or *affine k-scheme* or just *affine scheme* (over k).

Definition 1.14. – Let k be a commutative unital ring and Alg_k be the category of k -algebras. A functor $G : \text{Alg}_k \rightarrow \text{Grp}$ is called an *affine k-group functor* or *affine group functor* or *affine k-group scheme* or *affine group scheme* if the underlying set valued functor, which is a k -functor, is representable – that is to say – an affine functor.

Remark 1.15. – We will mostly use the words *affine functors* and *affine group functors*, suppressing the symbol k which is always understood in all contexts, and here the word “affine” should indicate representability.

Definition 1.16. – We say that a functor $G : \text{Alg}_k \rightarrow \text{Set}$ is a *group functor* if there exists natural transformations

$$m : G \times G \rightarrow G, \quad e : h^k \rightarrow G, \quad \text{and} \quad i : G \rightarrow G$$

such that the following hold :

- $m \circ (\text{id} \times m) = m \circ (m \times \text{id})$
- $m \circ (e \times \text{id}) = c$
- $m \circ (i, \text{id}) = e \circ t$

where $\text{id} : G \rightarrow G$ denotes the identity transformation, $c : h^k \times G \xrightarrow{\sim} G$ denotes the canonical identification, and $t : G \rightarrow h^k$ denotes the trivial transformation.

Remark 1.17. – Thus if G is a group functor in the sense of the above definition then for each k -algebra A the set maps m_A , e_A , and i_A gives a group structure on $G(A)$. Hence the name group functor. For a set valued functor which is a group functor, we also write $G : \text{Alg}_k \rightarrow \text{Grp}$.

Example 1.18. – Consider the functor $\mu_n : \text{Alg}_k \rightarrow \text{Set}$ defined by $\mu_n(A) := \{x \in A \mid x^n = 1\}$. Then $\mu_n(A) \subset A^\times$ and has a group structure for each A . Thus μ_n is actually a group functor. We show that it is representable. Consider the pair $(R_n, \bar{t})$ where $R_n := k[t]/(t^n - 1)$. We claim that this is a universal pair for μ_n . Indeed, if A is another k -algebra and $a \in \mu_n(A)$ (i.e. $a^n = 1$) then we consider the map $k[t] \rightarrow A$ defined by $t \mapsto a$. This factors through to give a map $R_n \rightarrow A$ mapping $\bar{t} \mapsto a$. Obviously, this map $R_n \rightarrow A$ is the unique map sending $\bar{t} \mapsto a$ since $\bar{t}$ generates the algebra R_n . Thus μ_n is an affine group scheme.

Example 1.19. – Next, let p be a prime and consider the functor

$$H : \text{Alg}_k \rightarrow \text{Grp} \quad \text{defined by} \quad A \rightsquigarrow H(A) := \{a \in A \mid a^{p^n} = 1 \text{ for some } n \in \mathbf{N}\}.$$

We claim that H is not representable. Suppose for contradiction $H \simeq h^R$ for some algebra with (R, r) being a universal pair. Then $r \in H(R)$ and hence there exists an integer n such that $r^{p^n} = 1$. Now consider the k -algebra $A := k[t]/(t^{p^{n+1}} - 1)$ and take $\bar{t} \in H(A)$. Then by the universal property of (R, r) there is a unique k -algebra map $\varphi : R \rightarrow A$ such that $\varphi(r) = \bar{t}$. But this means $\bar{t}^{p^n} = 1$ i.e. $(t^{p^{n+1}} - 1) \mid (t^{p^n} - 1)$ which is not possible.

Example 1.20. – Now consider the functors μ_{p^n} for a prime p and $n \in \mathbf{N}$. Then $(\mu_{p^n} \mid n > 0)$ forms a directed system :

$$\mu_p \rightarrow \mu_{p^2} \rightarrow \mu_{p^3} \rightarrow \cdots \rightarrow \mu_{p^n} \rightarrow \mu_{p^{n+1}} \rightarrow \cdots.$$

Then we can consider the group functor

$$\varinjlim_{n>0} \mu_{p^n} : \text{Alg}_k \rightarrow \text{Grp}.$$

We show that its underlying set-valued functor is naturally isomorphic to H of the previous paragraph. Indeed, for each k -algebra A we have $\mu_p(A) \subset \mu_{p^2}(A) \subset \cdots$ and hence $\varinjlim_{n>0} \mu_{p^n}(A) = \bigcup_{n>0} \mu_{p^n}(A) = H(A)$. Thus *direct limit of an inductive family of representable functors need not be representable in general.*

Proposition 1.21. – *If $(G_i : \text{Alg}_k \rightarrow \text{Set} \mid i \in I)$ is a projective family of representable functors, then the projective limit $\varprojlim_{i \in I} G_i$ is representable.*

Proof. – Let $G_i = h^{R_i}$ and suppose $r_i \in G_i(R_i) = \text{Hom}(R_i, R_i)$ be the identity map. Thus (R_i, r_i) is the universal pair. We define $G(A) := \varprojlim_{i \in I} G_i(A)$ for any k -algebra A ; this gives a functor $G : \text{Alg}_k \rightarrow \text{Set}$ which we claim is representable. We set $R := \varinjlim_{i \in I} R_i$ and suppose $\iota_i : R_i \rightarrow R$ are the canonical maps. We wish to show that $G \simeq h^R$. Now the universal element, say r , must be an element of $G(R) = \varprojlim_{i \in I} G_i(R) = \varprojlim_{i \in I} \text{Hom}(R_i, R)$. We *canonically* set it as $r := (\iota_i)_{i \in I}$. So first we check that r is indeed an element of $\varprojlim_{i \in I} \text{Hom}(R_i, R)$. To see this, let $i \leq j$ and suppose $f_{ij} : R_i \rightarrow R_j$ be the map corresponding to $G_j \rightarrow G_i$. This gives us the maps

$$p_{ij} : \text{Hom}(R_j, R) \rightarrow \text{Hom}(R_i, R), \quad \phi \mapsto \phi \circ f_{ij}.$$

Thus $(\text{Hom}(R_i, R), (p_{ij} \mid i \leq j))$ forms a projective system. Now it is clear that

$$p_{ij}(\iota_j) = \iota_j \circ f_{ij} = \iota_i.$$

Thus $r := (\iota_i)_{i \in I}$ is indeed a coherent sequence. It remains to show that the candidate pair (R, r) has the universal property. So let (S, s) be a pair where S is a k -algebra and $s = (s_i)_{i \in I} \in G(S) = \varprojlim_{i \in I} \text{Hom}(R_i, S)$ satisfying $s_i = s_j \circ f_{ij}$ for all $i \leq j$. Thus the maps $s_i : R_i \rightarrow S$ are compatible with the maps f_{ij} . Hence, by

the universal property of direct limit there exists a unique k -algebra map $\Phi : R \rightarrow S$ such that $\Phi \circ \iota_i = s_i$ for all i . Then the map

$$G(\Phi) : G(R) = \varprojlim_{i \in I} \text{Hom}(R_i, R) \rightarrow G(S) = \varprojlim_{i \in I} \text{Hom}(R_i, S)$$

is given by $G(\Phi)((\phi_i)_{i \in I}) = (\Phi \circ \phi_i)_{i \in I}$. We then see that $G(\Phi)(r) = (\Phi \circ \iota_i)_i = (s_i)_i = s$. For uniqueness, suppose $\Psi : R \rightarrow S$ is another map such that $G(\Psi)(r) = s$ which means

$$\Psi \circ \iota_i = s_i = \Phi \circ \iota_i \quad \text{for all } i$$

and thus $\Psi = \Phi$ by the universal property of $R = \varinjlim_{i \in I} R_i$. We have now shown that (R, r) is indeed the universal pair for $G = \varprojlim_{i \in I} G_i$ as asserted. $\square$

Definition 1.22. – Suppose F_1, F_2 , and E are affine functors represented by universal pairs (R_1, r_1) , (R_2, r_2) , and (S, s) respectively. Let $\eta_1 : F_1 \rightarrow E$ and $\eta_2 : F_2 \rightarrow E$ be natural transformations. We define the *fiber product of F_1 and F_2 over E w.r.t η_1, η_2* as

$$(F_1 \times_E F_2)(A) := \{(x_1, x_2) \in F_1(A) \times F_2(A) \mid \eta_1(A)(x_1) = \eta_2(A)(x_2)\}$$

for all k -algebras A .

Proposition 1.23. – $F_1 \times_E F_2$ is representable by $R_1 \otimes_S R_2$.

Proof. – Let natural transformations $\eta_1 : F_1 \rightarrow E$, $\eta_2 : F_2 \rightarrow E$ correspond respectively to algebra maps $f_1 : S \rightarrow R_1$, $f_2 : S \rightarrow R_2$. Thus by Yoneda bijection (fact 1.11) we know

$$\varphi_{\eta_1(A)(x_1)} = \varphi_{x_1} \circ f_1, \quad \forall A \text{ and } \forall x_1 \in F_1(A),$$

$$\varphi_{\eta_2(A)(x_2)} = \varphi_{x_2} \circ f_2, \quad \forall A \text{ and } \forall x_2 \in F_2(A).$$

Let $R := R_1 \otimes_S R_2$ and consider the canonical maps $i_1 : R_1 \rightarrow R$ and $i_2 : R_2 \rightarrow R$. Now define the element

$$r := (F_1(i_1)(r_1), F_2(i_2)(r_2)) \in F_1(R) \times F_2(R)$$

and we show that $r \in (F_1 \times_E F_2)(R)$ i.e. $\eta_1(R)(F_1(i_1)(r_1)) = \eta_2(R)(F_2(i_2)(r_2))$. First note that

$$\varphi_{F_1(i_1)(r_1)} = i_1 \quad \text{and} \quad \varphi_{F_2(i_2)(r_2)} = i_2.$$

Thus we get

$$\varphi_{\eta_1(R)(F_1(i_1)(r_1))} = \varphi_{F_1(i_1)(r_1)} \circ f_1 = i_1 \circ f_1$$

$$\varphi_{\eta_2(R)(F_2(i_2)(r_2))} = \varphi_{F_2(i_2)(r_2)} \circ f_2 = i_2 \circ f_2.$$

But recall that R_1 and R_2 are S -modules via f_1 and f_2 respectively. Hence by balancedness of $-\otimes_S -$ we have $i_1 \circ f_1 = i_2 \circ f_2$ and thus

$$\varphi_{\eta_1(R)(F_1(i_1)(r_1))} = \varphi_{\eta_2(R)(F_2(i_2)(r_2))} \Rightarrow \eta_1(R)(F_1(i_1)(r_1)) = \eta_2(R)(F_2(i_2)(r_2)).$$

Thus we have shown $r := (F_1(i_1)(r_1), F_2(i_2)(r_2)) \in (F_1 \times_E F_2)(R)$.

We claim that $(R, r) = (R_1 \otimes_S R_2, (F_1(i_1)(r_1), F_2(i_2)(r_2)))$ is a universal pair for $F_1 \times_E F_2$. So let (A, x) be any pair with $x = (x_1, x_2) \in (F_1 \times_E F_2)(A)$. Then $\eta_1(A)(x_1) = \eta_2(A)(x_2)$. Let $\varphi_{x_1} : R_1 \rightarrow A$ and $\varphi_{x_2} : R_2 \rightarrow A$ be the unique algebra maps such that $F_1(\varphi_{x_1})(r_1) = x_1$ and $F_2(\varphi_{x_2})(r_2) = x_2$. Now we have

$$\varphi_{\eta_1(A)(x_1)} = \varphi_{\eta_2(A)(x_2)} \Rightarrow \varphi_{x_1} \circ f_1 = \varphi_{x_2} \circ f_2 : S \rightarrow A$$

by fact 1.11. Thus the map $(a_1, a_2) \mapsto \varphi_{x_1}(a_1)\varphi_{x_2}(a_2) : R_1 \times R_2 \rightarrow A$ is bilinear and S -balanced and thus induces

$$\Phi : R_1 \otimes_S R_2 \rightarrow A$$

satisfying

$$\Phi \circ i_1 = \varphi_{x_1} \quad \text{and} \quad \Phi \circ i_2 = \varphi_{x_2}.$$

We then compute :

$$(F_1 \times_E F_2)(\Phi)(r) = (F_1(\Phi)(F_1(i_1)(r_1)), F_2(\Phi)(F_2(i_2)(r_2))) = (F_1(\varphi_{x_1})(r_1), F_2(\varphi_{x_2})(r_2)) = (x_1, x_2).$$

Finally we show that Φ is the unique such map with $(F_1 \times_E F_2)(\Phi)(r) = (x_1, x_2) = x$. Indeed, if $\Psi : R_1 \otimes_S R_2 \rightarrow A$ also has the property : $(F_1 \times_E F_2)(\Psi)(r) = (x_1, x_2)$. This implies

$$F_1(\Psi)(F_1(i_1)(r_1)) = x_1 = F_1(\varphi_{x_1})(r_1) \quad \text{and} \quad F_2(\Psi)(F_2(i_2)(r_2)) = x_2 = F_2(\varphi_{x_2})(r_2).$$

Hence by uniqueness of φ_{x_1} and φ_{x_2} we have

$$\Psi \circ i_1 = \varphi_{x_1} \quad \text{and} \quad \Psi \circ i_2 = \varphi_{x_2}.$$

But then by uniqueness of Φ with the property $\Phi \circ i_1 = \varphi_{x_1}$ and $\Phi \circ i_2 = \varphi_{x_2}$ we have $\Phi = \Psi$ as expected. This completes the proof that $F_1 \times_E F_2$ is representable. $\square$

Remark 1.24. – When $E = h^k$ we write $F_1 \times F_2$ in place of $F_1 \times_{h^k} F_2$ and we call it the *Cartesian product* or simply *product* of the representable functors F_1 and F_2 . Thus $F_1 \times F_2$ is represented by $R_1 \otimes R_2$. Note that for any algebra A and any $(z_1, z_2) \in F_1(A) \times F_2(A)$ we have a unique map $\varphi_{(z_1, z_2)} : R_1 \otimes R_2 \rightarrow A$ with the property

$$(F_1 \times F_2)(\varphi_{(z_1, z_2)})(F_1(i_1)(r_1), F_2(i_2)(r_2)) = (z_1, z_2)$$

or equivalently $\varphi_{z_1} = \varphi_{(z_1, z_2)} \circ i_1$ and $\varphi_{z_2} = \varphi_{(z_1, z_2)} \circ i_2$.

Proposition 1.25. – Let $\eta_1 : E_1 \rightarrow F_1$ and $\eta_2 : E_2 \rightarrow F_2$ be two natural transformations between representable functors E_i and F_i with universal pairs (R_i, r_i) and (S_i, s_i) respectively for $i = 1, 2$. Suppose they correspond to algebra maps $f_i : S_i \rightarrow R_i$ for $i = 1, 2$. Then $\eta_1 \times \eta_2 : E_1 \times E_2 \rightarrow F_1 \times F_2$ corresponds to $f_1 \otimes f_2$.

Proof. – Let $i = 1, 2$. Then we know that for any algebra A and $x_i \in E_i(A)$ with representing map $\varphi_{x_i} : R_i \rightarrow A$, we have, by fact 1.11 :

$$\varphi_{\eta_i(A)(x_i)} = \varphi_{x_i} \circ f_i.$$

Now $E_1 \times E_2$ is represented by $(R_1 \otimes R_2, (E_1(i_1)(r_1), E_2(i_2)(r_2)))$ where $i_1 : R_1 \rightarrow R_1 \otimes R_2$ and $i_2 : R_2 \rightarrow R_1 \otimes R_2$ are canonical maps. Thus

$$\varphi_{E_1(i_1)(r_1)} = i_1 \quad \text{and} \quad \varphi_{E_2(i_2)(r_2)} = i_2.$$

Similarly, $F_1 \times F_2$ is represented by $(S_1 \otimes S_2, (F_1(j_1)(s_1), F_2(j_2)(s_2)))$ where $j_1 : S_1 \rightarrow S_1 \otimes S_2$ and $j_2 : S_2 \rightarrow S_1 \otimes S_2$ are canonical and so we have

$$\varphi_{F_1(j_1)(s_1)} = j_1 \quad \text{and} \quad \varphi_{F_2(j_2)(s_2)} = j_2.$$

Now by recipe of fact 1.11 we consider

$$f = \varphi_{(\eta_1 \times \eta_2)(R_1 \otimes R_2)(E_1(i_1)(r_1), E_2(i_2)(r_2))}$$

which is the algebra map corresponding to the natural transformation $\eta_1 \times \eta_2$. We need to show that $f = f_1 \otimes f_2$. But first we set $R := R_1 \otimes R_2$ and compute

$$(\eta_1 \times \eta_2)(R)(E_1(i_1)(r_1), E_2(i_2)(r_2)) = (\eta_1(R)(E_1(i_1)(r_1)), \eta_2(R)(E_2(i_2)(r_2))).$$

We also set $z_1 := \eta_1(R)(E_1(i_1)(r_1))$ and $z_2 := \eta_2(R)(E_2(i_2)(r_2))$, and so $z := (z_1, z_2) \in F_1(R) \times F_2(R)$. By remark 1.24 we know $f = \varphi_z$ is the unique map that satisfies

$$f \circ j_1 = \varphi_{z_1} \quad \text{and} \quad f \circ j_2 = \varphi_{z_2}.$$

Now we compute φ_{z_1} and φ_{z_2} . First, using the fact 1.11 applied to η_1 we get

$$\varphi_{z_1} = \varphi_{\eta_1(R)(E_1(i_1)(r_1))} = \varphi_{E_1(i_1)(r_1)} \circ f_1 = i_1 \circ f_1.$$

Similarly we get

$$\varphi_{z_2} = i_2 \circ f_2.$$

Thus $f : S_1 \otimes S_2 \rightarrow R_1 \otimes R_2$ is the unique map satisfying : $f \circ j_1 = i_1 \circ f_1$ and $f \circ j_2 = i_2 \circ f_2$. However, it is clear that $f_1 \otimes f_2 : S_1 \otimes S_2 \rightarrow R_1 \otimes R_2$ satisfies

$$(f_1 \otimes f_2) \circ j_1 = i_1 \circ f_1 \quad \text{and} \quad (f_1 \otimes f_2) \circ j_2 = i_2 \circ f_2.$$

Thus $f = f_1 \otimes f_2$ as asserted. $\square$

Proposition 1.26. – Let E, F_1, F_2 be representable functors represented by universal pairs (R, r) , (S_1, s_1) and (S_2, s_2) . Suppose $\eta_1 : E \rightarrow F_1$ and $\eta_2 : E \rightarrow F_2$ be natural transformations corresponding to algebra maps $f_1 : S_1 \rightarrow R$ and $f_2 : S_2 \rightarrow R$ respectively. Then $(\eta_1, \eta_2) : E \rightarrow F_1 \times F_2$ corresponds to $(f_1, f_2) : S_1 \otimes S_2 \rightarrow R$ given by $a_1 \otimes a_2 \mapsto f_1(a_1)f_2(a_2)$.

Proof. – By Yoneda bijection 1.11, we have : for every algebra A and every $x \in E(A)$

$$\varphi_{\eta_1(A)}(x) = \varphi_x \circ f_1 \quad \text{and} \quad \varphi_{\eta_2(A)}(x) = \varphi_x \circ f_2$$

where $\varphi_x : R \rightarrow A$ is the unique map corresponding to $x \in E(A)$.

Also, $F_1 \times F_2$ is represented by $(S_1 \otimes S_2, (F_1(j_1)(s_1), F_2(j_2)(s_2)))$ where $j_1 : S_1 \rightarrow S_1 \otimes S_2$ and $j_2 : S_2 \rightarrow S_1 \otimes S_2$ are canonical and so we have

$$\varphi_{F_1(j_1)(s_1)} = j_1 \quad \text{and} \quad \varphi_{F_2(j_2)(s_2)} = j_2.$$

Again, by fact 1.11, the algebra map corresponding to (η_1, η_2) is

$$g := \varphi_{(\eta_1, \eta_2)(R)(r)} = \varphi_{(\eta_1(R)(r), \eta_2(R)(r))}.$$

We set $x_1 := \eta_1(R)(r)$ and $x_2 := \eta_2(R)(r)$. Then we note that

$$\varphi_{x_1} = \varphi_{\eta_1(R)(r)} = \varphi_r \circ f_1 = f_1$$

$$\varphi_{x_2} = \varphi_{\eta_2(R)(r)} = \varphi_r \circ f_2 = f_2$$

since $\varphi_r : R \rightarrow R$ is the representing map of the universal element r and so $\varphi_r = \text{id}_R$.

Now $g = \varphi_{(x_1, x_2)} : S_1 \otimes S_2 \rightarrow R$ is the unique map (representing map of (x_1, x_2)) satisfying the property

$$\varphi_{(x_1, x_2)} \circ j_1 = \varphi_{x_1} \quad \text{and} \quad \varphi_{(x_1, x_2)} \circ j_2 = \varphi_{x_2}.$$

Thus $g : S_1 \otimes S_2 \rightarrow R$ is the unique map with the property

$$g \circ j_1 = f_1 \quad \text{and} \quad g \circ j_2 = f_2.$$

However, $(f_1, f_2) := [a_1 \otimes a_2 \mapsto f_1(a_1)f_2(a_2)] : S_1 \otimes S_2 \rightarrow R$ also has the same property as g and thus $g = (f_1, f_2)$. $\square$

Proposition 1.27. – Let E_1, E_2, E_3 be representable functors given by universal pairs $(R_1, r_1), (R_2, r_2), (R_3, r_3)$ respectively. Suppose $\eta_1 : E_1 \rightarrow E_2$ and $\eta_2 : E_2 \rightarrow E_3$ be natural transformations corresponding to algebra maps $f_1 : R_2 \rightarrow R_1$ and $f_2 : R_3 \rightarrow R_2$ respectively. Then $\eta_2 \circ \eta_1 : E_1 \rightarrow E_3$ corresponds to $f_1 \circ f_2 : R_3 \rightarrow R_1$.

Proof. – First note that for any algebra A and any $x \in E_1(A)$ we have

$$\varphi_{(\eta_2 \circ \eta_1)(A)}(x) = \varphi_{\eta_2(A)}(\eta_1(A)(x)) = \varphi_{\eta_1(A)}(x) \circ f_2 = \varphi_x \circ f_1 \circ f_2$$

using the Yoneda bijection 1.11. Take $A = R_1$ and $x = r_1$. Then the algebra map corresponding to $\eta_2 \circ \eta_1$ is $\varphi_{(\eta_2 \circ \eta_1)(R_1)}(r_1)$ and we have

$$\varphi_{(\eta_2 \circ \eta_1)(R_1)}(r_1) = \varphi_{r_1} \circ f_1 \circ f_2 = \text{id}_{R_1} \circ f_1 \circ f_2 = f_1 \circ f_2.$$

$\square$

Proposition 1.28. – Let G be an affine group functor represented by (R, r) . Then the canonical natural isomorphism $c : h^k \times G \xrightarrow{\sim} G$ corresponds to the canonical algebra isomorphism $C : R \xrightarrow{\sim} k \otimes R$. Similarly, the trivial natural map $t : G \rightarrow h^k$ corresponds to the structure map $T : k \rightarrow R$ on R .

Proof. – We know that $h^k \times G$ is represented by $(k \otimes R, (h^k(i_1)(\text{id}_k), G(i_2)(r)))$ where $i_1 : k \rightarrow k \otimes R$ and $i_2 : R \rightarrow k \otimes R$ are canonical. First, we consider $c : h^k \times G \xrightarrow{\sim} G$. Then by 1.11 we have its corresponding algebra map as $\varphi_{c(k)(i_1, G(i_2)(r))} = \varphi_{G(i_2)(r)} = i_2$. But $i_2 : R \xrightarrow{\sim} k \otimes R$ is actually the canonical isomorphism which we denote by C .

Now for the trivial transformation $t : G \rightarrow h^k$ the corresponding algebra map is $\varphi_{t(R)(r)} = \varphi_T = T$ since $t(R) : G(R) \rightarrow \text{Hom}(k, R) = \{T_R\}$. $\square$

Theorem 1.29. – If $G : \text{Alg}_k \rightarrow \text{Set}$ is a representable group functor represented by the algebra R , then there are k -algebra morphisms

$$\begin{cases} \Delta : R \rightarrow R \otimes R & \text{called comultiplication} \\ \epsilon : R \rightarrow k & \text{called counit or augmentation} \\ S : R \rightarrow R & \text{called coinverse or antipode} \end{cases}$$

which satisfy the following rules :

- $(\text{id} \otimes \Delta) \circ \Delta = (\Delta \otimes \text{id}) \circ \Delta$.
- $(\epsilon \otimes \text{id}) \circ \Delta = C$.

- $(S, \text{id}) \circ \Delta = T \circ \epsilon$.

where $C : R \xrightarrow{\sim} k \otimes R$ corresponds to the canonical isomorphism $h^k \times G \xrightarrow{\sim} G$ and $T : k \rightarrow R$ corresponds to the trivial natural transformation $t : h^R \rightarrow h^k$.

Proof. – If $G : \text{Alg}_k \rightarrow \text{Set}$ is a group functor then there exists natural transformations

$$m : G \times G \rightarrow G, \quad e : h^k \rightarrow G, \quad \text{and} \quad i : G \rightarrow G$$

such that the following hold :

- $m \circ (\text{id} \times m) = m \circ (m \times \text{id})$
- $m \circ (e \times \text{id}) = c$
- $m \circ (i, \text{id}) = e \circ t$

where $\text{id} : G \rightarrow G$ denotes the identity transformation, $c : h^k \times G \rightarrow G$ denotes the canonical identification, and $t : G \rightarrow h^k$ denotes the trivial transformation.

Furthermore, if G is representable by some algebra R then the natural transformations m, e, i correspond to Δ, ϵ, S respectively, and c, t correspond to C, T respectively. The composition laws of m, e, i, c, t above yield the required composition laws with $\Delta, \epsilon, S, S, T$ as in the statement of the proposition. All these follow directly from the previous propositions. $\square$

Remark 1.30. – We consider the map $(S, \text{id}) : R \otimes R \rightarrow R$ corresponding to the natural map $(i, \text{id}) : G \rightarrow G \times G$. This map can be written as a composite

$$G \xrightarrow{d} G \times G \xrightarrow{i \times \text{id}} G \times G$$

where $d : G \rightarrow G \times G$ is the *diagonal map* given by $g \mapsto (g, g) : G(A) \rightarrow G(A) \times G(A)$ for each algebra A . We show that d corresponds to $M : R \otimes R \rightarrow R$ the multiplication map on R . First, we note that $G \times G$ is represented by $(R \otimes R, (G(i_1)(r), G(i_2)(r)))$ where $r \in G(R)$ is the universal element and $i_1 : R \rightarrow R \otimes R$, $i_2 : R \rightarrow R \otimes R$ are $a \mapsto a \otimes 1$ and $b \mapsto 1 \otimes b$ respectively. Then for any algebra A and any $(x, y) \in G(A) \times G(A)$ we consider the algebra map $\varphi_{(x,y)} : R \otimes R \rightarrow A$ which is the unique map satisfying

$$\varphi_{(x,y)} \circ i_1 = \varphi_x \quad \text{and} \quad \varphi_{(x,y)} \circ i_2 = \varphi_y.$$

Now consider the algebra map corresponding to d . It is $f = \varphi_{d(R)(r)} = \varphi_{(r,r)}$. Thus, taking $A = R$ and $x = y = r$ and noting that $\varphi_r = \text{id}_R$, we have that $f = \varphi_{(r,r)}$ is the unique algebra map satisfying

$$f \circ i_1 = \text{id}_R \quad \text{and} \quad f \circ i_2 = \text{id}_R.$$

But $M : R \otimes R \rightarrow R$ also satisfies $M \circ i_1 = \text{id}_R = M \circ i_2$. Hence $f = M$ as claimed. Consequently, the identity $(S, \text{id}) \circ \Delta = T \circ \epsilon$ is equivalent to

$$T \circ \epsilon = M \circ (S \otimes \text{id}) \circ \Delta.$$

Definition 1.31. – An algebra R equipped with algebra maps Δ, ϵ, S as in the previous theorem satisfying the given composition rules is called a *Hopf algebra*. We will denote a Hopf algebra by the quadruple (R, Δ, ϵ, S) .

Let G be an affine group functor represented by (R, r) with Hopf structure (Δ, ϵ, S) . For each k -algebra A and each $x \in G(A)$ we denote $\varphi_x : R \rightarrow A$ the unique k -algebra map such that $G(\varphi_x)(r) = x$. Recall that $x \mapsto \varphi_x : G(A) \xrightarrow{\sim} \text{Hom}(R, A)$ is actually given by the Yoneda bijection.

Proposition 1.32. – Fix a k -algebra A and consider the group structure on $G(A)$. We have the following :

- (1) If $m_A : G(A) \times G(A) \rightarrow G(A)$ is the multiplication and $x, y \in G(A)$, then the element

$$z := xy \in G(A)$$

corresponds to the algebra map given by

$$\varphi_z = M_A \circ (\varphi_x \otimes \varphi_y) \circ \Delta$$

where $M_A : A \otimes A \rightarrow A$ denotes the ring multiplication of A .

- (2) If $e_A : \text{Hom}(k, A) \rightarrow G(A)$ has image the identity element denoted again by $e_A \in G(A)$, then we have

$$\varphi_{e_A} = T_A \circ \epsilon$$

where $T_A : k \rightarrow A$ is the structure map.

(3) For $i_A : G(A) \rightarrow G(A)$ and any $x \in G(A)$ denote $x^{-1} := i_A(x)$, then we have

$$\varphi_{x^{-1}} = \varphi_x \circ S.$$

Proof. – We recall that (see remark 1.24) for any algebra A and any $(x, y) \in (G \times G)(A)$ the algebra map $\varphi_{(x,y)} : R \otimes R \rightarrow A$ is the unique map satisfying the properties : $\varphi_{(x,y)} \circ i_1 = \varphi_x$ and $\varphi_{(x,y)} \circ i_2 = \varphi_y$. Now let $z := m(A)(x, y)$ and by 1.11 we know

$$\varphi_{m(A)(x,y)} = \varphi_{(x,y)} \circ \Delta$$

since m corresponds to Δ . So it remains to show that $\varphi_{(x,y)} = M_A \circ (\varphi_x \otimes \varphi_y)$. Indeed, for any simple tensor $a \otimes b \in R \otimes R$ we have

$$\varphi_{(x,y)}(a \otimes b) = \varphi_{(x,y)}(i_1(a)i_2(b)) = \varphi_x(a)\varphi_y(b) = (M_A \circ (\varphi_x \otimes \varphi_y))(a \otimes b).$$

This proves (1).

Now consider the natural map $e : h^k \rightarrow G$ where for each algebra A the image of $e(A)$ is denoted by $e_A \in G(A)$ which is the identity of the group $G(A)$. Now by 1.11 we have

$$\varphi_{e_A} = \varphi_{e(A)(T_A)} = \varphi_{T_A} \circ \epsilon$$

where $\varphi_{T_A} : k \rightarrow A$ is an algebra map and thus $\varphi_{T_A} = T_A$ and hence $\varphi_{e_A} = T_A \circ \epsilon$, proving (2).

Finally for (3), consider $i : G \rightarrow G$. Take any algebra A and any $x \in G(A)$. We have

$$\varphi_{i(A)(x)} = \varphi_x \circ S$$

straightaway by 1.11. The proposition is proved. $\square$

Proposition 1.33. – *Let (R, Δ, ϵ, S) be a Hopf algebra. Then h^R can be made into a group functor : for each algebra A we define a group structure on $h^R(A) = \text{Hom}(R, A)$ as follows*

- For $\varphi_1, \varphi_2 \in \text{Hom}(R, A)$ we define $\varphi_1 * \varphi_2 := M_A \circ (\varphi_1 \otimes \varphi_2) \circ \Delta$.
- Given $T_A : k \rightarrow A$ as the structure map we define $e_A := T_A \circ \epsilon$.
- For $\varphi \in \text{Hom}(R, A)$ we define $i_A(\varphi) := \varphi \circ S$; we will conveniently write $i_A(\varphi)$ as φ^{-1} .

Proof. – First we check associativity : $\varphi_1 * (\varphi_2 * \varphi_3) = (\varphi_1 * \varphi_2) * \varphi_3$. We take $t \in R$ and write

$$\Delta(t) = \sum_j u_j \otimes v_j.$$

But by coassociativity we have $((\Delta \otimes \text{id}) \circ \Delta)(t) = ((\text{id} \otimes \Delta) \circ \Delta)(t)$. Writing

$$\Delta(u_j) = \sum_\ell a_{j\ell} \otimes b_{j\ell}$$

we get the following equality :

$$\sum_{j,\ell} a_{j\ell} \otimes b_{j\ell} \otimes v_j = \sum_j u_j \otimes \Delta(v_j).$$

Now we evaluate $(\varphi_1 * \varphi_2) * \varphi_3$ at t to get

$$\begin{aligned} ((\varphi_1 * \varphi_2) * \varphi_3)(t) &= (M_A \circ ((\varphi_1 * \varphi_2) \otimes \varphi_3) \circ \Delta)(t) \\ &= M_A \left(\sum_j (\varphi_1 * \varphi_2)(u_j) \otimes \varphi_3(v_j) \right) \\ &= M_A \left(\sum_j (M_A \circ (\varphi_1 \otimes \varphi_2) \circ \Delta)(u_j) \otimes \varphi_3(v_j) \right) \\ &= M_A \left(\sum_{j,\ell} M_A(\varphi_1(a_{j\ell}) \otimes \varphi_2(b_{j\ell})) \otimes \varphi_3(v_j) \right) \\ &= ((M_A \circ (M_A \otimes \text{id})) \circ (\varphi_1 \otimes \varphi_2 \otimes \varphi_3)) \left(\sum_{j,\ell} a_{j\ell} \otimes b_{j\ell} \otimes v_j \right). \end{aligned}$$

Similarly, we compute $\varphi_1 * (\varphi_2 * \varphi_3)$ at t to obtain

$$\begin{aligned}
(\varphi_1 * (\varphi_2 * \varphi_3))(t) &= (M_A \circ (\varphi_1 \otimes (\varphi_2 * \varphi_3))) (\Delta(t)) \\
&= M_A \left(\sum_j \varphi_1(u_j) \otimes (\varphi_2 * \varphi_3)(v_j) \right) \\
&= M_A \left(\sum_j \varphi_1(u_j) \otimes M_A((\varphi_2 \otimes \varphi_3)(\Delta(v_j))) \right) \\
&= ((M_A \circ (\text{id} \otimes M_A)) \circ (\varphi_1 \otimes \varphi_2 \otimes \varphi_3)) \left(\sum_j u_j \otimes \Delta(v_j) \right).
\end{aligned}$$

Now the assertion of associativity follows from the fact that $M_A \circ (M_A \otimes \text{id}) = M_A \circ (\text{id} \otimes M_A)$ which is associativity of the ring multiplication of A .

Next, we show that $e_A := T_A \circ \epsilon \in \text{Hom}(R, A)$ is the identity element. For each $\varphi \in \text{Hom}(R, A)$ we show that $e_A * \varphi = \varphi$. Let $t \in R$ and write $\Delta(t) = \sum_j u_j \otimes v_j$. Then we get

$$\begin{aligned}
(e_A * \varphi)(t) &= (M_A \circ (e_A \otimes \varphi)) \left(\sum_j u_j \otimes v_j \right) \\
&= M_A \left(\sum_j (T_A \circ \epsilon)(u_j) \otimes \varphi(v_j) \right) \\
&= (M_A \circ (T_A \otimes \varphi)) ((\epsilon \otimes \text{id}) \circ \Delta)(t).
\end{aligned}$$

Now $(\epsilon \otimes \text{id}) \circ \Delta = C$ the canonical identification $R \rightarrow k \otimes R$ given by $r \mapsto 1 \otimes r$. Thus we get

$$(e_A * \varphi)(t) = (M_A \circ (T_A \otimes \varphi)) (1 \otimes t) = \varphi(t).$$

Finally, we show that $\varphi^{-1} := i_A(\varphi) = \varphi \circ S$ is left inverse (w.r.t. $*$) to φ for any algebra A and $\varphi \in \text{Hom}(R, A)$. Indeed, fix $t \in R$ and write $\Delta(t) = \sum_j u_j \otimes v_j$. Then we have

$$\begin{aligned}
((\varphi \circ S) * \varphi)(t) &= \sum_j M_A(\varphi(S(u_j)) \otimes \varphi(v_j)) \\
&= \sum_j \varphi(S(u_j)) \varphi(v_j) \\
&= \sum_j \varphi(S(u_j) v_j) \\
&= \varphi((M_R \circ (S \otimes \text{id}) \circ \Delta)(t)) \\
&= \varphi(T_R \circ \epsilon)(t) \\
&= \varphi(\epsilon(t) \cdot 1_R) \\
&= T_A(\epsilon(t)) \varphi(1_R) \\
&= e_A(t).
\end{aligned}$$

Hence we have $(\varphi \circ S) * \varphi = e_A$ as asserted. $\square$

Example 1.34. – Consider the functor $\mathbf{G}_a : \mathbf{Alg}_k \rightarrow \mathbf{Grp}$ given by $A \rightsquigarrow (A, +)$. The pair $(k[X], X)$ is a universal pair making $\mathbf{G}_a$ a representable functor. Indeed, given pair (A, a) of k -algebra and an element $a \in \mathbf{G}_a(A) = A$, there is a unique k -algebra map $k[X] \rightarrow A$ with $X \mapsto a$. Now we will compute the algebra maps Δ, ϵ , and S .

First we consider the comultiplication corresponding to the natural transformation given by the additive group structure on each algebra $\mathbf{G}_a(A) = (A, +)$, which is denoted by

$$m(A) : \mathbf{G}_a(A) \times \mathbf{G}_a(A) \rightarrow \mathbf{G}_a(A)$$

and written explicitly as $(a, b) \mapsto a + b$. Now $\mathbf{G}_a \times \mathbf{G}_a$ is represented by

$$(k[X] \otimes k[X], (\mathbf{G}_a(i_1)(X), \mathbf{G}_a(i_2)(X)))$$

where $i_1 : k[X] \rightarrow k[X] \otimes k[X]$ and $i_2 : k[X] \rightarrow k[X] \otimes k[X]$ are canonical. By 1.11 we know

$$\Delta = \varphi_{m(k[X] \otimes k[X])}(\mathbf{G}_a(i_1)(X), \mathbf{G}_a(i_2)(X)) = \varphi_{X \otimes 1 + 1 \otimes X}.$$

But for any algebra A and any $y \in \mathbf{G}_a(A)$ we know that $\varphi_y : k[X] \rightarrow A$ is the unique map such that $\mathbf{G}_a(\varphi_y)(X) = \varphi_y(X) = y$. Thus we have

$$\Delta : k[X] \rightarrow k[X] \otimes k[X]$$

is the unique algebra map such that $\Delta(X) = X \otimes 1 + 1 \otimes X$.

Next, we consider the natural transformation $e : h^k \rightarrow \mathbf{G}_a$. This corresponds to the algebra map

$$\epsilon := \varphi_{e(k)}(\text{id}_k) = \varphi_0$$

since $e(k) : \{\text{id}_k\} \rightarrow k$ has image the identity element of $\mathbf{G}_a(k) = (k, +)$, which is 0. Thus ϵ is the unique algebra map $k[X] \rightarrow k$ such that $\epsilon(X) = 0$.

Finally, we consider $i : \mathbf{G}_a \rightarrow \mathbf{G}_a$ which corresponds to

$$S = \varphi_{i(k[X])}(X) = \varphi_{-X}$$

and thus $S : k[X] \rightarrow k[X]$ is the unique map such that $S(X) = -X$.

Example 1.35. – We consider the functor $\mathbf{G}_m : \mathbf{Alg}_k \rightarrow \mathbf{Grp}$ given by $A \rightsquigarrow A^\times$. Then one can show that $\mathbf{G}_m$ is representable with universal pair $(k[X, Y]/(XY - 1), \bar{X})$. So now we compute the algebra maps Δ, ϵ, S .

First, we consider the comultiplication which corresponds to $m : \mathbf{G}_m \times \mathbf{G}_m \rightarrow \mathbf{G}_m$. Now we know that $\mathbf{G}_m \times \mathbf{G}_m$ is represented by the universal pair

$$\left(\frac{k[X, Y]}{(XY - 1)} \otimes \frac{k[X, Y]}{(XY - 1)}, (\mathbf{G}_m(i_1)(\bar{X}), \mathbf{G}_m(i_2)(\bar{X})) \right)$$

where $i_1 : \frac{k[X, Y]}{(XY - 1)} \rightarrow \frac{k[X, Y]}{(XY - 1)} \otimes \frac{k[X, Y]}{(XY - 1)}$ and $i_2 : \frac{k[X, Y]}{(XY - 1)} \rightarrow \frac{k[X, Y]}{(XY - 1)} \otimes \frac{k[X, Y]}{(XY - 1)}$ are usual canonical maps. Then we have

$$\Delta = \varphi_m\left(\frac{k[X, Y]}{(XY - 1)} \otimes \frac{k[X, Y]}{(XY - 1)}\right)(\mathbf{G}_m(i_1)(\bar{X}), \mathbf{G}_m(i_2)(\bar{X})) = \varphi_{\bar{X} \otimes \bar{X}}.$$

Thus $\Delta : \frac{k[X, Y]}{(XY - 1)} \rightarrow \frac{k[X, Y]}{(XY - 1)} \otimes \frac{k[X, Y]}{(XY - 1)}$ is the unique algebra map such that $\Delta(\bar{X}) = \bar{X} \otimes \bar{X}$.

Next, we consider the transformation $e : h^k \rightarrow \mathbf{G}_m$ which corresponds to

$$\epsilon := \varphi_{e(k)}(1_k) = \varphi_1$$

and thus $\epsilon : \frac{k[X, Y]}{(XY - 1)} \rightarrow k$ is the unique algebra map such that $\epsilon(\bar{X}) = 1$.

Finally, we consider $i : \mathbf{G}_m \rightarrow \mathbf{G}_m$ corresponding to

$$S := \varphi_i\left(\frac{k[X, Y]}{(XY - 1)}\right)(\bar{X}) = \varphi_{\bar{Y}}$$

and thus $S : \frac{k[X, Y]}{(XY - 1)} \rightarrow \frac{k[X, Y]}{(XY - 1)}$ is the unique algebra map carrying $\bar{X} \mapsto \bar{Y}$.

Example 1.36. – Now we consider $\mu_n : \mathbf{Alg}_k \rightarrow \mathbf{Grp}$ which is represented by the universal pair $(\frac{k[X]}{(X^n - 1)}, \bar{X})$. We compute Δ, ϵ, S .

We first consider the natural map $m : \mu_n \times \mu_n \rightarrow \mu_n$ which corresponds to

$$\Delta := \varphi_m\left(\frac{k[X]}{(X^n - 1)} \otimes \frac{k[X]}{(X^n - 1)}\right)(\mu_n(i_1)(\bar{X}), \mu_n(i_2)(\bar{X})) = \varphi_{\bar{X} \otimes \bar{X}}$$

and thus $\Delta : \frac{k[X]}{(X^n - 1)} \rightarrow \frac{k[X]}{(X^n - 1)} \otimes \frac{k[X]}{(X^n - 1)}$ is the unique algebra map such that $\Delta(\bar{X}) = \bar{X} \otimes \bar{X}$.

Similarly, the counit $e : h^k \rightarrow \mu_n$ corresponds to

$$\epsilon = \varphi_{e(k)}(1_k) = \varphi_1$$

and $\epsilon : \frac{k[X]}{(X^n - 1)} \rightarrow k$ is hence the unique algebra map such that $\epsilon(\bar{X}) = 1$.

Finally, the natural map $i : \mu_n \rightarrow \mu_n$ corresponds to

$$S := \varphi_i\left(\frac{k[X]}{(X^n-1)}\right)(\bar{X}) = \varphi_{\bar{X}^{-1}}.$$

Thus $S : \frac{k[X]}{(X^n-1)} \rightarrow \frac{k[X]}{(X^n-1)}$ is the unique algebra map such that $S(\bar{X}) = \bar{X}^{-1}$.

Example 1.37. – We now consider the functor $\mathbf{GL}_n : \mathbf{Alg}_k \rightarrow \mathbf{Grp}$ defined by $A \rightsquigarrow \mathbf{GL}_n(A)$, the matrices with unit determinants. First, we show that it is represented by the universal pair

$$\left(R := \frac{k[X_{11}, \dots, X_{nn}, T]}{(\det(X_{ij})T - 1)}, \mathcal{X} \right)$$

where $\mathcal{X} = (\overline{X_{ij}})_{i,j=1}^n \in \mathbf{GL}_n(R)$. Indeed, if (A, x) is any other pair with $x = (x_{ij}) \in \mathbf{GL}_n(A)$ then we define the map $\phi : k[X_{11}, \dots, X_{nn}, T] \rightarrow A$ by $X_{ij} \mapsto x_{ij}$ and $T \mapsto (\det x)^{-1}$. This factors through R to give a map $\varphi : R \rightarrow A$ such that $\overline{X_{ij}} \mapsto x_{ij}$, and so $\bar{T} \mapsto (\det x)^{-1}$. Thus $\mathbf{GL}_n(\varphi)(\mathcal{X}) = x$. This φ is unique in the sense that if $\psi : R \rightarrow A$ is another map such that $\mathbf{GL}_n(\psi)(\mathcal{X}) = x$ then we have $\psi(\overline{X_{ij}}) = x_{ij}$. But $\psi(\det(\overline{X_{ij}})\bar{T}) = 1$ and thus $\psi(\bar{T}) = (\det x)^{-1}$. Hence $\psi = \varphi$. Also for each algebra A the bijection $\omega_{X,A} : \mathbf{Hom}(R, A) \xrightarrow{\sim} \mathbf{GL}_n(A)$ is given by

$$\varphi \mapsto \mathbf{GL}_n(\varphi)(\mathcal{X}) = (\varphi(\overline{X_{ij}}))_{i,j}.$$

Now we compute the algebra map $\Delta : R \rightarrow R \otimes R$ corresponding to $m : \mathbf{GL}_n \times \mathbf{GL}_n \rightarrow \mathbf{GL}_n$. We know by 1.11 that

$$\Delta = \varphi_{m(R \otimes R)(\mathbf{GL}_n(i_1)(\mathcal{X}), \mathbf{GL}_n(i_2)(\mathcal{X}))} = \varphi_{(\overline{X_{ij}} \otimes 1)_{i,j} (1 \otimes \overline{X_{ij}})_{i,j}}.$$

Thus $\Delta : R \rightarrow R \otimes R$ is the unique algebra map such that

$$\mathbf{GL}_n(\Delta)((\overline{X_{ij}})_{i,j}) = (\overline{X_{ij}} \otimes 1)_{i,j} (1 \otimes \overline{X_{ij}})_{i,j} = \left(\sum_l \overline{X_{il}} \otimes \overline{X_{lj}} \right)_{i,j}$$

which implies that

$$\Delta(\overline{X_{ij}}) = \sum_l \overline{X_{il}} \otimes \overline{X_{lj}}$$

Now for the $e : h^k \rightarrow \mathbf{GL}_n$ the corresponding algebra map is

$$\epsilon = \varphi_{e(k)(1_k)} = \varphi_I$$

and hence $\epsilon : R \rightarrow k$ is the unique algebra map such that $\mathbf{GL}_n(\epsilon)(\overline{X_{ij}})_{i,j} = I$ which means

$$\epsilon(\overline{X_{ij}}) = \delta_{ij}.$$

Finally, the map $i : \mathbf{GL}_n \rightarrow \mathbf{GL}_n$ corresponds to

$$S = \varphi_{i(R)(\mathcal{X})} = \varphi_{\mathcal{X}^{-1}}$$

and thus it is the unique map such that $\mathbf{GL}_n(S)(\overline{X_{ij}})_{i,j} = (\overline{X_{ij}})_{i,j}^{-1}$ which is equivalent to saying

$$S(\mathcal{X}_{ij}) = (\mathcal{X}^{-1})_{ij}.$$

Example 1.38. – Next, suppose k has characteristic $p > 0$. We consider the functor α_p which is defined as

$$A \rightsquigarrow \alpha_p(A) := \{x \in A \mid x^p = 0\} : \mathbf{Alg}_k \rightarrow \mathbf{Grp}.$$

By binomial theorem $(x+y)^p = x^p + y^p$ for all $x, y \in A$ and thus $\alpha_p(A)$ is a subgroup of $(A, +)$. Then α_p is represented by the universal pair $(R := \frac{k[X]}{(X^p)}, \bar{X} := X + (X^p))$. Indeed, if (A, a) is another pair with $a \in A$ and $a^p = 0$ then consider the map $k[X] \rightarrow A$ carrying $X \mapsto a$. This factors to give $\varphi : R \rightarrow A$ with $\bar{X} \mapsto a$, and is the unique such map since $\bar{X}$ generates R . Here Δ, ϵ, S can be computed in the same way as $\mathbf{G}_a$. We just mention the maps. Indeed, $\Delta : R \rightarrow R \otimes R$ is the unique map such that $\Delta(\bar{X}) = \bar{X} \otimes 1 + 1 \otimes \bar{X}$. Also $\epsilon : R \rightarrow k$ is the unique map such that $\epsilon(\bar{X}) = 0$. Finally, $S : R \rightarrow R$ is the unique map such that $S(\bar{X}) = -\bar{X}$.

2. CLOSED SUBFUNCTORS

Definition 2.1. – Let F be a k -functor. A *subfunctor* of F is a k -functor G such that $G(A) \subset F(A)$ and $G(f) = F(f)|_{G(A)}$ for every k -algebras A, A' and all $f \in \text{Hom}_{k\text{-alg}}(A, A')$.

Remark 2.2. – Let G be an association $A \rightsquigarrow G(A)$. Then G can be made into a subfunctor of F if $G(A) \subset F(A)$ and $F(f)G(A) \subset G(A')$ for all algebras A, A' and all $f : A \rightarrow A'$: we simply set $Gf := Ff|_{G(A)}$ for every A .

Example 2.3. – Note that $\mathbf{SL}_n$ is a subfunctor of $\mathbf{GL}_n$; similarly μ_n is a subfunctor of $\mathbf{G}_m$. When characteristic of k is $p > 0$ then α_p is a subfunctor of $\mathbf{G}_a$.

Definition 2.4. – For a given family $(F_i)_{i \in I}$ of subfunctors of a functor F , the *intersection* $\bigcap_i F_i$ is defined as $A \rightsquigarrow (\bigcap_i F_i)(A) := \bigcap_i F_i(A)$ for every k -algebra A .

Remark 2.5. It is clear that $\bigcap_i F_i$ can be made into a subfunctor of F since for any algebra map $f : A \rightarrow B$ we have

$$Ff \left(\left(\bigcap_i F_i \right) (A) \right) = Ff \left(\bigcap_i F_i(A) \right) \subset \bigcap_i (Ff(F_i(A))) \subset \bigcap_i F_i(B) = \left(\bigcap_i F_i \right) (B)$$

thus we set $(\bigcap_i F_i)(f) := Ff|_{(\bigcap_i F_i)(A)}$.

Definition 2.6. – Let F, F' be two k -functors and $\eta : F \rightarrow F'$ be a natural transformation. For any subfunctor G' of F' we define the *inverse image* $\eta^{-1}(G')$ of G' under η by

$$A \rightsquigarrow \eta^{-1}(G')(A) := \eta(A)^{-1}(G'(A)) = \{x \in F(A) \mid \eta(A)(x) \in G'(A)\}$$

for all k -algebras A .

Proposition 2.7. – The association $A \rightsquigarrow \eta^{-1}(G')(A)$ makes $\eta^{-1}(G')$ a subfunctor of F .

Proof. – Write $H := \eta^{-1}(G')$. Note that $H(A) \subset F(A)$ for all algebras A by definition. We have to show that given $f : A \rightarrow B$ an algebra map, $F(f)(H(A)) \subset H(B)$. Let $x \in H(A)$ and consider $F(f)(x)$. We have to show that $\eta(B)(Ff(x)) \in G'(B)$. But

$$\eta(B)(Ff(x)) = F'f(\eta(A)(x)).$$

Since $\eta(A)(x) \in G'(A)$ and G' is a subfunctor of F' we have $F'f(\eta(A)(x)) \in G'(B)$. □

Proposition 2.8. – Let $\eta : F \rightarrow F'$ be a natural transformation, and $(G'_i)_{i \in I}$ be a family of subfunctors of F' . Then

$$\eta^{-1} \left(\bigcap_i G'_i \right) = \bigcap_i \eta^{-1}(G'_i).$$

Proof. – Fix any algebra A . Then we have

$$\begin{aligned} x \in \eta^{-1} \left(\bigcap_i G'_i \right) (A) &\iff \eta(A)(x) \in \bigcap_i G'_i(A) \iff \eta(A)(x) \in G'_i(A) \quad \text{for all } i \\ &\iff x \in \bigcap_i (\eta^{-1}(G'_i))(A) \iff x \in \left(\bigcap_i \eta^{-1}(G'_i) \right) (A). \end{aligned}$$

□

Now recall that for three k -functors F_1, F_2, S and two natural transformations $\eta_1 : F_1 \rightarrow S, \eta_2 : F_2 \rightarrow S$, the fiber product $F_1 \times_S F_2$ (relative to η_1 and η_2) is defined by

$$A \rightsquigarrow (F_1 \times_S F_2)(A) := \{(x_1, x_2) \in F_1(A) \times F_2(A) \mid \eta_1(A)(x_1) = \eta_2(A)(x_2)\}.$$

Also we have the *diagonal subfunctor* D_S of $S \times S$ defined by

$$A \rightsquigarrow D_S(A) := \{(s, s) \mid s \in S(A)\}.$$

Proposition 2.9. – $F_1 \times_S F_2$ is the inverse image of the diagonal subfunctor D_S under $\eta_1 \times \eta_2 : F_1 \times F_2 \rightarrow S \times S$ i.e. $F_1 \times_S F_2 = (\eta_1 \times \eta_2)^{-1}(D_S)$.

Proof. – Let A be a k -algebra. Then we have

$$(\eta_1 \times \eta_2)^{-1}(D_S)(A) = \{(x_1, x_2) \in F_1(A) \times F_2(A) \mid (\eta_1 \times \eta_2)(A)(x_1, x_2) \in D_S(A)\}.$$

But the condition $(\eta_1 \times \eta_2)(A)(x_1, x_2) \in D_S(A)$ is equivalent to saying $\eta_1(A)(x_1) = \eta_2(A)(x_2) \in S(A)$. Thus $(\eta_1 \times \eta_2)^{-1}(D_S)(A) = (F_1 \times_S F_2)(A)$. $\square$

Proposition 2.10. – Let $p_i : F_1 \times_S F_2 \rightarrow F_i$ for $i = 1, 2$ be the canonical projections. Let G_i be subfunctors of F_i for $i = 1, 2$. Then

$$G_1 \times_S G_2 := [A \rightsquigarrow \{(x_1, x_2) \in G_1(A) \times G_2(A) \mid \eta_1(A)(x_1) = \eta_2(A)(x_2)\}]$$

is a subfunctor of $F_1 \times_S F_2$ and $G_1 \times_S G_2 = p_1^{-1}(G_1) \cap p_2^{-1}(G_2)$.

Proof. – We first note that $H := G_1 \times_S G_2$ is indeed a subfunctor of $F_1 \times_S F_2$. Clearly, $H(A) \subset (F_1 \times_S F_2)(A)$ for every algebra A . Now let $\psi : A \rightarrow B$ be an algebra map and take $(x_1, x_2) \in H(A)$. Then $x_i \in G_i(A)$ for $i = 1, 2$. Since G_i is a subfunctor of F_i we have

$$F_i(\psi)(x_i) \in G_i(B) \quad \text{for } i = 1, 2.$$

Using naturality of η_i ,

$$\eta_1(B)(F_1(\psi)(x_1)) = S(\psi)(\eta_1(A)(x_1)), \quad \eta_2(B)(F_2(\psi)(x_2)) = S(\psi)(\eta_2(A)(x_2)).$$

But $\eta_1(A)(x_1) = \eta_2(A)(x_2)$ and thus

$$\eta_1(B)(F_1(\psi)(x_1)) = \eta_2(B)(F_2(\psi)(x_2)).$$

Therefore,

$$(F_1 \times_S F_2)(\psi)(x_1, x_2) = (F_1(\psi)(x_1), F_2(\psi)(x_2)) \in H(B)$$

and so H is a subfunctor of $F_1 \times_S F_2$.

Next, for $i = 1, 2$ note that

$$\begin{aligned} p_i^{-1}(G_i)(A) &= \{(x_1, x_2) \in (F_1 \times_S F_2)(A) \mid p_i(A)(x_1, x_2) \in G_i(A)\} \\ &= \{(x_1, x_2) \in F_1(A) \times F_2(A) \mid \eta_1(A)(x_1) = \eta_2(A)(x_2) \text{ and } x_i \in G_i(A)\}. \end{aligned}$$

Thus we get

$$\begin{aligned} (p_1^{-1}(G_1) \cap p_2^{-1}(G_2))(A) &= p_1^{-1}(G_1)(A) \cap p_2^{-1}(G_2)(A) \\ &= \{(x_1, x_2) \in F_1(A) \times F_2(A) \mid \eta_1(A)(x_1) = \eta_2(A)(x_2) \text{ and } (x_1, x_2) \in G_1(A) \times G_2(A)\} \\ &= \{(x_1, x_2) \in G_1(A) \times G_2(A) \mid \eta_1(A)(x_1) = \eta_2(A)(x_2)\} \\ &= (G_1 \times_S G_2)(A) \end{aligned}$$

for every algebra A . $\square$

We now consider subfunctors of affine k -functors. Let F be an affine k -functor represented by a universal pair (R, r) with $r \in F(R)$. Now fix a k -algebra A . Then, for each $x \in FA$ the universal mapping property of (R, r) yields a unique map $\varphi_x : R \rightarrow A$ such that $F(\varphi_x)(r) = x$.

Definition 2.11. – For any subset $I \subset R$ we define $V(I)$ by the association

$$A \rightsquigarrow V(I)(A) := \{x \in FA \mid \varphi_x(I) = 0\}.$$

Notation 2.12. – For any $f \in I$ and $x \in FA$, it is conventional to write $f(x) := \varphi_x(f)$.

Remark 2.13. – $V(I)$ depends only on the ideal generated by I . Indeed, $\varphi_x(I) = 0 \iff \varphi_x(\langle I \rangle) = 0$.

Proposition 2.14. – The association $A \rightsquigarrow V(I)(A)$ makes $V(I)$ into a subfunctor of F .

Proof. – For each algebra $V(I)(A) \subset FA$ by definition. Thus it remains to check that for any algebra map $\psi : A \rightarrow A'$ we have $F(\psi)(V(I)(A)) \subset V(I)(A')$. So let $x \in V(I)(A) \subset FA$, thus $\varphi_x(I) = 0$. Consider the equation

$$F(\varphi_x)(r) = x \quad \text{with } \varphi_x : R \rightarrow A,$$

and apply $F(\psi)$ to obtain

$$F(\psi \circ \varphi_x)(r) = F(\psi)(x) \in FA'.$$

Then we have the unique algebra map $\varphi_{F(\psi)(x)} : R \rightarrow A'$ such that $F(\varphi_{F(\psi)(x)})(r) = F(\psi)(x)$. Hence we have

$$\psi \circ \varphi_x = \varphi_{F(\psi)(x)}.$$

Now we have to show that $F(\psi)(x) \in V(I)(A')$. This means we have to show that $\varphi_{F(\psi)(x)}(I) = 0$. But note that

$$\varphi_{F(\psi)(x)}(I) = \psi(\varphi_x(I)) = \psi(0) = 0.$$

Thus we can set $V(I)(\psi) := F(\psi)|_{V(I)(A)}$. The proof is complete. $\square$

Proposition 2.15. – *Let $I \subset R$ be an ideal. Then we have*

$$I = \{f \in R \mid f(V(I)(A)) = 0 \text{ for all } A\}.$$

Proof. – First we make the condition “ $f(V(I)(A)) = 0$ for all A ” explicit. Precisely, this means

$$\text{for every } k\text{-algebra } A \text{ and every } x \in V(I)(A) \text{ we have } \varphi_x(f) = 0.$$

Now we write

$$J := \{f \in R \mid \forall A, \forall x \in V(I)(A), \varphi_x(f) = 0\}.$$

Then clearly $I \subset J$ by definition of $V(I)$. For the reverse inclusion consider the canonical k -algebra map $\pi : R \rightarrow R/I$ and let $\bar{r} := F(\pi)(r) \in F(R/I)$. Then by construction $\pi = \varphi_{\bar{r}}$ since $\varphi_{\bar{r}} : R \rightarrow R/I$ is the unique algebra map such that $F(\varphi_{\bar{r}})(r) = \bar{r}$. Moreover, $\bar{r} \in V(I)(R/I)$ since $\varphi_{\bar{r}}(I) = \pi(I) = 0$. Now let $f \in J$. Then for $A = R/I$ and $\bar{r} \in V(I)(A) = V(I)(R/I)$ we have $\varphi_{\bar{r}}(f) = \pi(f) = 0$. Thus $f \in \text{Ker } \pi = I$, and since $f \in J$ was arbitrary, we have $J \subset I$. This completes the proof. $\square$

Corollary 2.16. – *The map $I \mapsto V(I)$ from the collection of ideals of R to the collection of subfunctors of F is injective.*

Proof. – If $V(I) = V(J)$ then we have

$$\begin{aligned} I &= \{f \in R \mid \forall A, \forall x \in V(I)(A), \varphi_x(f) = 0\} \\ &= \{f \in R \mid \forall A, \forall x \in V(J)(A), \varphi_x(f) = 0\} \\ &= J. \end{aligned}$$

$\square$

Corollary 2.17. – *For ideals I, I' we have*

$$I \subset I' \iff V(I') \text{ is a subfunctor of } V(I).$$

Proof. – Let $I \subset I'$. Suppose A is any k -algebra and $x \in V(I')(A)$. Then $\varphi_x(I') = 0$ and so $\varphi_x(I) = 0$. Hence $x \in V(I)(A)$. Therefore $V(I')(A) \subset V(I)(A)$ for all A , which means that $V(I')$ is a subfunctor of $V(I)$. Conversely, suppose $V(I')$ is a subfunctor of $V(I)$, so that for all k -algebras A we have $V(I')(A) \subset V(I)(A)$. But then by the proposition we have

$$I = \{f \in R \mid \forall A, \forall x \in V(I)(A), \varphi_x(f) = 0\} \subset \{f \in R \mid \forall A, \forall x \in V(I')(A), \varphi_x(f) = 0\} = I'.$$

$\square$

Definition 2.18. – We call a subfunctor G of F closed if it is of the form $G = V(I)$ for some ideal $I \subset R$ of the algebra R representing F .

Proposition 2.19. – *A closed subfunctor $V(I)$ is an affine k -functor. More precisely,*

$$V(I) \simeq h^{R/I}.$$

Proof. – In fact, we show that $(R/I, \bar{r})$ where $\bar{r} := F(\pi)(r) \in V(I)(R/I)$ and $\pi : R \rightarrow R/I$ is the canonical map, is a universal pair. So let (A, a) be another pair with $a \in V(I)(A)$. Then, there is a unique k -algebra map $\varphi_a : R \rightarrow A$ such that $F(\varphi_a)(r) = a$. Since $a \in V(I)(A)$ by definition we have $\varphi_a(I) = 0$. This induces $\Psi : R/I \rightarrow A$ given by $x + I \mapsto \varphi_a(x)$. Now $\Psi \circ \pi = \varphi_a$. Also, check that

$$V(I)(\Psi)(\bar{r}) = F(\Psi)(\bar{r}) = F(\Psi \circ \pi)(r) = F(\varphi_a)(r) = a.$$

Thus we have shown $V(I)(\Psi)(\bar{r}) = a$ and it remains to check that Ψ is the unique map with this property. Indeed, if $\Psi' : R/I \rightarrow A$ is such that $V(I)(\Psi')(\bar{r}) = F(\Psi')(\bar{r}) = a$, then

$$F(\Psi' \circ \pi)(r) = a = F(\varphi_a)(r).$$

Thus the uniqueness of φ_a implies $\Psi' \circ \pi = \varphi_a = \Psi \circ \pi$. As $\pi : R \rightarrow R/I$ is surjective, this implies $\Psi = \Psi'$ as required. $\square$

Remark 2.20. – F itself and the “empty functor” (mapping each $A \neq 0$ to $\emptyset$) are closed subfunctors of F . We check that $V(\{0\}) = F$. Indeed, $V(\{0\})(A) = F(A)$ for every algebra A , since $\varphi_x(\{0\}) = 0$ for every $x \in F(A)$. Similarly, we can show that $V(R)(A) = \emptyset$ if $A \neq 0$. Indeed, if $V(R)(A) \neq \emptyset$ for some algebra, then take some $x \in V(R)(A)$. Then, we know $\varphi_x(R) = 0$ and in particular $1_A = \varphi_x(1_R) = 0$. Thus $A = 0$.

Proposition 2.21. – *For an affine functor F represented by (R, r) as above, every $x \in F(k)$ defines a closed subfunctor of F as follows : for each k -algebra A consider the structure map $T_A : k \rightarrow A$ and set $x_A := F(T_A)(x)$. Then $A \rightsquigarrow x_A$ defines a closed subfunctor of F .*

Proof. – Indeed, let $\varphi_x : R \rightarrow k$ be the unique algebra map such that $F(\varphi_x)(r) = x$, and put $I := \text{Ker } \varphi_x$.

First, we check that the association $A \rightsquigarrow x_A$ is indeed a subfunctor of F . For this we take $\psi : A \rightarrow B$ to be any algebra map, and then we have

$$F(\psi)(x_A) = F(\psi)(F(T_A)x) = F(\psi \circ T_A)(x) = F(T_B)(x) = x_B.$$

Thus $A \rightsquigarrow x_A$ is indeed a subfunctor of F .

Now $x_A := F(T_A)(x) \in F(A)$ and $\varphi_{x_A} : R \rightarrow A$ is the unique map such that $F(\varphi_{x_A})(r) = x_A$. But also note that $F(T_A \circ \varphi_x)(r) = F(T_A)(F\varphi_x(r)) = F(T_A)(x) = x_A$. Thus we have

$$T_A \circ \varphi_x = \varphi_{x_A}.$$

Thus it remains to show $V(I)(A) = \{x_A\}$ for every algebra A . Indeed, $\varphi_{x_A}(I) = T_A(\varphi_x(I)) = T_A(0) = 0$ which means $\{x_A\} \subset V(I)(A)$. For the reversed inclusion, take any $y \in V(I)(A)$. Let $\varphi_y : R \rightarrow A$ be the unique map such that $F(\varphi_y)(r) = y$. By hypothesis $\varphi_y(I) = 0$. Now note that $\varphi_x : R \rightarrow k$ being a k -algebra map is surjective. Consider the map $\theta : k \rightarrow A$ defined by $\theta(\varphi_x(a)) := \varphi_y(a)$ for any $a \in R$. This is well defined, since $\varphi_x(a) = \varphi_x(a')$ implies $a - a' \in I = \text{Ker } \varphi_x$ and thus $\varphi_y(a - a') = 0$ and $\varphi_y(a) = \varphi_y(a')$. Also note that θ is additive, unital and k -linear : indeed, the first two conditions are easy to see; lastly, for $\lambda \in k$ and $a \in R$ we have $\theta(\lambda\varphi_x(a)) = \theta(\varphi_x(\lambda \cdot a)) = \varphi_y(\lambda \cdot a) = \lambda \cdot \theta(\varphi_x(a))$. Thus $\theta : k \rightarrow A$ is a k -algebra map and hence $\theta = T_A$. Now for any $a \in R$ we have

$$(T_A \circ \varphi_x)(a) = \theta(\varphi_x(a)) = \varphi_y(a)$$

which means $\varphi_y = T_A \circ \varphi_x$. Finally, we get

$$y = F(\varphi_y)(r) = F(T_A \circ \varphi_x)(r) = F(T_A)(x) = x_A.$$

Thus we have shown $V(I)(A) = \{x_A\}$ for every algebra A . Hence $(A \rightsquigarrow x_A) = V(\text{Ker } \varphi_x)$ as asserted. $\square$

Proposition 2.22. – *For a family of ideals $(I_j)_{j \in \Lambda}$ in R we have*

$$\bigcap_{j \in \Lambda} V(I_j) = V\left(\sum_{j \in \Lambda} I_j\right).$$

Proof. – Fix an algebra A . Then we have

$$\begin{aligned} x \in V\left(\sum_j I_j\right)(A) &\iff \varphi_x\left(\sum_j I_j\right) = 0 \\ &\iff \varphi_x(I_j) = 0 \text{ for all } j \\ &\iff x \in V(I_j)(A) \text{ for all } j \\ &\iff x \in \left(\bigcap_j V(I_j)\right)(A). \end{aligned}$$

□

Proposition 2.23. – Let $F' \simeq h^{R'}$ and $F \simeq h^R$ be affine k -functors, and $\eta : F' \rightarrow F$ be a natural transformation corresponding to the algebra map $f : R \rightarrow R'$. For any ideal I of R we have

$$\eta^{-1}(V(I)) = V(\langle f(I) \rangle).$$

Proof. – Fix an algebra A . We write the map $\eta(A) : F'(A) \rightarrow F(A)$ as the composite

$$F'(A) \xrightarrow{\sim} \text{Hom}(R', A) \xrightarrow{\omega_{f,A}} \text{Hom}(R, A) \xrightarrow{\sim} F(A)$$

which is given by the following chase of elements :

$$x' \mapsto \varphi_{x'} \mapsto \varphi_{x'} \circ f \mapsto \eta(A)(x').$$

Thus $\varphi_{\eta(A)(x')} = \varphi_{x'} \circ f$ for every $x' \in F'(A)$.

Now fix any ideal $I \subset R$ and observe that

$$\begin{aligned} \eta^{-1}(V(I))(A) &= \{x' \in F'(A) \mid \eta(A)(x') \in V(I)(A)\} \\ &= \{x' \in F'(A) \mid \varphi_{\eta(A)(x')}(I) = 0\} \\ &= \{x' \in F'(A) \mid \varphi_{x'}(f(I)) = 0\} \\ &= \{x' \in F'(A) \mid \varphi_{x'}(\langle f(I) \rangle) = 0\} \\ &= V(\langle f(I) \rangle)(A). \end{aligned}$$

This completes the proof. □

Proposition 2.24. – Let F_1, F_2 be affine functors represented by R_1 and R_2 respectively. If S is another affine functor represented by B and if morphisms $F_1 \rightarrow S, F_2 \rightarrow S$ are fixed, then for ideals $I_1 \subset R_1, I_2 \subset R_2$ we have

$$V(I_1) \times_S V(I_2) = V(I_1 \otimes_B R_2 + R_1 \otimes_B I_2).$$

Proof. – By Proposition 2.10, $V(I_1) \times_S V(I_2) = p_1^{-1}(V(I_1)) \cap p_2^{-1}(V(I_2))$. Now note that $p_i : F_1 \times_S F_2 \rightarrow F_i$ corresponds to the canonical map $\iota_i : R_i \rightarrow R_1 \otimes_B R_2$ for $i = 1, 2$. Using Proposition 2.23 we have

$$p_1^{-1}(V(I_1)) = V(\langle \iota_1(I_1) \rangle) \quad \text{and} \quad p_2^{-1}(V(I_2)) = V(\langle \iota_2(I_2) \rangle).$$

However observe that $\langle \iota_1(I_1) \rangle = I_1 \otimes_B R_2$ and $\langle \iota_2(I_2) \rangle = R_1 \otimes_B I_2$. Also, by Proposition 2.22, we get

$$p_1^{-1}(V(I_1)) \cap p_2^{-1}(V(I_2)) = V(\langle \iota_1(I_1) \rangle + \langle \iota_2(I_2) \rangle).$$

Thus we finally obtain

$$V(I_1) \times_S V(I_2) = V(I_1 \otimes_B R_2 + R_1 \otimes_B I_2).$$

□

Proposition 2.25. – Let F_1, F_2 be affine k -functors represented by algebras R_1, R_2 respectively. Suppose $\eta : F_1 \rightarrow F_2$ is a natural transformation with corresponding algebra map $f : R_2 \rightarrow R_1$. The following statements are equivalent :

- (1) f is surjective.
- (2) η is an isomorphism onto a closed subfunctor of F_2 .

Proof. – (1 $\implies$ 2) Let (R_1, r_1) represent F_1 and (R_2, r_2) represent F_2 . It is given that f is surjective. Set $I := \text{Ker } f$. For any algebra A and any $z \in F_i(A)$ we write $\varphi_z : R_i \rightarrow A$ the unique k -algebra map such that $F_i(\varphi_z)(r_i) = z$. Fix an algebra A .

We now show that $\eta(A)(F_1(A)) \subset V(I)(A)$. Indeed, take $x \in F_1(A)$ and consider $\eta(A)(x) \in F_2(A)$. Then as in the proof of Proposition 2.23, we have

$$\varphi_{\eta(A)(x)} = \varphi_x \circ f : R_2 \rightarrow A.$$

But $\varphi_{\eta(A)(x)}(I) = \varphi_x(f(I)) = 0$ and thus $\eta(A)(x) \in V(I)(A)$.

Next, we show that $\eta(A) : F_1(A) \rightarrow V(I)(A)$ is surjective. Let $y \in V(I)(A) \subset F_2(A)$ and $\varphi_y : R_2 \rightarrow A$ be the unique k -algebra map such that $F_2(\varphi_y)(r_2) = y$. Also by definition of $\varphi_y(I) = 0$. Now we define a map $\psi : R_1 \rightarrow A$ as follows : for $a_1 \in R_1$ we can choose $a_2 \in R_2$ such that $f(a_2) = a_1$ since f is assumed to be

surjective; then we set $\psi(a_1) := \varphi_y(a_2)$. Then ψ is well-defined since if a_2, a'_2 map to a_1 then $a_2 - a'_2 \in I$ and so $\varphi_y(a_2) = \varphi_y(a'_2)$. Also ψ is a k -algebra map as sums, products and scalars lift and are respected by φ_y . Now we set $x := F_1(\psi)(r_1) \in F_1(A)$. Then by definition of φ_x we have $\psi = \varphi_x$. Thus we have

$$\varphi_{\eta(A)(x)} = \varphi_x \circ f = \psi \circ f.$$

For any $a_2 \in R_2$ set $a_1 := f(a_2)$. By definition we have $\psi(a_1) = \varphi_y(a_2)$. Thus we get

$$(\psi \circ f)(a_2) = \psi(a_1) = \varphi_y(a_2).$$

Hence $\varphi_y = \varphi_{\eta(A)(x)}$. But then we have

$$y = F_2(\varphi_y)(r_2) = F_2(\varphi_{\eta(A)(x)})(r_2) = \eta(A)(x).$$

Thus we have shown that $\eta(A)$ is surjective.

Finally, we show that $\eta(A)$ is injective. For if $\eta(A)(x) = \eta(A)(x')$ for $x, x' \in F_1(A)$ then we have

$$\varphi_x \circ f = \varphi_{\eta(A)(x)} = \varphi_{\eta(A)(x')} = \varphi_{x'} \circ f.$$

But since f is surjective this implies $\varphi_x = \varphi_{x'}$ whence $x = x'$.

Combining the above assertions we obtain that for each algebra A we have a bijection

$$\eta(A) : F_1(A) \xrightarrow{\sim} V(I)(A)$$

which is natural in A since η itself is a natural transformation. This completes the proof of $1 \Rightarrow 2$.

($2 \Rightarrow 1$) Now suppose $\eta : F_1 \rightarrow F_2$ is an isomorphism onto a closed subfunctor $V(I)$. This means that there exists a natural isomorphism $\Phi : F_1 \xrightarrow{\sim} V(I)$ such that $\eta = j \circ \Phi$ where $j : V(I) \hookrightarrow F_2$ is just the inclusion transformation. At first we show that j corresponds to the canonical projection $\pi : R_2 \rightarrow R_2/I$. Indeed, fix any algebra A and consider the composite

$$\text{Hom}(R_2/I, A) \xrightarrow{\sim} V(I)(A) \xrightarrow{j_A} F_2(A) \xrightarrow{\sim} \text{Hom}(R_2, A)$$

which is obtained by the following chase of elements :

$$g \mapsto z = F_2(g)(\bar{r}_2) \mapsto z \mapsto \varphi_z$$

where $\bar{r}_2 := F_2(\pi)(r_2)$ is the universal element for the universal pair $(R_2/I, \bar{r}_2)$ of $V(I)$. Now we have

$$F_2(\varphi_z)(r_2) = z = F_2(g)(\bar{r}_2) = F_2(g)(F_2(\pi)(r_2)) = F_2(g \circ \pi)(r_2)$$

and hence $\varphi_z = g \circ \pi$. Thus the composite sends $g \mapsto g \circ \pi$. Hence j corresponds to ω_π . But now since $\eta = j \circ \Phi$ we have $f_2 = \phi \circ \pi$ where $\phi : R_2/I \xrightarrow{\sim} R_1$ is an isomorphism of k -algebras corresponding to the natural isomorphism Φ . Thus f_2 is a surjection. Therefore, $2 \Rightarrow 1$. $\square$

3. CLOSED SUBGROUPS AND HOMOMORPHISMS

We now consider group functors.

Definition 3.1. – Let G be a k -group functor. A subfunctor H of G is called a *subgroup functor* of G if $H(A)$ is a subgroup of $G(A)$ for each k -algebra A .

Definition 3.2. – A *Hopf ideal* in a Hopf k -algebra (R, Δ, ϵ, S) is an ideal $I \subset R$ such that

$$\Delta(I) \subset R \otimes I + I \otimes R, \quad \epsilon(I) = 0, \quad S(I) \subset I.$$

Proposition 3.3. – Let G be an affine group functor represented by a Hopf algebra (R, Δ, ϵ, S) . Then a closed affine subfunctor $V(I)$ is a subgroup functor of G if and only if I is a Hopf ideal.

Proof. – First assume that I is a Hopf ideal satisfying

$$\Delta(I) \subset R \otimes I + I \otimes R, \quad \epsilon(I) = 0, \quad S(I) \subset I.$$

Fix an algebra A . We wish to show that $V(I)(A)$ is a subgroup of $G(A)$.

First we show closure under multiplication. Take $x, y \in V(I)(A)$. Then $\varphi_x(I) = 0 = \varphi_y(I)$. Put $z := m_A(x, y)$ and we will simply write $z = xy \in G(A)$. We wish to show $z \in V(I)(A)$ or equivalently $\varphi_z(I) = 0$. But recall by Proposition 1.32 we know

$$\varphi_z = M_A \circ (\varphi_x \otimes \varphi_y) \circ \Delta.$$

So take $t \in I$ and write $\Delta(t) = \sum u_j \otimes v_j + \sum u'_k \otimes v'_k \in R \otimes I + I \otimes R$ with $v_j, u'_k \in I$. Then we get

$$(\varphi_x \otimes \varphi_y)(\Delta(t)) = \sum \varphi_x(u_j) \otimes \varphi_y(v_j) + \sum \varphi_x(u'_k) \otimes \varphi_y(v'_k) = 0$$

since $\varphi_y(v_j) = 0 = \varphi_x(u'_k)$. Thus $\varphi_z(t) = 0$ and since $t \in I$ was arbitrary we have $\varphi_z(I) = 0$ as asserted.

Next, we show that $e_A \in G(A)$ also lies in $V(I)(A)$ or equivalently $\varphi_{e_A}(I) = 0$. But recall that $\varphi_{e_A} = T_A \circ \epsilon$ by Proposition 1.32, and by hypothesis $\epsilon(I) = 0$. Thus we have $\varphi_{e_A}(I) = 0$. Hence $e_A \in V(I)(A)$.

Finally, let $x \in V(I)(A)$ and we wish to show that $x^{-1} := i_A(x) \in V(I)(A)$ or equivalently $\varphi_{x^{-1}}(I) = 0$. But again by Proposition 1.32, we know $\varphi_{x^{-1}} = \varphi_x \circ S$. But $S(I) \subset I$ and thus $\varphi_{x^{-1}}(I) = \varphi_x(S(I)) \subset \varphi_x(I) = 0$. Hence $x^{-1} \in V(I)(A)$.

Now for the converse, we assume that $V(I)(A)$ is a subgroup of $G(A)$ for every k -algebra A . We then wish to show that I is a Hopf ideal.

We first show that $\epsilon(I) = 0$. For this take $A = k$. Now the identity $e_k \in G(k)$ also belongs to $V(I)(k)$ and thus $\varphi_{e_k}(I) = 0$. But $\varphi_{e_k} = T_k \circ \epsilon = \text{id}_k \circ \epsilon = \epsilon$ and so $\epsilon(I) = 0$ as required.

Next, we show that $S(I) \subset I$. For this we take $A = R/I$ and let $\pi : R \rightarrow R/I$ be the quotient map. Set $y := G(\pi)(r) \in G(R/I)$. Then by definition $\varphi_y = \pi$. But $\varphi_y(I) = \pi(I) = 0$ and so $y \in V(I)(R/I)$. Since $V(I)(R/I)$ is a subgroup of $G(R/I)$ we know $y^{-1} = i_{R/I}(y) \in V(I)(R/I)$. But $\varphi_{y^{-1}} = \varphi_y \circ S$ and $\varphi_{y^{-1}}(I) = \varphi_y(S(I)) = 0$ implies $S(I) \subset \text{Ker } \varphi_y = \text{Ker } \pi = I$.

Finally, we show $\Delta(I) \subset R \otimes I + I \otimes R$. For this we take $A = (R \otimes R)/J$ where $J := R \otimes I + I \otimes R$ and let $\rho : R \otimes R \rightarrow (R \otimes R)/J$ be the canonical quotient. We define two algebra maps $\alpha, \beta : R \rightarrow A$ by $\alpha(t) := \rho(t \otimes 1)$ and $\beta(t) := \rho(1 \otimes t)$ for all $t \in R$. Now set

$$x := G(\alpha)(r) \in G(A) \text{ and } y := G(\beta)(r) \in G(A).$$

Then by definition $\varphi_x = \alpha$ and $\varphi_y = \beta$. Since $\alpha(I) = 0 = \beta(I)$ we have $x, y \in V(I)(A)$. But then we get $z := xy \in V(I)(A)$ and thus $\varphi_z(I) = 0$. But we know that $\varphi_z = M_A \circ (\varphi_x \otimes \varphi_y) \circ \Delta$. Take $t \in I$ and write $\Delta(t) = \sum u_j \otimes v_j + \sum u'_k \otimes v'_k \in R \otimes I + I \otimes R$ with $v_j, u'_k \in I$. Then we have

$$\begin{aligned} 0 &= M_A((\varphi_x \otimes \varphi_y)(\Delta(t))) = M_A \left(\sum \varphi_x(u_j) \otimes \varphi_y(v_j) + \sum \varphi_x(u'_k) \otimes \varphi_y(v'_k) \right) \\ &= \sum \alpha(u_j) \beta(v_j) + \sum \alpha(u'_k) \beta(v'_k) \\ &= \sum \rho(u_j \otimes 1) \rho(1 \otimes v_j) + \sum \rho(u'_k \otimes 1) \rho(1 \otimes v'_k) \\ &= \sum \rho(u_j \otimes v_j) + \sum \rho(u'_k \otimes v'_k) \\ &= \rho(\Delta(t)) \end{aligned}$$

and thus $\Delta(t) \in \text{Ker } \rho = J = R \otimes I + I \otimes R$. This completes the proof of the fact that $\Delta(I) \subset R \otimes I + I \otimes R$ as asserted, and also that of the proposition. $\square$

Example 3.4. – Let G be an affine group functor, and H be the trivial subgroup functor i.e. H is given by $A \rightsquigarrow H(A) := \{e_A\} \subset G(A)$. Suppose G is represented by the universal pair (R, r) with Hopf algebra structure (Δ, ϵ, S) . We take $e_k \in G(k)$. Now we see by Proposition 2.21 that $H = (A \rightsquigarrow e_A = G(T_A)(e_k))$ is a closed subfunctor, and is of the form $V(\text{Ker } \varphi_{e_k})$ where $\varphi_{e_k} : R \rightarrow k$ is the unique k -algebra map such that $G(\varphi_{e_k})(r) = e_k$. Now consider the composite

$$\text{Hom}(k, k) \rightarrow G(k) \rightarrow \text{Hom}(R, k)$$

which is $\text{id}_k \mapsto e_k \mapsto \varphi_{e_k}$. But this composite is given by right composition by ϵ . Thus we have $\epsilon = \varphi_{e_k}$. Hence we have shown that $H = V(\text{Ker } \epsilon)$. Therefore $\text{Ker } \epsilon$ is a Hopf ideal by Proposition 3.3 above. This ideal $I_G := \text{Ker } \epsilon$ is called the *augmentation ideal*.

We now proceed to find equivalent ‘‘Hopf conditions’’ for a closed subgroup functor to be normal.

Definition 3.5. – Let G be a group functor. A subgroup functor H of G is said to be *normal* if $H(A)$ is a normal subgroup of $G(A)$ for each algebra A .

Now let G be an affine group functor with Hopf structure (R, Δ, ϵ, S) and (R, r) being a universal pair.

Definition 3.6. – We define the *coadjoint* operator $\text{Ad} : R \rightarrow R \otimes R$ by

$$\text{Ad} := (M_R \otimes \text{id}) \circ (\text{id} \otimes S \otimes \text{id}) \circ (\text{id} \otimes \tau) \circ (\Delta \otimes \text{id}) \circ \Delta$$

where $M_R : R \otimes R \rightarrow R$ is the ring multiplication, and $\tau : R \otimes R \rightarrow R \otimes R$ is the map that swaps factors.

Remark 3.7. – Consider the following composite

$$G \times G \xrightarrow{d \times \text{id}} G \times G \times G \xrightarrow{\text{id} \times i \times \text{id}} G \times G \times G \xrightarrow{\text{id} \times (\text{swap})} G \times G \times G \xrightarrow{m \times \text{id}} G \times G \xrightarrow{m} G$$

which given given by the element chase

$$(g, x) \mapsto (g, g, x) \mapsto (g, g^{-1}, x) \mapsto (g, x, g^{-1}) \mapsto (gx, g^{-1}) \mapsto gxg^{-1}.$$

Observe that the algebra map corresponding to this conjugation map is precisely Ad .

Lemma 3.8. – Let $t \in R$ and write

$$\Delta(t) = \sum_j u_j \otimes v_j \quad (u_j, v_j \in R),$$

and for each j write

$$\Delta(u_j) = \sum_\ell a_{j\ell} \otimes b_{j\ell} \quad (a_{j\ell}, b_{j\ell} \in R).$$

Then we have

$$\text{Ad}(t) = \sum_{j,\ell} (a_{j\ell} S(v_j)) \otimes b_{j\ell}.$$

Proof. – This is a direct verification using the defining formula for Ad . □

Lemma 3.9. – Let G be an affine group functor represented by Hopf structure (R, Δ, ϵ, S) and with universal pair (R, r) , and let A any algebra. Suppose $g, x \in G(A)$ and $t \in R$. Then

$$\varphi_{gxg^{-1}}(t) = (M_A \circ (\varphi_g \otimes \varphi_x) \circ \text{Ad})(t).$$

Proof. – Using Proposition 1.32 we have

$$\begin{aligned} \varphi_{gxg^{-1}} &= M_A \circ (\varphi_{gx} \otimes \varphi_{g^{-1}}) \circ \Delta \\ &= M_A \circ ((M_A \circ (\varphi_g \otimes \varphi_x) \circ \Delta) \otimes (\varphi_g \circ S)) \circ \Delta. \end{aligned}$$

Now we write

$$\Delta(t) = \sum_j u_j \otimes v_j \quad \text{and} \quad \Delta(u_j) = \sum_\ell a_{j\ell} \otimes b_{j\ell}.$$

Thus we get

$$\begin{aligned} \varphi_{gxg^{-1}}(t) &= (M_A \circ (\varphi_{gx} \otimes \varphi_{g^{-1}})) \left(\sum_j u_j \otimes v_j \right) \\ &= M_A \left(\sum_j (M_A \circ (\varphi_g \otimes \varphi_x) \circ \Delta)(u_j) \otimes \varphi_g(Sv_j) \right) \\ &= \sum_{j,\ell} \varphi_g(a_{j\ell}) \varphi_x(b_{j\ell}) \varphi_g(Sv_j) \\ &= \sum_{j,\ell} \varphi_x(b_{j\ell}) \varphi_g(a_{j\ell} Sv_j) \\ &= M_A \left(\sum_{j,\ell} \varphi_g(a_{j\ell} Sv_j) \otimes \varphi_x(b_{j\ell}) \right) \\ &= (M_A \circ (\varphi_g \otimes \varphi_x))(\text{Ad}(t)). \end{aligned}$$

□

Proposition 3.10. – Let G be an affine group functor, and $H = V(I)$ be a closed subgroup functor. Then H is normal if and only if $\text{Ad}(I) \subset R \otimes I$.²

Proof. – Assume $\text{Ad}(I) \subset R \otimes I$. We fix an algebra A and we wish to show $V(I)(A)$ is normal subgroup of $G(A)$. Let $g \in G(A)$ and $x \in V(I)(A)$. We must show that $gxg^{-1} \in V(I)(A)$ or equivalently $\varphi_{gxg^{-1}}(I) = 0$. So take $t \in I$ and since $\text{Ad}(I) \subset R \otimes I$ we can write

$$\text{Ad}(t) = \sum_j r_j \otimes a_j \quad \text{with } r_j \in R, a_j \in I.$$

Since $\varphi_x(a_j) = 0$ for all j , using the previous lemma we can easily see that $\varphi_{gxg^{-1}}(t) = 0$. Thus H is normal.

Conversely, assume H is normal. Now consider the product $G \times H$ which is represented by $R \otimes (R/I)$. First observe that $R \otimes (R/I) \simeq (R \otimes R)/(R \otimes I) =: S$. Indeed, if $\pi : R \rightarrow R/I$ is the canonical quotient then we obtain

$$R \otimes I \xrightarrow{\text{id} \otimes \iota} R \otimes R \xrightarrow{\text{id} \otimes \pi} R \otimes (R/I) \rightarrow 0$$

from the short exact sequence $0 \rightarrow I \xrightarrow{\iota} R \xrightarrow{\pi} R/I \rightarrow 0$ by tensoring $R \otimes -$. Since the map $\text{id} \otimes \pi$ is surjective we have $(R \otimes R)/(\text{Im}(\text{id} \otimes \iota)) \simeq R \otimes (R/I)$ and we write $R \otimes I$ in place of $\text{Im}(\text{id} \otimes \iota)$ by abuse of notation (see footnote 2). Now fix $t \in I$. Let $\Pi : R \otimes R \rightarrow (R \otimes R)/(R \otimes I)$ be the canonical quotient map. We first show that for any algebra A and any map $\psi : S \rightarrow A$ we have

$$\psi(\Pi(\text{Ad}(t))) = 0.$$

Indeed, given $\psi : S \rightarrow A$ we define $\varphi_g, \varphi_x : R \rightarrow A$ by

$$\varphi_g(u) := \psi(\overline{u \otimes 1}), \quad \varphi_x(v) := \psi(\overline{1 \otimes v}) \quad (u, v \in R).$$

Then $\varphi_x(I) = 0$ and so $x \in H(A) = V(I)(A)$. Also note that

$$\psi(\overline{u \otimes v}) = \psi(\overline{u \otimes 1})\psi(\overline{1 \otimes v}) = \varphi_g(u)\varphi_x(v) = (M_A \circ (\varphi_g \otimes \varphi_x))(u \otimes v).$$

Now since $gxg^{-1} \in H(A)$ we get using Lemma 3.9

$$0 = \varphi_{gxg^{-1}}(t) = (M_A \circ (\varphi_g \otimes \varphi_x))(\text{Ad}(t)) = \psi(\Pi(\text{Ad}(t)))$$

and we have thus shown $\psi(\Pi(\text{Ad}(t))) = 0$ for any algebra A and any map $\psi : S \rightarrow A$. To finish the proof we need to show that $\text{Ad}(t) \in R \otimes I$. But now we just take $A = S$ and $\psi = \text{id}_S$ to get

$$\Pi(\text{Ad}(t)) = 0$$

whence $\text{Ad}(t) \in \text{Ker } \Pi = R \otimes I$. □

Definition 3.11. – Let G and H be group functors. A natural transformation $\eta : G \rightarrow H$ is called *homomorphism of group functors* if $\eta(A) : G(A) \rightarrow H(A)$ is a group homomorphism for each k -algebra A .

Definition 3.12. – Let (R, Δ, ϵ, S) and $(R', \Delta', \epsilon', S')$ be Hopf algebras. A *homomorphism of Hopf algebras* $f : R \rightarrow R'$ is k -algebra homomorphism such that $(f \otimes f) \circ \Delta = \Delta' \circ f$.

Proposition 3.13. – Let $G \simeq h^{R_1}$ and $H \simeq h^{R_2}$ be affine group functors. A natural transformation $\eta : G \rightarrow H$ is a homomorphism of group functors if and only if the corresponding algebra map $f : R_2 \rightarrow R_1$ is a homomorphism of Hopf algebras.

Proof. – We identify $G = h^{R_1}$ and $H = h^{R_2}$ and $\eta : h^{R_1} \rightarrow h^{R_2}$ with $h^{R_i}(A)$ having the group law as in Proposition 1.33 for every k -algebra A . Then $\eta = \omega_f$.

Now suppose η is a homomorphism of group functors. Then for all A and all $\varphi_1, \varphi_2 \in \text{Hom}(R_1, A)$ we have

$$\eta(A)(\varphi_1 *_1 \varphi_2) = \eta(A)(\varphi_1) *_2 \eta(A)(\varphi_2).$$

The left hand side is

$$\eta(A)(\varphi_1 *_1 \varphi_2) = (\varphi_1 *_1 \varphi_2) \circ f = (M_A \circ (\varphi_1 \otimes \varphi_2)) \circ (\Delta_1 \circ f).$$

²Here $R \otimes I$ denotes the k -submodule of $R \otimes R$ by pure tensors of the form $r \otimes i$ with $r \in R$ and $i \in I$. It is not the abstract k -module $R \otimes I$. Note that this is really an abuse of notation since $\text{Ad}(R) \subset R \otimes R$. What we mean by $R \otimes I$ is really the image of the map $R \otimes I \xrightarrow{\text{id} \otimes \iota} R \otimes R$ where $\iota : I \rightarrow R$ is the inclusion.

The right hand side is

$$\begin{aligned}\eta(A)(\varphi_1) * \eta(A)(\varphi_2) &= M_A \circ ((\varphi_1 \circ f) \otimes (\varphi_2 \circ f)) \circ \Delta_2 \\ &= M_A \circ ((\varphi_1 \otimes \varphi_2) \circ (f \otimes f)) \circ \Delta_2 \\ &= (M_A \circ (\varphi_1 \otimes \varphi_2)) \circ ((f \otimes f) \circ \Delta_2).\end{aligned}$$

Thus we get

$$(M_A \circ (\varphi_1 \otimes \varphi_2)) \circ (\Delta_1 \circ f) = (M_A \circ (\varphi_1 \otimes \varphi_2)) \circ ((f \otimes f) \circ \Delta_2)$$

for all k -algebras A and all $\varphi_1, \varphi_2 \in \text{Hom}(R_1, A)$. We take $A := R_1 \otimes R_1$ and set

$$\varphi_1 := \iota_1 : R_1 \rightarrow R_1 \otimes R_1, \quad a \mapsto a \otimes 1$$

and

$$\varphi_2 := \iota_2 : R_1 \rightarrow R_1 \otimes R_1, \quad b \mapsto 1 \otimes b.$$

Then note that

$$(M_{R_1 \otimes R_1} \circ (\iota_1 \otimes \iota_2))(a \otimes b) = M_A((a \otimes 1) \otimes (1 \otimes b)) = (a \otimes 1)(1 \otimes b) = a \otimes b$$

i.e. $M_{R_1 \otimes R_1} \circ (\iota_1 \otimes \iota_2) = \text{id}_{R_1 \otimes R_1}$. Consequently, we have $\Delta_1 \circ f = (f \otimes f) \circ \Delta_2$.

Conversely, assume $\Delta_1 \circ f = (f \otimes f) \circ \Delta_2$. We wish to show that $\eta = \omega_f : h^{R_1} \rightarrow h^{R_2}$ is a homomorphism of group functors. Indeed, we have

$$\begin{aligned}\eta(A)(\varphi_1 * \varphi_2) &= (\varphi_1 * \varphi_2) \circ f \\ &= M_A \circ (\varphi_1 \otimes \varphi_2) \circ \Delta_1 \circ f \\ &= M_A \circ (\varphi_1 \otimes \varphi_2) \circ (f \otimes f) \circ \Delta_2 \\ &= M_A \circ ((\varphi_1 \circ f) \otimes (\varphi_2 \circ f)) \circ \Delta_2 \\ &= \eta(A)(\varphi_1) * \eta(A)(\varphi_2).\end{aligned}$$

This completes the proof. $\square$

Remark 3.14. – For every homomorphism $f : R \rightarrow R'$ of Hopf algebras we have

$$\begin{cases} \epsilon' \circ f = \epsilon \\ f \circ S = S' \circ f \end{cases}$$

Indeed, $\eta = \omega_f : h^{R'} \rightarrow h^R$ is a homomorphism of group functors. Now for any algebra A we know $\omega_{f,A}$ is a group homomorphism, and so have $\omega_{f,A}(e'_A) = e_A$ where $e'_A = T_A \circ \epsilon' \in h^{R'}(A)$ and $e_A = T_A \circ \epsilon \in h^R(A)$ are the identities. Taking $A = k$ we have $e'_k = T_k \circ \epsilon' = \text{id}_k \circ \epsilon' = \epsilon'$ and also $e_k = \epsilon$. Thus we get

$$\omega_{f,k}(\epsilon') = \epsilon$$

whence $\epsilon' \circ f = \epsilon$.

Also for each $\alpha \in \text{Hom}(R', A)$ we know $\omega_{f,A}(\alpha^{-1}) = \omega_{f,A}(\alpha)^{-1}$. This is same as

$$\alpha \circ S' \circ f = \alpha \circ f \circ S.$$

Taking $A = R'$ and $\alpha = \text{id}_{R'}$ we get $S' \circ f = f \circ S$.

Proposition 3.15. – Let $G_1 \simeq h^{R_1}$ and $G_2 \simeq h^{R_2}$ be affine group functors and let $\eta : G_1 \rightarrow G_2$ be a homomorphism of group functors corresponding to the homomorphism of Hopf algebras $f : R_2 \rightarrow R_1$. Then η is an isomorphism onto a closed subgroup functor if and only if f is surjective.

Proof. – This follows from Proposition 2.25. Indeed if η is an isomorphism onto a closed (subgroup) functor then f is surjective. Conversely, if f is surjective then η is isomorphism onto a closed subfunctor : $\eta : G_1 \xrightarrow{\sim} V(I) \subset G_2$. However, it is given as a part of the hypothesis that η is a homomorphism of group functors. Thus $V(I)(A) = \eta(A)(G_1(A))$ is a subgroup of $G_2(A)$ for every algebra A . $\square$

Definition 3.16. – Let $\eta : G \rightarrow H$ be a homomorphism of group functors. The association

$$A \rightsquigarrow N(A) := \text{Ker}(\eta(A))$$

is called the *kernel* of η .

Remark 3.17. – If $\psi : A \rightarrow B$ is any algebra map, then $G(\psi)N(A) \subset N(B)$. Indeed, if $x \in N(A)$ then $\eta(A)(x) = e_{H(A)}$. But then by using naturality of η we have $H(\psi)\eta(A) = \eta(B)G(\psi)$. Evaluating at x we get $H(\psi)(e_{HA}) = e_{HB} = \eta(B)(G(\psi)(x))$. Thus $G(\psi)(x) \in N(B)$. Also note that N is normal in G .

Proposition 3.18. – $N \simeq G \times_H h^k$ where the fiber product is taken with respect to the maps $\eta : G \rightarrow H$ and $e : h^k \rightarrow H$.

Proof. – First, note that

$$\begin{aligned} (G \times_H h^k)(A) &= \{(g, u) \in G(A) \times h^k(A) \mid \eta(A)(g) = e_A(T_A) = e_{HA}\} \\ &= \{(g, u) \in G(A) \times h^k(A) \mid g \in N(A)\} \\ &= N(A) \times \{T_A\}. \end{aligned}$$

Thus we have a bijection $\Phi_A : N(A) \rightarrow (G \times_H h^k)(A)$ given by $g \mapsto (g, T_A)$. We show that $\Phi = (\Phi_A)_A$ is natural. For this let $\psi : A \rightarrow B$ be any algebra map. We wish to show that

$$\Phi_B \circ N(\psi) = (G \times_H h^k)(\psi) \circ \Phi_A.$$

So let $x \in N(A)$ and consider

$$(G \times_H h^k)(\psi)(x, T_A) = (G(\psi)(x), h^k(\psi)(T_A)) = (N(\psi)(x), T_B).$$

On the other hand we have

$$\Phi_B(N(\psi)(x)) = (N(\psi)(x), T_B)$$

by definition of Φ . Thus $\Phi_B \circ N(\psi) = (G \times_H h^k)(\psi) \circ \Phi_A$. $\square$

Proposition 3.19. – Suppose G and H are affine group functors represented by algebra R and R' respectively. Let $\eta : G \rightarrow H$ be a homomorphism of group functors. Then its kernel N is a closed normal subgroup functor of G .

Proof. – We already know that N is a normal subgroup functor of G and $N \simeq G \times_H h^k$. Thus N is represented by $R \otimes_{R'} k$ using Proposition 1.23. Now let $\epsilon' : R' \rightarrow k$ be the counit and set $I_{R'} := \text{Ker } \epsilon'$. Consider the short exact sequence

$$0 \rightarrow I_{R'} \xrightarrow{\iota} R' \xrightarrow{\epsilon'} k \rightarrow 0$$

and tensor it with $R \otimes_{R'}$ – to obtain

$$R \otimes_{R'} I_{R'} \xrightarrow{\alpha} R \otimes_{R'} R' \xrightarrow{\beta} R \otimes_{R'} k \rightarrow 0.$$

Now let $\mu : R \otimes_{R'} R' \xrightarrow{\sim} R$ be $r \otimes r' \mapsto rf(r')$. Transporting the above sequence via μ we obtain

$$R \otimes_{R'} I_{R'} \xrightarrow{\mu \circ \alpha} R \xrightarrow{\beta \circ \mu^{-1}} R \otimes_{R'} k \rightarrow 0.$$

It is easy to check that this is exact at R . Now set $J := \text{Im}(\mu \circ \alpha)$. Explicitly, and element $r \otimes i$ maps to $\mu(r \otimes i) = rf(i)$ and thus we have

$$J = R \cdot f(I_{R'}) = \langle f(I_{R'}) \rangle.$$

Consequently, we have

$$R \otimes_{R'} k \simeq R/J = R/\langle f(I_{R'}) \rangle.$$

Hence $N \simeq h^{R \otimes_{R'} k} \simeq h^{R/J}$ and so N is closed by Proposition 2.19. $\square$

Example 3.20. – The n -th power map $\mathbf{G}_m \rightarrow \mathbf{G}_m$ has kernel μ_n and hence μ_n is a closed normal subgroup of $\mathbf{G}_m$. Similarly, for the determinant homomorphism $\det : \mathbf{GL}_n \rightarrow \mathbf{G}_m$ the kernel $\mathbf{SL}_n$ is a closed normal subgroup of $\mathbf{GL}_n$.

Definition 3.21. – Homomorphisms $\chi : G \rightarrow \mathbf{G}_m$ are called *characters* of G .

Proposition 3.22. – Let G be an affine group functor represented by a universal pair (R, r) with Hopf structure $(R, \Delta_R, \epsilon_R, S_R)$. If $\chi : G \rightarrow \mathbf{G}_m$ is a character of G , then the element

$$b := \chi(R)(r) \in R^\times$$

has the property

$$\Delta_R(b) = b \otimes b.$$

Proof. – For any algebra A and $z \in G(A)$ we have the unique algebra map $\varphi_z : R \rightarrow A$ such that $G(\varphi_z)(r) = z$. Now set $A = R \otimes R$. Take $i_1 : R \rightarrow R \otimes R$ be $t \mapsto t \otimes 1$ and $i_2 : R \rightarrow R \otimes R$ be $t \mapsto 1 \otimes t$. We set $x := G(i_1)(r)$ and $y := G(i_2)(r)$. Then we have

$$\varphi_x = i_1 \quad \text{and} \quad \varphi_y = i_2.$$

Thus, using Proposition 1.32, we have

$$\varphi_{xy} = M_{R \otimes R} \circ (i_1 \otimes i_2) \circ \Delta_R.$$

Since $\chi(R \otimes R)$ is a group homomorphism we have

$$\chi(R \otimes R)(xy) = \chi(R \otimes R)(x) \cdot \chi(R \otimes R)(y).$$

We have

$$\chi(R \otimes R)(x) = \chi(R \otimes R)(G(i_1)(r)) = \mathbf{G}_m(i_1)(\chi(R)(r)) = i_1(b) = b \otimes 1.$$

Similarly, $\chi(R \otimes R)(y) = 1 \otimes b$. Thus we get

$$\chi(R \otimes R)(x) \cdot \chi(R \otimes R)(y) = b \otimes b.$$

On the other hand

$$\chi(R \otimes R)(xy) = \chi(R \otimes R)(G(\varphi_{xy})(r)) = \mathbf{G}_m(\varphi_{xy})(\chi(R)(r)) = \varphi_{xy}(b).$$

But we have

$$\varphi_{xy}(b) = (M_{R \otimes R} \circ (i_1 \otimes i_2) \circ \Delta_R)(b) = \Delta_R(b)$$

since $(M_{R \otimes R} \circ (i_1 \otimes i_2))(a \otimes b) = (a \otimes 1) \cdot (1 \otimes b) = a \otimes b$ i.e. $M_{R \otimes R} \circ (i_1 \otimes i_2) = \text{id}_{R \otimes R}$. We have thus shown that

$$\Delta_R(b) = b \otimes b.$$

□

Remark 3.23. – If (R, Δ, ϵ, S) is a Hopf algebra and $b \in R^\times$ is such that $\Delta(b) = b \otimes b$ then

$$\epsilon(b) = 1 \quad \text{and} \quad S(b) = b^{-1}.$$

To show this, we first recall that $(\epsilon \otimes \text{id}_R) \circ \Delta = C$. Evaluating at b we get

$$\epsilon(b) \otimes b = 1 \otimes b \in k \otimes R.$$

Now applying $\text{id}_k \otimes \epsilon$ we get

$$\epsilon(b) \otimes \epsilon(b) = 1 \otimes \epsilon(b) \in k \otimes k.$$

Finally, applying the canonical isomorphism $k \otimes k \xrightarrow{\sim} k$, we obtain $\epsilon(b)^2 = \epsilon(b)$ and since $b \in R^\times$ we know $\epsilon(b) \in k^\times$. Hence we get $\epsilon(b) = 1$.

Also by Remark 1.30 we know

$$T_R \circ \epsilon = M_R \circ (S \otimes \text{id}) \circ \Delta.$$

Applying this to b we obtain

$$\epsilon(b) \cdot 1_R = S(b)b.$$

By the previous paragraph $\epsilon(b) = 1_R$ and thus $S(b) = b^{-1}$.

Remark 3.24. – We should also note that if $b \in R$ has the property that $\Delta(b) = b \otimes b$ and $\epsilon(b) = 1$, then b is invertible i.e. $b \in R^\times$. Indeed, $(S, \text{id}) \circ \Delta = T \circ \epsilon$ implies

$$1 = T(\epsilon(b)) = (S, \text{id})(b \otimes b) = S(b)b$$

that is to say, $b \in R^\times$ and $S(b) = b^{-1}$.

Definition 3.25. – Let (R, Δ, ϵ, S) be a Hopf algebra. An element $b \in R^\times$ such that $\Delta(b) = b \otimes b$ is called a *group-like element* of R .

Remark 3.26. – Group-like elements of a Hopf algebra R forms a subgroup of $R^\times$.

For an affine group functor G we can define a group structure on the *set* of characters of G denoted

$$\text{Hom}(G, \mathbf{G}_m)$$

by pointwise multiplication. More precisely, for two characters $\chi_1, \chi_2 \in \text{Hom}(G, \mathbf{G}_m)$ we define

$$\chi_1 \cdot \chi_2 : G \rightarrow \mathbf{G}_m$$

by

$$(\chi_1 \cdot \chi_2)(A)(x) := \chi_1(A)(x)\chi_2(A)(x), \quad \text{for all algebra } A \text{ and all } x \in G(A).$$

One can then verify easily that $\chi_1 \cdot \chi_2$ is indeed a natural transformation. Also the identity element of $\text{Hom}(G, \mathbf{G}_m)$ is written $\mathbf{1} : G \rightarrow \mathbf{G}_m$ defined by $\mathbf{1}(A)(x) := 1_A \in A^\times$ for all algebras A and all $x \in G(A)$. Finally, inverse of $\chi : G \rightarrow \mathbf{G}_m$ is denoted by $\chi^{-1} : G \rightarrow \mathbf{G}_m$ and for each algebra A and $x \in G(A)$ we have $\chi^{-1}(A)(x) := (\chi(A)(x))^{-1}$. Thus $\text{Hom}(G, \mathbf{G}_m)$ is actually a commutative group (written multiplicatively).

Proposition 3.27. – *Let G be an affine group functor with Hopf structure $(R, \Delta_R, \epsilon_R, S_R)$ and universal pair (R, r) . Then the subgroup (of $R^\times$) of group-like elements is isomorphic to $\text{Hom}(G, \mathbf{G}_m)$.*

Proof. – Let $\text{Gp}(R) \subset R^\times$ be the subgroup of group-like elements. We show that the map

$$\Phi : [\chi \mapsto \chi(R)(r)] : \text{Hom}(G, \mathbf{G}_m) \rightarrow \text{Gp}(R)$$

is a group isomorphism. It is clearly a group homomorphism. Thus it remains to show that it is in fact a bijection. We construct its inverse

$$\Psi := [b \mapsto \chi_b] : \text{Gp}(R) \rightarrow \text{Hom}(G, \mathbf{G}_m)$$

as follows : let $b \in \text{Gp}(R)$ i.e. $b \in R^\times$ and $\Delta_R(b) = b \otimes b$; define $\chi_b(A) : G(A) \rightarrow A^\times$ for each algebra A by

$$\chi_b(A)(x) := \varphi_x(b) \in A^\times.$$

We check that $\chi_b(A)$ is a group homomorphism. Indeed, we have

$$\chi_b(A)(xy) = \varphi_{xy}(b) = (M_A \circ (\varphi_x \otimes \varphi_y))(\Delta_R(b)) = \varphi_x(b)\varphi_y(b) = \chi_b(A)(x)\chi_b(A)(y).$$

Next, we show that $\chi_b(A)$ is natural in A . So take any algebra map $\psi : A \rightarrow B$. Then we know that

$$G(\psi)(x) = G(\psi \circ \varphi_x)(r)$$

and hence

$$\varphi_{G(\psi)(x)} = \psi \circ \varphi_x.$$

Therefore, we obtain

$$(\chi_b(B) \circ G(\psi))(x) = \varphi_{G(\psi)(x)}(b) = \psi(\varphi_x(b)) = (\mathbf{G}_m(\psi) \circ \chi_b(A))(x)$$

for all $x \in G(A)$ i.e. χ_b is a natural transformation.

Finally, we check that $\Phi \circ \Psi$ and $\Psi \circ \Phi$ are identities. First we show that

$$\Psi \circ \Phi = \text{id}_{\text{Hom}(G, \mathbf{G}_m)}.$$

We take any character $\chi \in \text{Hom}(G, \mathbf{G}_m)$. Set $b = \Phi(\chi) = \chi(R)(r) \in R^\times$. Now we must show that $\Psi(b) = \chi_b = \chi$. For this we take any algebra A and any $x \in G(A)$. Then using naturality of χ we have

$$\chi(A)(x) = \chi(A)(G(\varphi_x)(r)) = \mathbf{G}_m(\varphi_x)(\chi(R)(r)) = \varphi_x(b) = \chi_b(A)(x)$$

thus proving $\chi = \chi_b$. Hence $\Psi \circ \Phi = \text{id}_{\text{Hom}(G, \mathbf{G}_m)}$ as asserted.

We show now that

$$\Phi \circ \Psi = \text{id}_{\text{Gp}(R)}.$$

So let $b \in \text{Gp}(R)$. We observe that

$$\Phi(\Psi(b)) = \Phi(\chi_b) = \chi_b(R)(r) = \varphi_r(b) = b$$

since $\varphi_r : R \rightarrow R$ is the unique map such that $G(\varphi_r)(r) = r$ and hence $\varphi_r = \text{id}_R$. This completes the proof. $\square$

Definition 3.28. – Let (R, Δ, ϵ, S) be a Hopf algebra. An element $b \in R$ is called *primitive* if

$$\Delta(b) = b \otimes 1 + 1 \otimes b.$$

Remark 3.29. – If $b \in R$ is a primitive element, then $\epsilon(b) = 0$ and $S(b) = -b$. Indeed, evaluating $(\epsilon \otimes \text{id}) \circ \Delta = C$ on b we get

$$\epsilon(b) \otimes 1 + 1 \otimes b = 1 \otimes b$$

and thus $\epsilon(b) \otimes 1 = 0 \in k \otimes R$ and hence taking image under $k \otimes R \xrightarrow{\sim} R$ we have

$$\epsilon(b) = 0.$$

On the other hand we evaluate $M_R \circ (S \otimes \text{id}) \circ \Delta = T_R \circ \epsilon$ on b to obtain

$$S(b) + b = T_R(\epsilon(b)) = 0$$

and thus $S(b) = -b$. Also it is easy to check that primitive elements form a subgroup of $(R, +)$.

Now we consider the set of homomorphisms

$$\text{Hom}(G, \mathbf{G}_a)$$

for an affine group functor G . This can also be made into a group : for $\kappa_1, \kappa_2 \in \text{Hom}(G, \mathbf{G}_a)$ we set

$$(\kappa_1 + \kappa_2)(A)(x) := \kappa_1(A)(x) + \kappa_2(A)(x), \quad \text{for all algebras } A \text{ and all } x \in G(A).$$

Here the identity is $\mathbf{0} : G \rightarrow \mathbf{G}_a$ defined by

$$\mathbf{0}(A)(x) := 0, \quad \text{for all algebras } A \text{ and all } x \in G(A).$$

Similarly, the inverse of $\kappa \in \text{Hom}(G, \mathbf{G}_a)$ is denoted $-\kappa$ and defined as

$$(-\kappa)(A)(x) := -\kappa(A)(x), \quad \text{for all algebras } A \text{ and all } x \in G(A).$$

Thus $\text{Hom}(G, \mathbf{G}_a)$ becomes an abelian group under “pointwise addition”.

Proposition 3.30. – *Let G be an affine group functor with representing Hopf algebra $(R, \Delta_R, \epsilon_R, S_R)$ and universal pair (R, r) . Suppose $\text{Pr}(R)$ is the subgroup of primitive elements. Then $\text{Pr}(R)$ is isomorphic to the group $\text{Hom}(G, \mathbf{G}_a)$.*

Proof. – We show that the map

$$\Phi : \text{Hom}(G, \mathbf{G}_a) \rightarrow \text{Pr}(R)$$

defined by

$$\kappa \mapsto \kappa(R)(r)$$

is a group isomorphism. First we check that $b := \kappa(R)(r) \in \text{Pr}(R)$. Then it is clear that Φ is a group homomorphism. Take $i_1 : R \rightarrow R \otimes R$ be $t \mapsto t \otimes 1$ and $i_2 : R \rightarrow R \otimes R$ be $t \mapsto 1 \otimes t$. We set $x := G(i_1)(r)$ and $y := G(i_2)(r)$. Then we have

$$\varphi_x = i_1 \quad \text{and} \quad \varphi_y = i_2.$$

Thus, using Proposition 1.32, we have

$$\varphi_{xy} = M_{R \otimes R} \circ (i_1 \otimes i_2) \circ \Delta_R.$$

Since $\kappa(R \otimes R)$ is a group homomorphism we have

$$\kappa(R \otimes R)(xy) = \kappa(R \otimes R)(x) + \kappa(R \otimes R)(y).$$

We have

$$\kappa(R \otimes R)(x) = \kappa(R \otimes R)(G(i_1)(r)) = \mathbf{G}_a(i_1)(\kappa(R)(r)) = i_1(b) = b \otimes 1.$$

Similarly, $\kappa(R \otimes R)(y) = 1 \otimes b$. Thus we get

$$\kappa(R \otimes R)(x) + \kappa(R \otimes R)(y) = b \otimes 1 + 1 \otimes b.$$

On the other hand

$$\kappa(R \otimes R)(xy) = \kappa(R \otimes R)(G(\varphi_{xy})(r)) = \mathbf{G}_a(\varphi_{xy})(\kappa(R)(r)) = \varphi_{xy}(b).$$

However, we have

$$\varphi_{xy}(b) = (M_{R \otimes R} \circ (i_1 \otimes i_2) \circ \Delta_R)(b) = \Delta_R(b)$$

since $M_{R \otimes R} \circ (i_1 \otimes i_2) = \text{id}_{R \otimes R}$. Hence we have proved that

$$\Delta_R(b) = b \otimes 1 + 1 \otimes b$$

i.e. $b := \kappa(R)(r)$ is indeed primitive.

Now we show that

$$\Psi : \text{Pr}(\mathbf{R}) \rightarrow \text{Hom}(\mathbf{G}, \mathbf{G}_a)$$

defined by

$$b \mapsto \kappa_b$$

is inverse of Φ , where $\kappa_b \in \text{Hom}(\mathbf{G}, \mathbf{G}_a)$ is given as

$$\kappa_b(\mathbf{A})(x) := \varphi_x(b), \quad \text{for all algebra } \mathbf{A} \text{ and all } x \in \mathbf{G}(\mathbf{A}).$$

We check that $\kappa_b(\mathbf{A})$ is a group homomorphism. Indeed, we have

$$\kappa_b(\mathbf{A})(xy) = \varphi_{xy}(b) = (M_A \circ (\varphi_x \otimes \varphi_y))(\Delta_R(b)) = \varphi_x(b) + \varphi_y(b) = \kappa_b(\mathbf{A})(x) + \kappa_b(\mathbf{A})(y).$$

Naturality of κ_b can be checked by the exact same arguments of Proposition 3.27. Finally, $\Phi \circ \Psi = \text{id}$ and $\Psi \circ \Phi = \text{id}$ can be proved exactly as in Proposition 3.27. $\square$

4. DIAGONALIZABLE GROUP FUNCTORS

Let $\mathbf{M}$ be any abelian group with the group law written multiplicatively. We form the group algebra $k[\mathbf{M}]$: this is a free k -module with the elements of $\mathbf{M}$ as basis and the ring multiplication is given by the k -linear extension of the group multiplication of $\mathbf{M}$.

Proposition 4.1. – *The k -algebra $k[\mathbf{M}]$ can be made into a Hopf algebra by setting*

$$\Delta(m) := m \otimes m, \quad \epsilon(m) := 1, \quad S(m) := m^{-1}$$

for all $m \in \mathbf{M}$.

Proof. – The defining axioms for a Hopf algebra can be directly verified on the basis elements $m \in \mathbf{M}$. $\square$

Definition 4.2. – An affine group functor $\mathbf{G}$ is called *diagonalizable* if its associated Hopf algebra is of the form $k[\mathbf{M}]$.

Proposition 4.3. – *If $k[\mathbf{M}]$ is a finitely generated k -algebra, then $\mathbf{M}$ is a finitely generated abelian group.*

Proof. – Suppose $a_1, \dots, a_n$ are algebra generators of $k[\mathbf{M}]$. Then there exists a finite set $\{m_1, \dots, m_l\}$ of elements of $\mathbf{M}$ such that each $a_i \in k\text{-span}\{m_1, \dots, m_l\}$. Let $\mathbf{M}'$ be the subgroup of $\mathbf{M}$ generated by $\{m_1, \dots, m_l\}$. Then since each $a_i \in k[\mathbf{M}']$ we have $k[\mathbf{M}] = k[\mathbf{M}']$. But since $\mathbf{M}$ is a k -basis we conclude that $\mathbf{M} = \mathbf{M}'$. Hence $\mathbf{M}$ is a finitely generated abelian group. $\square$

Proposition 4.4. – *Let $\mathbf{M}_1, \mathbf{M}_2$ be subgroups of $\mathbf{M}$ and $\mathbf{M} = \mathbf{M}_1 \oplus \mathbf{M}_2$. Then $k[\mathbf{M}] \simeq k[\mathbf{M}_1] \otimes k[\mathbf{M}_2]$.*

Proof. – We have $\mathbf{M} = \mathbf{M}_1 \oplus \mathbf{M}_2$, which means each $m \in \mathbf{M}$ can be uniquely written as $m = m_1 m_2$ with $m_1 \in \mathbf{M}_1$ and $m_2 \in \mathbf{M}_2$. We define $\Phi : k[\mathbf{M}_1] \otimes k[\mathbf{M}_2] \rightarrow k[\mathbf{M}]$ first on pure tensors by

$$\Phi(m_1 \otimes m_2) := m_1 m_2, \quad m_1 \in \mathbf{M}_1, m_2 \in \mathbf{M}_2$$

and then extend k -linearly. By the UMP of tensor products, Φ is well-defined, and it can be checked that it is a k -algebra homomorphism.

Similarly, we define $\Psi : k[\mathbf{M}] \rightarrow k[\mathbf{M}_1] \otimes k[\mathbf{M}_2]$ by first writing an $m \in \mathbf{M}$ uniquely as $m = m_1 m_2$ with $m_1 \in \mathbf{M}_1$ and $m_2 \in \mathbf{M}_2$ and then setting

$$\Psi(m) := m_1 \otimes m_2$$

and extending k -linearly. Again, this is well-defined as $\mathbf{M} = \mathbf{M}_1 \oplus \mathbf{M}_2$ and it can be easily checked that Ψ is indeed a k -algebra homomorphism.

Finally, we can also check that Φ and Ψ are inverses of each other. $\square$

Proposition 4.5. – *$k[\mathbf{Z}] \simeq \frac{k[X,Y]}{(XY-1)}$ and $k[\mathbf{Z}/n\mathbf{Z}] \simeq \frac{k[X]}{(X^n-1)}$.*

Proof. – First, we write a basis of $k[\mathbf{Z}]$ by giving a multiplicative description of $\mathbf{Z}$ as follows :

$$e_n, \quad n \in \mathbf{Z}$$

with multiplication

$$e_n e_m := e_{n+m}, \quad e_0 := 1.$$

Thus every element of $k[\mathbf{Z}]$ is a finite linear combination

$$\sum_{n \in \mathbf{Z}} a_n e_n.$$

Now we define the k -algebra map

$$\Phi : k[X, Y] \rightarrow k[\mathbf{Z}], \quad X \mapsto e_1, \quad Y \mapsto e_{-1}.$$

Then clearly $\Phi(XY - 1) = 0$. Since every basis element e_n lies in the image we get an induced surjection

$$\bar{\Phi} : \frac{k[X, Y]}{(XY - 1)} \rightarrow k[\mathbf{Z}], \quad \bar{X} \mapsto e_1, \quad \bar{Y} \mapsto e_{-1}.$$

Now we define the inverse of $\bar{\Phi}$ as

$$\Psi : k[\mathbf{Z}] \rightarrow \frac{k[X, Y]}{(XY - 1)}$$

given by

$$\Psi(e_n) := \begin{cases} X^n + (XY - 1), & n \geq 0 \\ Y^{-n} + (XY - 1), & n < 0 \end{cases}$$

extended k -linearly. Then it is easy to check that Ψ is also a k -algebra map. Finally, by evaluating on the generators, we can check that $\bar{\Phi} \circ \Psi$ and $\Psi \circ \bar{\Phi}$ are identity maps.

Next, we consider $\mathbf{Z}/n\mathbf{Z}$ and write it multiplicatively as $\{e_0, e_1, \dots, e_{n-1}\}$ corresponding to the classes $\{\bar{0}, \bar{1}, \dots, \overline{n-1}\}$. Then we have

$$e_0 = 1, \quad e_i e_j = e_{i+j \bmod n}.$$

Thus e_1 is invertible and $e_1^n = e_0 = 1$ and so every element is a linear combination of $\{1, e_1, \dots, e_1^{n-1}\}$. Now $\Phi : k[X] \rightarrow k[\mathbf{Z}/n\mathbf{Z}]$ defined by $X \mapsto e_1$ descends to a surjection

$$\bar{\Phi} : \frac{k[X]}{(X^n - 1)} \rightarrow k[\mathbf{Z}/n\mathbf{Z}]$$

which has inverse

$$\Psi : k[\mathbf{Z}/n\mathbf{Z}] \rightarrow \frac{k[X]}{(X^n - 1)}$$

given by $e_j \mapsto X^j + (X^n - 1)$. Both are k -algebra maps as can be easily checked. $\square$

Theorem 4.6. – *Let G be a diagonalizable affine group functor represented by a finitely generated k -algebra R . Then G is isomorphic to a finite product of copies of $\mathbf{G}_m$ and various μ_n .*

Proof. – We are given that $R \simeq k[M]$ which is a finitely generated k -algebra. Thus, M is a finitely generated abelian group. Thus, R is isomorphic to a tensor product of finitely many copies of $k[\mathbf{Z}]$ and $k[\mathbf{Z}/n\mathbf{Z}]$ which are isomorphic as k -algebras to $\frac{k[X, Y]}{(XY - 1)}$ and $\frac{k[X]}{(X^n - 1)}$ respectively. But we can check that the k -algebra isomorphisms $\bar{\Phi}$ defined in the proof of Proposition 4.5 are indeed Hopf algebra maps i.e. they commute with the comultiplications. For instance, in case of $k[\mathbf{Z}]$ we have :

$$(\bar{\Phi} \otimes \bar{\Phi})(\Delta(\bar{X})) = (\bar{\Phi} \otimes \bar{\Phi})(\bar{X} \otimes \bar{X}) = \bar{\Phi}(\bar{X}) \otimes \bar{\Phi}(\bar{X}) = e_1 \otimes e_1 = \Delta(e_1) = \Delta(\bar{\Phi}(\bar{X})).$$

Thus, by Proposition 3.13, $k[\mathbf{Z}]$ corresponds to $\mathbf{G}_m$ and $k[\mathbf{Z}/n\mathbf{Z}]$ corresponds to μ_n . This proves the statement of the theorem. $\square$

Lemma 4.7. – *Let R be a Hopf algebra over a field k . Then, the group-like elements in R are linearly independent.*

Proof. – Suppose b and $\{b_i\}$ are group-like elements with $b = \sum_i \lambda_i b_i$ with $\lambda_i \in k$. We may assume that the b_i are independent. Then $1 = \epsilon(b) = \sum_i \lambda_i \epsilon(b_i) = \sum_i \lambda_i$ by Remark 3.23. But $\Delta(b) = b \otimes b = \sum_{i,j} \lambda_i \lambda_j b_i \otimes b_j$ and on the other hand $\Delta(b) = \sum_i \lambda_i \Delta(b_i) = \sum_i \lambda_i (b_i \otimes b_i)$. However, the elements $b_i \otimes b_j$ are linearly independent. Indeed, consider a k -basis of R extending the linearly independent set $\{b_i\}$ and then consider the dual basis elements $\{b_i^*\}$ where $b_i^*(b_j) = \delta_{ij}$; then apply $b_i^* \otimes b_j^*$ to a relation among $\{b_i \otimes b_j\}_{i,j}$ to obtain the desired fact. Hence, we can compare coefficients to obtain that $\lambda_i \lambda_j = 0$ for all $i \neq j$ and $\lambda_i^2 = \lambda_i$. Thus $\lambda_i \in \{0,1\}$. As $\sum_i \lambda_i = 1$, there is at least some non-zero λ_i and so this $\lambda_i = 1$. Consequently, $\lambda_j = 0$ for all $j \neq i$. This implies $\sum_j \lambda_j b_j$ equals b_i . $\square$

Theorem 4.8. – *Let k be a field. An affine group functor is diagonalizable if and only if its representing algebra is spanned by group-like elements. Furthermore, there is an anti-equivalence between diagonalizable group functors and abelian groups, where a diagonalizable group functor G corresponds to its group of characters.*

Proof. – First, suppose G is an affine group functor with representing algebra R being spanned by group-likes. By the previous lemma, we know they form a basis of R . Now recall that the group-likes form a subgroup of $R^\times$. Denote this subgroup by $X_G \subset R^\times$ and recall that $X_G \simeq \text{Hom}(G, \mathbf{G}_m)$ by Proposition 3.27. Thus, we have a bijection $k[X_G] \rightarrow R$ that canonically preserves the multiplication and Hopf algebra structure on the basis elements. So this bijection is a Hopf-algebra map. Thus, G is diagonalizable.

Conversely, suppose G is diagonalizable with $R \simeq k[M]$ for some abelian group M . By definition of the Hopf algebra structure on $k[M]$ we know that M consists of group-like elements, and obviously spans $k[M]$. In fact, if $b \in k[M]^\times$ is group-like that $b \in M$. To show this, write $b = \sum_i \lambda_i m_i$ with $\lambda_i \in k$ and $m_i \in M$. Again, $m_i \otimes m_j$ are linearly independent, so $\Delta(b) = b \otimes b = \sum_{i,j} \lambda_i \lambda_j (m_i \otimes m_j)$ and $\Delta(b) = \sum_i \lambda_i (m_i \otimes m_i)$ together yield $\lambda_i = 1$ for precisely one i since $b \neq 0$, and $\lambda_j = 0$ for all $j \neq i$. Thus $b = m_i$ and we have shown that $X_G \simeq M$.

Finally, consider the association $G \rightsquigarrow X_G = \text{Hom}(G, \mathbf{G}_m)$ from the category of diagonalizable group functors to that of abelian groups. Thus, a homomorphism of group functors $G \rightarrow H$ induces a group homomorphism $X_H \rightarrow X_G$. We then have a Hopf algebra map $k[X_H] \rightarrow k[X_G]$. Also, given a Hopf algebra map $k[X_H] \rightarrow k[X_G]$ we get a group homomorphism $X_H \rightarrow X_G$ since group-likes are mapped to group-likes. Thus, the group homomorphism $X_H \rightarrow X_G$ determines the Hopf algebra map $k[X_H] \rightarrow k[X_G]$ and hence the morphism $G \rightarrow H$ and we obtain that $\text{Hom}(G, H) \simeq \text{Hom}(X_H, X_G)$. $\square$

5. COMMUTATIVE ALGEBRA 1 : PROJECTIVE MODULES

5.1. Properties of Hom functor. Let k be a commutative unital ring. Consider k -modules M, M_1, N , and N_1 , and homomorphisms $f : M_1 \rightarrow M$ and $g : N \rightarrow N_1$. We define the following homomorphisms :

$$\begin{aligned} f^* : \text{Hom}_k(M, N) &\rightarrow \text{Hom}_k(M_1, N), & f^*(\varphi) &:= \varphi \circ f \\ g_* : \text{Hom}_k(M, N) &\rightarrow \text{Hom}_k(M, N_1), & g_*(\psi) &:= g \circ \psi. \end{aligned}$$

Lemma 5.1. – *The following hold :*

- (1) *Let $M_1 \xrightarrow{f} M \xrightarrow{g} M_2 \rightarrow 0$ be an exact sequence of k -modules. Then for any k -module N the following sequence*

$$0 \rightarrow \text{Hom}_k(M_2, N) \xrightarrow{g^*} \text{Hom}_k(M, N) \xrightarrow{f^*} \text{Hom}_k(M_1, N)$$

is exact.

- (2) *Let $0 \rightarrow N_1 \xrightarrow{f} N \xrightarrow{g} N_2$ be an exact sequence of k -modules. Then for any k -module M the following sequence*

$$0 \rightarrow \text{Hom}_k(M, N_1) \xrightarrow{f_*} \text{Hom}_k(M, N) \xrightarrow{g_*} \text{Hom}_k(M, N_2)$$

is exact.

Proof. – 1. We first prove that g^* is injective. Consider $\varphi \in \text{Hom}_k(M_2, N)$ with $g^*(\varphi) = 0$. This means $\varphi \circ g = 0$. But since g is surjective by hypothesis, it follows that $\varphi = 0$.

Next, we show that $\text{Im } g^* = \text{Ker } f^*$. Since $f^* \circ g^* = (g \circ f)^* = 0$, we have $\text{Im } g^* \subset \text{Ker } f^*$. On the other hand, take $\psi \in \text{Ker } f^*$. Since g is surjective, we define $\varphi \in \text{Hom}_k(M_2, N)$ by $\varphi(g(m)) := \psi(m)$ for all

$m \in M$. This is well-defined since if $g(m) = g(n)$, then $m - n \in \text{Ker } g = \text{Im } f$, and $f^*(\psi) = \psi \circ f = 0$ implies $\psi(m - n) = 0$ i.e. $\psi(m) = \psi(n)$. Thus we get $g^*(\varphi) = \psi$ as required i.e. $\text{Im } g^* \supset \text{Ker } f^*$.

2. To show that f_* is injective, suppose $f_*(\varphi) = f \circ \varphi = 0$. Since f is injective, $\text{Im } \varphi \subset \text{Ker } f = 0$ means $\varphi = 0$.

Now, we show $\text{Im } f_* = \text{Ker } g_*$. First, note that $g_* \circ f_* = (g \circ f)_* = 0$ implies $\text{Im } f_* \subset \text{Ker } g_*$. We show the opposite inclusion. So take $\psi \in \text{Ker } g_*$ i.e. $g \circ \psi = 0$. Define $\varphi \in \text{Hom}_k(M, N_1)$ by $\varphi(m) := f^{-1}(\psi(m))$ for any $m \in M$. This is well-defined since f is injective and $\text{Im } \psi \subset \text{Ker } g = \text{Im } f$. Automatically, we get $f \circ \varphi = \psi$ i.e. $f_*(\varphi) = \psi \in \text{Im } f_*$. $\square$

We note in the following fact that the converse of the above statements are also true.

Fact 5.2. – *The following hold :*

(1) Let $M_1 \xrightarrow{f} M \xrightarrow{g} M_2 \rightarrow 0$ be a sequence of k -modules such that the sequence

$$0 \rightarrow \text{Hom}_k(M_2, N) \xrightarrow{g^*} \text{Hom}_k(M, N) \xrightarrow{f^*} \text{Hom}_k(M_1, N)$$

is exact for any k -module N . Then,

$$M_1 \xrightarrow{f} M \xrightarrow{g} M_2 \rightarrow 0$$

is exact.

(2) Let $0 \rightarrow N_1 \xrightarrow{f} N \xrightarrow{g} N_2$ be a sequence of k -modules such that the sequence

$$0 \rightarrow \text{Hom}_k(M, N_1) \xrightarrow{f_*} \text{Hom}_k(M, N) \xrightarrow{g_*} \text{Hom}_k(M, N_2)$$

is exact for any k -module M . Then,

$$0 \rightarrow N_1 \xrightarrow{f} N \xrightarrow{g} N_2$$

is exact.

Proof. – 1. We first show g is surjective. For this take $N = M_2/\text{Im } g$. Now $\pi : M_2 \rightarrow N$ has image under g^* as $g^*(\pi) = \pi \circ g = 0$. Thus, since g^* is injective, $\pi = 0$ i.e. $M_2 = \text{Im } g$.

Next, we show $\text{Im } f = \text{Ker } g$. First, set $N = M_2$. Then, since $(g \circ f)^* = f^* \circ g^* = 0$, we have

$$0 = (g \circ f)^*(\text{id}_{M_2}) = \text{id}_{M_2} \circ g \circ f = g \circ f$$

i.e. $\text{Im } f \subset \text{Ker } g$. To prove the opposite inclusion, we take $N = M/\text{Im } f$. Then $\pi : M \rightarrow M/\text{Im } f$ is the canonical projection, and $f^*(\pi) = 0$ so that $\pi \in \text{Ker } f^* = \text{Im } g^*$. Thus there exists $\varphi \in \text{Hom}_k(M_2, N)$ such that $\varphi \circ g = g^*(\varphi) = \pi$. Hence, $\text{Ker } g \subset \text{Ker } \pi = \text{Im } f$.

2. We show f is injective. Choose $M = \text{Ker } f$ and consider $i : \text{Ker } f \hookrightarrow N_1$. Now, we have $f_*(i) = f \circ i = 0$, and since f_* is injective, we get $i = 0$ i.e. $\text{Ker } f = 0$ i.e. f is injective.

Next, we show $\text{Im } f = \text{Ker } g$. First, we take $M = N_1$ and then since $(g \circ f)_* = g_* \circ f_* = 0$, we get

$$0 = (g_* \circ f_*)(\text{id}_{N_1}) = g \circ f \circ \text{id}_{N_1} = g \circ f$$

i.e. $\text{Im } f \subset \text{Ker } g$. To prove the opposite inclusion, take $M = \text{Ker } g$ and consider $i : \text{Ker } g \hookrightarrow N$. Then $g_*(i) = 0$ i.e. $i \in \text{Ker } g_* = \text{Im } f_*$. So, there exists $\varphi \in \text{Hom}_k(\text{Ker } g, N_1)$ such that $f_*(\varphi) = f \circ \varphi = i$. Thus $\text{Im } i = \text{Ker } g \subset \text{Im } f$ as required. $\square$

5.2. Projective modules. Given two k -modules M and N , the definition of direct sum of modules yields an exact sequence

$$0 \rightarrow M \xrightarrow{i_M} M \oplus N \xrightarrow{\pi_N} N \rightarrow 0,$$

where $i_M(m) = (m, 0)$ and $\pi_N(m, n) = n$.

In general, a short exact sequence

$$0 \rightarrow M \xrightarrow{f} N \xrightarrow{g} P \rightarrow 0$$

does not provide a decomposition of N in terms of M and P .

Lemma 5.3. – *Let $0 \rightarrow M \xrightarrow{f} N \xrightarrow{g} P \rightarrow 0$ be an exact sequence of k -modules. The following statements are equivalent :*

(1) There exists an isomorphism $N \xrightarrow{\varphi} M \oplus P$ that makes the following diagram commutative

$$\begin{array}{ccccccc} 0 & \longrightarrow & M & \xrightarrow{f} & N & \xrightarrow{g} & P \longrightarrow 0 \\ & & \text{id}_M \downarrow & & \varphi \downarrow & & \text{id}_P \downarrow \\ 0 & \longrightarrow & M & \xrightarrow{i_M} & M \oplus P & \xrightarrow{\pi_M} & P \longrightarrow 0 \end{array}$$

(2) There exists a homomorphism $r : N \rightarrow M$ such that $r \circ f = \text{id}_M$.

(3) There exists a homomorphism $s : P \rightarrow N$ such that $g \circ s = \text{id}_P$.

If any of the above equivalent conditions hold, we say that the sequence *splits*, that r is a *retraction* of f , and s is a *section* of g .

Proof. – Consider the following diagram :

$$\begin{array}{ccccccc} 0 & \longrightarrow & M & \xrightarrow{f} & N & \xrightarrow{g} & P \longrightarrow 0 \\ & & \text{id}_M \downarrow & & \varphi \downarrow & & \text{id}_P \downarrow \\ & & \pi_M \swarrow & & \varphi^{-1} \swarrow & & i_P \searrow \\ 0 & \longrightarrow & M & \xrightarrow{i_M} & M \oplus P & \xrightarrow{\pi_P} & P \longrightarrow 0 \end{array}$$

1 $\Rightarrow$ 2. We define $r := \pi_M \circ \varphi$. Then $r \circ f = \pi_M \circ \varphi \circ f = \pi_M \circ i_M \circ \text{id}_M = \text{id}_M$.

1 $\Rightarrow$ 3. We define $s := \varphi^{-1} \circ i_P$. Then $g \circ s = g \circ \varphi^{-1} \circ i_P = \text{id}_P \circ \pi_P \circ i_P = \text{id}_P$.

2 $\Rightarrow$ 1. Define $\varphi : N \rightarrow M \oplus P$ by $\varphi(n) := (r(n), g(n))$. Then φ is injective, since $r(n) = 0 = g(n)$ implies $n \in \text{Ker}(g) = \text{Im}(f)$ and also $n \in \text{Ker}(r)$, thus $n = 0$. Now suppose $(m, p) \in M \oplus P$. Since $g : N \rightarrow P$ is surjective, there is some $n_0 \in N$ such that $g(n_0) = p$. Consider the element

$$n := n_0 - f(r(n_0) - m) \in N.$$

Then $r(n) = m$ since $r(f(r(n_0))) = r(n_0)$ and $r(f(m)) = m$. Also, $g(n) = g(n_0) - g(f(\dots)) = g(n_0) = p$ since $g \circ f = 0$. Thus $\varphi(n) = (m, p)$ and so φ is surjective.

Now it remains to check that the diagram commutes. Indeed, we have

$$\varphi(f(m)) = (r(f(m)), g(f(m))) = (m, 0) = i_M(m)$$

and

$$\pi_P(\varphi(n)) = \pi_P(r(n), g(n)) = g(n).$$

3 $\Rightarrow$ 1. We define $\psi : M \oplus P \rightarrow N$ by $\psi(m, p) := f(m) + s(p)$. Clearly, we have

$$(\psi \circ i_M)(m) = \psi(m, 0) = f(m)$$

i.e. $\psi \circ i_M = f$. On the other hand,

$$(g \circ \psi)(m, p) = g(s(p)) = p = \pi_P(m, p)$$

i.e. $g \circ \psi = \pi_P$.

Now we show that ψ is an isomorphism. First, suppose $\psi(m, p) = f(m) + s(p) = 0$, then $g(f(m)) + g(s(p)) = p = 0$. Thus $f(m) = 0$ and so $m = 0$ since f is injective. Therefore, ψ is injective. Finally, let $n \in N$, and set $p := g(n) \in P$, $n' := n - s(p) \in N$. Then, $g(n') = g(n) - g(s(p)) = p - p = 0$ i.e. $n' \in \text{Ker}(g) = \text{Im}(f)$ and so $n' = f(m)$ for some $m \in M$. Hence $n = f(m) + s(p) = \psi(m, p)$ i.e. ψ is surjective. $\square$

Definition 5.4. – A k -module P is called *projective* if, for any pair of k -modules M and N , and any homomorphisms $g : M \rightarrow N$ and $f : P \rightarrow N$ with g surjective, there exists a homomorphism $\tilde{f} : P \rightarrow M$ which makes the following diagram commute :

$$\begin{array}{ccc} & P & \\ \tilde{f} \swarrow & \downarrow f & \\ M & \xrightarrow{g} & N \longrightarrow 0. \end{array} \quad \boxed{g \circ \tilde{f} = f}$$

Fact 5.5. – Every free module is projective.

Proof. – Let P be a free k -module, and let $\mathcal{B}$ be a k -basis of P . Consider the diagram

$$\begin{array}{ccccc} & & P & & \\ & \swarrow \tilde{f} & \downarrow f & & \\ M & \xrightarrow{g} & N & \longrightarrow & 0 \end{array}$$

with g surjective. It suffices to define $\tilde{f}$ on the basis elements $b \in \mathcal{B}$. So we define it via $g(\tilde{f}(b)) = f(b)$, and this is possible since g is surjective. $\square$

Lemma 5.6. – *Let P be a k -module. The following are equivalent :*

- (1) P is projective.
- (2) For any short exact sequence of k -modules

$$0 \rightarrow N_1 \xrightarrow{f} N \xrightarrow{g} N_2 \rightarrow 0,$$

the sequence

$$0 \rightarrow \text{Hom}_k(P, N_1) \xrightarrow{f_*} \text{Hom}_k(P, N) \xrightarrow{g_*} \text{Hom}_k(P, N_2) \rightarrow 0$$

is exact.

- (3) Every exact sequence $0 \rightarrow M \rightarrow N \rightarrow P \rightarrow 0$ splits.
- (4) P is a direct summand of every free module M that has a surjective map $\pi : M \rightarrow P$.
- (5) P is a direct summand of a free module.

Proof. – $1 \Rightarrow 2$. We just have to show that g_* is surjective, since we already know $\text{Hom}_k(P, \cdot)$ is left exact. But that follows directly from the definition of P being projective.

$2 \Rightarrow 1$. Take $N_1 = \text{Ker } g$ for any surjection $N \xrightarrow{g} N_2 \rightarrow 0$. Then surjectivity of g_* implies P is projective, since given $\varphi \in \text{Hom}_k(P, N_2)$ there exists $\tilde{f} \in \text{Hom}_k(P, N)$ such that $g_*(\tilde{f}) = g \circ \tilde{f} = \varphi$.

$1 \Rightarrow 3$. Consider an exact sequence $0 \rightarrow M \xrightarrow{\alpha} N \xrightarrow{\beta} P \rightarrow 0$. As P is projective, there exists a homomorphism s that makes the following diagram commutative :

$$\begin{array}{ccccccc} & & & & P & & \\ & & & & \downarrow \text{id}_P & & \\ & & & \swarrow s & & & \\ 0 & \longrightarrow & M & \longrightarrow & N & \xrightarrow{\beta} & P \longrightarrow 0. \end{array}$$

Therefore, s is a section of β and the sequence splits by Lemma 5.3.

$3 \Rightarrow 4$. If M is a k -module and $\pi : M \rightarrow P$ is a surjection, then $0 \rightarrow \text{Ker } \pi \rightarrow M \xrightarrow{\pi} P \rightarrow 0$ is exact, and by hypothesis splits. Thus, by Lemma 5.3, $M \simeq P \oplus \text{Ker } \pi$. Thus, P is (isomorphic to) a direct summand of M .

$4 \Rightarrow 5$. An module, in particular P , is a quotient of a free module. So this is obvious.

$5 \Rightarrow 1$. Let $f : P \rightarrow N$ and $g : M \rightarrow N$ be homomorphisms, where g is surjective. By assumption, there exists k -modules Q and F , with F free, such that $F = Q \oplus P$. Define $\varphi : F \rightarrow N$ by setting $\varphi|_Q = 0$ and $\varphi|_P = f$. Since F is free, it is also projective, thus there exists $\tilde{\varphi} : F \rightarrow M$ such that $\varphi = g \circ \tilde{\varphi}$. We have constructed a diagram

$$\begin{array}{ccccc} F & \xleftarrow{\iota_P} & P & & \\ \tilde{\varphi} \downarrow & \searrow \varphi & \downarrow f & & \\ M & \xrightarrow{g} & N & \longrightarrow & 0 \end{array}$$

where the two triangles commute. Now define $\tilde{f} := \tilde{\varphi} \circ \iota_P$ and clearly we have $g \circ \tilde{f} = f$ as required. $\square$

5.3. Dual modules.

Definition 5.7. – The dual of a k -module M is the k -module $M^D := \text{Hom}_k(M, k)$.

Explicitly, the action of k on M^D is given by $(\lambda \cdot \varphi)(m) := \lambda \varphi(m) = \varphi(\lambda \cdot m)$ for all $\lambda \in k$, $\varphi \in \text{Hom}_k(M, k)$, and all $m \in M$.

Lemma 5.8. – *If M is a finitely generated projective k -module, then so is M^D .*

Proof. – Let F be a free module of finite rank such that $\pi : F \rightarrow M$ is a surjection. Then, by Lemma 5.6, $F \simeq M \oplus N$ for some k -module N . Thus,

$$F^D = \text{Hom}_k(F, k) \simeq \text{Hom}_k(M, k) \oplus \text{Hom}_k(N, k) = M^D \oplus N^D.$$

Since F is free of finite rank, so is F^D : just take the “dual basis” of any finite basis of F . Therefore, M^D is also finitely generated (being quotient of a finitely generated free module) and projective. $\square$

Now suppose $\varphi \in \text{Hom}_k(M, N)$. Then we have an induced map $\varphi^D \in \text{Hom}_k(N^D, M^D)$ defined by $\varphi^D(\alpha) := \alpha \circ \varphi$ for all $\alpha \in N^D$. Thus, we have an induced map $\varphi^{DD} \in \text{Hom}_k(M^{DD}, N^{DD})$, where for a k -module M , its *double dual* is denoted by $M^{DD} := (M^D)^D$. Also, it is easy to see that for composable homomorphisms φ, ψ we have $(\varphi \circ \psi)^D = \psi^D \circ \varphi^D$ and so $(\varphi \circ \psi)^{DD} = \varphi^{DD} \circ \psi^{DD}$.

Also, for every k -module M , we have a canonical evaluation map $\varepsilon_M : M \rightarrow M^{DD}$ given by $\varepsilon_M(m)(\alpha) := \alpha(m)$ for all $m \in M, \alpha \in M^D$.

Fact 5.9. – *The maps ε_M are natural in M i.e. for any k -module homomorphism $\varphi : M \rightarrow N$ the following diagram*

$$\begin{array}{ccc} M & \xrightarrow{\varepsilon_M} & M^{DD} \\ \varphi \downarrow & & \downarrow \varphi^{DD} \\ N & \xrightarrow{\varepsilon_N} & N^{DD} \end{array}$$

is commutative.

Proof. – We have to show that $\varepsilon_N \circ \varphi = \varphi^{DD} \circ \varepsilon_M$. Fix $m \in M$, and we show that $\varepsilon_N(\varphi(m)) = \varphi^{DD}(\varepsilon_M(m)) \in \text{Hom}_k(N^D, k)$. So take $\alpha \in N^D = \text{Hom}_k(N, k)$. We evaluate the left hand side at α to get :

$$\varepsilon_N(\varphi(m))(\alpha) = \alpha(\varphi(m)).$$

The right hand side evaluated at α gives :

$$\varphi^{DD}(\varepsilon_M(m))(\alpha) = (\varepsilon_M(m) \circ \varphi^D)(\alpha) = \varepsilon_M(m)(\alpha \circ \varphi) = \alpha(\varphi(m)).$$

$\square$

Lemma 5.10. – *If M is a finitely generated projective k -module, then the evaluation homomorphism $M \rightarrow M^{DD}$ is an isomorphism.*

Proof. – Let F be a free k -module of finite rank with a k -basis $(e_1, \dots, e_n)$. Then $\{e_i^* \mid i = 1, \dots, n\} \subset F^D$ defined by $e_i^*(e_j) := \delta_{ij}$ is also a k -basis of F^D . Similarly, $\{e_i^{**} \mid i = 1, \dots, n\}$ defined by $e_i^{**}(e_j^*) := \delta_{ij}$ is a k -basis of F^{DD} . It is then easy to see that $\varepsilon_F(e_i) = e_i^{**}$ and hence ε_F is a k -module isomorphism.

Now suppose M is a finitely generated projective k -module. Then there exists a free module F of finite rank and homomorphisms $\pi : F \rightarrow M, \iota : M \rightarrow F$ such that $\pi \circ \iota = \text{id}_M$. But then taking double duals we get $\pi^{DD} \circ \iota^{DD} = \text{id}_M^{DD} = \text{id}_{M^{DD}}$. By naturality of ε we have the following two squares

$$\begin{array}{ccc} M & \xrightarrow{\varepsilon_M} & M^{DD} \\ \iota \downarrow & & \downarrow \iota^{DD} \\ F & \xrightarrow{\varepsilon_F} & F^{DD} \end{array} \quad \begin{array}{ccc} M & \xrightarrow{\varepsilon_M} & M^{DD} \\ \pi \uparrow & & \uparrow \pi^{DD} \\ F & \xrightarrow{\varepsilon_F} & F^{DD} \end{array}$$

which commute. Hence $\iota^{DD} \circ \varepsilon_M = \varepsilon_F \circ \iota$ is injective since both ε_F and ι are injective. Thus ε_M is also injective. On the other hand, $\varepsilon_M \circ \pi = \pi^{DD} \circ \varepsilon_F$ is surjective since both ε_F and π^{DD} are surjective. Therefore, ε_M is surjective as well. $\square$

Lemma 5.11. – *Let M, N be finitely generated projective k -modules. Then, we have*

$$\text{Hom}_k(M, N) \simeq \text{Hom}_k(N^D, M^D).$$

Proof. – We exhibit an explicit isomorphism

$$\Phi : \text{Hom}_k(M, N) \xrightarrow{\sim} \text{Hom}_k(N^D, M^D)$$

defined by

$$\Phi(\varphi)(\beta) := \varphi^D(\beta) = \beta \circ \varphi, \quad \text{for } \varphi \in \text{Hom}_k(M, N) \text{ and } \beta \in N^D.$$

Now given $\psi \in \text{Hom}_k(N^D, M^D)$ we have its dual

$$\psi^D : M^{DD} \rightarrow N^{DD}, \quad \psi^D(F) := F \circ \psi.$$

We define

$$\Psi : \text{Hom}_k(N^D, M^D) \xrightarrow{\sim} \text{Hom}_k(M, N)$$

by

$$\Psi(\psi) := \varepsilon_N^{-1} \circ \psi^D \circ \varepsilon_M \in \text{Hom}_k(M, N)$$

where $\varepsilon_M : M \xrightarrow{\sim} M^{DD}$ and $\varepsilon_N : N \xrightarrow{\sim} N^{DD}$ are the canonical isomorphisms of Lemma 5.10.

We first show that $\Psi(\Phi(\varphi)) = \varphi$ for every $\varphi \in \text{Hom}_k(M, N)$. Thus, for every $m \in M$, we have to show

$$(\Psi(\Phi(\varphi)))(m) = m \in N.$$

We show that their images under ε_N are equal. Take any $\beta \in N^D$ and we evaluate

$$\begin{aligned} \varepsilon_N((\Psi(\Phi(\varphi)))(m))(\beta) &= \beta((\Psi(\Phi(\varphi)))(m)) = \beta(\Psi(\varphi^D)(m)) \\ &= \beta(\varepsilon_N^{-1} \circ \varphi^{DD} \circ \varepsilon_M(m)) \\ &= \beta(\varepsilon_N^{-1} \circ \varepsilon_N \circ \varphi(m)) \\ &= \beta(\varphi(m)) \\ &= \varepsilon_N(\varphi(m))(\beta) \end{aligned}$$

i.e. $\varepsilon_N((\Psi(\Phi(\varphi)))(m)) = \varepsilon_N(\varphi(m))$ and so $\Psi(\Phi(\varphi)) = \varphi$.

Next, we show that $\Phi(\Psi(\psi)) = \psi$ for all $\psi \in \text{Hom}_k(N^D, M^D)$. Thus, for every $\beta \in N^D$ and $m \in M$ we compute

$$\begin{aligned} (\Phi(\Psi(\psi)))(\beta)(m) &= (\beta \circ \Psi(\psi))(m) \\ &= \beta(\Psi(\psi)(m)) \\ &= \varepsilon_N(\Psi(\psi)(m))(\beta) \\ &= ((\varepsilon_N \circ \varepsilon_N^{-1} \circ \psi^D \circ \varepsilon_M)(m))(\beta) \\ &= (\psi^D(\varepsilon_M(m)))(\beta) \\ &= \psi(\beta)(m). \end{aligned}$$

This completes the proof. □

We now prove two crucial properties of tensor products.

Fact 5.12. – For any k -modules M, N , and P we have

$$\text{Hom}_k(M, \text{Hom}_k(N, P)) \simeq \text{Hom}_k(M \otimes_k N, P).$$

Proof. – We define a natural map

$$\Phi : \text{Hom}_k(M, \text{Hom}_k(N, P)) \rightarrow \text{Hom}_k(M \otimes_k N, P)$$

by

$$\Phi(\alpha)(m \otimes n) := \alpha(m)(n) \quad \text{for all } m \in M, n \in N.$$

Clearly, Φ is k -linear. We define its inverse

$$\Psi : \text{Hom}_k(M \otimes_k N, P) \rightarrow \text{Hom}_k(M, \text{Hom}_k(N, P))$$

by

$$\Psi(\beta)(m)(n) := \beta(m \otimes n).$$

It is easy to check that Φ and Ψ are inverses of each other. □

Fact 5.13. – Let $M_1 \xrightarrow{\varphi_1} M_2 \xrightarrow{\varphi_2} M_3 \rightarrow 0$ be an exact sequence of k -modules. Then, for any k -module N the sequence

$$M_1 \otimes_k N \xrightarrow{\varphi_1 \otimes \text{id}_N} M_2 \otimes_k N \xrightarrow{\varphi_2 \otimes \text{id}_N} M_3 \otimes_k N \rightarrow 0$$

is exact as well.

Proof. – Let N, P be any k -modules. Using Lemma 5.1 we obtain an exact sequence

$$0 \rightarrow \text{Hom}_k(M_3, \text{Hom}_k(N, P)) \rightarrow \text{Hom}_k(M_2, \text{Hom}_k(N, P)) \rightarrow \text{Hom}_k(M_1, \text{Hom}_k(N, P)).$$

By the previous fact this is equivalent to the sequence

$$0 \rightarrow \text{Hom}_k(M_3 \otimes_k N, P) \rightarrow \text{Hom}_k(M_2 \otimes_k N, P) \rightarrow \text{Hom}_k(M_1 \otimes_k N, P)$$

which is exact for all P . Now using Fact 5.2 we conclude that

$$M_1 \otimes_k N \rightarrow M_2 \otimes_k N \rightarrow M_3 \otimes_k N \rightarrow 0$$

is exact. $\square$

We now return again to duals. The following result is important.

Proposition 5.14. – *Let M and N be k -modules. There is a homomorphism of k -modules $\zeta : M^D \otimes_k N \rightarrow \text{Hom}_k(M, N)$, which is natural in M , such that $\zeta(\alpha \otimes n)(m) = \alpha(m)n$ for all $m \in M, n \in N$, and $\alpha \in M^D$. If M is finitely generated and projective, then ζ is an isomorphism.*

Proof. – The map $M^D \times N \rightarrow \text{Hom}_k(M, N)$ given by $(\alpha, n) \mapsto [m \mapsto \alpha(m)n]$ is bilinear and k -balanced. Thus we have the induced map $\zeta : M^D \otimes_k N \rightarrow \text{Hom}_k(M, N)$. We verify that this is natural in M . Suppose $\varphi : M' \rightarrow M$ is a k -module homomorphism, and we have to show that the following square

$$\begin{array}{ccc} M^D \otimes_k N & \xrightarrow{\zeta_M} & \text{Hom}_k(M, N) \\ \varphi^D \otimes \text{id}_N \downarrow & & \downarrow \varphi^* \\ M'^D \otimes_k N & \xrightarrow{\zeta_{M'}} & \text{Hom}_k(M', N) \end{array}$$

is commutative. Take a pure tensor $\alpha \otimes n$ and we see that

$$\varphi^* (\zeta_M(\alpha \otimes n)) = \varphi^* ([m \mapsto \alpha(m)n]) = [m' \mapsto \alpha(\varphi(m'))n]$$

and on the other hand we have

$$\zeta_{M'} ((\varphi^D \otimes \text{id}_N) (\alpha \otimes n)) = \zeta_{M'} ((\alpha \circ \varphi) \otimes n) = [m' \mapsto \alpha(\varphi(m'))n].$$

This completes the proof of naturality in M .

First, suppose M is free with a finite basis $(e_i)_{i \in I}$, so that M^D is also free with dual basis $(e_i^*)_{i \in I}$, and every element of $M^D \otimes_k N$ can be written as $\sum_{i \in I} e_i^* \otimes n_i$ for some unique³ $n_i \in N$. Then we have

$$\zeta \left(\sum_{i \in I} e_i^* \otimes n_i \right) (e_j) = \sum_{i \in I} (e_i^*(e_j) n_i) = n_j.$$

Now since M is free, we know $\text{Hom}_k(M, N) \simeq \bigoplus_{i \in I} N$, where an element $(n_i)_{i \in I}$ corresponds to the unique homomorphism $M \rightarrow N$ that takes e_i to n_i . Thus, $(n_i)_{i \in I}$ corresponds to $\zeta(\sum_{i \in I} e_i^* \otimes n_i)$. This proves that ζ is a bijection.

Now assume that M is finitely generated and projective. So there exists a finitely generated free module F and homomorphisms $\pi : F \rightarrow M$, $\iota : M \rightarrow F$ such that $\pi \circ \iota = \text{id}_M$. By naturality we get the following two commuting squares :

$$\begin{array}{ccc} M^D \otimes_k N & \xrightarrow{\zeta_M} & \text{Hom}_k(M, N) \\ \pi^D \otimes \text{id}_N \downarrow & & \downarrow \pi^* \\ F^D \otimes_k N & \xrightarrow{\zeta_F} & \text{Hom}_k(F, N) \end{array} \quad \begin{array}{ccc} M^D \otimes_k N & \xrightarrow{\zeta_M} & \text{Hom}_k(M, N) \\ \iota^D \otimes \text{id}_N \uparrow & & \uparrow \iota^* \\ F^D \otimes_k N & \xrightarrow{\zeta_F} & \text{Hom}_k(F, N) \end{array}$$

Now $\pi \circ \iota = \text{id}_M$ implies $\iota^D \circ \pi^D = \text{id}_{F^D}$. Thus π^D is injective and ι^D is surjective. Also since $(\iota^D \otimes \text{id}_N) \circ (\pi^D \otimes \text{id}_N) = (\iota^D \circ \pi^D) \otimes \text{id}_N = \text{id} \otimes \text{id}$ we conclude that $\pi^D \otimes \text{id}_N$ is injective and $\iota^D \otimes \text{id}_N$ is surjective. Finally, $(\pi \circ \iota)^* = \iota^* \circ \pi^* = \text{id}$ which implies π^* is injective and ι^* is surjective. So, the commutativity of the first square gives $\pi^* \circ \zeta_M = \zeta_F \circ (\pi^D \otimes \text{id})$ where the right side is injective as ζ_F is an isomorphism. Thus

³Note that for a free k -module F with basis $(e_i)_{i \in I}$, we have $F \otimes_k N \simeq \bigoplus_{i \in I} N$ where the isomorphism is given by $(\sum_i \lambda_i e_i) \otimes n \mapsto (\lambda_i n)_{i \in I}$, and its inverse is $(n_i)_{i \in I} \mapsto \sum_i e_i \otimes n_i$.

ζ_M is injective. The second square gives $\zeta_M \circ (\iota^D \otimes \text{id}) = \iota^* \circ \zeta_F$, where the right side is surjective, and so ζ_M is surjective. This completes the proof that ζ_M is an isomorphism when M is finitely generated and projective. $\square$

There are important corollaries to the above proposition.

Corollary 5.15. – *Let M be a finitely generated projective k -module. There is an isomorphism $M \otimes_k N \simeq \text{Hom}_k(M^D, N)$.*

Proof. – Since $\varepsilon_M : M \xrightarrow{\sim} M^{DD}$ is an isomorphism, we have

$$M \otimes_k N \simeq M^{DD} \otimes_k N \simeq \text{Hom}_k(M^D, N).$$

$\square$

Corollary 5.16. – *Let M, N be finitely generated projective k -modules. There is an isomorphism $M^D \otimes_k N^D \simeq (M \otimes_k N)^D$.*

Proof. – By the above proposition we have

$$M^D \otimes_k N^D \simeq \text{Hom}_k(M, N^D) = \text{Hom}_k(M, \text{Hom}_k(N, k)) \simeq \text{Hom}_k(M \otimes_k N, k) = (M \otimes_k N)^D$$

where the last isomorphism follows from Fact 5.12. $\square$

We finally show that duals commute with forming base change.

Corollary 5.17. – *Let M be a finitely generated projective k -module and k' is a k -algebra. Then we have*

$$(M \otimes_k k')^D = \text{Hom}_{k'}(M \otimes_k k', k') \simeq M^D \otimes_k k'$$

as k' -modules.

Proof. – First we show that

$$\text{Hom}_{k'}(M \otimes_k k', k') \simeq \text{Hom}_k(M, k').$$

Define a map

$$\Phi : \text{Hom}_{k'}(M \otimes_k k', k') \rightarrow \text{Hom}_k(M, k')$$

by

$$\Phi(F)(m) := F(m \otimes 1) \quad \text{for all } m \in M.$$

For a given $F \in \text{Hom}_{k'}(M \otimes_k k', k')$ this map $\Phi(F)$ is k -linear. Indeed, if $a \in k$ then

$$\Phi(F)(am) = F(am \otimes 1) = F(m \otimes a) = aF(m \otimes 1) = a\Phi(F)(m).$$

It is also easy to see that Φ is k' -linear where the k' -action on $\text{Hom}_{k'}(M \otimes_k k', k')$ and $\text{Hom}_k(M, k')$ are natural. Define the inverse

$$\Psi : \text{Hom}_k(M, k') \rightarrow \text{Hom}_{k'}(M \otimes_k k', k')$$

by

$$\Psi(g)(m \otimes \lambda) := \lambda g(m) \quad \text{for all } m \in M.$$

Now we check that $(\Phi \circ \Psi)(g)(m) = \Psi(g)(m \otimes 1) = g(m)$ for all $m \in M$. Also, $(\Psi \circ \Phi)(F)(m \otimes \lambda) = \lambda \Phi(F)(m) = \lambda F(m \otimes 1) = F(m \otimes \lambda)$ for all $m \in M$ and $\lambda \in k'$. Thus Φ and Ψ are inverses of each other.

Now using the proposition above, we have $M^D \otimes_k k' \simeq \text{Hom}_k(M, k')$ as k -modules. Here also it is easy to check that this isomorphism is k' -linear with the natural k' -action on $M^D \otimes_k k'$. Composing the above two k' -linear isomorphisms, we get the required isomorphism

$$M^D \otimes_k k' \simeq (M \otimes_k k')^D.$$

$\square$

6. FINITE GROUP FUNCTORS AND CARTIER DUALITY

Let Γ be a finite group. Can there be a representable functor F with the underlying association $A \rightsquigarrow \Gamma : \mathbf{Alg}_k \rightarrow \mathbf{Set}$? Not really!

Example 6.1. – There can not be a representable functor F with $F(A)$ having two elements for every algebra A . Indeed, suppose $F \simeq h^R$ for some k -algebra R . Then for $A = k \times k$ with the usual k -algebra structure satisfies $\mathrm{Hom}(R, k \times k) \simeq \mathrm{Hom}(R, k) \times \mathrm{Hom}(R, k)$. But this is a contradiction.

So let $R := k^\Gamma$ denote the k -algebra of all set maps from $\Gamma \rightarrow k$. For a $\sigma \in \Gamma$ we define the element $e_\sigma \in R$ by

$$e_\sigma(\tau) := \begin{cases} 1, & \tau = \sigma \\ 0, & \tau \neq \sigma \end{cases}.$$

Then $\{e_\sigma \mid \sigma \in \Gamma\}$ is a k -basis of R . Obviously, as k -algebras $R \simeq k \times \cdots \times k$ ($|\Gamma|$ -times). We also have $e_\sigma^2 = e_\sigma$, $e_\sigma e_\tau = 0$ for $\sigma \neq \tau$, and $\sum_{\sigma \in \Gamma} e_\sigma = 1_R$.

We now define $\Delta : R \rightarrow R \otimes R$ by $\Delta(e_\rho) := \sum_{\rho=\sigma\tau} (e_\sigma \otimes e_\tau)$. Also, define $S : R \rightarrow R$ by setting $S(e_\sigma) := e_{\sigma^{-1}}$ and $\epsilon : R \rightarrow k$ by $\epsilon(e_\sigma) = \begin{cases} 1, & \sigma = 1_\Gamma \\ 0, & \text{otherwise} \end{cases}$.

Fact 6.2. – The above defined maps Δ, S, ϵ are k -algebra maps.

Proof. – This is easy to verify for S and ϵ . So we check only for Δ . First, we check that Δ is unital. Indeed, $\Delta(1) = \Delta(\sum_\rho e_\rho) = \sum_\rho \sum_{\rho=\sigma\tau} e_\sigma \otimes e_\tau = \sum_{\sigma, \tau} e_\sigma \otimes e_\tau = (\sum_\sigma e_\sigma) \otimes (\sum_\tau e_\tau) = 1 \otimes 1$. Next, we check that Δ is multiplicative. It suffices to check this on basis elements i.e. we show that $\Delta(e_\rho e_{\rho'}) = \Delta(e_\rho) \Delta(e_{\rho'})$ for all ρ, ρ' . If $\rho \neq \rho'$ then $\Delta(e_\rho e_{\rho'}) = 0$. On the other hand,

$$\Delta(e_\rho) \Delta(e_{\rho'}) = \left(\sum_{\rho=\sigma\tau} e_\sigma \otimes e_\tau \right) \left(\sum_{\rho'=\sigma'\tau'} e_{\sigma'} \otimes e_{\tau'} \right) = \sum_{\rho=\sigma\tau, \rho'=\sigma'\tau'} e_\sigma e_{\sigma'} \otimes e_\tau e_{\tau'} = 0,$$

since if $\rho \neq \rho'$ then for each term of the sum involving $\sigma, \sigma', \tau, \tau'$, we have $\sigma \neq \sigma'$ or $\tau \neq \tau'$. If $\rho = \rho'$ then $\Delta(e_\rho^2) = \Delta(e_\rho)$ and $\Delta(e_\rho) \Delta(e_\rho) = \sum_{\rho=\sigma\tau} e_\sigma^2 \otimes e_\tau^2 = \sum_{\rho=\sigma\tau} e_\sigma \otimes e_\tau$. $\square$

Proposition 6.3. – The maps Δ, S, ϵ make $R = k^\Gamma$ a Hopf algebra.

Proof. – We verify the coassociativity axiom $(\mathrm{id} \otimes \Delta) \circ \Delta = (\Delta \otimes \mathrm{id}) \circ \Delta$ on the basis elements. Indeed, we have

$$((\mathrm{id} \otimes \Delta) \circ \Delta)(e_\rho) = \sum_{\rho=\sigma\tau} e_\sigma \otimes \Delta(e_\tau) = \sum_{\rho=\sigma\tau} e_\sigma \otimes \left(\sum_{\tau=\mu\nu} e_\mu \otimes e_\nu \right) = \sum_{\rho=\sigma\mu\nu} e_\sigma \otimes e_\mu \otimes e_\nu,$$

and on the other hand, we have

$$((\Delta \otimes \mathrm{id}) \circ \Delta)(e_\rho) = \sum_{\rho=\delta\gamma} \Delta(e_\delta) \otimes e_\gamma = \sum_{\rho=\delta\gamma} \left(\sum_{\delta=\alpha\beta} e_\alpha \otimes e_\beta \right) \otimes e_\gamma = \sum_{\rho=\alpha\beta\gamma} e_\alpha \otimes e_\beta \otimes e_\gamma.$$

Next, we verify the counit axiom $(\epsilon \otimes \mathrm{id}) \circ \Delta = C$, where $C : R \xrightarrow{\sim} k \otimes R$ is the canonical isomorphism. Evaluating on the basis element we get

$$((\epsilon \otimes \mathrm{id}) \circ \Delta)(e_\rho) = \sum_{\rho=\sigma\tau} \epsilon(e_\sigma) \otimes e_\tau = 1 \otimes e_\rho = C(e_\rho)$$

as required.

Finally, we verify the antipode axiom : $T \circ \epsilon = M \circ (S \otimes \mathrm{id}) \circ \Delta$ where $T : k \rightarrow R$ is the structure map and $M : R \otimes R \rightarrow R$ is induced by the ring multiplication. Again, we evaluate the right hand side on a basis element to get

$$(M \circ (S \otimes \mathrm{id}) \circ \Delta)(e_\rho) = M \left(\sum_{\rho=\sigma\tau} S(e_\sigma) \otimes e_\tau \right) = \sum_{\rho=\sigma\tau} e_{\sigma^{-1}} e_\tau.$$

Now if $\rho \neq 1_\Gamma$ then each term in the above sum is zero since $e_{\sigma^{-1}}e_\tau = 1$ precisely when $\tau = \sigma^{-1}$ i.e. $\rho = \sigma\tau = 1$. If $\rho = 1$ then for each $\sigma \in \Gamma$ we have a summand with $\tau = \sigma^{-1}$, whence the sum is $\sum_\sigma e_\sigma^2 = \sum_\sigma e_\sigma = 1_R$. Thus

$$(M \circ (S \otimes \text{id}) \circ \Delta)(e_\rho) = \delta_{\rho, 1_\Gamma} \cdot 1_R.$$

On the other hand, we have

$$T(\epsilon(e_\rho)) = \epsilon(e_\rho) \cdot 1_R = \delta_{\rho, 1_\Gamma} \cdot 1_R.$$

The proof is complete. $\square$

Definition 6.4. – The affine group functor corresponding to k^Γ is called *constant group functor* for Γ , which is denoted again by Γ if there is no scope of confusion.

Now we justify the use of the word “constant”.

Proposition 6.5. – Let A be a k -algebra with no idempotents other than 0 and 1. Then we have an isomorphism between Γ and $\text{Hom}_{k\text{-Alg}}(R, A)$.

Proof. – Define a function $\Gamma \rightarrow \text{Hom}_{k\text{-Alg}}(R, A)$ by $\sigma \mapsto \varphi_\sigma$ where $\varphi_\sigma(f) := f(\sigma) \cdot 1_A$ for $f \in R := k^\Gamma$. Clearly, φ_σ is a k -algebra map. The map $\sigma \mapsto \varphi_\sigma$ is injective since for $\sigma \neq \tau$ we have $\varphi_\sigma(e_\sigma) = 1_A \neq 0_A = \varphi_\tau(e_\sigma)$ and so $\varphi_\sigma \neq \varphi_\tau$. Now suppose $\psi : R \rightarrow A$ is a k -algebra map. Set $a_\sigma := \psi(e_\sigma)$. Then for each σ we have $a_\sigma^2 = a_\sigma$. Thus $a_\sigma = 0$ or 1. But if $a_\sigma = 1$ and $a_\tau = 1$ for some $\sigma \neq \tau$ then, since $a_\sigma a_\tau = 0$ for all $\sigma \neq \tau$, we have a contradiction. Thus is some $a_\sigma = 1$ then for all $\tau \neq \sigma$ we have $a_\tau = 0$. Also, some a_σ must be 1 since $\sum_\sigma a_\sigma = \psi(\sum_\sigma e_\sigma) = \psi(1) = 1$. So for this σ we have $\psi = \varphi_\sigma$.

Now we show that $\sigma \mapsto \varphi_\sigma$ is a group homomorphism. We compute

$$(\varphi_\sigma * \varphi_\tau)(e_\rho) = (M_A \circ (\varphi_\sigma \otimes \varphi_\tau) \circ \Delta_R)(e_\rho) = \sum_{\rho=\alpha\beta} \varphi_\sigma(e_\alpha) \varphi_\tau(e_\beta) = \sum_{\rho=\alpha\beta} \delta_{\sigma,\alpha} \delta_{\tau,\beta} 1_A = \begin{cases} 1_A & \text{if } \rho = \sigma\tau \\ 0 & \text{otherwise} \end{cases}.$$

Thus we have shown $\varphi_\sigma * \varphi_\tau = \varphi_{\sigma\tau}$. Hence $\Gamma \simeq \text{Hom}_{k\text{-Alg}}(k^\Gamma, A)$ as groups if A has no idempotents other than 0 and 1. $\square$

Remark 6.6. – If k is a field and Γ is a finite group, then if G is the functor corresponding to the Hopf algebra k^Γ , we have $\Gamma \simeq G(k)$. This is the analog of the classical dictum “finite groups are algebraic groups”.

Next we study a slight generalization of the notion of constant group functors.

Definition 6.7. – Let (R, Δ, ϵ, S) be a Hopf algebra over k . If R is a finitely generated projective k -module, then the corresponding affine group functor is called *finite group functor*.

Remark 6.8. – Thus constant group functor for a finite group Γ is obviously a finite group functor.

Now we fix a Hopf algebra (R, Δ, ϵ, S) such that R is a finitely generated projective k -module. Assume also that the corresponding group functor G is abelian, which means Δ is cocommutative i.e. $\Delta = \tau \circ \Delta$ where $\tau : R \otimes R \rightarrow R \otimes R$ is the swap homomorphism.

We consider the k -module R^D and at first put a k -algebra structure on it. First, chasing through the chain of isomorphisms in the proof of Corollary 5.16 we note that the canonical isomorphism

$$\alpha_2 : R^D \otimes_k R^D \xrightarrow{\sim} (R \otimes_k R)^D$$

is given by

$$\alpha_2(\varphi \otimes \psi)(x \otimes y) = \varphi(x)\psi(y) \quad \text{for all } x, y \in R \text{ and } \varphi, \psi \in R^D.$$

Similarly the canonical isomorphism

$$\alpha_3 : R^D \otimes_k R^D \otimes_k R^D \xrightarrow{\sim} (R \otimes_k R \otimes_k R)^D$$

is given by

$$\alpha_3(\varphi \otimes \psi \otimes \chi)(x \otimes y \otimes z) = \varphi(x)\psi(y)\chi(z) \quad \text{for all } x, y, z \in R \text{ and } \varphi, \psi, \chi \in R^D.$$

Now we define the ring multiplication on R^D by

$$M_{R^D} := \Delta^D \circ \alpha_2 : R^D \otimes_k R^D \rightarrow R^D.$$

For $r \in R$, writing $\Delta(r) = \sum_i u_i \otimes v_i$, and $\varphi, \psi \in R^D$ this is explicitly given by

$$(\varphi \star \psi)(r) := M_{R^D}(\varphi \otimes \psi)(r) = (\alpha_2(\varphi \otimes \psi) \circ \Delta)(r) = \sum_i \varphi(u_i) \psi(v_i).$$

Clearly, $\varphi \star \psi : R \rightarrow k$ is k -linear and hence $\varphi \star \psi \in R^D$. Similarly, we define the unit element of R^D as ϵ . Indeed, for $\varphi \in R^D$ we have

$$(\varphi \star \epsilon)(r) = \sum_i \varphi(u_i) \epsilon(v_i) = \varphi \left(\sum_i \epsilon(v_i) u_i \right),$$

and the counit axiom $(\text{id} \otimes \epsilon) \circ \Delta = C$ applied to r gives $\sum_i u_i \otimes \epsilon(v_i) = r \otimes 1$. Note that $C : R \xrightarrow{\sim} R \otimes_k k$ is the canonical isomorphism, and if we apply C^{-1} to the above equality, we obtain

$$\sum_i \epsilon(v_i) u_i = r,$$

and so we have

$$(\varphi \star \epsilon)(r) = \varphi(r) \quad \text{for all } r \in R.$$

Thus $\varphi \star \epsilon = \varphi$ and similarly we can show that $\epsilon \star \varphi = \varphi$.

We will now show that $\star$ is associative. This means we have to show that

$$M_{R^D} \circ (M_{R^D} \otimes \text{id}) = M_{R^D} \circ (\text{id} \otimes M_{R^D}).$$

First, fix a $r \in R$ and write

$$\Delta(r) = \sum_p s_p \otimes t_p$$

and then we also write

$$\Delta(s_p) = \sum_q a_{pq} \otimes b_{pq} \quad \text{and} \quad \Delta(t_p) = \sum_{q'} c_{pq'} \otimes d_{pq'}.$$

Fix a triple tensor $\varphi \otimes \psi \otimes \chi$ and compute the left hand side

$$\begin{aligned} (M_{R^D} (M_{R^D} (\varphi \otimes \psi) \otimes \chi))(r) &= \sum_p M_{R^D}(\varphi \otimes \psi)(s_p) \chi(t_p) \\ &= \sum_p \left(\sum_q \varphi(a_{pq}) \psi(b_{pq}) \right) \chi(t_p) \\ &= \sum_{p,q} \varphi(a_{pq}) \psi(b_{pq}) \chi(t_p). \end{aligned}$$

Whereas the right hand side is

$$\begin{aligned} (M_{R^D} (\varphi \otimes M_{R^D} (\psi \otimes \chi)))(r) &= \sum_p \varphi(s_p) M_{R^D}(\psi \otimes \chi)(t_p) \\ &= \sum_p \varphi(s_p) \sum_{q'} \psi(c_{pq'}) \chi(d_{pq'}) \\ &= \sum_{p,q'} \varphi(s_p) \psi(c_{pq'}) \chi(d_{pq'}). \end{aligned}$$

Now by coassociativity of Δ we know

$$(\Delta \otimes \text{id}) \circ \Delta = (\text{id} \otimes \Delta) \circ \Delta$$

and applying this to r we obtain the equality

$$\begin{aligned} \sum_p \Delta(s_p) \otimes t_p &= \sum_p \left(\sum_q a_{pq} \otimes b_{pq} \right) \otimes t_p = \sum_{p,q} a_{pq} \otimes b_{pq} \otimes t_p \\ &= \sum_p s_p \otimes \Delta(t_p) = \sum_{p,q'} s_p \otimes c_{pq'} \otimes d_{pq'}. \end{aligned}$$

Applying $\alpha_3(\varphi \otimes \psi \otimes \chi)$ to the above equality, we obtain

$$\sum_{p,q} \varphi(a_{pq})\psi(b_{pq})\chi(t_p) = \sum_{p,q'} \varphi(s_p)\psi(c_{pq'})\chi(d_{pq'})$$

and this completes the proof of associativity of the ring multiplication $\star$. Consequently, we have the k -algebra (R^D, M_{R^D}, ϵ) .

Next, we show that R^D is a commutative ring i.e. $\varphi \star \psi = \psi \star \varphi$ for all $\varphi, \psi \in R^D$. Let $r \in R$ and write $\Delta(r) = \sum_i u_i \otimes v_i = \tau(\Delta(r)) = \sum_i v_i \otimes u_i$. Applying $\varphi \star \psi$ and $\psi \star \varphi$ to r using both the presentations of $\Delta(r)$ we obtain the desired equality.

Now we define a Hopf structure on R^D . First, we define the comultiplication on R^D to be

$$\Delta_{R^D} := \alpha_2^{-1} \circ M_R^D : R^D \rightarrow R^D \otimes_k R^D.$$

We first note a defining property of Δ_{R^D} . Suppose $\varphi \in R^D$, and write

$$\Delta_{R^D}(\varphi) = \sum_i f_i \otimes g_i.$$

Thus, for any $u, v \in R$ we have

$$\alpha_2(\Delta_{R^D}(\varphi))(u \otimes v) = M_R^D(\varphi)(u \otimes v)$$

which gives

$$\sum_i f_i(u)g_i(v) = \varphi(uv).$$

Now we check that Δ_{R^D} is a k -algebra homomorphism. We already know that it is k -linear, so it remains to check that it is unital and respects the ring multiplication $\star$ on R^D . For $\varphi, \psi \in R^D$ we first show that

$$\Delta_{R^D}(\varphi \star \psi) = \Delta_{R^D}(\varphi) \cdot \Delta_{R^D}(\psi).$$

So it suffices to show that their images under α_2 are same. Fix $x, y \in R$, and write

$$\Delta(x) = \sum_a p_a \otimes q_a, \quad \text{and} \quad \Delta(y) = \sum_b r_b \otimes s_b$$

so that

$$\Delta(xy) = \Delta(x)\Delta(y) = \sum_{a,b} p_a r_b \otimes q_a s_b.$$

Using $\alpha_2 \circ \Delta_{R^D} = M_R^D$, we get

$$\begin{aligned} \alpha_2(\Delta_{R^D}(\varphi \star \psi))(x \otimes y) &= (\varphi \star \psi)(xy) \\ &= \alpha_2(\varphi \otimes \psi)\Delta(xy) \\ &= \sum_{a,b} \varphi(p_a r_b) \psi(q_a s_b). \end{aligned}$$

On the other hand write

$$\Delta_{R^D}(\varphi) = \sum_i f_i \otimes g_i \quad \text{and} \quad \Delta_{R^D}(\psi) = \sum_j c_j \otimes d_j,$$

so that

$$\Delta_{R^D}(\varphi) \cdot \Delta_{R^D}(\psi) = \sum_{i,j} (f_i \star c_j) \otimes (g_i \star d_j),$$

and thus for all $x, y \in R$ we have

$$\begin{aligned}
\alpha_2 (\Delta_{R^D}(\varphi) \cdot \Delta_{R^D}(\psi)) (x \otimes y) &= \sum_{i,j} (f_i \star c_j)(x) \cdot (g_i \star d_j)(y) \\
&= \sum_{i,j} \left(\sum_a f_i(p_a) c_j(q_a) \right) \left(\sum_b g_i(r_b) d_j(s_b) \right) \\
&= \sum_{a,b} \left(\sum_i f_i(p_a) g_i(r_b) \right) \left(\sum_j c_j(q_a) d_j(s_b) \right).
\end{aligned}$$

Now by the defining relation we know that

$$\sum_i f_i(p_a) g_i(r_b) = \varphi(p_a r_b) \quad \text{and} \quad \sum_j c_j(q_a) d_j(s_b) = \psi(q_a s_b).$$

Thus, we obtain

$$\alpha_2 (\Delta_{R^D}(\varphi) \cdot \Delta_{R^D}(\psi)) (x \otimes y) = \sum_{a,b} \left(\sum_i f_i(p_a) g_i(r_b) \right) \left(\sum_j c_j(q_a) d_j(s_b) \right) = \sum_{a,b} \varphi(p_a r_b) \psi(q_a s_b)$$

as expected. This completes the proof that Δ_{R^D} respects the ring multiplication on R^D .

Now we show that Δ_{R^D} is unital i.e. $\Delta_{R^D}(\epsilon) = \epsilon \otimes \epsilon$, where $\epsilon \in R^D$ is the multiplicative unit w.r.t. $\star$. Thus it suffices to show that $\alpha_2(\epsilon \otimes \epsilon) = M_R^D(\epsilon) = \epsilon \circ M_R$. So take any $x, y \in R$ and we obtain

$$\alpha_2(\epsilon \otimes \epsilon)(x \otimes y) = \epsilon(x)\epsilon(y) = \epsilon(xy) = \epsilon(M_R(x \otimes y)).$$

This completes the proof that Δ_{R^D} is a ring homomorphism.

Next we will verify that Δ_{R^D} is coassociative i.e.

$$(\Delta_{R^D} \otimes \text{id}) \circ \Delta_{R^D} = (\text{id} \otimes \Delta_{R^D}) \circ \Delta_{R^D}.$$

As in the previous paragraph, we fix a $\varphi \in R^D$ and show that

$$\alpha_3 (((\Delta_{R^D} \otimes \text{id}) \circ \Delta_{R^D})(\varphi)) = \alpha_3 (((\text{id} \otimes \Delta_{R^D}) \circ \Delta_{R^D})(\varphi)).$$

So fix arbitrary $x, y, z \in R$, write $\Delta_{R^D}(\varphi) = \sum_i f_i \otimes g_i$, and first compute the left hand side :

$$\begin{aligned}
\alpha_3 (((\Delta_{R^D} \otimes \text{id}) \circ \Delta_{R^D})(\varphi)) (x \otimes y \otimes z) &= \alpha_3 ((\Delta_{R^D} \otimes \text{id})(\Delta_{R^D}(\varphi))) (x \otimes y \otimes z) \\
&= \alpha_3 \left((\Delta_{R^D} \otimes \text{id}) \left(\sum_i f_i \otimes g_i \right) \right) (x \otimes y \otimes z) \\
&= \alpha_3 \left(\sum_i \Delta_{R^D}(f_i) \otimes g_i \right) (x \otimes y \otimes z) \\
&= \sum_i (\alpha_2 (\Delta_{R^D}(f_i))) (x \otimes y) \cdot g_i(z) \\
&= \sum_i f_i(xy) g_i(z) \\
&= \varphi(xyz).
\end{aligned}$$

On the other hand, we can similarly check that

$$\begin{aligned}
\alpha_3(((\text{id} \otimes \Delta_{\mathbf{R}^D}) \circ \Delta_{\mathbf{R}^D})(\varphi))(x \otimes y \otimes z) &= \alpha_3\left(\sum_i f_i \otimes \Delta_{\mathbf{R}^D}(g_i)\right)(x \otimes y \otimes z) \\
&= \sum_i f_i(x) \cdot (\alpha_2(\Delta_{\mathbf{R}^D}(g_i)))(y \otimes z) \\
&= \sum_i f_i(x) g_i(yz) \\
&= \varphi(xyz).
\end{aligned}$$

Note that in both of the above verifications we have crucially used the defining relation mentioned in the beginning of the previous paragraph. This completes the proof of coassociativity of $\Delta_{\mathbf{R}^D}$.

Now we define the counit and coinverse of $\mathbf{R}^D$ as

$$\epsilon_{\mathbf{R}^D} := \beta \circ \mathbf{T}_{\mathbf{R}}^D : \mathbf{R}^D \rightarrow k \quad \text{and} \quad \mathbf{S}_{\mathbf{R}^D} := \mathbf{S}^D : \mathbf{R}^D \rightarrow \mathbf{R}^D$$

where $\beta : k^D = \text{Hom}_k(k, k) \xrightarrow{\sim} k$ is the canonical k -module isomorphism which is given by $\beta(f) = f(1)$. These two maps are k -linear and thus it remains to show that they are ring homomorphisms. First, we consider $\epsilon_{\mathbf{R}^D}$. Note that for any $\varphi \in \mathbf{R}^D$ we have

$$\epsilon_{\mathbf{R}^D}(\varphi) = \beta(\varphi \circ \mathbf{T}_{\mathbf{R}}) = (\varphi \circ \mathbf{T}_{\mathbf{R}})(1) = \varphi(1_{\mathbf{R}}).$$

Thus, for $\varphi, \psi \in \mathbf{R}^D$, using $\Delta(1_{\mathbf{R}}) = 1_{\mathbf{R}} \otimes 1_{\mathbf{R}}$, we get

$$\epsilon_{\mathbf{R}^D}(\varphi \star \psi) = (\varphi \star \psi)(1_{\mathbf{R}}) = \varphi(1_{\mathbf{R}})\psi(1_{\mathbf{R}}) = \epsilon_{\mathbf{R}^D}(\varphi)\epsilon_{\mathbf{R}^D}(\psi).$$

The fact that $\epsilon_{\mathbf{R}^D}$ is unital is clear : $\epsilon_{\mathbf{R}^D}(\epsilon) = \epsilon(1) = 1$.

Now we consider $\mathbf{S}^D$. We will first show that $\mathbf{S}^D$ is unital. Indeed, we have

$$\mathbf{S}^D(\epsilon) = \epsilon \circ \mathbf{S} = \epsilon$$

since interpreting this at the level of groups it is equivalent to $i \circ e = e$ which simply says for each algebra A the inverse of the identity in $G(A)$ is the identity itself. Now take $\varphi, \psi \in \mathbf{R}^D$ and we want to show that

$$\mathbf{S}^D(\varphi \star \psi) = \mathbf{S}^D(\varphi) \star \mathbf{S}^D(\psi) \iff (\varphi \star \psi) \circ \mathbf{S} = (\varphi \circ \mathbf{S}) \star (\psi \circ \mathbf{S}).$$

So we fix any $r \in \mathbf{R}$ and write

$$\Delta(r) = \sum_i u_i \otimes v_i.$$

Now note that $\Delta \circ \mathbf{S} = (\mathbf{S} \otimes \mathbf{S}) \circ \tau \circ \Delta$, where $\tau : \mathbf{R} \otimes \mathbf{R} \rightarrow \mathbf{R} \otimes \mathbf{R}$ is the swapping map. This follows from the fact that the left hand side corresponds to the composite $(x, y) \mapsto xy \mapsto (xy)^{-1} = y^{-1}x^{-1}$ which can also be written as the composite $(x, y) \mapsto (x^{-1}, y^{-1}) \mapsto (y^{-1}, x^{-1}) \mapsto y^{-1}x^{-1}$. Thus we obtain

$$\Delta(\mathbf{S}(r)) = (\mathbf{S} \otimes \mathbf{S})(\tau(\Delta(r))) = \sum_i \mathbf{S}(v_i) \otimes \mathbf{S}(u_i).$$

Consequently, we get

$$\mathbf{S}^D(\varphi \star \psi)(r) = \sum_i \varphi(\mathbf{S}(v_i))\psi(\mathbf{S}(u_i)).$$

On the other hand, since G is abelian, we know $\tau \circ \Delta = \Delta$, and so

$$\Delta(r) = \sum_i u_i \otimes v_i = \sum_i v_i \otimes u_i.$$

Using the later presentation of $\Delta(r)$, we have

$$(\mathbf{S}^D(\varphi) \star \mathbf{S}^D(\psi))(r) = \sum_i \mathbf{S}^D(\varphi)(v_i)\mathbf{S}^D(\psi)(u_i) = \sum_i \varphi(\mathbf{S}(v_i))\psi(\mathbf{S}(u_i)).$$

Thus, we have shown that $\mathbf{S}^D$ is a ring homomorphism.

Now we will check that the counit and coinverse axioms hold i.e.

$$(\epsilon_{\mathbf{R}^D} \otimes \text{id}_{\mathbf{R}^D}) \circ \Delta_{\mathbf{R}^D} = C_{\mathbf{R}^D} \quad \text{and} \quad \mathbf{T}_{\mathbf{R}^D} \circ \epsilon_{\mathbf{R}^D} = M_{\mathbf{R}^D} \circ (\mathbf{S}_{\mathbf{R}^D} \otimes \text{id}_{\mathbf{R}^D}) \circ \Delta_{\mathbf{R}^D}.$$

We first note that the structure map $T_{R^D} : k \rightarrow R^D$ is equal to $\epsilon^D \circ \beta^{-1}$. Indeed, for any $\lambda \in k$, we get

$$(\epsilon^D \circ \beta^{-1})(\lambda) = \beta^{-1}(\lambda) \circ \epsilon$$

and so for any $r \in R$ we obtain $(\beta^{-1}(\lambda) \circ \epsilon)(r) = \lambda \epsilon(r) = (\lambda \cdot \epsilon)(r)$ i.e. $(\epsilon^D \circ \beta^{-1})(\lambda) = \lambda \cdot \epsilon = T_{R^D}(\lambda)$.

Now we will verify the counit axiom i.e. $(\epsilon_{R^D} \otimes \text{id}_{R^D}) \circ \Delta_{R^D} = C_{R^D}$. Take any $\varphi \in R^D$ and write $\Delta_{R^D}(\varphi) = \sum_i f_i \otimes g_i$. So we get

$$((\epsilon_{R^D} \otimes \text{id}_{R^D}) \circ \Delta_{R^D})(\varphi) = \sum_i \epsilon_{R^D}(f_i) \otimes g_i = \sum_i f_i(1) \otimes g_i \in k \otimes R^D.$$

Now any $\lambda \otimes \psi \in k \otimes R^D$ corresponds to the k -linear map $[r \mapsto \lambda \psi(r)] \in R^D$ and thus the above sum corresponds to $[r \mapsto \sum_i f_i(1)g_i(r)]$, which is the map $\alpha_2(\sum_i f_i \otimes g_i)$ applied on $1 \otimes r$. But we know

$$\alpha_2\left(\sum_i f_i \otimes g_i\right) = (\alpha_2 \circ \Delta_{R^D})(\varphi) = M_R^D(\varphi)$$

and so $M_R^D(\varphi)(1 \otimes r) = \varphi(r)$. Thus $\sum_i f_i(1) \otimes g_i$ corresponds to φ .

Next, we will verify the coinverse axiom i.e. $(T_{R^D} \circ \epsilon_{R^D})(\varphi) = (M_{R^D} \circ (S_{R^D} \otimes \text{id}_{R^D}) \circ \Delta_{R^D})(\varphi)$ for every $\varphi \in R^D$. So we fix an $r \in R$ and we compute the action of the left hand side :

$$(T_{R^D}(\epsilon_{R^D}(\varphi)))(r) = (T_{R^D}(\varphi(1_R)))(r) = (\varphi(1_R) \cdot \epsilon)(r) = \varphi(1_R)\epsilon(r).$$

Next, we compute the action of right hand side on r . But first, we observe that for any $\theta \otimes \eta$ one has

$$(\alpha_2(S^D \otimes \text{id})(\theta \otimes \eta))(x \otimes y) = (\alpha_2(S^D(\theta) \otimes \eta))(x \otimes y) = \theta(S(x))\eta(y) = (\alpha_2(\theta \otimes \eta))(S(x) \otimes y)$$

for every $x \otimes y$. Equivalently, for every $\Psi \in (R \otimes R)^D$ we have

$$\alpha_2((S^D \otimes \text{id})(\alpha_2^{-1}(\Psi))) = \Psi \circ (S \otimes \text{id}).$$

Now set $\Psi := \alpha_2(\Delta_{R^D}(\varphi))$, and noting that $M_{R^D} = \Delta^D \circ \alpha_2$, we obtain

$$(M_{R^D} \circ (S_{R^D} \otimes \text{id}_{R^D}) \circ \Delta_{R^D})(\varphi) = \Delta^D(\Psi \circ (S \otimes \text{id})).$$

Writing $\Delta(r) = \sum_j u_j \otimes v_j$, and evaluating the right hand side of the above equation at r we obtain

$$\sum_j \Psi(S(u_j) \otimes v_j) = \sum_j \varphi(S(u_j)v_j) = \varphi\left(\sum_j S(u_j)v_j\right).$$

But the antipode axiom for R , which is $M_R \circ (S \otimes \text{id}) \circ \Delta = T_R \circ \epsilon$ applied to r yields

$$\sum_j S(u_j)v_j = \epsilon(r) \cdot 1_R.$$

Therefore $\varphi\left(\sum_j S(u_j)v_j\right) = \epsilon(r)\varphi(1_R)$. This proves the antipode axiom.

Finally, we show that R^D is also a cocommutative Hopf algebra, which is equivalent to showing that $G^D := \text{Hom}(R^D, -)$ is an abelian group functor. First note that for any $\Phi \in R^D \otimes R^D$ writing $\Phi = \sum_j f_j \otimes g_j$, we have

$$(\alpha_2(\tau_{R^D}(\Phi)))(x \otimes y) = \left(\alpha_2\left(\sum_j g_j \otimes f_j\right)\right)(x \otimes y) = \sum_j g_j(x)f_j(y)$$

for all $x \otimes y$. On the other hand

$$(\alpha_2(\Phi) \circ \tau_{R \otimes R})(x \otimes y) = \alpha_2(\Phi)(y \otimes x) = \sum_j f_j(y)g_j(x).$$

Thus, we have shown

$$(\alpha_2 \circ \tau_{R^D})(\Phi) = \alpha_2(\Phi) \circ \tau_{R \otimes R}$$

for all Φ , and since α_2 is an isomorphism, the above is equivalent to saying

$$\alpha_2 \circ \tau_{R^D} \circ \alpha_2^{-1} = (\cdot) \circ \tau_{R \otimes R}$$

as operators on $(R \otimes R)^D$.

Now take an arbitrary $\varphi \in R^D$ and compute

$$\begin{aligned} (\alpha_2 \circ \tau_{R^D} \circ \alpha_2^{-1} \circ M_R^D)(\varphi) &= M_R^D(\varphi) \circ \tau_{R \otimes R} \\ &= \varphi \circ (M_R \circ \tau_{R \otimes R}) \\ &= \varphi \circ M_R \\ &= M_R^D(\varphi) \end{aligned}$$

as R is a commutative ring which means $M_R = M_R \circ \tau_{R \otimes R}$. So we conclude that

$$\alpha_2 \circ \tau_{R^D} \circ \alpha_2^{-1} \circ M_R^D = M_R^D.$$

Apply α_2^{-1} to obtain the equality

$$\tau_{R^D} \circ \alpha_2^{-1} \circ M_R^D = \alpha_2^{-1} \circ M_R^D.$$

But $\Delta_{R^D} = \alpha_2^{-1} \circ M_R^D$ be definition, thus we finally get

$$\tau_{R^D} \circ \Delta_{R^D} = \Delta_{R^D}$$

i.e. Δ_{R^D} is cocommutative.

We summarize all the above work in the following theorem.

Theorem 6.9. – *Let G be a finite abelian group functor represented by the Hopf algebra (R, Δ, ϵ, S) . Then $(R^D, \Delta_{R^D}, \epsilon_{R^D}, S_{R^D})$ represents another (dual) finite abelian group functor G^D . Furthermore, $(G^D)^D \simeq G$ and $\text{Hom}(G, H) \simeq \text{Hom}(H^D, G^D)$.*

Proof. – Only the second statement requires a proof. We first show that $G^{DD} \simeq G$ by showing that

$$\varepsilon_R : R \rightarrow R^{DD}$$

is a Hopf isomorphism. Since it is already known that it is a k -linear bijection, it only remains to show that ε_R is a ring homomorphism. Take $r, s \in R$, and we must show that

$$\varepsilon_R(rs) = \varepsilon_R(r) \star \varepsilon_R(s).$$

So consider any $\varphi \in R^D$ and write $\Delta_{R^D}(\varphi) = \sum_i f_i \otimes g_i$; we evaluate both sides of the above equality at φ . The left side gives

$$\varepsilon_R(rs)(\varphi) = \varphi(rs).$$

Whereas the right side gives

$$\begin{aligned} (\varepsilon_R(r) \star \varepsilon_R(s))(\varphi) &= \sum_i \varepsilon_R(r)(f_i) \varepsilon_R(s)(g_i) \\ &= \sum_i f_i(r) g_i(s) \\ &= \varphi(rs). \end{aligned}$$

Next we show that ε_R is unital. The unit in R^{DD} is ϵ_{R^D} . We have to show $\varepsilon_R(1) = \epsilon_{R^D}$. Take any φ and evaluate both sides at φ . The left side gives $\varepsilon_R(1)(\varphi) = \varphi(1)$, whereas the right side gives $\epsilon_{R^D}(\varphi) = \varphi(1)$ by definition. This shows that $R \simeq R^{DD}$ as k -algebras. It remains to show that ε_R preserves comultiplication i.e.

$$\Delta_{R^{DD}} \circ \varepsilon_R = (\varepsilon_R \otimes \varepsilon_R) \circ \Delta_R.$$

So for every $r \in R$ we must show that $(\Delta_{R^{DD}} \circ \varepsilon_R)(r) = ((\varepsilon_R \otimes \varepsilon_R) \circ \Delta_R)(r) \in R^{DD} \otimes R^{DD}$. Consider $\alpha_2 : R^{DD} \otimes R^{DD} \xrightarrow{\sim} (R^D \otimes R^D)^D$ and it suffices to show that the images of the above two expressions under α_2 are equal. Take $\varphi \otimes \psi \in R^D \otimes R^D$ and write $\Delta_R(r) = \sum_j u_j \otimes v_j$, and evaluate :

$$\alpha_2(\Delta_{R^{DD}}(\varepsilon_R(r)))(\varphi \otimes \psi) = M_{R^D}^D(\varepsilon_R(r))(\varphi \otimes \psi) = \varepsilon_R(r)(\varphi \star \psi) = (\varphi \star \psi)(r) = \sum_j \varphi(u_j) \psi(v_j).$$

On the other hand the right hand side yields

$$\alpha_2 \left(\sum_j \varepsilon_R(u_j) \otimes \varepsilon_R(v_j) \right) (\varphi \otimes \psi) = \sum_j \varepsilon_R(u_j)(\varphi) \varepsilon_R(v_j)(\psi) = \sum_j \varphi(u_j) \psi(v_j)$$

as required. This completes the proof that $R \simeq R^{\text{DD}}$ as Hopf algebras, and thus $G \simeq G^{\text{DD}}$.

Next, we show that $\text{Hom}(G, H) \simeq \text{Hom}(H^{\text{D}}, G^{\text{D}})$ where H is another finite abelian group functor represented by a Hopf algebra R' . But note that $\text{Hom}(G, H)$ corresponds to $\text{Hom}_{\text{Hopf}}(R', R)$ whereas $\text{Hom}(H^{\text{D}}, G^{\text{D}})$ corresponds to $\text{Hom}_{\text{Hopf}}(R^{\text{D}}, R'^{\text{D}})$. So we just show that

$$\text{Hom}_{\text{Hopf}}(R', R) \simeq \text{Hom}_{\text{Hopf}}(R^{\text{D}}, R'^{\text{D}}).$$

Consider a map $f \in \text{Hom}_{\text{Hopf}}(R', R)$, and its dual $f^{\text{D}} : R^{\text{D}} \rightarrow R'^{\text{D}}$. We first show that f^{D} is a Hopf algebra homomorphism.

We begin by first showing that f^{D} is a ring homomorphism. So let $\varphi, \psi \in R^{\text{D}}$, and we must show that

$$f^{\text{D}}(\varphi \star \psi) = f^{\text{D}}(\varphi) \star f^{\text{D}}(\psi).$$

Take any $r' \in R'$ and write $\Delta_{R'}(r') = \sum_i u_i \otimes v_i$. Then we have

$$(f^{\text{D}}(\varphi) \star f^{\text{D}}(\psi))(r') = \sum_i f^{\text{D}}(\varphi)(u_i) f^{\text{D}}(\psi)(v_i) = \sum_i \varphi(f(u_i)) \psi(f(v_i)).$$

On the other hand, since f is a Hopf algebra map, we know $\Delta_R \circ f = (f \otimes f) \circ \Delta_{R'}$. Applying this to r' we get

$$\Delta_R(f(r')) = \sum_i f(u_i) \otimes f(v_i).$$

Thus we have

$$f^{\text{D}}(\varphi \star \psi)(r') = (\varphi \star \psi)(f(r')) = \sum_i \varphi(f(u_i)) \psi(f(v_i)).$$

Consequently, $f^{\text{D}}(\varphi \star \psi) = f^{\text{D}}(\varphi) \star f^{\text{D}}(\psi)$ as asserted. The fact that f^{D} is unital is rather easy to see : $f^{\text{D}}(\epsilon_R) = \epsilon_R \circ f = \epsilon_{R'}$ since f is a Hopf algebra map (see Remark 3.14). This concludes the proof that f^{D} is a ring homomorphism, and since f^{D} is k -linear it is a k -algebra map. So it remains to show that f^{D} preserves comultiplication i.e.

$$\Delta_{R'^{\text{D}}} \circ f^{\text{D}} = (f^{\text{D}} \otimes f^{\text{D}}) \circ \Delta_{R^{\text{D}}}.$$

Apply $\alpha_2^{R'} : R'^{\text{D}} \otimes R'^{\text{D}} \xrightarrow{\sim} (R' \otimes R')^{\text{D}}$ to both sides. We first show that $\alpha_2^{R'} \circ (f^{\text{D}} \otimes f^{\text{D}}) \circ \Delta_{R^{\text{D}}} = ((f \otimes f)^{\text{D}}) \circ (\Delta_R)^{\text{D}}$. Since $\Delta_{R^{\text{D}}} = (\alpha_2^R)^{-1} \circ \Delta_R$, it suffices to show

$$\alpha_2^{R'} \circ (f^{\text{D}} \otimes f^{\text{D}}) \circ (\alpha_2^R)^{-1} = (f \otimes f)^{\text{D}}.$$

So we take any arbitrary $T \in (R \otimes R)^{\text{D}}$ and we have to show $(\alpha_2^{R'} \circ (f^{\text{D}} \otimes f^{\text{D}}) \circ (\alpha_2^R)^{-1})(T) = (f \otimes f)^{\text{D}}(T)$ as elements of $(R' \otimes R')^{\text{D}}$. So take any $r' \otimes s' \in R' \otimes R'$ and evaluate the two sides of the previous equality at $r' \otimes s'$. Now we have

$$(f \otimes f)^{\text{D}}(T)(r' \otimes s') = T(f(r') \otimes f(s')).$$

On the other hand, writing $(\alpha_2^R)^{-1}(T) = \sum_l \varphi_l \otimes \psi_l$, we get

$$(\alpha_2^{R'} \circ (f^{\text{D}} \otimes f^{\text{D}}) \circ (\alpha_2^R)^{-1})(T) = \alpha_2^{R'} \left(\sum_l f^{\text{D}}(\varphi_l) \otimes f^{\text{D}}(\psi_l) \right).$$

Therefore, we have

$$\left(\alpha_2^{R'} \left(\sum_l f^{\text{D}}(\varphi_l) \otimes f^{\text{D}}(\psi_l) \right) \right)(r' \otimes s') = \sum_l f^{\text{D}}(\varphi_l)(r') f^{\text{D}}(\psi_l)(s') = \sum_l \varphi_l(f(r')) \psi_l(f(s')).$$

But also we have

$$T(f(r') \otimes f(s')) = \alpha_2^R \left(\sum_l \varphi_l \otimes \psi_l \right)(f(r') \otimes f(s')) = \sum_l \varphi_l(f(r')) \psi_l(f(s')).$$

This concludes the proof of the equality : $\alpha_2^R \circ (f^D \otimes f^D) \circ \Delta_{R^D} = ((f \otimes f)^D) \circ (M_R)^D$.

Now we return to the proof of $\Delta_{R^D} \circ f^D = (f^D \otimes f^D) \circ \Delta_{R^D}$. Applying α_2^R to both sides, we have to show that

$$M_{R'}^D \circ f^D = ((f \otimes f)^D) \circ M_R^D.$$

Thus, we have to show that $M_{R'}^D(f^D(\varphi)) = (f \otimes f)^D(M_R^D(\varphi))$ for any $\varphi \in R^D$. The equality is supposed to hold in $(R' \otimes R')^D$. So take any $r' \otimes s' \in R' \otimes R'$ and evaluate at it. The left hand side gives

$$M_{R'}^D(f^D(\varphi))(r' \otimes s') = f^D(\varphi)(r's') = \varphi(f(r's')).$$

The right hand side gives

$$(f \otimes f)^D(M_R^D(\varphi))(r' \otimes s') = M_R^D(\varphi)(f(r') \otimes f(s')) = \varphi(f(r')f(s')) = \varphi(f(r's'))$$

since f is a k -algebra map, in particular a ring homomorphism. This completes the proof of the fact that f^D is a Hopf algebra homomorphism as claimed.

Thus we get a legitimate map

$$f \mapsto f^D : \text{Hom}_{\text{Hopf}}(R', R) \rightarrow \text{Hom}_{\text{Hopf}}(R^D, R'^D).$$

We must show that this is a bijection. Since the evaluation map ε_R is natural in R we have

$$\varepsilon_R^{-1} \circ f^{DD} \circ \varepsilon_{R'} = f.$$

This already shows that $f \mapsto f^D$ is an injection, since if $g \mapsto g^D = f^D$, then $f^{DD} = g^{DD}$ and by the above equality we get $f = g$. For showing surjectivity, take any $h \in \text{Hom}_{\text{Hopf}}(R^D, R'^D)$ and consider $h^D : R'^{DD} \rightarrow R^{DD}$ which is again a Hopf algebra map. Define

$$f := \varepsilon_R^{-1} \circ h^D \circ \varepsilon_{R'} : R' \rightarrow R.$$

We have to show that $f^D = h$. Take arbitrary $\varphi \in R^D$ and $r' \in R'$, and first note that

$$f^D(\varphi)(r') = \varphi(f(r')) = \varphi(\varepsilon_R^{-1}(h^D(\varepsilon_{R'}(r')))).$$

Now note that by definition

$$\varepsilon_R(r)(\psi) = \psi(r), \quad \text{for all } r \in R \text{ and } \psi \in R^D.$$

Replacing $\varepsilon_R(r)$ by $\theta \in R^{DD}$ we have the equivalent defining property

$$\theta(\psi) = \psi(\varepsilon_R^{-1}(\theta)), \quad \text{for all } \theta \in R^{DD} \text{ and } \psi \in R^D.$$

So we have

$$f^D(\varphi)(r') = \varphi(\varepsilon_R^{-1}(h^D(\varepsilon_{R'}(r')))) = h^D(\varepsilon_{R'}(r'))(\varphi) = \varepsilon_{R'}(r')(h(\varphi)) = h(\varphi)(r').$$

Consequently, $f^D = h$ as required. This completes the proof of the theorem. $\square$

Proposition 6.10. – Let G be a finite abelian group functor with Hopf algebra R . Then $G^D(k)$ is isomorphic to the group-like elements of R , and hence isomorphic to $\text{Hom}(G, \mathbf{G}_m)$.

Proof. – Note that $G^D(k) \simeq \text{Hom}_{k\text{-Alg}}(R^D, k) \subset \text{Hom}_k(R^D, k) = R^{DD}$. So any $\varphi \in \text{Hom}_{k\text{-Alg}}(R^D, k)$ can be written as $\varepsilon_R(b)$ for a unique $b \in R$. We will show that b is group-like. But first we observe the following two facts :

Fact 6.11. – If M is a finitely generated projective k -module, then M^D separates points of M i.e. given $m \neq 0$, there exists $\varphi \in M^D$ such that $\varphi(m) \neq 0$.

Proof. – Since ε_M is an isomorphism, $\varepsilon_M(m) \neq 0$ and so there exists $\varphi \in M^D$ such that $\varepsilon_R(m)(\varphi) = \varphi(m) \neq 0$. $\square$

Fact 6.12. – If M is a finitely generated projective k -module, so is $M \otimes_k M$.

Proof. – Let F be a free k -module of finite rank with a surjection $\pi : F \rightarrow M$ having a section $\iota : M \rightarrow F$ i.e. $\pi \circ \iota = \text{id}_M$. Now if $(e_i)_{i=1}^n$ is a k -basis of F , we know $(e_i \otimes e_j)_{i,j}$ is a k -basis of $F \otimes_k F$ (it certainly spans; and if $\sum_{i,j} c_{ij}(e_i \otimes e_j) = 0$, then for each p, q we apply $\alpha_2(e_p^* \otimes e_q^*)$ to the relation to obtain $c_{p,q} = 0$). Then $\pi \otimes \pi : F \otimes_k F \rightarrow M \otimes_k M$ is a surjection with section $\iota \otimes \iota$. Hence $M \otimes_k M$ is a projective k -module. $\square$

Now we return to the main argument. It is given that $\varphi = \varepsilon_R(b)$ is a k -algebra homomorphism, and we have to show that b is group-like. Now $\varepsilon_R(b) : R^D \rightarrow k$ is a k -algebra homomorphism and in particular a ring homomorphism, thus, for $\psi, \chi \in R^D$, we have

$$(\psi \star \chi)(b) = \varepsilon_R(b)(\psi \star \chi) = \varepsilon_R(b)(\psi)\varepsilon_R(b)(\chi) = \psi(b)\chi(b).$$

Writing $\Delta(b) = \sum_j s_j \otimes t_j$ the right hand side gives

$$(\psi \star \chi)(b) = \sum_j \psi(s_j)\chi(t_j).$$

Thus we have shown that for every pure tensor $\psi \otimes \chi$ we have

$$\alpha_2(\psi \otimes \chi)(\Delta(b)) = \alpha_2(\psi \otimes \chi)(b \otimes b)$$

which implies

$$\Phi(\Delta(b)) = \Phi(b \otimes b), \quad \text{for all } \Phi \in (R \otimes R)^D.$$

But since $(R \otimes R)^D$ separates $R \otimes R$, the above equality implies

$$\Delta(b) = b \otimes b.$$

Also $\varphi(\epsilon_R) = 1$ since φ is a ring homomorphism and so it is unital. But this means $\epsilon_R(b) = 1$. Hence, by Remark 3.24 we conclude that b is group-like.

Conversely, suppose $b \in R^\times$ is group-like i.e. $\Delta(b) = b \otimes b$. We have to show that $\varepsilon_R(b)$ is a k -algebra homomorphism. By definition it is k -linear. So it remains to show that it is a unital ring homomorphism. Using the presentation $\Delta(b) = b \otimes b$ we obtain

$$\varepsilon_R(b)(\psi \star \chi) = \psi(b)\chi(b) = \varepsilon_R(b)(\psi)\varepsilon_R(b)(\chi).$$

Finally, $\varepsilon_R(b)(\epsilon_R) = \epsilon_R(b) = 1$ by Remark 3.23. This completes the proof of the proposition. $\square$

Example 6.13. – Let Γ be a finite abelian group. Set $R := k^\Gamma$ with a Hopf algebra structure where the comultiplication $\Delta : R \rightarrow R \otimes R$ is defined by

$$\Delta(e_\rho) := \sum_{\rho=\sigma\tau} e_\sigma \otimes e_\tau,$$

where the basis elements e_σ is given by

$$e_\sigma(\tau) = \begin{cases} 1, & \tau = \sigma \\ 0, & \tau \neq \sigma \end{cases}.$$

First we show that R is cocommutative i.e. $\tau \circ \Delta = \Delta$. Evaluating on basis elements we have :

$$\tau(\Delta(e_\rho)) = \tau\left(\sum_{\rho=\sigma\tau} e_\sigma \otimes e_\tau\right) = \sum_{\rho=\sigma\tau} e_\tau \otimes e_\sigma = \sum_{\rho=\tau\sigma} e_\tau \otimes e_\sigma = \Delta(e_\rho).$$

Note that the penultimate equality follows from the fact that Γ is abelian.

Now we want to show that R^D is isomorphic to $k[\Gamma]$ as a Hopf algebra. So we define

$$\Phi : k[\Gamma] \rightarrow R^D$$

on the basis elements by

$$\Phi(g) := \text{ev}_g,$$

where $\text{ev}_g : R \rightarrow k$ is the evaluation map given by $\text{ev}_g(f) := f(g)$ for all $f \in R = k^\Gamma$. We show that Φ is a ring homomorphism. Take $g, h \in \Gamma$. For any basis element $e_\rho \in R$, writing $\Delta(e_\rho) = \sum_{\rho=\sigma\tau} e_\sigma \otimes e_\tau$, we compute :

$$(\text{ev}_g \star \text{ev}_h)(e_\rho) = \sum_{\rho=\sigma\tau} \text{ev}_g(e_\sigma)\text{ev}_h(e_\tau) = \sum_{\rho=\sigma\tau} \delta_{\sigma,g}\delta_{\tau,h} = \delta_{\rho,gh} = \text{ev}_{gh}(e_\rho).$$

Note that the sum $\sum_{\rho=\sigma\tau} \delta_{\sigma,g} \delta_{\tau,h}$ equals 1 if and only if $\sigma = g, \tau = h$, and $\sigma\tau = gh = \rho$. Hence $\Phi(g) \star \Phi(h) = \Phi(gh)$ on generators and by linearity on all of $k[\Gamma]$.

We check that Φ is unital. Indeed, $\Phi(1_\Gamma) = \text{ev}_{1_\Gamma}$ and recall that $\epsilon : R \rightarrow k$ was defined by

$$\epsilon(e_\sigma) = \begin{cases} 1, & \sigma = 1_\Gamma \\ 0, & \text{otherwise} \end{cases}.$$

Thus $\text{ev}_{1_\Gamma}(e_\sigma) = e_\sigma(1_\Gamma) = \delta_{\sigma,1} = \epsilon(e_\sigma)$ i.e. $\text{ev}_{1_\Gamma} = \epsilon$.

Next, to show that Φ is a bijection it suffices to show that ev_g are linearly independent since both $R = k^\Gamma$ and R^D are free k -modules of rank $|\Gamma|$. Indeed, suppose we have

$$0 = \sum_{g \in \Gamma} a_g \text{ev}_g \quad (a_g \in k).$$

Evaluating this at the basis element e_h we get $a_h = 0$, and since h is arbitrary we have $a_g = 0$ for all $g \in \Gamma$.

Finally we show that Φ is a Hopf algebra homomorphism. Recall that the comultiplications are given by

$$\Delta_{k[\Gamma]}(g) = g \otimes g \quad (g \in \Gamma)$$

and

$$\Delta_{R^D} = \alpha_2^{-1} \circ M_R^D.$$

So we have to show that

$$\Delta_{R^D}(\text{ev}_g) = \Delta_{R^D}(\Phi(g)) = (\Phi \otimes \Phi)(\Delta_{k[\Gamma]}(g)) = \Phi(g) \otimes \Phi(g) = \text{ev}_g \otimes \text{ev}_g.$$

Since both sides are elements of $R^D \otimes R^D$ we apply α_2 to check the equality in $(R \otimes R)^D$. Thus we have

$$\alpha_2(\Delta_{R^D}(\text{ev}_g))(e_\sigma \otimes e_\tau) = \text{ev}_g(e_\sigma e_\tau).$$

On the other hand, we have

$$\alpha_2(\text{ev}_g \otimes \text{ev}_g)(e_\sigma \otimes e_\tau) = \text{ev}_g(e_\sigma) \text{ev}_g(e_\tau) = \text{ev}_g(e_\sigma e_\tau).$$

In conclusion, if Γ is a finite abelian group, then the dual of the constant group functor k^Γ is the finite abelian diagonalizable group functor $k[\Gamma]$.

Example 6.14. – If $\Gamma = \mathbf{Z}/n\mathbf{Z}$ then $k[\Gamma]$ is isomorphic as Hopf algebra to $\frac{k[X]}{(X^n-1)}$ (see Proposition 4.5 and the proof of Theorem 4.6) and thus the dual of $\mathbf{Z}/n\mathbf{Z}$ is μ_n .

Consequently, if Γ is a finite abelian group, then it is a product of various $\mathbf{Z}/n\mathbf{Z}$, and its dual is a diagonalizable group functor which is a product of various μ_n corresponding to the groups $\mathbf{Z}/n\mathbf{Z}$.

7. BASE CHANGE

Let G be an affine group functor with representing Hopf algebra (R, Δ, ϵ, S) . Fix any k -algebra A and we will construct a Hopf structure on the A -algebra $R \otimes_k A$. Note that the structure map $T_{R \otimes A} : A \rightarrow R \otimes_k A$ is given by $a \mapsto 1_R \otimes a$. Now it can be easily checked that the canonical map

$$\Phi : (R \otimes_k R) \otimes_k A \rightarrow (R \otimes_k A) \otimes_A (R \otimes_k A), \quad (r \otimes s) \otimes a \mapsto (r \otimes 1_R) \otimes (s \otimes a)$$

is an isomorphism of A -algebras, with inverse given by

$$\Psi : (R \otimes_k A) \otimes_A (R \otimes_k A) \rightarrow (R \otimes_k R) \otimes_k A, \quad (r \otimes a) \otimes (s \otimes b) \mapsto (r \otimes s) \otimes (ab).$$

We now define the comultiplication on $R \otimes_k A$ as

$$\Delta_{R \otimes A} : R \otimes_k A \rightarrow (R \otimes_k A) \otimes_A (R \otimes_k A)$$

given by

$$\Delta_{R \otimes A} := \Phi \circ (\Delta \otimes \text{id}_A).$$

Since both $\Delta \otimes \text{id}_A$ and Φ are A -linear, so is $\Delta_{R \otimes A}$. So it remains to prove that it is a ring homomorphism. But this again follows from the fact that Φ and $\Delta \otimes \text{id}_A$ are both ring homomorphisms. Now we check the coassociativity of $\Delta_{R \otimes A}$. Fix $r \in R$, $a \in A$ and write

$$\Delta(r) = \sum u_i \otimes v_i.$$

Then by definition

$$\Delta_{R \otimes A}(r \otimes a) = \sum_i (u_i \otimes 1) \otimes (v_i \otimes a).$$

Now writing

$$\Delta(u_i) = \sum_j u_{ij}^{(1)} \otimes u_{ij}^{(2)}$$

and then applying $\Delta_{R \otimes A} \otimes \text{id}_{R \otimes A}$ to the expression $\Delta_{R \otimes A}(r \otimes a)$ we obtain

$$((\Delta_{R \otimes A} \otimes \text{id}) \circ \Delta_{R \otimes A})(r \otimes a) = \sum_{i,j} (u_{ij}^{(1)} \otimes 1) \otimes (u_{ij}^{(2)} \otimes 1) \otimes (v_i \otimes a).$$

Next, after writing

$$\Delta(v_i) = \sum_k v_{ik}^{(1)} \otimes v_{ik}^{(2)},$$

we compute the action of $\text{id}_{R \otimes A} \otimes \Delta_{R \otimes A}$ on the same to obtain

$$((\text{id}_{R \otimes A} \otimes \Delta_{R \otimes A}) \circ \Delta_{R \otimes A})(r \otimes a) = \sum_{i,k} (u_i \otimes 1) \otimes (v_{ik}^{(1)} \otimes 1) \otimes (v_{ik}^{(2)} \otimes a).$$

However, coassociativity of Δ yields the equality

$$\sum_{i,j} u_{ij}^{(1)} \otimes u_{ij}^{(2)} \otimes v_i = \sum_{i,k} u_i \otimes v_{ik}^{(1)} \otimes v_{ik}^{(2)}.$$

Now apply the k -linear map

$$R \otimes_k R \otimes_k R \rightarrow (R \otimes_k A) \otimes_A (R \otimes_k A) \otimes_A (R \otimes_k A), x \otimes y \otimes z \mapsto (x \otimes 1) \otimes (y \otimes 1) \otimes (z \otimes a)$$

to the equality $\sum_{i,j} u_{ij}^{(1)} \otimes u_{ij}^{(2)} \otimes v_i = \sum_{i,k} u_i \otimes v_{ik}^{(1)} \otimes v_{ik}^{(2)}$ we get the equality

$$((\Delta_{R \otimes A} \otimes \text{id}) \circ \Delta_{R \otimes A})(r \otimes a) = ((\text{id}_{R \otimes A} \otimes \Delta_{R \otimes A}) \circ \Delta_{R \otimes A})(r \otimes a).$$

Thus coassociativity of $\Delta_{R \otimes A}$ is established.

Next, we define the counit on $R \otimes_k A$ as

$$\epsilon_{R \otimes A} : R \otimes_k A \rightarrow A, \quad r \otimes a \mapsto \epsilon_R(r) \cdot a.$$

It is easy to see that $\epsilon_{R \otimes A}$ is an A -algebra homomorphism. We verify

$$(\epsilon_{R \otimes A} \otimes \text{id}) \circ \Delta_{R \otimes A} = C.$$

Take any $r \otimes a$ and after writing $\Delta(r) = \sum_i u_i \otimes v_i$, apply the left hand side to get

$$\begin{aligned} (\epsilon_{R \otimes A} \otimes \text{id}) \left(\sum_i (u_i \otimes 1) \otimes (v_i \otimes a) \right) &= \sum_i (\epsilon_R(u_i) \cdot 1) \otimes (v_i \otimes a) \\ &= \sum_i 1_A \otimes ((\epsilon_R(u_i) \cdot v_i) \otimes a) \\ &= 1_A \otimes \left(\sum_i (\epsilon_R(u_i) \cdot v_i) \otimes a \right). \end{aligned}$$

But $\sum_i \epsilon_R(u_i) \cdot v_i = r$ by the counit axiom for R which gives the equality $\sum_i \epsilon_R(u_i) \otimes v_i = 1 \otimes R$ in $k \otimes R$, and then applying the canonical map $k \otimes R \rightarrow R$ we get the desired equality in R . Thus we have

$$((\epsilon_{R \otimes A} \otimes \text{id}) \circ \Delta_{R \otimes A})(r \otimes a) = 1_A \otimes (r \otimes a) = C(r \otimes a),$$

where $C : R \otimes_k A \xrightarrow{\sim} A \otimes_k (R \otimes_k A)$ is the canonical isomorphism. We have proved the counit axiom.

Lastly, we define the coinverse

$$S_{R \otimes A} : R \otimes A \rightarrow R \otimes A$$

by

$$r \otimes a \mapsto S(r) \otimes a \text{ i.e. } S_{R \otimes A} := S \otimes \text{id}_A.$$

Here also it is clear that $S_{R \otimes A}$ is A -linear. We check that the coinverse axiom holds i.e. :

$$M_{R \otimes A} \circ (S_{R \otimes A} \otimes \text{id}) \circ \Delta_{R \otimes A} = T_{R \otimes A} \circ \epsilon_{R \otimes A}.$$

So take any $r \otimes a \in R \otimes_k A$ and write $\Delta(r) = \sum_i u_i \otimes v_i$. Then we know that

$$\Delta_{R \otimes A}(r \otimes a) = \sum_i (u_i \otimes 1_A) \otimes (v_i \otimes a).$$

So the left hand side gives

$$(M_{R \otimes A} \circ (S_{R \otimes A} \otimes \text{id}) \circ \Delta_{R \otimes A})(r \otimes a) = \sum_i (S(u_i)v_i) \otimes a = \left(\sum_i S(u_i)v_i \right) \otimes a.$$

Now the coinverse axiom for R yields the equality $\sum_i S(u_i)v_i = \epsilon_R(r) \cdot 1_R$. Hence we have

$$\left(\sum_i S(u_i)v_i \right) \otimes a = (\epsilon_R(r) \cdot 1_R) \otimes a = 1_R \otimes (\epsilon_R(r) \cdot a).$$

On the other hand we have

$$(T_{R \otimes A} \circ \epsilon_{R \otimes A})(r \otimes a) = T_{R \otimes A}(\epsilon_R(r) \cdot a) = 1_R \otimes (\epsilon_R(r) \cdot a).$$

Thus the coinverse axiom is proved. We summarize this in the following fact.

Fact 7.1. – *Let (R, Δ, ϵ, S) be a Hopf algebra, and let A be any fixed k -algebra. Then $(R \otimes_k A, \Delta_{R \otimes A}, \epsilon_{R \otimes A}, S_{R \otimes A})$ is a Hopf algebra over A . The corresponding affine group functor is denoted by G_A which is the functor $B \rightsquigarrow \text{Hom}_{A\text{-Alg}}(R \otimes_k A, B)$.*

Proposition 7.2. – *There is an isomorphism of groups*

$$\text{Hom}_{A\text{-Alg}}(R \otimes_k A, B) \simeq \text{Hom}_{k\text{-Alg}}(R, B).$$

Proof. – We define $\Phi : \text{Hom}_{A\text{-Alg}}(R \otimes_k A, B) \rightarrow \text{Hom}_{k\text{-Alg}}(R, B)$ by $\varphi \mapsto \Phi(\varphi) := [r \mapsto \varphi(r \otimes 1_A)]$. If φ is an A -algebra homomorphism it can be easily checked that $\Phi(\varphi)$ is a k -algebra homomorphism. We define the inverse of Φ as the map $\Psi : \text{Hom}_{k\text{-Alg}}(R, B) \rightarrow \text{Hom}_{A\text{-Alg}}(R \otimes_k A, B)$ defined by $\psi \mapsto \Psi(\psi) := [r \otimes a \mapsto \psi(r)T_B(a)]$. Again it can be easily checked that $\Psi(\psi)$ is an A -algebra homomorphism. Finally, it is also straightforward to show that Φ and Ψ are inverses to each other.

We show that Φ is a group homomorphism. So given $\varphi_1, \varphi_2 \in \text{Hom}_{A\text{-Alg}}(R \otimes_k A, B)$ we have to show

$$\Phi(\varphi_1 * \varphi_2) = \Phi(\varphi_1) * \Phi(\varphi_2).$$

Take any $r \in R$ and write $\Delta(r) = \sum_i u_i \otimes v_i$; we compute the action of the left hand side at r first :

$$\begin{aligned} \Phi(\varphi_1 * \varphi_2)(r) &= (\varphi_1 * \varphi_2)(r \otimes 1) \\ &= M_B((\varphi_1 \otimes \varphi_2) \Delta_{R \otimes A}(r \otimes 1)) \\ &= M_B\left((\varphi_1 \otimes \varphi_2) \sum_i (u_i \otimes 1) \otimes_A (v_i \otimes 1)\right) \\ &= \sum_i \varphi_1(u_i \otimes 1) \varphi_2(v_i \otimes 1) \\ &= \sum_i \Phi(\varphi_1)(u_i) \Phi(\varphi_2)(v_i). \end{aligned}$$

Now the action of the right hand side :

$$(\Phi(\varphi_1) * \Phi(\varphi_2))(r) = M_B((\Phi(\varphi_1) \otimes \Phi(\varphi_2)) \Delta_R(r)) = \sum_i \Phi(\varphi_1)(u_i) \Phi(\varphi_2)(v_i).$$

This completes the proof of the asserted isomorphism. $\square$

As an application of the idea of base change, we again consider a finite *abelian* group functor G with representing Hopf algebra (R, Δ, ϵ, S) with R being a finitely generated k -module. We want to compute $G^D(A)$ for an arbitrary k -algebra A generalizing Proposition 6.10. First consider the group functor G_A which is represented by the A -algebra $R \otimes_k A$. Note that $R \otimes_k A$ is a finitely generated A -module, since if $r_1, \dots, r_n$ generate R over k , then $r_1 \otimes 1, \dots, r_n \otimes 1$ generate $R \otimes_k A$ over A . We also observe the following.

Fact 7.3. – *If G is finite abelian so is G_A .*

Proof. – This follows directly from Proposition 7.2, since $G_A(B) = \text{Hom}_{A\text{-Alg}}(R \otimes A, B) \simeq \text{Hom}_{k\text{-Alg}}(R, B) \simeq G(B)$.

However let us also see a Hopf algebraic proof. Let τ_A denote the flip map on $(R \otimes_k A) \otimes_A (R \otimes_k A)$. We have to show that

$$\Delta_{R \otimes A} = \tau_A \circ \Delta_{R \otimes A}.$$

Take any $r \otimes a \in R \otimes_k A$, and write $\Delta(r) = \sum_i u_i \otimes v_i$. Now since G is abelian we know $\Delta = \tau \circ \Delta$ and thus $\Delta(r) = \sum_i v_i \otimes u_i$. Consequently, we have

$$\Delta_{R \otimes A}(r \otimes a) = \sum_i (u_i \otimes 1) \otimes (v_i \otimes a) = \sum_i (v_i \otimes 1) \otimes (u_i \otimes a).$$

But since the tensor products above are in $(R \otimes_k A) \otimes_A (R \otimes_k A)$ by A -balancedness we have

$$(v_i \otimes 1) \otimes (u_i \otimes a) = (v_i \otimes a) \otimes (u_i \otimes 1)$$

and so we see that

$$\tau_A(\Delta_{R \otimes A}(r \otimes a)) = \Delta_{R \otimes A}(r \otimes a).$$

□

Thus we know that $(G_A)^D$ is represented by $(R \otimes_k A)^D = \text{Hom}_A(R \otimes_k A, A)$ which is a Hopf algebra over A by the constructions of the previous section. Now by Corollary 5.17 we know

$$\text{Hom}_A(R \otimes_k A, A) \simeq R^D \otimes_k A$$

as A -modules. We consider the canonical A -module homomorphism

$$\Phi : R^D \otimes_k A \rightarrow \text{Hom}_A(R \otimes_k A, A)$$

defined on pure tensors by

$$\Phi(\varphi \otimes \alpha)(r \otimes a) = \varphi(r) \cdot (\alpha a),$$

where $\varphi \in R^D$, $r \in R$, and $a, \alpha \in A$. We will prove that Φ is an A -module isomorphism. We need the following lemma.

Lemma 7.4. – *Let R be a finitely generated projective k -module. Then there exists a finite family $\{r_1, \dots, r_n\} \subset R$ and linear functionals $\{\rho_1, \dots, \rho_n\} \subset R^D$ such that*

$$\forall r \in R : \sum_{i=1}^n \rho_i(r) r_i = r.$$

Proof. – Suppose F is a finitely generated free k -module of rank n such that we have k -module homomorphisms

$$\pi : F \rightarrow R \text{ and } \iota : R \rightarrow F$$

satisfying $\pi \circ \iota = \text{id}_R$ i.e. π is a split surjection. Let $(e_i)_{i=1}^n$ be a k -basis of F and $(e_i^*)_{i=1}^n$ be its dual basis i.e. $e_i^*(e_j) = \delta_{ij}$ for all i, j . Define

$$r_i := \pi(e_i) \in R, \quad \rho_i := e_i^* \circ \iota \in R^D.$$

Now for any $r \in R$ we have

$$\sum_i \rho_i(r) r_i = \sum_i \rho_i(r) \pi(e_i) = \pi \left(\sum_i \rho_i(r) e_i \right) = \pi \left(\sum_i e_i^*(\iota(r)) e_i \right) = \pi(\iota(r)) = r$$

since for any $x \in F$ we know that $\sum_i e_i^*(x) e_i = x$. This completes the proof. □

Proposition 7.5. – *The canonical A -module homomorphism*

$$\Phi : R^D \otimes_k A \rightarrow \text{Hom}_A(R \otimes_k A, A)$$

defined on pure tensors by

$$\Phi(\varphi \otimes \alpha)(r \otimes a) = \varphi(r) \cdot (\alpha a),$$

where $\varphi \in R^D$, $r \in R$, and $a, \alpha \in A$, is an isomorphism.

Proof. – We construct the inverse of Φ written

$$\Psi : \text{Hom}_A(\mathbf{R} \otimes_k A, A) \rightarrow \mathbf{R}^D \otimes_k A$$

as follows : fix a dual system $(r_i, \rho_i)_{i=1}^n$ coming from the previous lemma, and for each $F \in \text{Hom}_A(\mathbf{R} \otimes_k A, A)$ we set

$$\Psi(F) := \sum_{i=1}^n \rho_i \otimes F(r_i \otimes 1_A).$$

Note that this particular Ψ depends on the dual system $(r_i, \rho_i)_{i=1}^n$.

Now we show that $\Phi \circ \Psi = \text{id}$. So take any $F \in \text{Hom}_A(\mathbf{R} \otimes_k A, A)$ and any $r \otimes a \in \mathbf{R} \otimes_k A$. Then we compute :

$$\begin{aligned} \Phi(\Psi(F))(r \otimes a) &= \Phi\left(\sum_{i=1}^n \rho_i \otimes F(r_i \otimes 1_A)\right)(r \otimes a) \\ &= \sum_i \rho_i(r) \cdot (F(r_i \otimes 1_A)a) \\ &= a \left(\sum_i \rho_i(r) \cdot F(r_i \otimes 1_A) \right) \\ &= aF\left(\sum_i \rho_i(r) \cdot (r_i \otimes 1_A)\right) \\ &= aF(r \otimes 1_A) \\ &= F(r \otimes a), \end{aligned}$$

where in the last equality the A -linearity of F is used and in the penultimate equality the k -linear map $\mathbf{R} \rightarrow \mathbf{R} \otimes_k A$ given by $r \mapsto r \otimes 1_A$ is applied to the equality $r = \sum_i \rho_i(r) \cdot r_i$.

Next, we show that $\Psi \circ \Phi = \text{id}$. So take any element $X \in \mathbf{R}^D \otimes_k A$ and write it as a finite sum $X = \sum_j \varphi_j \otimes \alpha_j$. Then

$$\begin{aligned} \Psi(\Phi(X)) &= \sum_{i=1}^n \rho_i \otimes \Phi(X)(r_i \otimes 1_A) \\ &= \sum_{i=1}^n \rho_i \otimes \left(\sum_j \varphi_j(r_i) \cdot \alpha_j \right) \\ &= \sum_j \left(\sum_{i=1}^n \varphi_j(r_i) \cdot \rho_i \right) \otimes \alpha_j. \end{aligned}$$

Now for any $r \in \mathbf{R}$ we have

$$\left(\sum_{i=1}^n \varphi_j(r_i) \cdot \rho_i \right) (r) = \sum_{i=1}^n \rho_i(r) \varphi_j(r_i) = \varphi_j \left(\sum_{i=1}^n \rho_i(r) r_i \right) = \varphi_j(r).$$

Thus $\sum_{i=1}^n \varphi_j(r_i) \cdot \rho_i = \varphi_j$. Hence $\Psi(\Phi(X)) = \sum_j \varphi_j \otimes \alpha_j = X$. This completes the proof that Φ is an isomorphism. $\square$

Proposition 7.6. – *The A -module isomorphism Φ in the previous proposition is a Hopf algebra homomorphism. Consequently, $(G_A)^D \simeq (G^D)_A$.*

Proof. – We first show that Φ is a ring homomorphism. First note that on $\mathbf{R}^D \otimes_k A$ the ring multiplication is defined by

$$(\varphi \otimes \alpha) \cdot (\psi \otimes \beta) = (\varphi \star \psi) \otimes (\alpha\beta)$$

where $\star$ is the ring multiplication on R^D from dualizing Δ_R . On the other hand, the ring multiplication on $(R \otimes A)^D := \text{Hom}_A(R \otimes_k A, A)$ is given by

$$(F \star_A G)(x) = \sum_i F(x_i^{(1)})G(x_i^{(2)}), \quad \forall F, G \in (R \otimes A)^D$$

where $\Delta_{R \otimes A}(x) = \sum_i x_i^{(1)} \otimes x_i^{(2)}$, and $\Delta_{R \otimes A}$ is the base changed comultiplication on the Hopf A -algebra $R \otimes_k A$. Take any $r \otimes a$ and write $\Delta_R(r) = \sum_i u_i \otimes v_i$ so that we have

$$\Delta_{R \otimes A}(r \otimes a) = \sum_i (u_i \otimes 1) \otimes (v_i \otimes a).$$

Then at first we compute

$$\begin{aligned} \Phi((\varphi \otimes \alpha) \cdot (\psi \otimes \beta))(r \otimes a) &= \Phi((\varphi \star \psi) \otimes (\alpha\beta))(r \otimes a) \\ &= (\varphi \star \psi)(r) \cdot (\alpha\beta a) \\ &= \left(\sum_i \varphi(u_i) \psi(v_i) \right) \cdot (\alpha\beta a). \end{aligned}$$

On the other hand we have

$$\begin{aligned} (\Phi(\varphi \otimes \alpha) \star_A \Phi(\psi \otimes \beta))(r \otimes a) &= \sum_i \Phi(\varphi \otimes \alpha)(u_i \otimes 1) \Phi(\psi \otimes \beta)(v_i \otimes a) \\ &= \sum_i (\varphi(u_i) \cdot \alpha) (\psi(v_i) \cdot (\beta a)) \\ &= \sum_i (\varphi(u_i) \psi(v_i)) \cdot (\alpha\beta a). \end{aligned}$$

Thus we have shown Φ respects ring multiplication. Also Φ is unital since the unity of $R^D \otimes_k A$ is $\epsilon_R \otimes 1_A$ and $\Phi(\epsilon_R \otimes 1_A)(r \otimes a) = \epsilon_R(r) \cdot a = \epsilon_{R \otimes A}(r \otimes a)$ i.e. $\Phi(\epsilon_R \otimes 1_A) = \epsilon_{R \otimes A}$.

Next, we show that Φ respects comultiplication i.e.

$$(\Phi \otimes \Phi) \circ \Delta_{R^D \otimes A} = \Delta_{(R \otimes A)^D} \circ \Phi.$$

Thus for any $\varphi \otimes \alpha \in R^D \otimes_k A$ we have to show

$$((\Phi \otimes \Phi) \circ \Delta_{R^D \otimes A})(\varphi \otimes \alpha) = (\Delta_{(R \otimes A)^D} \circ \Phi)(\varphi \otimes \alpha) \in (R \otimes_k A)^D \otimes_A (R \otimes_k A)^D.$$

Now writing $\alpha_2^{(R \otimes A)^D} : (R \otimes_k A)^D \otimes_A (R \otimes_k A)^D \xrightarrow{\sim} ((R \otimes_k A) \otimes_A (R \otimes_k A))^D$ as the canonical isomorphism, it suffices to show

$$\alpha_2^{(R \otimes A)^D} (((\Phi \otimes \Phi) \circ \Delta_{R^D \otimes A})(\varphi \otimes \alpha)) = \alpha_2^{(R \otimes A)^D} ((\Delta_{(R \otimes A)^D} \circ \Phi)(\varphi \otimes \alpha)).$$

So write $\Delta_{R^D}(\varphi) = \sum_i \varphi_i^{(1)} \otimes \varphi_i^{(2)}$. First we apply $\Phi \otimes \Phi$ to

$$\Delta_{R^D \otimes A}(\varphi \otimes \alpha) = \sum_i (\varphi_i^{(1)} \otimes 1) \otimes (\varphi_i^{(2)} \otimes \alpha)$$

to obtain

$$((\Phi \otimes \Phi) \circ \Delta_{R^D \otimes A})(\varphi \otimes \alpha) = \sum_i \Phi(\varphi_i^{(1)} \otimes 1) \otimes \Phi(\varphi_i^{(2)} \otimes \alpha).$$

Now take any $(r \otimes a) \otimes (s \otimes b) \in (R \otimes_k A) \otimes_A (R \otimes_k A)$ and compute :

$$\begin{aligned} \left(\alpha_2^{(R \otimes A)^D} (((\Phi \otimes \Phi) \circ \Delta_{R^D \otimes A})(\varphi \otimes \alpha)) \right) ((r \otimes a) \otimes (s \otimes b)) &= \sum_i \Phi(\varphi_i^{(1)} \otimes 1)(r \otimes a) \Phi(\varphi_i^{(2)} \otimes \alpha)(s \otimes b) \\ &= \sum_i \left(\varphi_i^{(1)}(r) \cdot a \right) \left(\varphi_i^{(2)}(s) \cdot (\alpha b) \right) \\ &= \left(\sum_i \varphi_i^{(1)}(r) \varphi_i^{(2)}(s) \right) \cdot (a \alpha b) \\ &= \varphi(rs) \cdot (a \alpha b). \end{aligned}$$

Now we evaluate $\alpha_2^{(R \otimes A)^D} ((\Delta_{(R \otimes A)^D} \circ \Phi)(\varphi \otimes \alpha))$ on $(r \otimes a) \otimes (s \otimes b)$. This gives

$$\begin{aligned} \left(\alpha_2^{(R \otimes A)^D} (\Delta_{(R \otimes A)^D} (\Phi(\varphi \otimes \alpha))) \right) ((r \otimes a) \otimes (s \otimes b)) &= (M_{R \otimes A}^D(\Phi(\varphi \otimes \alpha))) ((r \otimes a) \otimes (s \otimes b)) \\ &= \Phi(\varphi \otimes \alpha) (M_{R \otimes A}((r \otimes a) \otimes (s \otimes b))) \\ &= \Phi(\varphi \otimes \alpha) (rs \otimes ab) \\ &= \varphi(rs) \cdot (\alpha ab). \end{aligned}$$

This completes the proof. $\square$

Now we can compute $G^D(A)$ for an arbitrary algebra A when G is a finite abelian group functor.

Corollary 7.7. – *We have*

$$\begin{aligned} G^D(A) &\simeq \text{Hom}_{k\text{-Alg}}(R^D, A) \\ &\simeq \text{Hom}_{A\text{-Alg}}(R^D \otimes_k A, A) \\ &\simeq (G^D)_A(A) \\ &\simeq (G_A)^D(A) \\ &\simeq \{\text{group-like elements of } R \otimes A\}. \end{aligned}$$

8. REPRESENTATIONS

Definition 8.1. – Let G be a group functor and X be a set functor. An *action* of G on X is a natural transformation $G \times X \rightarrow X$ such that the maps $G(A) \times X(A) \rightarrow X(A)$ are group actions for every k -algebra A .

Now fix a k -module V , and let X be the functor $A \rightsquigarrow V \otimes_k A$. Then $X(A)$ is naturally an A -module for any k -algebra A . So, for any group functor G acting on X , if additionally the action of $G(A)$ on $X(A)$ is A -linear for every k -algebra A , then we say that we have a *linear representation* of G on V .

Now consider the assignment $GL_V : A \rightsquigarrow \text{Aut}_A(V \otimes_k A)$. We want to make this a functor, so we must understand what is a natural way to define $GL_V(\varphi)$ for any algebra map $\varphi : A \rightarrow B$. First observe that there is a canonical B -module isomorphism

$$\beta : (V \otimes_k A) \otimes_A B \rightarrow V \otimes_k B, \quad \beta((v \otimes a) \otimes b) = v \otimes (\varphi(a)b) = v \otimes (a \cdot b)$$

with inverse given by

$$\gamma : V \otimes_k B \rightarrow (V \otimes_k A) \otimes_A B, \quad \gamma(v \otimes b) = (v \otimes 1) \otimes b.$$

Now $GL_V(\varphi) : \text{Aut}_A(V \otimes A) \rightarrow \text{Aut}_B(V \otimes B)$ is set as follows : for any $T \in \text{Aut}_A(V \otimes A)$ we take

$$GL_V(\varphi)(T) = \beta \circ (T \otimes \text{id}_B) \circ \beta^{-1}.$$

Proposition 8.2. – $GL_V(\varphi)(T)$ is the unique B -linear automorphism satisfying

$$(\text{id}_V \otimes \varphi) \circ T = GL_V(\varphi)(T) \circ (\text{id}_V \otimes \varphi).$$

Also, GL_V respects function composition, and is therefore a group functor.

Proof. – We first show that $GL_V(\varphi)(T)$ indeed satisfies the above equation. Consider the canonical map

$$\iota : V \otimes_k A \rightarrow (V \otimes_k A) \otimes_A B, \quad x \mapsto x \otimes 1$$

and take any $v \otimes a \in V \otimes A$. Then note that

$$\beta(\iota(v \otimes a)) = \beta((v \otimes a) \otimes 1) = v \otimes \varphi(a) = (\text{id}_V \otimes \varphi)(v \otimes a) \quad \text{i.e.} \quad \beta \circ \iota = \text{id}_V \otimes \varphi.$$

Therefore

$$\begin{aligned} GL_V(\varphi)(T) ((\text{id}_V \otimes \varphi)(v \otimes a)) &= GL_V(\varphi)(T) (\beta(\iota(v \otimes a))) \\ &= \beta ((T \otimes \text{id}_B) (\iota(v \otimes a))) \\ &= \beta ((T(v \otimes a)) \otimes 1) \\ &= \beta (\iota(T(v \otimes a))) \\ &= (\text{id}_V \otimes \varphi)(T(v \otimes a)) \end{aligned}$$

i.e. $\text{GL}_V(\varphi)(T) \circ (\text{id}_V \otimes \varphi) = (\text{id}_V \otimes \varphi) \circ T$.

To show that $\text{GL}_V(\varphi)(T)$ is the unique such map, suppose $S : V \otimes B \rightarrow V \otimes B$ be another B -linear automorphism satisfying

$$S \circ (\text{id}_V \otimes \varphi) = (\text{id}_V \otimes \varphi) \circ T.$$

Then for any $v \otimes 1 \in V \otimes A$ we have

$$\begin{aligned} S(v \otimes 1) &= S((\text{id}_V \otimes \varphi)(v \otimes 1)) \\ &= ((\text{id}_V \otimes \varphi) \circ T)(v \otimes 1) \\ &= (\text{GL}_V(\varphi)(T) \circ (\text{id}_V \otimes \varphi))(v \otimes 1) \\ &= \text{GL}_V(\varphi)(T)(v \otimes 1). \end{aligned}$$

By B -linearity we get $S = \text{GL}_V(\varphi)(T)$ on the whole of $V \otimes B$.

Finally, it remains to show that for $A \xrightarrow{\varphi} B \xrightarrow{\psi} C$ we have

$$\text{GL}_V(\psi \circ \varphi) = \text{GL}_V(\psi) \circ \text{GL}_V(\varphi).$$

Let $T \in \text{Aut}_A(V \otimes A)$, then by definition $\text{GL}_V(\psi \circ \varphi)(T)$ is the unique C -linear automorphism of $V \otimes C$ satisfying

$$\text{GL}_V(\psi \circ \varphi)(T) \circ (\text{id}_V \otimes (\psi \circ \varphi)) = (\text{id}_V \otimes (\psi \circ \varphi)) \circ T.$$

So it is enough to show that

$$S := \text{GL}_V(\psi)(\text{GL}_V(\varphi)(T))$$

satisfies

$$S \circ (\text{id}_V \otimes (\psi \circ \varphi)) = (\text{id}_V \otimes (\psi \circ \varphi)) \circ T.$$

Since $\text{GL}_V(\varphi)(T) \in \text{Aut}_B(V \otimes B)$ by definition we have

$$(\text{id}_V \otimes \varphi) \circ T = \text{GL}_V(\varphi)(T) \circ (\text{id}_V \otimes \varphi).$$

Also $\text{GL}_V(\psi)(\text{GL}_V(\varphi)(T)) = S$ satisfies :

$$(\text{id}_V \otimes \psi) \circ \text{GL}_V(\varphi)(T) = S \circ (\text{id}_V \otimes \psi).$$

Composing the two we get

$$\begin{aligned} (\text{id}_V \otimes (\psi \circ \varphi)) \circ T &= (\text{id}_V \otimes \psi) \circ (\text{id}_V \otimes \varphi) \circ T \\ &= (\text{id}_V \otimes \psi) \circ \text{GL}_V(\varphi)(T) \circ (\text{id}_V \otimes \varphi) \\ &= S \circ (\text{id}_V \otimes \psi) \circ (\text{id}_V \otimes \varphi) \\ &= S \circ (\text{id}_V \otimes (\psi \circ \varphi)). \end{aligned}$$

□

We will now show that a linear representation of G on a k -module V gives a homomorphism of group functors $G \rightarrow \text{GL}_V$. Indeed, for each k -algebra A let the group action be denoted as

$$\mu_A : G(A) \times (V \otimes_k A) \rightarrow V \otimes_k A.$$

For any $g \in G(A)$, $x \in V \otimes_k A$ we may write $\mu_A(g, x)$ as $g \cdot_A x$. We define a map $\Phi_A : G(A) \rightarrow \text{GL}_V(A) = \text{Aut}_A(V \otimes_k A)$ by $\Phi_A(g)(x) := \mu_A(g, x) =: g \cdot_A x$. It is also clear that $\Phi_A(g)$ is A -linear automorphism of $V \otimes_k A$, with the inverse of $\Phi_A(g)$ being $\Phi_A(g^{-1})$. So it remains to show that Φ_A is a group homomorphism. Take any $g, h \in G(A)$ and note that

$$\Phi_A(gh)(x) = (gh) \cdot_A x = g \cdot_A (h \cdot_A x) = \Phi_A(g)(\Phi_A(h)(x)) = (\Phi_A(g) \circ \Phi_A(h))(x)$$

for all $x \in V \otimes_k A$ i.e. $\Phi_A(gh) = \Phi_A(g) \circ \Phi_A(h)$.

We now check that these Φ_A for all k -algebras A assemble to give a natural transformation $G \rightarrow \text{GL}_V$. In other words, if $\varphi : A \rightarrow B$ is a k -algebra map and $g \in G(A)$ we must show that

$$\text{GL}_V(\varphi)(\Phi_A(g)) = \Phi_B(G(\varphi)(g)).$$

But for any $x \in V \otimes A$ we have

$$\begin{aligned}\Phi_B(G(\varphi)(g))((\text{id}_V \otimes \varphi)(x)) &= \mu_B(G(\varphi)(g), (\text{id}_V \otimes \varphi)(x)) \\ &= (\text{id}_V \otimes \varphi)(\mu_A(g, x)) \\ &= (\text{id}_V \otimes \varphi)(\Phi_A(g)(x))\end{aligned}$$

by the naturality of the action $\mu : G \times X \rightarrow X$. Thus we get

$$\Phi_B(G(\varphi)(g)) \circ (\text{id}_V \otimes \varphi) = (\text{id}_V \otimes \varphi) \circ \Phi_A(g)$$

and by Proposition 8.2 we obtain

$$\text{GL}_V(\varphi)(\Phi_A(g)) = \Phi_B(G(\varphi)(g)).$$

Conversely, given any homomorphism of group functors $\Phi : G \rightarrow \text{GL}_V$ we get a linear representation of G on V . For any k -algebra A and $g \in G(A)$, $x \in V \otimes_k A$, we set $g \cdot_A x := \Phi_A(g)(x)$. It follows that this gives an A -linear action of $G(A)$ on $V \otimes_k A$. Also the naturality of Φ implies the naturality of this action $G \times X \rightarrow X$.

Theorem 8.3. – *Let G be an affine group functor represented by (R, Δ, S, ϵ) . Then given a linear representation Φ of G on V we obtain a k -linear map $\rho : V \rightarrow V \otimes R$ such that*

$$\begin{array}{ccc} V & \xrightarrow{\rho} & V \otimes R \\ \rho \downarrow & & \downarrow \text{id} \otimes \Delta \\ V \otimes R & \xrightarrow{\rho \otimes \text{id}} & V \otimes R \otimes R \end{array} \quad \text{and} \quad \begin{array}{ccc} V & \xrightarrow{\rho} & V \otimes R \\ \downarrow & \swarrow \text{id} \otimes \epsilon & \\ V \otimes k & & \end{array}$$

commute. Conversely, if there exists such a ρ satisfying the above commutativity conditions then we can obtain a representation Φ .

Proof. – First we observe a naturality condition. Since $\Phi : G \rightarrow \text{GL}_V$ corresponds to a natural action $\mu : G \times X \rightarrow X$, where X is the functor $A \rightsquigarrow V \otimes A$, thus for any algebra map $\varphi : A \rightarrow B$ the following square commutes :

$$\begin{array}{ccc} G(A) \times (V \otimes A) & \xrightarrow{\mu_A} & V \otimes A \\ G(\varphi) \times X(\varphi) \downarrow & & \downarrow X(\varphi) \\ G(B) \times (V \otimes B) & \xrightarrow{\mu_B} & V \otimes B \end{array}$$

Thus for $g \in G(A)$ and for all $x \in V \otimes A$, we have

$$X(\varphi)(\mu_A(g, x)) = \mu_B(G(\varphi)(g), X(\varphi)(x))$$

i.e.

$$X(\varphi)(\Phi_A(g)(x)) = \Phi_B(G(\varphi)(g))(X(\varphi)(x)).$$

Noting that $X(\varphi) = \text{id}_V \otimes \varphi$, we obtain the equality

$$(\text{id}_V \otimes \varphi) \circ \Phi_A(g) = \Phi_B(G(\varphi)(g)) \circ (\text{id}_V \otimes \varphi).$$

Now take $u \in G(R)$ to be the universal element so that (R, u) is the universal pair. Recall that u corresponds to $\text{id}_R : R \rightarrow R$ under the Yoneda bijection $G(R) \xrightarrow{\sim} \text{Hom}_{k\text{-Alg}}(R, R)$. Also recall that for any algebra A and any element $g \in G(A)$ we have $G(\varphi_g)(u) = g$. If $\psi : A \rightarrow B$ is another algebra map then applying $G(\psi)$ to the equality $G(\varphi_g)(u) = g$ we get

$$G(\psi \circ \varphi_g)(u) = G(\psi)(g).$$

Thus the algebra map $\varphi_{G(\psi)(g)} : R \rightarrow B$ corresponding to $G(\psi)(g)$ satisfies

$$\varphi_{G(\psi)(g)} = \psi \circ \varphi_g.$$

We define $\rho : V \rightarrow V \otimes_k R$ by

$$\rho := \Phi_R(u) \circ \iota$$

where $\iota : V \rightarrow V \otimes_k R$ is the canonical map $v \mapsto v \otimes 1$.

First we will take $\varphi = \epsilon : R \rightarrow k$ and $g = u$ to obtain

$$(\text{id}_V \otimes \epsilon) \circ \Phi_A(u) = \Phi_B(G(\epsilon)(u)) \circ (\text{id}_V \otimes \epsilon).$$

Here note that $G(\epsilon)(u)$ corresponds to $\epsilon \circ \varphi_u = \epsilon \circ \text{id}_R = \epsilon$. The identity element of the group $\text{Hom}(R, k)$ is $T_k \circ \epsilon = \text{id}_k \circ \epsilon = \epsilon$. Thus $G(\epsilon)(u) \in G(k)$ is actually the identity element of the group $G(k)$. Therefore $\Phi_k(G(\epsilon)(u)) = \text{id}_{V \otimes k}$ and we get

$$(\text{id}_V \otimes \epsilon) \circ \Phi_A(u) = \text{id}_{V \otimes k} \circ (\text{id}_V \otimes \epsilon).$$

Composing on the right by ι and noting that $(\text{id}_V \otimes \epsilon) \circ \iota : V \rightarrow V \otimes k$ is just the canonical isomorphism, we get

$$(\text{id}_V \otimes \epsilon) \circ \rho = [V \xrightarrow{\sim} V \otimes k]$$

which is the second commutativity condition.

Next we take $\varphi = \Delta : R \rightarrow R \otimes R$ and $g = u \in G(R)$. Then we have

$$(\text{id}_V \otimes \Delta) \circ \Phi_R(u) = \Phi_{R \otimes R}(G(\Delta)(u)) \circ (\text{id}_V \otimes \Delta).$$

We compose the canonical map $V \rightarrow V \otimes R$ on the right of the above equality and then apply the resulting equality on an arbitrary element $v \in V$ to obtain

$$(\text{id}_V \otimes \Delta)(\rho(v)) = \Phi_{R \otimes R}(G(\Delta)(u))((\text{id}_V \otimes \Delta)(v \otimes 1_R)) = \Phi_{R \otimes R}(G(\Delta)(u))(v \otimes 1_R \otimes 1_R).$$

Now consider the canonical map $\iota_1 : R \rightarrow R \otimes R, r \mapsto r \otimes 1$ and $\iota_2 : R \rightarrow R \otimes R, r \mapsto 1 \otimes r$. Let $u_1, u_2 \in G(R \otimes R)$ be the elements corresponding to ι_1, ι_2 respectively. We claim that

$$G(\Delta)(u) = u_1 u_2.$$

Indeed, first note that $G(\Delta)(u)$ corresponds to $\Delta \circ \varphi_u = \Delta \circ \text{id}_R = \Delta$. On the other hand, $u_1 u_2$ corresponds to $\iota_1 * \iota_2 = M_{R \otimes R} \circ (\iota_1 \otimes \iota_2) \circ \Delta$. But we can check that

$$M_{R \otimes R} \circ (\iota_1 \otimes \iota_2) \circ \Delta = \Delta.$$

To see this write $\Delta(r) = \sum_i a_i \otimes b_i$ and apply $M_{R \otimes R} \circ (\iota_1 \otimes \iota_2) \circ \Delta$ to get

$$M_{R \otimes R} \left(\sum_i (a_i \otimes 1) \otimes (1 \otimes b) \right) = \sum_i (a_i \otimes 1)(1 \otimes b) = \sum_i a_i \otimes b_i = \Delta(r).$$

Thus we have shown that $G(\Delta)(u) = u_1 u_2$. Therefore we have obtained the equality

$$(\text{id}_V \otimes \Delta)(\rho(v)) = \Phi_{R \otimes R}(u_1 u_2)(v \otimes 1_R \otimes 1_R) = \Phi_{R \otimes R}(u_1)(\Phi_{R \otimes R}(u_2)(v \otimes 1_R \otimes 1_R)).$$

Proving the following claim will finish the proof.

Claim. – $\Phi_{R \otimes R}(u_1)(\Phi_{R \otimes R}(u_2)(v \otimes 1_R \otimes 1_R)) = (\rho \otimes \text{id}_R)(\rho(v))$.

Proof of the claim. – Now we will use the naturality condition with $\varphi = \iota_2 : R \rightarrow R \otimes R, g = u$. We have

$$(\text{id}_V \otimes \iota_2) \circ \Phi_R(u) = \Phi_{R \otimes R}(G(\iota_2)(u)) \circ (\text{id}_V \otimes \iota_2) = \Phi_{R \otimes R}(u_2) \circ (\text{id}_V \otimes \iota_2).$$

Applying this to $v \otimes 1$ we get

$$(\text{id}_V \otimes \iota_2)(\rho(v)) = \Phi_{R \otimes R}(u_2)(v \otimes 1_R \otimes 1_R).$$

Now writing $\rho(v) = \sum_j v_j \otimes r_j$ the left hand side becomes $\sum_j v_j \otimes 1_R \otimes r_j$ and thus we have

$$\Phi_{R \otimes R}(u_2)(v \otimes 1_R \otimes 1_R) = \sum_j v_j \otimes 1_R \otimes r_j.$$

Similarly substituting $\varphi = \iota_1$ and $g = u$ in the naturality condition we obtain

$$(\text{id}_V \otimes \iota_1) \circ \Phi_R(u) = \Phi_{R \otimes R}(G(\iota_1)(u)) \circ (\text{id}_V \otimes \iota_1) = \Phi_{R \otimes R}(u_1) \circ (\text{id}_V \otimes \iota_1).$$

Thus, applying the above to $v_j \otimes 1$ for each j we get

$$\Phi_{R \otimes R}(u_1)(v_j \otimes 1_R \otimes 1_R) = (\text{id}_V \otimes \iota_1)(\rho(v_j)).$$

On the other hand, by $R \otimes R$ -linearity of $\Phi_{R \otimes R}(u_1)$ we have

$$\Phi_{R \otimes R}(u_1)(v_j \otimes 1_R \otimes r_j) = (1_R \otimes r_j) \cdot \Phi_{R \otimes R}(u_1)(v_j \otimes 1_R \otimes 1_R) = (1_R \otimes r_j) \cdot (\text{id}_V \otimes \iota_1)(\rho(v_j)).$$

Writing $\rho(v_j) = \sum_l w_l \otimes s_l$ we note that the right hand side above is

$$(1_R \otimes r_j) \cdot (\text{id}_V \otimes \iota_1)(\rho(v_j)) = \sum_l w_l \otimes s_l \otimes r_j$$

and so we get

$$\Phi_{R \otimes R}(u_1)(v_j \otimes 1_R \otimes r_j) = \sum_l w_l \otimes s_l \otimes r_j = \rho(v_j) \otimes r_j.$$

Summing over j we have

$$\Phi_{R \otimes R}(u_1) \left(\sum_j v_j \otimes 1_R \otimes r_j \right) = \sum_j \rho(v_j) \otimes r_j = (\rho \otimes \text{id}_R) \rho(v).$$

We conclude that

$$\begin{aligned} \Phi_{R \otimes R}(u_1) (\Phi_{R \otimes R}(u_2)(v \otimes 1_R \otimes 1_R)) &= \Phi_{R \otimes R}(u_1) \left(\sum_j v_j \otimes 1_R \otimes r_j \right) \\ &= (\rho \otimes \text{id}_R) \rho(v) \end{aligned}$$

completing the claim. $\square$

The proof of the first part of the proposition is complete.

Now we will proceed to show the converse. Let $\rho : V \rightarrow V \otimes_k R$ be a k -linear map satisfying the commutativity conditions in the statement of the theorem. We construct a linear representation $\Phi : G \rightarrow GL_V$. Fix any k -algebra A and any $g \in G(A)$, and let $\varphi_g : R \rightarrow A$ be the corresponding algebra map. Then, for any $v \otimes a \in V \otimes A$ we define

$$\Phi_A(g)(v \otimes a) := a \cdot ((\text{id}_V \otimes \varphi_g)(\rho(v))).$$

To see the above expression explicitly, write $\rho(v) = \sum_i v_i \otimes r_i$ and then compute to obtain

$$\Phi_A(g)(v \otimes a) = \sum_i v_i \otimes (a \varphi_g(r_i)).$$

It is then easy to see that $\Phi_A(g)$ is A -linear. To show that $\Phi_A(g)$ is an automorphism it suffices to show that

- $\Phi_A(e_A) = \text{id}_{V \otimes A}$ and
- $\Phi_A(gh) = \Phi_A(g) \circ \Phi_A(h)$

since then we have $\Phi_A(gg^{-1}) = \Phi_A(g) \circ \Phi_A(g^{-1}) = \text{id}_{V \otimes A}$ and similarly $\Phi_A(g^{-1}) \circ \Phi_A(g) = \text{id}_{V \otimes A}$. So we next show the above two equalities.

For the first one, recall that e_A corresponds to $\varphi_{e_A} = T_A \circ \epsilon$ (see Proposition 1.32) and thus we have

$$\Phi_A(e_A)(v \otimes a) = \sum_i v_i \otimes (T_A(\epsilon(r_i))a).$$

Now using $(\text{id}_V \otimes \epsilon)\rho(v) = v \otimes 1 \in V \otimes k$ we get

$$\sum_i v_i \otimes \epsilon(r_i) = v \otimes 1$$

and then applying the map $\text{id}_V \otimes T_A : V \otimes k \rightarrow V \otimes A$ we get

$$\sum_i v_i \otimes (T_A(\epsilon(r_i))) = v \otimes 1_A$$

which after being acted upon by a gives

$$\Phi_A(e_A)(v \otimes a) = v \otimes a$$

i.e. $\Phi_A(e_A) = \text{id}_{V \otimes A}$.

For the second one, take two elements $g, h \in G(A)$ and suppose $\varphi_g, \varphi_h : R \rightarrow A$ be the corresponding algebra maps. We prove

$$\Phi_A(gh) = \Phi_A(g) \circ \Phi_A(h)$$

by evaluating both sides on the pure tensor $v \otimes a$. We first write

$$\rho(v) = \sum_i v_i \otimes r_i$$

and for each i we write

$$\rho(v_i) = \sum_j w_{ij} \otimes s_{ij}.$$

Now by definition we have

$$\Phi_A(h)(v \otimes a) = \sum_i v_i \otimes \varphi_h(r_i)a.$$

Applying $\Phi_A(g)$ we get

$$\Phi_A(g)(\Phi_A(h)(v \otimes a)) = \sum_i \Phi_A(g)(v_i \otimes \varphi_h(r_i)a) = \sum_{i,j} w_{ij} \otimes \varphi_g(s_{ij})\varphi_h(r_i)a.$$

Now we use the coassociativity : $(\rho \otimes \text{id}_R) \circ \rho = (\text{id}_V \otimes \Delta) \circ \rho$. Applying the left hand side to v we get

$$(\rho \otimes \text{id}_R)(\rho(v)) = \sum_i \rho(v_i) \otimes r_i = \sum_{i,j} w_{ij} \otimes s_{ij} \otimes r_i.$$

Writing $\Delta(r_i) = \sum_\ell p_{i\ell} \otimes q_{i\ell}$ for each i we get

$$(\text{id}_V \otimes \Delta)(\rho(v)) = \sum_i v_i \otimes \Delta(r_i) = \sum_{i,\ell} v_i \otimes p_{i\ell} \otimes q_{i\ell}$$

by applying the right hand side to v . Therefore we have the equality

$$\sum_{i,j} w_{ij} \otimes s_{ij} \otimes r_i = \sum_{i,\ell} v_i \otimes p_{i\ell} \otimes q_{i\ell}.$$

Then applying the map $[x \otimes r \otimes s \mapsto x \otimes \varphi_g(r)\varphi_h(s)a] : V \otimes R \otimes R \rightarrow V \otimes A$ to the above equality we obtain

$$\sum_{i,j} w_{ij} \otimes \varphi_g(s_{ij})\varphi_h(r_i)a = \sum_{i,\ell} v_i \otimes \varphi_g(p_{i\ell})\varphi_h(q_{i\ell})a.$$

Thus we have

$$\Phi_A(g)(\Phi_A(h)(v \otimes a)) = \sum_{i,\ell} v_i \otimes \varphi_g(p_{i\ell})\varphi_h(q_{i\ell})a.$$

On the other hand, recall the group law on $\text{Hom}(R, A)$ gives

$$\varphi_{gh} = M_A \circ (\varphi_g \otimes \varphi_h) \circ \Delta$$

which when applied to each r_i yields

$$\varphi_{gh}(r_i) = \sum_\ell \varphi_g(p_{i\ell})\varphi_h(q_{i\ell}).$$

Thus we get

$$\sum_{i,\ell} v_i \otimes \varphi_g(p_{i\ell})\varphi_h(q_{i\ell})a = \sum_i v_i \otimes \left(\sum_\ell \varphi_g(p_{i\ell})\varphi_h(q_{i\ell}) \right) a = \sum_i v_i \otimes \varphi_{gh}(r_i)a.$$

Combining all these reductions we get

$$\begin{aligned} \Phi_A(gh)(v \otimes a) &= \sum_i v_i \otimes \varphi_{gh}(r_i)a \\ &= \sum_{i,\ell} v_i \otimes \varphi_g(p_{i\ell})\varphi_h(q_{i\ell})a \\ &= \Phi_A(g)(\Phi_A(h)(v \otimes a)). \end{aligned}$$

Thus we have proved $\Phi_A(gh) = \Phi_A(g) \circ \Phi_A(h)$.

It remains to show that $(\Phi_A \mid A \text{ is a } k\text{-algebra})$ is a natural transformation. Let $\varphi : A \rightarrow B$ be an algebra map, $g \in G(A)$, and $\varphi_g : R \rightarrow A$ be the corresponding algebra map. Now $G(\varphi)(g) \in G(B)$ and we know $\varphi_{G(\varphi)(g)} = \varphi \circ \varphi_g : R \rightarrow B$. For $v \otimes a \in V \otimes A$ we have

$$\begin{aligned} (\text{id}_V \otimes \varphi)(\Phi_A(g)(v \otimes a)) &= (\text{id}_V \otimes \varphi) \left(\sum_i v_i \otimes \varphi_g(r_i) a \right) \\ &= \sum_i v_i \otimes \varphi(\varphi_g(r_i) a) \\ &= \sum_i v_i \otimes \varphi_{G(\varphi)(g)}(r_i) \varphi(a) \\ &= \Phi_B(G(\varphi)(g))(v \otimes \varphi(a)) \\ &= (\Phi_B(G(\varphi)(g)) \circ (\text{id}_V \otimes \varphi))(v \otimes a) \end{aligned}$$

and thus we have proved

$$(\text{id}_V \otimes \varphi) \circ \Phi_A(g) = \Phi_B(G(\varphi)(g)) \circ (\text{id}_V \otimes \varphi)$$

which is precisely the naturality condition.

It remains to check that the above two constructions are inverse of each other. Thus we must first show that starting with a representation $\Phi : G \rightarrow \text{GL}_V$, the comodule map ρ can be used to recover Φ . To be precise, suppose we define

$$\Phi'_A(g)(v \otimes a) := a \cdot ((\text{id}_V \otimes \varphi_g)(\rho(v))),$$

then we must show that $\Phi_A = \Phi'_A$ for every algebra A . First recall the naturality condition

$$\begin{aligned} (\text{id}_V \otimes \varphi_g) \circ \Phi_R(u) &= \Phi_A(G(\varphi_g)(u)) \circ (\text{id}_V \otimes \varphi_g) \\ &= \Phi_A(g) \circ (\text{id}_V \otimes \varphi_g). \end{aligned}$$

Applying this to $v \otimes 1$ and noting that $\rho(v) = \Phi_R(u)(v \otimes 1)$ we have

$$(\text{id}_V \otimes \varphi_g)(\rho(v)) = \Phi_A(g)((\text{id}_V \otimes \varphi_g)(v \otimes 1)) = \Phi_A(g)(v \otimes 1).$$

Therefore, using A -linearity of the map $\Phi_A(g)$, we get

$$\Phi'_A(g)(v \otimes a) = a \cdot \Phi_A(g)(v \otimes 1) = \Phi_A(g)(v \otimes a)$$

for every $v \otimes a \in V \otimes A$.

On the other hand, starting with some comodule (ρ, V) one obtains a representation Φ as in the previous paragraph. Now we need to show that ρ' defined by

$$\rho'(v) := \Phi_R(u)(v \otimes 1)$$

is actually equal to ρ . But since $\varphi_u = \text{id}_R$, we simply note that

$$\rho'(v) = \Phi_R(u)(v \otimes 1) = 1 \cdot ((\text{id}_V \otimes \varphi_u)(\rho(v))) = (\text{id}_V \otimes \text{id}_R)(\rho(v)) = \rho(v)$$

for every $v \in V$. This completes the proof of the theorem. $\square$

Definition 8.4. – Let (R, Δ, S, ϵ) be a Hopf algebra. A k -module V equipped with a k -linear map $\rho : V \rightarrow V \otimes R$ satisfying $(\text{id}_R \otimes \epsilon)\rho = [V \xrightarrow{\sim} V \otimes k]$ and $(\text{id}_V \otimes \Delta) \circ \rho = (\rho \otimes \text{id}_R) \circ \rho$ is called an R -comodule.

Example 8.5. – When $V = R$ and $\rho = \Delta$ we call the corresponding representation obtained from ρ as the *regular representation*.

We now introduce the analogue of the notion of intertwiners between group representations.

Definition 8.6. – Let G be a group functor, and $\Phi_V : G \rightarrow \text{GL}_V$, $\Phi_W : G \rightarrow \text{GL}_W$ be two linear representations of G . A k -linear map $T : V \rightarrow W$ is called an *intertwiner* if for every k -algebra A and $g \in G(A)$:

$$(T \otimes \text{id}_A) \circ \Phi_{V,A}(g) = \Phi_{W,A}(g) \circ (T \otimes \text{id}_A).$$

Definition 8.7. – Let (R, Δ, S, ϵ) be a Hopf algebra over k , and let (ρ, V) and (σ, W) be two R -comodules. A k -linear map $T : V \rightarrow W$ is called a *homomorphism of comodules* if

$$\sigma \circ T = (T \otimes \text{id}_R) \circ \rho.$$

Proposition 8.8. – Let G be an affine group functor with representing Hopf algebra (R, Δ, S, ϵ) . Then intertwiners are homomorphism of R -comodules and vice-versa.

Proof. – Suppose (ρ, V) and (σ, W) are two R -comodules and let $\Phi_V : G \rightarrow GL_V$, $\Phi_W : G \rightarrow GL_W$ be the corresponding representations. Assume $T : V \rightarrow W$ is a homomorphism of comodules i.e. $\sigma \circ T = (T \otimes \text{id}_R) \circ \rho$. Take any algebra A and any $v \otimes a \in V \otimes A$, and write $\rho(v) = \sum_i v_i \otimes r_i$. Then $\Phi_{V,A}(g)(v \otimes a) = \sum_i v_i \otimes (\varphi_g(r_i)a)$. Applying $T \otimes \text{id}_A$ we get

$$((T \otimes \text{id}_A) \circ \Phi_{V,A}(g))(v \otimes a) = \sum_i T(v_i) \otimes (\varphi_g(r_i)a).$$

On the other hand, $\sigma(T(v)) = (T \otimes \text{id}_R)(\rho(v)) = \sum_i T(v_i) \otimes r_i$. Thus we have

$$(\Phi_{W,A}(g) \circ (T \otimes \text{id}_A))(v \otimes a) = \Phi_{W,A}(g)(T(v) \otimes a) = \sum_i T(v_i) \otimes (\varphi_g(r_i)a).$$

We have thus shown $(T \otimes \text{id}_A) \circ \Phi_{V,A}(g) = \Phi_{W,A}(g) \circ (T \otimes \text{id}_A)$.

Now assume $T : V \rightarrow W$ is an intertwiner. We have to show that it is also a homomorphism of comodules. Now T satisfies

$$(T \otimes \text{id}_A) \circ \Phi_{V,A}(g) = \Phi_{W,A}(g) \circ (T \otimes \text{id}_A) \quad \forall A, \forall g \in G(A).$$

We take $A = R$ and $g = u \in G(R)$ the universal element. Thus we have

$$(T \otimes \text{id}_R) \circ \Phi_{V,R}(u) = \Phi_{W,R}(u) \circ (T \otimes \text{id}_R).$$

Applying both sides to $v \otimes 1$ and using $\rho(v) = \Phi_{V,R}(u)(v \otimes 1)$, $\sigma(w) = \Phi_{W,R}(u)(w \otimes 1)$ we get

$$(T \otimes \text{id}_R)(\rho(v)) = \Phi_{W,R}(u)(T(v) \otimes 1) = \sigma(T(v)), \quad \forall v \in V.$$

Thus we have shown $(T \otimes \text{id}_R) \circ \rho = \sigma \circ T$. □

Proposition 8.9. – Let (ρ, V) and (σ, W) be two R -comodules where (R, Δ, S, ϵ) is a Hopf algebra. We define

$$\rho \oplus \sigma : V \oplus W \rightarrow (V \oplus W) \otimes_k R$$

by

$$(\rho \oplus \sigma)(v, w) = \Theta(\rho(v), \sigma(w))$$

where $\Theta : (V \otimes R) \oplus (W \otimes R) \xrightarrow{\sim} (V \oplus W) \otimes R$ is given by $(v \otimes r, w \otimes s) \mapsto (v, 0) \otimes r + (0, w) \otimes s$. This makes $V \oplus W$ into an R -comodule, called the direct sum of V and W .

Proof. – First note that the inverse of the map Θ in the above definition is given by

$$(v, w) \otimes r \mapsto (v \otimes r, w \otimes r).$$

We verify that $\rho \oplus \sigma$ satisfies the comodule axioms.

First we will check that $(\text{id}_{V \oplus W} \otimes \epsilon) \circ (\rho \oplus \sigma) = [V \oplus W \xrightarrow{\sim} (V \oplus W) \otimes k]$. Take $(v, w) \in V \oplus W$ and write $\rho(v) = \sum_i v_i \otimes r_i$, $\sigma(w) = \sum_j w_j \otimes s_j$. Then we have

$$(\rho \oplus \sigma)(v, w) = \sum_i (v_i, 0) \otimes r_i + \sum_j (0, w_j) \otimes s_j.$$

Applying $\text{id}_{V \oplus W} \otimes \epsilon$ to this equality we get

$$\begin{aligned} (\text{id}_{V \oplus W} \otimes \epsilon)((\rho \oplus \sigma)(v, w)) &= \sum_i (v_i, 0) \otimes \epsilon(r_i) + \sum_j (0, w_j) \otimes \epsilon(s_j) \\ &= \sum_i (\epsilon(r_i)v_i, 0) \otimes 1 + \sum_j (0, \epsilon(s_j)w_j) \otimes 1 \\ &= \left(\sum_i \epsilon(r_i)v_i, 0 \right) \otimes 1 + \left(0, \sum_j \epsilon(s_j)w_j \right) \otimes 1 \\ &= \left(\sum_i \epsilon(r_i)v_i, \sum_j \epsilon(s_j)w_j \right) \otimes 1. \end{aligned}$$

But since ρ and σ satisfy $(\text{id}_V \otimes \epsilon) \circ \rho = [V \xrightarrow{\sim} V \otimes k]$ and $(\text{id}_W \otimes \epsilon) \circ \sigma = [W \xrightarrow{\sim} W \otimes k]$ respectively, applying these to v and w respectively we get

$$\sum_i \epsilon(r_i) v_i \otimes 1 = \sum_i v_i \otimes \epsilon(r_i) = v \otimes 1 \quad \text{and} \quad \sum_j \epsilon(s_j) w_j \otimes 1 = \sum_j w_j \otimes \epsilon(s_j) = w \otimes 1$$

which give : $\sum_i \epsilon(r_i) v_i = v$ and $\sum_j \epsilon(s_j) w_j = w$ respectively. Therefore, we have

$$(\text{id}_{V \oplus W} \otimes \epsilon)((\rho \oplus \sigma)(v, w)) = (v, w) \otimes 1$$

as required.

Next we will check that $(\text{id}_{V \oplus W} \otimes \Delta) \circ (\rho \oplus \sigma) = ((\rho \oplus \sigma) \otimes \text{id}_R) \circ (\rho \oplus \sigma)$. As before, we first write $(\rho \oplus \sigma)(v, w) = \sum_i (v_i, 0) \otimes r_i + \sum_j (0, w_j) \otimes s_j$, and then we write $\Delta(r_i) = \sum_\alpha a_{\alpha i} \otimes b_{\alpha i}$ and $\Delta(s_j) = \sum_\beta c_{\beta j} \otimes d_{\beta j}$. Then we evaluate $(\text{id}_{V \oplus W} \otimes \Delta) \circ (\rho \oplus \sigma)$ on (v, w) and obtain

$$\begin{aligned} (\text{id}_{V \oplus W} \otimes \Delta)((\rho \oplus \sigma)(v, w)) &= \sum_i (v_i, 0) \otimes \Delta(r_i) + \sum_j (0, w_j) \otimes \Delta(s_j) \\ &= \sum_{i, \alpha} (v_i, 0) \otimes a_{\alpha i} \otimes b_{\alpha i} + \sum_{j, \beta} (0, w_j) \otimes c_{\beta j} \otimes d_{\beta j}. \end{aligned}$$

Similarly, evaluating $((\rho \oplus \sigma) \otimes \text{id}_R) \circ (\rho \oplus \sigma)$ on (v, w) we get

$$\begin{aligned} ((\rho \oplus \sigma) \otimes \text{id}_R)((\rho \oplus \sigma)(v, w)) &= \sum_i (\rho \oplus \sigma)(v_i, 0) \otimes r_i + \sum_j (\rho \oplus \sigma)(0, w_j) \otimes s_j \\ &= \sum_i \Theta(\rho(v_i), 0) \otimes r_i + \sum_j \Theta(0, \sigma(w_j)) \otimes s_j. \end{aligned}$$

Then we write $\rho(v_i) = \sum_m u_{im} \otimes t_{im}$ and $\sigma(w_j) = \sum_n z_{jn} \otimes q_{jn}$; the above becomes

$$\begin{aligned} ((\rho \oplus \sigma) \otimes \text{id}_R)((\rho \oplus \sigma)(v, w)) &= \sum_i \Theta(\rho(v_i), 0) \otimes r_i + \sum_j \Theta(0, \sigma(w_j)) \otimes s_j \\ &= \sum_i \Theta \left(\sum_m u_{im} \otimes t_{im}, 0 \right) \otimes r_i + \sum_j \Theta \left(0, \sum_n z_{jn} \otimes q_{jn} \right) \otimes s_j \\ &= \sum_{i, m} (u_{im}, 0) \otimes t_{im} \otimes r_i + \sum_{j, n} (0, z_{jn}) \otimes q_{jn} \otimes s_j. \end{aligned}$$

Finally, using $(\text{id}_V \otimes \Delta)(\rho(v)) = (\rho \otimes \text{id}_R)(\rho(v))$ and $(\text{id}_W \otimes \Delta)(\sigma(w)) = (\sigma \otimes \text{id}_R)(\sigma(w))$ we get

$$\sum_{i, \alpha} v_i \otimes a_{\alpha i} \otimes b_{\alpha i} = \sum_{i, m} u_{im} \otimes t_{im} \otimes r_i$$

and

$$\sum_{j, \beta} w_j \otimes c_{\beta j} \otimes d_{\beta j} = \sum_{j, n} z_{jn} \otimes q_{jn} \otimes s_j.$$

We obtain the required equality after applying the maps $V \otimes R \otimes R \rightarrow (V \oplus W) \otimes R \otimes R$, $v \otimes r \otimes s \mapsto (v, 0) \otimes r \otimes s$ and $W \otimes R \otimes R \rightarrow (V \oplus W) \otimes R \otimes R$, $w \otimes r \otimes s \mapsto (0, w) \otimes r \otimes s$. $\square$

Proposition 8.10. – *Let (ρ, V) and (σ, W) be R -comodules. We define a map*

$$\rho \bar{\otimes} \sigma : V \otimes W \rightarrow (V \otimes W) \otimes R$$

by the composite

$$V \otimes W \xrightarrow{\rho \otimes \sigma} (V \otimes R) \otimes (W \otimes R) \xrightarrow{\sim} V \otimes R \otimes W \otimes R \xrightarrow{\text{id}_V \otimes (\text{swap}) \otimes \text{id}_R} V \otimes W \otimes R \otimes R \xrightarrow{\text{id}_V \otimes W \otimes M_R} (V \otimes W) \otimes R.$$

Then this makes $(\rho \bar{\otimes} \sigma, V \otimes W)$ an R -comodule, called the tensor product of the comodules V and W .

Proof. – We first verify that $(\text{id}_{V \otimes W} \otimes \epsilon) \circ (\rho \bar{\otimes} \sigma) = [V \otimes W \xrightarrow{\sim} V \otimes W \otimes k]$. Take $v \in V, w \in W$ and write $\rho(v) = \sum_i v_i \otimes r_i$, $\sigma(w) = \sum_j w_j \otimes s_j$. Then we have

$$(\text{id}_{V \otimes W} \otimes \epsilon)((\rho \bar{\otimes} \sigma)(v \otimes w)) = \sum_{i, j} v_i \otimes w_j \otimes \epsilon(r_i s_j) = \sum_{i, j} \epsilon(r_i) v_i \otimes \epsilon(s_j) w_j \otimes 1.$$

On the other hand applying $(\text{id}_V \otimes \epsilon) \circ \rho = [V \xrightarrow{\sim} V \otimes k]$ and $(\text{id}_V \otimes \epsilon) \circ \sigma = [W \xrightarrow{\sim} W \otimes k]$ to v and w respectively, we get

$$\sum_i \epsilon(r_i) v_i \otimes 1 = v \otimes 1 \quad \text{and} \quad \sum_j \epsilon(s_j) w_j \otimes 1 = w \otimes 1$$

which is equivalent to $\sum_i \epsilon(r_i) v_i = v$ and $\sum_j \epsilon(s_j) w_j = w$ respectively. Thus we have

$$(\text{id}_{V \otimes W} \otimes \epsilon)((\rho \bar{\otimes} \sigma)(v \otimes w)) = \sum_{i,j} \epsilon(r_i) v_i \otimes \epsilon(s_j) w_j \otimes 1 = v \otimes w \otimes 1.$$

Next, we check that $(\text{id}_{V \otimes W} \otimes \Delta) \circ (\rho \bar{\otimes} \sigma) = ((\rho \bar{\otimes} \sigma) \otimes \text{id}_R) \circ (\rho \bar{\otimes} \sigma)$. As before, we fix some $v \otimes w$ and apply $(\text{id}_{V \otimes W} \otimes \Delta) \circ (\rho \bar{\otimes} \sigma)$ to get

$$(\text{id}_{V \otimes W} \otimes \Delta)((\rho \bar{\otimes} \sigma)(v \otimes w)) = \sum_{i,j} v_i \otimes w_j \otimes \Delta(r_i s_j).$$

Now write $\Delta(r_i) = \sum_\alpha a_{i\alpha} \otimes b_{i\alpha}$ and $\Delta(s_j) = \sum_\beta c_{j\beta} \otimes d_{j\beta}$. Thus, we obtain

$$(\text{id}_{V \otimes W} \otimes \Delta)((\rho \bar{\otimes} \sigma)(v \otimes w)) = \sum_{i,j,\alpha,\beta} v_i \otimes w_j \otimes a_{i\alpha} c_{j\beta} \otimes b_{i\alpha} d_{j\beta}.$$

Also writing $\rho(v_i) = \sum_\gamma v_{i\gamma} \otimes t_{i\gamma}$ and $\sigma(w_j) = \sum_\delta w_{j\delta} \otimes u_{j\delta}$ we obtain

$$(\rho \bar{\otimes} \sigma)(v_i \otimes w_j) = \sum_{\gamma,\delta} (v_{i\gamma} \otimes w_{j\delta}) \otimes t_{i\gamma} u_{j\delta}$$

whence we get

$$\begin{aligned} ((\rho \bar{\otimes} \sigma) \otimes \text{id}_R)((\rho \bar{\otimes} \sigma)(v \otimes w)) &= ((\rho \bar{\otimes} \sigma) \otimes \text{id}_R) \left(\sum_{i,j} v_i \otimes w_j \otimes r_i s_j \right) \\ &= \sum_{i,j,\gamma,\delta} (v_{i\gamma} \otimes w_{j\delta}) \otimes t_{i\gamma} u_{j\delta} \otimes r_i s_j. \end{aligned}$$

Finally, we use $(\rho \otimes \text{id}_R)(\rho(v)) = (\text{id}_V \otimes \Delta)(\rho(v))$ and $(\sigma \otimes \text{id}_R)(\sigma(w)) = (\text{id}_W \otimes \Delta)(\sigma(w))$ to get

$$\sum_{i,\gamma} v_{i\gamma} \otimes t_{i\gamma} \otimes r_i = \sum_{i,\alpha} v_i \otimes a_{i\alpha} \otimes b_{i\alpha}$$

and

$$\sum_{j,\delta} w_{j\delta} \otimes u_{j\delta} \otimes s_j = \sum_{j,\beta} w_j \otimes c_{j\beta} \otimes d_{j\beta}.$$

Now tensoring the above two equalities and then applying the canonical map

$$(V \otimes R \otimes R) \otimes (W \otimes R \otimes R) \rightarrow (V \otimes W) \otimes R \otimes R$$

given by $(v \otimes r \otimes r') \otimes (w \otimes s \otimes s') \mapsto (v \otimes w) \otimes rs \otimes r's'$ we get the desired equality

$$\sum_{i,j,\alpha,\beta} v_i \otimes w_j \otimes a_{i\alpha} c_{j\beta} \otimes b_{i\alpha} d_{j\beta} = \sum_{i,j,\gamma,\delta} (v_{i\gamma} \otimes w_{j\delta}) \otimes t_{i\gamma} u_{j\delta} \otimes r_i s_j$$

whence we have $(\text{id}_{V \otimes W} \otimes \Delta)((\rho \bar{\otimes} \sigma)(v \otimes w)) = ((\rho \bar{\otimes} \sigma) \otimes \text{id}_R)((\rho \bar{\otimes} \sigma)(v \otimes w))$. $\square$

Proposition 8.11. – Let (ρ, V) be a comodule corresponding to the representation $\Phi : G \rightarrow \text{GL}_V$ of an affine group functor represented by (R, Δ, S, ϵ) . A submodule W of V is called a subcomodule if $\rho(W)$ is contained in the image of $W \otimes R$ in $V \otimes R$ ⁴. Suppose k is a field, then W is a subcomodule if and only if $G(A)$ maps $W \otimes A$ to itself⁵.

⁴Note that the inclusion $W \hookrightarrow V$ gives rise to a map $W \otimes R \rightarrow V \otimes R$ which is not necessarily injective.

⁵For this reason we need $W \otimes A \hookrightarrow V \otimes A$ whence the assumption : k is a field.

Proof. – Suppose k is a field; then since all k -modules are in fact vector spaces, and thus flat, we have $W \otimes R \hookrightarrow V \otimes R$. Now assume W is a subcomodule. Then $\rho(W) \subset W \otimes R$. Fix an algebra A and a group element $g \in G(A)$. Recall that

$$\Phi_A(g)(v \otimes a) := a \cdot ((\text{id}_V \otimes \varphi_g)(\rho(v))), \quad \forall v \otimes a \in V \otimes A.$$

Now take any $w \otimes a \in W \otimes A \subset V \otimes A$ and write $\rho(w) = \sum_i w_i \otimes r_i$ with $w_i \in W$ and $r_i \in R$ since $\rho(W) \subset W \otimes R$. Thus we obtain

$$\Phi_A(g)(w \otimes a) = \sum_i w_i \otimes (\varphi_g(r_i) \cdot a) \in W \otimes A.$$

Conversely, suppose $\Phi_A(g)(W \otimes A) \subset W \otimes A$ for every A and $g \in G(A)$. Now for any $w \in W$ we have $\rho(w) = \Phi_R(u)(w \otimes 1) \in W \otimes R$. $\square$

Proposition 8.12. – Let W be a subcomodule of an R -comodule (ρ, V) . Suppose $q : V \rightarrow V/W$ be the canonical quotient map. Then, the composite

$$V \xrightarrow{\rho} V \otimes R \xrightarrow{q \otimes \text{id}_R} (V/W) \otimes R$$

induces a map $\bar{\rho} : V/W \rightarrow (V/W) \otimes R$ and makes $(\bar{\rho}, V/W)$ into an R -comodule.

Proof. – We first verify the following composition rules which will be used without reference :

- $(\text{id}_{V/W} \otimes \epsilon) \circ (q \otimes \text{id}_R) = (q \otimes \text{id}_k) \circ (\text{id}_V \otimes \epsilon)$;
- $(\text{id}_{V/W} \otimes \Delta) \circ (q \otimes \text{id}_R) = (q \otimes \text{id}_R \otimes \text{id}_R) \circ (\text{id}_V \otimes \Delta)$;
- $(\bar{\rho} \otimes \text{id}_R) \circ (q \otimes \text{id}_R) = (q \otimes \text{id}_R \otimes \text{id}_R) \circ (\rho \otimes \text{id}_R)$.

The first two are easy to check directly. We verify the third one. Take any $v \otimes r$ and note that by definition $\bar{\rho} \circ q = (q \otimes \text{id}_R) \circ \rho$. Evaluating on $v \otimes r$ we get

$$\begin{aligned} (\bar{\rho} \otimes \text{id}_R)((q \otimes \text{id}_R)(v \otimes r)) &= (\bar{\rho} \otimes \text{id}_R)(q(v) \otimes r) \\ &= (\bar{\rho} \circ q)(v) \otimes r \\ &= (q \otimes \text{id}_R)(\rho(v)) \otimes r \\ &= ((q \otimes \text{id}_R \otimes \text{id}_R) \circ (\rho \otimes \text{id}_R))(v \otimes r). \end{aligned}$$

Now, using $(\text{id}_V \otimes \epsilon) \circ \rho = [V \xrightarrow{\sim} V \otimes k]$, we check that

$$\begin{aligned} (\text{id}_{V/W} \otimes \epsilon) \circ \bar{\rho} \circ q &= (\text{id}_{V/W} \otimes \epsilon) \circ (q \otimes \text{id}_R) \circ \rho \\ &= (q \otimes \text{id}_k) \circ (\text{id}_V \otimes \epsilon) \circ \rho \\ &= (q \otimes \text{id}_k) \circ [V \xrightarrow{\sim} V \otimes k] \\ &= [V/W \xrightarrow{\sim} (V/W) \otimes k] \circ q, \end{aligned}$$

and since q is surjective we have $(\text{id}_{V/W} \otimes \epsilon) \circ \bar{\rho} = [V/W \xrightarrow{\sim} (V/W) \otimes k]$.

Next, we check that $(\text{id}_{V/W} \otimes \Delta) \circ \bar{\rho} = (\bar{\rho} \otimes \text{id}_R) \circ \bar{\rho}$. We compose q on the right, and use $(\text{id}_V \otimes \Delta) \circ \rho = (\rho \otimes \text{id}_R) \circ \rho$, to get

$$\begin{aligned} (\text{id}_{V/W} \otimes \Delta) \circ \bar{\rho} \circ q &= (\text{id}_{V/W} \otimes \Delta) \circ (q \otimes \text{id}_R) \circ \rho \\ &= (q \otimes \text{id}_R \otimes \text{id}_R) \circ (\text{id}_V \otimes \Delta) \circ \rho \\ &= (q \otimes \text{id}_R \otimes \text{id}_R) \circ (\rho \otimes \text{id}_R) \circ \rho \\ &= (\bar{\rho} \otimes \text{id}_R) \circ (q \otimes \text{id}_R) \circ \rho \\ &= (\bar{\rho} \otimes \text{id}_R) \circ \bar{\rho} \circ q, \end{aligned}$$

and again, since q is surjective, we get $(\text{id}_{V/W} \otimes \Delta) \circ \bar{\rho} = (\bar{\rho} \otimes \text{id}_R) \circ \bar{\rho}$. $\square$

Lemma 8.13. – Let (ρ, V) be an R -comodule and suppose V is free over k and of finite rank with basis $\{v_i\}_1^n$. Write $\rho(v_j) = \sum_i v_i \otimes r_{ij}$. If G is the affine group functor corresponding to R , and $\Phi : G \rightarrow \text{GL}_V \simeq \mathbf{GL}_n$ is

the representation corresponding to ρ , then the Hopf algebra map $f : k[\mathbf{GL}_n] \rightarrow R$ corresponding to Φ maps $\overline{X_{ij}}$ to r_{ij} . Also, we have

$$\Delta(r_{ij}) = \sum_l r_{il} \otimes r_{lj}.$$

Here we recall that $k[\mathbf{GL}_n] := \frac{k[X_{11}, \dots, X_{nn}, T]}{(\det(X_{ij})T - 1)}$ is the Hopf algebra for $\mathbf{GL}_n$ with the universal element being $\mathcal{X} = (\overline{X_{ij}})_{i,j=1}^n \in \mathbf{GL}_n(k[\mathbf{GL}_n])$, and if $\Delta_{\mathbf{GL}_n}$ denotes the comultiplication map for $\mathbf{GL}_n$ then we know that $\Delta_{\mathbf{GL}_n}(\overline{X_{ij}}) = \sum_l \overline{X_{il}} \otimes \overline{X_{lj}}$ (see Example 1.37).

Proof. – Since $\rho(v_j) = \sum_i v_i \otimes r_{ij}$, for any algebra A and $g \in G(A)$, we get

$$\Phi_A(g)(v_j \otimes 1) = (\text{id}_V \otimes \varphi_g)\rho(v_j) = \sum_i v_i \otimes \varphi_g(r_{ij}).$$

Thus, in the A -basis $\{v_j \otimes 1\}_{j=1}^n$ of $V \otimes A$, the matrix of $\Phi_A(g)$ is precisely $(\varphi_g(r_{ij}))_{i,j=1}^n$.

Now recall from Example 1.37 that for any algebra A , any matrix $M = (m_{ij}) \in \mathbf{GL}_n(A)$ corresponds to the unique algebra map $\varphi_M : k[\mathbf{GL}_n] \rightarrow A$ such that $\varphi_M(\overline{X_{ij}}) = m_{ij}$. In particular, for $\Phi_A(g) \equiv (\varphi_g(r_{ij})) \in \mathbf{GL}_n(A)$, we have

$$\varphi_{\Phi_A(g)}(\overline{X_{ij}}) = \varphi_g(r_{ij}).$$

On the other hand, by Yoneda bijection (see Fact 1.11) we know that

$$\varphi_{\Phi_A(g)} = \varphi_g \circ f.$$

Evaluating this at $\overline{X_{ij}}$ we get

$$\varphi_g(f(\overline{X_{ij}})) = \varphi_{\Phi_A(g)}(\overline{X_{ij}}) = \varphi_g(r_{ij})$$

for every algebra A and $g \in G(A)$. In particular, taking $A = R$ and $g = u \in G(R)$ the universal element, and noting that $\varphi_u = \text{id}_R$ we get

$$f(\overline{X_{ij}}) = r_{ij}$$

as desired.

Finally, to show $\Delta(r_{ij}) = \sum_l r_{il} \otimes r_{lj}$, we evaluate the condition $(\text{id}_V \otimes \Delta) \circ \rho = (\rho \otimes \text{id}_R) \circ \rho$ on the basis vector v_j . The left hand side gives

$$(\text{id}_V \otimes \Delta)(\rho(v_j)) = \sum_i v_i \otimes \Delta(r_{ij}),$$

whereas the right hand side gives

$$(\rho \otimes \text{id}_R)(\rho(v_j)) = \sum_l \rho(v_l) \otimes r_{lj} = \sum_{l,i} v_i \otimes r_{il} \otimes r_{lj}.$$

Now we note that $\{v_i \otimes 1_R \otimes 1_R\}$ is an $R \otimes R$ -basis of $V \otimes R \otimes R$, and so by comparing coefficients we get $\Delta(r_{ij}) = \sum_l r_{il} \otimes r_{lj}$. $\square$

Theorem 8.14. – Let k be a field, (R, Δ, S, ϵ) be a Hopf algebra over k . Every R -comodule V is a directed union of finite dimensional subcomodules.

Proof. – First note that the family of all finite dimensional subcomodules of V is *directed* – that is to say – given any two finite dimensional subcomodules W_1, W_2 of V , there is a third, namely $W_1 + W_2$, containing both W_1 and W_2 which is also a finite dimensional subcomodule. Therefore, it suffices to show that given any $v \in V$ there exists a finite dimensional subcomodule W containing v .

So take a basis $\{r_i\}$ of R and write $\rho(v) = \sum_i v_i \otimes r_i$: a finite sum with all v_i non-zero. We also write $\Delta(r_i) = \sum \lambda_{ijl}(r_j \otimes r_l)$ with $\lambda_{ijl} \in k$. Then

$$\sum \rho(v_i) \otimes r_i = (\rho \otimes \text{id}_R)(\rho(v)) = (\text{id}_V \otimes \Delta)(\rho(v)) = \sum v_i \otimes \lambda_{ijl} r_j \otimes r_l.$$

Now we compare coefficients, or more precisely consider the dual basis $\{r_\ell^*\}_\ell$ of the k -vector space R and apply the linear map $\text{id}_{V \otimes R} \otimes r_\ell^* : V \otimes R \otimes R \rightarrow V \otimes R$ given by $v \otimes r \otimes s \mapsto r_\ell^*(s) \cdot (v \otimes r)$ to the above equality. This gives

$$(\text{id}_{V \otimes R} \otimes r_\ell^*) \left(\sum_i \rho(v_i) \otimes r_i \right) = \sum_i r_\ell^*(r_i) \cdot \rho(v_i) = \rho(v_\ell)$$

and

$$(\text{id}_{V \otimes R} \otimes r_\ell^*) \left(\sum_{i,j,l} v_i \otimes \lambda_{ijl} r_j \otimes r_l \right) = \sum_{i,j,l} r_\ell^*(r_l) (v_i \otimes \lambda_{ijl} r_j) = \sum_{i,j} v_i \otimes \lambda_{ij\ell} r_j.$$

Therefore, we obtain the equality $\rho(v_\ell) = \sum_{i,j} v_i \otimes \lambda_{ij\ell} r_j$. Hence the space W spanned by v and the v_i is a subcomodule, and it is obviously finite dimensional. $\square$

Theorem 8.15. – *Let k be a field and (R, Δ, S, ϵ) be a k -Hopf algebra. Then R is a directed union of Hopf subalgebras R_α which are finitely generated k -algebras.*

Proof. – First note that the family of all finitely generated Hopf subalgebras of R is directed. Suppose $R_\alpha = k[S_\alpha]$ and $R_\beta = k[S_\beta]$ are two finitely generated Hopf subalgebras with S_α and S_β being finite subsets of R . It suffices to show that any finite subset of R is contained in one such R_α , since then $S_\alpha \cup S_\beta$ is contained in some finitely generated Hopf subalgebra, and thus R_α and R_β are also contained in this Hopf subalgebra. Thus $R = \bigcup_\alpha R_\alpha$.

Now consider R itself as an R -comodule via the regular representation $\Delta : R \rightarrow R \otimes R$ (see Example 8.5). By the previous theorem any finite subset is contained in a finite dimensional subspace (rather subcomodule), say V of R , such that $\Delta(V) \subset V \otimes R$. Let $\{v_j\}_1^n$ be a basis of V , and write $\Delta(v_j) = \sum_i v_i \otimes r_{ij}$. Then by Lemma 8.13 we know that $\Delta(r_{ij}) = \sum_l r_{il} \otimes r_{lj}$. Now consider the span U of $\{v_j\}$ and $\{r_{ij}\}$, and note that $\Delta(U) \subset U \otimes U$.

Now note that for any $a \in R$ if $\Delta(a) = \sum_i b_i \otimes c_i$, then $\Delta(Sa) = \sum_i S(c_i) \otimes S(b_i)$ since $\Delta \circ S = (\text{swap}) \circ (S \otimes S) \circ \Delta$. Consider the finite dimensional subspace $L := U + S(U)$. Then clearly $\Delta(L) \subset L \otimes L$. Also noting $S \circ S = \text{id}_R$, we conclude that $S(L) \subset L$. Now take $R_\alpha = k[L]$: this is a finitely generated subalgebra, generated by a (finite) basis of L . It is a Hopf algebra since $\Delta(L) \subset L \otimes L$ and $S(L) \subset L$ implies $\Delta(R_\alpha) \subset R_\alpha \otimes R_\alpha$ and $S(R_\alpha) \subset R_\alpha$. Finally, $R_\alpha \supset L \supset U \supset V$ and hence contains the finite set we started with. $\square$

We call an affine group functor *algebraic* if its representing algebra is finitely generated.

Corollary 8.16. – *Every affine group functor G over a field is an inverse limit of algebraic affine group functors.*

Proof. – Let (R, Δ, S, ϵ) be the Hopf algebra for G , and let G_α be the group functors corresponding to the R_α . For any algebra A , we will show that

$$\text{Hom}_{k\text{-Alg}}(R, A) \rightarrow \varprojlim_\alpha \text{Hom}_{k\text{-Alg}}(R_\alpha, A), \quad \varphi \mapsto (\varphi|_{R_\alpha})_\alpha$$

is an isomorphism of groups and natural in A .

We first show that the above map is bijective. If φ, ψ are such that $\varphi|_{R_\alpha} = \psi|_{R_\alpha}$ for all α , then since $R = \bigcup_\alpha R_\alpha$ we have $\varphi = \psi$, i.e. the above map is injective. On the other hand, given a compatible family $(\varphi_\alpha)_\alpha \in \varprojlim_\alpha \text{Hom}_{k\text{-Alg}}(R_\alpha, A)$ we define $\varphi : R \rightarrow A$ by $\varphi|_{R_\alpha} := \varphi_\alpha$. Then well-definedness follows because if $r \in R_\alpha \cap R_\beta$ then by directedness there exists γ such that $R_\alpha, R_\beta \subset R_\gamma$ and thus $\varphi(r) = \varphi_\gamma(r) = \varphi_\alpha(r) = \varphi_\beta(r)$ by compatibility since $\varphi_\gamma|_{R_\alpha} = \varphi_\alpha$ and $\varphi_\gamma|_{R_\beta} = \varphi_\beta$. Also it is then clear that φ is an algebra map, and obviously $\varphi \mapsto (\varphi_\alpha)_\alpha$.

Next, we show that $\varphi \mapsto (\varphi|_{R_\alpha})_\alpha$ is a group homomorphism. Take $\varphi, \psi \in \text{Hom}_{k\text{-Alg}}(R, A)$ and note that $(\varphi * \psi)|_{R_\alpha} = (M_A \circ (\varphi \otimes \psi) \circ \Delta)|_{R_\alpha} = M_A \circ (\varphi|_{R_\alpha} \otimes \psi|_{R_\alpha}) \circ \Delta_{R_\alpha} = \varphi|_{R_\alpha} * \psi|_{R_\alpha}$. Thus we have $\varphi * \psi \mapsto ((\varphi * \psi)|_{R_\alpha})_\alpha = (\varphi|_{R_\alpha} * \psi|_{R_\alpha})_\alpha = (\varphi|_{R_\alpha})_\alpha * (\psi|_{R_\alpha})_\alpha$. Hence the map $\varphi \mapsto (\varphi|_{R_\alpha})_\alpha$ is actually a group isomorphism.

It remains to show the naturality in the algebra A . In fact, given any algebra map $\Phi : A \rightarrow B$, we have to show that the following square

$$\begin{array}{ccc} \text{Hom}_{k\text{-Alg}}(R, A) & \longrightarrow & \varprojlim_\alpha \text{Hom}_{k\text{-Alg}}(R_\alpha, A) \\ \downarrow & & \downarrow \\ \text{Hom}_{k\text{-Alg}}(R, B) & \longrightarrow & \varprojlim_\alpha \text{Hom}_{k\text{-Alg}}(R_\alpha, B) \end{array}$$

commutes. Take any $\varphi \in \text{Hom}_{k\text{-Alg}}(\mathbf{R}, \mathbf{A})$. Commutativity of the above square is equivalent to showing $(\Phi \circ \varphi|_{\mathbf{R}_\alpha})_\alpha = ((\Phi \circ \varphi)|_{\mathbf{R}_\alpha})_\alpha$ which is clear. Thus we have shown $\mathbf{G} \simeq \varprojlim_\alpha \mathbf{G}_\alpha$. $\square$

Theorem 8.17. – *Every algebraic affine group functor over a field is isomorphic to a closed subgroup functor of some $\mathbf{GL}_n$.*

Proof. – Let $\mathbf{G}$ be a given algebraic affine group functor over a field k , represented by the finitely generated k -Hopf algebra $(\mathbf{R}, \Delta, S, \epsilon)$. As before, consider $\mathbf{R}$ itself as an $\mathbf{R}$ -comodule via $\Delta : \mathbf{R} \rightarrow \mathbf{R} \otimes \mathbf{R}$ (i.e. the *regular representation*). By Theorem 8.14 there exists a finite dimensional subcomodule $\mathbf{V}$ of $\mathbf{R}$ containing the finite set of algebra generators of $\mathbf{R}$. Let $\{v_j\}_{j=1}^n$ be a basis of $\mathbf{V}$, and write $\Delta(v_j) = \sum_i v_i \otimes r_{ij}$. Let $\Phi : \mathbf{G} \rightarrow \mathbf{GL}_n$ be the representation corresponding to the comodule $(\Delta|_{\mathbf{V}}, \mathbf{V})$. By Lemma 8.13 the corresponding Hopf algebra map $k[\mathbf{GL}_n] \rightarrow \mathbf{R}$ has r_{ij} in its image. But also note that $1 \otimes v_j = (\epsilon \otimes \text{id}_{\mathbf{R}})(\Delta(v_j)) = \sum_i \epsilon(v_i) \otimes r_{ij}$ and so $v_j = \sum_i \epsilon(v_i) r_{ij}$ are also contained in the image of Φ . Therefore, all the algebra generators of $\mathbf{R}$ are also contained in $\text{Im}(\Phi)$ and hence Φ is surjective. By Proposition 3.15 we know Φ is thus an isomorphism onto a closed subfunctor. $\square$

Lemma 8.18. – *Let $\mathbf{G}$ be an affine group functor over a field k . Every finite dimensional representation of $\mathbf{G}$ embeds in a finite sum of copies of the regular representation.*

Proof. – Let $\mathbf{V}$ be a finite dimensional $\mathbf{R}$ -comodule where $\mathbf{R}$ is the Hopf algebra corresponding to $\mathbf{G}$. Let $\mathbf{M} := \mathbf{V} \otimes \mathbf{R}$. Then clearly $\mathbf{M} \simeq \mathbf{R}^n$ if $\dim_k \mathbf{V} = n$. Consider the map $\sigma := \text{id}_{\mathbf{V}} \otimes \Delta : \mathbf{V} \otimes \mathbf{R} \rightarrow \mathbf{V} \otimes \mathbf{R} \otimes \mathbf{R}$. We verify that $(\sigma, \mathbf{M})$ is an $\mathbf{R}$ -comodule. Indeed, we have to show that

$$(\text{id}_{\mathbf{M}} \otimes \epsilon) \circ \sigma = [\mathbf{M} \xrightarrow{\sim} \mathbf{M} \otimes k] \quad \text{and} \quad (\text{id}_{\mathbf{M}} \otimes \Delta) \circ \sigma = (\sigma \otimes \text{id}_{\mathbf{R}}) \circ \sigma.$$

So take any $v \otimes r \in \mathbf{M}$ and by definition $\sigma(v \otimes r) = v \otimes \Delta(r)$. Hence

$$\begin{aligned} ((\text{id}_{\mathbf{M}} \otimes \epsilon) \circ \sigma)(v \otimes r) &= (\text{id}_{\mathbf{V}} \otimes \text{id}_{\mathbf{R}} \otimes \epsilon)(v \otimes \Delta(r)) \\ &= v \otimes (\text{id}_{\mathbf{R}} \otimes \epsilon)(\Delta(r)). \end{aligned}$$

But $(\text{id}_{\mathbf{R}} \otimes \epsilon) \circ \Delta = [\mathbf{R} \xrightarrow{\sim} \mathbf{R} \otimes k]$ and thus $((\text{id}_{\mathbf{M}} \otimes \epsilon) \circ \sigma)(v \otimes r) = v \otimes r \otimes 1$ i.e. $(\text{id}_{\mathbf{M}} \otimes \epsilon) \circ \sigma = [\mathbf{M} \xrightarrow{\sim} \mathbf{M} \otimes k]$. For the other condition, again we take $v \otimes r \in \mathbf{M}$ and using coassociativity $(\text{id}_{\mathbf{R}} \otimes \Delta) \circ \Delta = (\Delta \otimes \text{id}_{\mathbf{R}}) \circ \Delta$ we get

$$\begin{aligned} ((\text{id}_{\mathbf{M}} \otimes \Delta) \circ \sigma)(v \otimes r) &= (\text{id}_{\mathbf{V}} \otimes \text{id}_{\mathbf{R}} \otimes \Delta)(v \otimes \Delta(r)) \\ &= v \otimes ((\text{id}_{\mathbf{R}} \otimes \Delta) \circ \Delta)(r) \\ &= v \otimes ((\Delta \otimes \text{id}_{\mathbf{R}}) \circ \Delta)(r) \\ &= (\text{id}_{\mathbf{V}} \otimes \Delta \otimes \text{id}_{\mathbf{R}})(v \otimes \Delta(r)) \\ &= ((\sigma \otimes \text{id}_{\mathbf{R}}) \circ \sigma)(v \otimes r). \end{aligned}$$

Hence $(\text{id}_{\mathbf{M}} \otimes \Delta) \circ \sigma = (\sigma \otimes \text{id}_{\mathbf{R}}) \circ \sigma$.

Next we check that $\rho : \mathbf{V} \rightarrow \mathbf{M} = \mathbf{V} \otimes \mathbf{R}$ is an injective homomorphism of comodules. But the condition $\sigma \circ \rho = (\rho \otimes \text{id}_{\mathbf{R}}) \circ \rho$ is equivalent to $(\text{id}_{\mathbf{V}} \otimes \Delta) \circ \rho = (\rho \otimes \text{id}_{\mathbf{R}}) \circ \rho$ which is one of the defining conditions for ρ being a comodule. Finally, we use the fact that $(\text{id}_{\mathbf{V}} \otimes \epsilon) \circ \rho = [\mathbf{V} \xrightarrow{\sim} \mathbf{V} \otimes k]$ is an isomorphism, and hence the precomposed map ρ must be injective. $\square$

9. SCHEME THEORY 1 : PREASHEAVES, SHEAVES, AND SHEAFIFICATION

Let X be a topological space. We denote $\mathbf{Opn}(X)$ to be the category whose objects are open sets in X and for two open sets U, V the set of morphisms $\text{Hom}(U, V) := \{U \hookrightarrow V\}$ when $U \subset V$ and $\text{Hom}(U, V) := \emptyset$ if $U \not\subset V$.

By a *presheaf* of sets, rings, abelian groups,... on X we mean a contravariant functor from $\mathbf{Opn}(X)$ to sets, rings, abelian groups,... respectively. Given any open set U of X and a presheaf $\mathcal{F}$ on X , the elements of $\mathcal{F}(U)$ will be called *sections* on U . Given $V \subset U$ open sets, the image of a section $f \in \mathcal{F}(U)$ in $\mathcal{F}(V)$ will be written as $f|_V$, indicating that sections are to be thought of naively as functions on open sets.

A presheaf $\mathcal{F}$ on X is a *sheaf* if for every open set U and every open covering $U = \bigcup_{\lambda \in \Lambda} U_\lambda$ the following two conditions hold :

- (1) If $f, g \in \mathcal{F}(U)$ satisfy $f|_{U_\lambda} = g|_{U_\lambda}$ for all λ , then $f = g$. (*Locality axiom*)
- (2) If $f_\lambda \in \mathcal{F}(U_\lambda)$, $\lambda \in \Lambda$, satisfy $f_\lambda|_{U_\lambda \cap U_{\lambda'}} = f_{\lambda'}|_{U_\lambda \cap U_{\lambda'}}$ for all $\lambda, \lambda' \in \Lambda$, then there exists $f \in \mathcal{F}(U)$ such that $f|_{U_\lambda} = f_\lambda$ for all $\lambda \in \Lambda$. (*Gluing axiom*)

Definition 9.1. – Let X be a topological space equipped with a presheaf $\mathcal{F}$. Then for any point $x \in X$, we call

$$\mathcal{F}_x := \varinjlim_{x \in U} \mathcal{F}(U)$$

the *stalk* of $\mathcal{F}$ at x . Here, the limit is taken over all open neighborhoods U containing x . For any open U containing x , the image of some $f \in \mathcal{F}(U)$ in $\mathcal{F}_x$ is denoted by f_x .

Lemma 9.2. – Let X be a topological space and $\mathcal{F}$ be a presheaf satisfying the locality axiom. Then, for any open subset $U \subset X$, the map

$$\mathcal{F}(U) \rightarrow \prod_{x \in U} \mathcal{F}_x, \quad f \mapsto (f_x)_{x \in U}$$

is injective.

Proof. – Suppose $f, g \in \mathcal{F}(U)$ be such that $f_x = g_x$ for all $x \in U$. Then for each $x \in U$ there exists an open set $V_x \subset U$ containing x such that $f|_{V_x} = g|_{V_x}$. Since $U = \bigcup_{x \in U} V_x$, the hypothesis directly yields $f = g$. $\square$

Lemma 9.3. – Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of presheaves on a topological space X and consider any point $x \in X$. Then φ induces a morphism of stalks $\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$ in such a way that for all open neighborhoods $U \subset X$ of x the canonical diagram

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\varphi(U)} & \mathcal{G}(U) \\ \downarrow & & \downarrow \\ \mathcal{F}_x & \xrightarrow{\varphi_x} & \mathcal{G}_x \end{array}$$

is commutative.

Proof. – Let $x \in X$. For open sets $V \subset U$ of X containing x , consider the following commutative diagram :

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\text{res}} & \mathcal{F}(V) \\ \downarrow \varphi(U) & & \downarrow \varphi(V) \\ \mathcal{G}(U) & \xrightarrow{\text{res}} & \mathcal{G}(V) \\ & \searrow & \swarrow \\ & \mathcal{G}_x & \end{array}$$

which shows that $\mathcal{G}_x$ is a co-cone below the diagram $(\mathcal{F}(U))_{x \in U}$. Thus, by the universal property of $\mathcal{F}_x := \varinjlim_{x \in U} \mathcal{F}(U)$ we get a unique morphism $\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$ making all such squares (for opens $U \ni x$) in the statement of the lemma commutative. $\square$

Lemma 9.4. – Let $\mathcal{F}$ and $\mathcal{G}$ be presheaves on a topological space X , and $\varphi, \psi : \mathcal{F} \rightarrow \mathcal{G}$ be morphisms of presheaves such that $\varphi_x = \psi_x$ for all $x \in X$. If $\mathcal{G}$ is a sheaf, then $\varphi = \psi$.

Proof. – Fix an open set $U \subset X$. We have to show that $\varphi(U) = \psi(U)$. So take a section $s \in \mathcal{F}(U)$. To show that $\varphi(U)(s) = \psi(U)(s)$ it suffices to show that $\varphi(U)(s)_x = \psi(U)(s)_x$ for all $x \in U$. But, by the previous lemma we have

$$\varphi(U)(s)_x = \varphi_x(s_x) = \psi_x(s_x) = \psi(U)(s)_x$$

as required. $\square$

Proposition 9.5. – Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of sheaves on a topological space X .

- (1) φ is injective — in the sense that $\varphi(U) : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ is injective for each open subset $U \subset X$ — if and only if $\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$ is injective for all $x \in X$.

- (2) φ is locally surjective — in the sense that for any section $g \in \mathcal{G}(U)$ over an open subset $U \subset X$ and a point $x \in U$, there is an open neighborhood $U_x \subset U$ of x such that $g|_{U_x}$ admits a preimage with respect to $\varphi(U_x) : \mathcal{F}(U_x) \rightarrow \mathcal{G}(U_x)$ — if and only if $\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$ is surjective for all $x \in X$.
- (3) φ is an isomorphism if and only if $\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$ is bijective for all $x \in X$.

Proof. — (1) First, we prove the only if direction. Let $x \in X$ and suppose $\varphi_x(s_x) = \varphi_x(t_x)$ for $s_x, t_x \in \mathcal{F}_x$. We choose representatives :

$$s_x = [s_U], \quad t_x = [t_V]$$

where $x \in U \cap V$ and $s_U \in \mathcal{F}(U), t_V \in \mathcal{F}(V)$ for two open sets U, V containing x . Now, by Lemma 9.3, the equality $\varphi_x(s_x) = \varphi_x(t_x)$ can be written as $[\varphi(U)(s_U)] = [\varphi(V)(t_V)]$. Thus there exists an open set $W \subset U \cap V$ containing x such that $\varphi(U)(s_U)|_W = \varphi(V)(t_V)|_W$. Since φ is a natural transformation we know that it is compatible with restrictions, hence $\varphi(W)(s_U|_W) = \varphi(W)(t_V|_W)$. Now using injectivity of $\varphi(W)$ we get $s_U|_W = t_V|_W$ and therefore, $s_x = t_x$.

Next, for the if part, assume that φ_x is injective for all $x \in X$. For any open set $U \subset X$, we have to show $\varphi(U) : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ is injective as well. So take $f, f' \in \mathcal{F}(U)$ with $g = \varphi(U)(f)$ and $g' = \varphi(U)(f')$ such that $g = g'$. Then by Lemma 9.3, we have

$$\varphi_x(f_x) = g_x = g'_x = \varphi_x(f'_x)$$

and hence $f_x = f'_x$ for all $x \in U$, since φ_x is injective for all $x \in X$. But this implies $f = f'$ by Lemma 9.2. Thus $\varphi(U)$ is injective for all open set $U \subset X$.

(2) Now suppose φ is locally surjective. Take $g_x \in \mathcal{G}_x$ for some fixed $x \in X$. Choose a representative $g_U \in \mathcal{G}(U)$ for some open subset U containing x . Now by local surjectivity we choose an $f \in \mathcal{F}(V)$ for some open set $V \subset U$ containing x such that $\varphi(V)(f) = g|_V$. Then, by Lemma 9.3, we have

$$\varphi_x(f_x) = \varphi(V)(f)_x = (g|_V)_x = g_x$$

and hence φ_x is surjective.

Conversely, assume φ_x is surjective for all $x \in X$ and consider an element $g \in \mathcal{G}(U)$ over some open set $U \subset X$. Fixing $x \in U$, there exists an element $f_x \in \mathcal{F}_x$ such that $\varphi_x(f_x) = g_x$. Now let $U_x \subset U$ be an open set containing x such that the germ f_x is induced by some element $f_{U_x} \in \mathcal{F}(U_x)$. Thus $(\varphi(U_x)(f_{U_x}))_x = g_x = (g|_{U_x})_x$ by Lemma 9.3. Hence there exists another open neighborhood of x , say $V_x \subset U_x$ such that $\varphi(U_x)(f_{U_x})|_{V_x} = g|_{V_x}$ and then since φ is natural, we have $\varphi(V_x)(f_{U_x}|_{V_x}) = g|_{V_x}$ i.e. φ is locally surjective.

(3) If φ is an isomorphism, then $\varphi(U)$ is a bijection for each open set $U \subset X$, hence by the first two items we have that φ_x is also a bijection for each $x \in X$.

Conversely, suppose φ_x is a bijection for each $x \in X$. Then, φ is injective and locally surjective. Thus it remains to show that $\varphi(U)$ is a surjection for every open set $U \subset X$. So consider some $g \in \mathcal{G}(U)$ on some open set $U \subset X$. For every $x \in U$, there exists an open set $U_x \subset U$ containing x together with a section $f_{U_x} \in \mathcal{F}(U_x)$ such that $\varphi(U_x)(f_{U_x}) = g|_{U_x}$. Since the map

$$\varphi(U_x \cap U_{x'}) : \mathcal{F}(U_x \cap U_{x'}) \rightarrow \mathcal{G}(U_x \cap U_{x'})$$

is injective for any points $x, x' \in U$, we have

$$f_{U_x}|_{U_x \cap U_{x'}} = f_{U_{x'}}|_{U_x \cap U_{x'}}$$

since both map to $g|_{U_x \cap U_{x'}}$ under $\varphi(U_x \cap U_{x'})$. Using the fact that $\mathcal{F}$ is a sheaf, there is a unique $f \in \mathcal{F}(U)$ such that $f|_{U_x} = f_{U_x}$ for all $x \in U$. But then again using the sheaf property of $\mathcal{G}$, we see that $\varphi(U)$ maps f to g since $\varphi(U)(f)|_{U_x} = \varphi(U_x)(f_{U_x}) = g|_{U_x}$ for each $x \in U$. $\square$

Let X be a topological space and $\mathcal{F}$ be a presheaf on it. Furthermore, consider an open set $U \subset X$ and an open covering $\mathfrak{U} = (U_\lambda)_{\lambda \in \Lambda}$ of U by open sets $U_\lambda \subset U$ i.e. $U = \bigcup_{\lambda \in \Lambda} U_\lambda$. Then, we set

$$H^0(\mathfrak{U}, \mathcal{F}) := \text{Ker} \left(\prod_{\lambda \in \Lambda} \mathcal{F}(U_\lambda) \rightrightarrows \prod_{(\lambda, \lambda') \in \Lambda \times \Lambda'} \mathcal{F}(U_\lambda \cap U_{\lambda'}) \right)$$

which, written explicitly, is just all the sequences of sections on the open sets U_λ that agree on each of the pairwise intersections $U_\lambda \cap U_{\lambda'}$. Thus there is a canonical map

$$\mathcal{F}(U) \rightarrow H^0(\mathfrak{U}, \mathcal{F}), \quad f \mapsto (f|_{U_\lambda})_{\lambda \in \Lambda}$$

which is a bijection if $\mathcal{F}$ is a sheaf.

Fact 9.6. – $H^0(\mathfrak{U}, \cdot)$ is a functor on the category of presheaves on X , since any morphism $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ gives rise to a canonical map $H^0(\mathfrak{U}, \varphi) : H^0(\mathfrak{U}, \mathcal{F}) \rightarrow H^0(\mathfrak{U}, \mathcal{G})$.

Proof. – Define $H^0(\mathfrak{U}, \varphi) ((f_\lambda)_{\lambda \in \Lambda}) := (\varphi(U_\lambda)(f_\lambda))_{\lambda \in \Lambda}$, and the fact that $(\varphi(U_\lambda)(f_\lambda))_{\lambda \in \Lambda} \in H^0(\mathfrak{U}, \mathcal{G})$ follows from naturality of φ i.e. its compatibility with restrictions. $\square$

Now if $\mathfrak{U} = (U_\lambda)_{\lambda \in \Lambda}$ and $\mathfrak{V} = (V_\mu)_{\mu \in M}$ are two open coverings of U , we call $\mathfrak{V}$ a *refinement* of $\mathfrak{U}$ if there exists a map $\tau : M \rightarrow \Lambda$ such that $V_\mu \subset U_{\tau(\mu)}$ for all $\mu \in M$. In words, every open set of $\mathfrak{V}$ is contained in some open set of $\mathfrak{U}$. Then there is a canonical map

$$H^0(\mathfrak{U}, \mathcal{F}) \rightarrow H^0(\mathfrak{V}, \mathcal{F}), \quad (f_\lambda)_{\lambda \in \Lambda} \mapsto (f_{\tau(\mu)}|_{V_\mu})_{\mu \in M}$$

and it can be checked that this map is independent of τ . Indeed, let $\tau' : M \rightarrow \Lambda$ be a second map such that $V_\mu \subset U_{\tau'(\mu)}$ for every $\mu \in M$. Then $V_\mu \subset U_{\tau(\mu)} \cap U_{\tau'(\mu)}$, and since any $(f_\lambda)_{\lambda \in \Lambda} \in H^0(\mathfrak{U}, \mathcal{F})$ satisfies

$$f_{\tau(\mu)}|_{U_{\tau(\mu)} \cap U_{\tau'(\mu)}} = f_{\tau'(\mu)}|_{U_{\tau(\mu)} \cap U_{\tau'(\mu)}},$$

we see that $f_{\tau(\mu)}|_{V_\mu} = f_{\tau'(\mu)}|_{V_\mu}$ for all $\mu \in M$.

Now fix an open set $U \subset X$ and consider the collection of all open coverings $\mathfrak{U}$ of U as an index set I_U , equipped with the partial order : $\mathfrak{U} \leq \mathfrak{V}$ if and only if $\mathfrak{V}$ is a refinement of $\mathfrak{U}$. Then I_U is a directed set⁶ since, for two open coverings $\mathfrak{U} = (U_\lambda)_{\lambda \in \Lambda}$ and $\mathfrak{V} = (V_\mu)_{\mu \in M}$ of U , we can always construct the *product covering* $\mathfrak{U} \times \mathfrak{V} := (U_\lambda \cap V_\mu)_{(\lambda, \mu) \in \Lambda \times M}$, which is a common refinement of $\mathfrak{U}$ and $\mathfrak{V}$. Hence, $(H^0(\mathfrak{U}, \mathcal{F}))_{\mathfrak{U} \in I_U}$ is an inductive system, and we consider the limit

$$\mathcal{F}^+(U) := \varinjlim_{\mathfrak{U} \in I_U} H^0(\mathfrak{U}, \mathcal{F}).$$

Fact 9.7. – As U varies over the open subsets of X , the assignment $\mathcal{F}^+$ defines a presheaf on X .

Proof. – Indeed, for open subsets $V \subset U \subset X$ and an open covering $\mathfrak{U} = (U_\lambda)_{\lambda \in \Lambda}$ of U consider the corresponding restriction $\mathfrak{U}|_V = (U_\lambda \cap V)_{\lambda \in \Lambda}$. Consider the morphisms of type

$$H^0(\mathfrak{U}, \mathcal{F}) \rightarrow H^0(\mathfrak{U}|_V, \mathcal{F}), \quad (f_\lambda)_{\lambda \in \Lambda} \mapsto (f_\lambda|_{U_\lambda \cap V})_{\lambda \in \Lambda}.$$

Then, if $\mathfrak{U}'$ is a refinement of $\mathfrak{U}$, we have the following diagram

$$\begin{array}{ccc} H^0(\mathfrak{U}, \mathcal{F}) & \longrightarrow & H^0(\mathfrak{U}', \mathcal{F}) \\ \downarrow & & \downarrow \\ H^0(\mathfrak{U}|_V, \mathcal{F}) & \longrightarrow & H^0(\mathfrak{U}'|_V, \mathcal{F}) \\ & \searrow & \swarrow \\ & \mathcal{F}^+(V) & \end{array}$$

in which the square and the triangle commute. Thus, $\mathcal{F}^+(V)$ forms a co-cone below the diagram $(H^0(\mathfrak{U}, \mathcal{F}))_{\mathfrak{U} \in I_U}$. So, we get an induced map

$$\mathcal{F}^+(U) \rightarrow \mathcal{F}^+(V).$$

⁶We are interpreting an open covering as a subset of the power set of U , and thus for an open covering $\mathfrak{U}$, another open covering in which the sets are the same as that of $\mathfrak{U}$ but with repetitions will not be considered distinct from $\mathfrak{U}$. Therefore, I_U is actually a set.

Concretely, this map sends a class of $((f_\lambda)_{\lambda \in \Lambda}, \mathfrak{U} = (U_\lambda)_{\lambda \in \Lambda})$ to the class of $((f_\lambda|_{V \cap U_\lambda})_{\lambda \in \Lambda}, \mathfrak{U}|_V)$. Also, a crucial fact is that the following square commutes :

$$\begin{array}{ccc} H^0(\mathfrak{U}, \mathcal{F}) & \xrightarrow{\text{res}} & H^0(\mathfrak{U}|_V, \mathcal{F}) \\ \downarrow & & \downarrow \\ \mathcal{F}^+(U) & \xrightarrow{\text{res}} & \mathcal{F}^+(V) \end{array}$$

for all opens $V \subset U \subset X$. □

Fact 9.8. – *Maps of type*

$$\mathcal{F}(U) \rightarrow H^0(\mathfrak{U}, \mathcal{F}), \quad f \mapsto (f|_{U_\lambda})_{\lambda \in \Lambda}$$

induce a morphism of presheaves $\mathcal{F} \rightarrow \mathcal{F}^+$. Furthermore, if $\mathcal{F}$ is a sheaf, then this is a natural isomorphism.

Proof. – Composing with the map $H^0(\mathfrak{U}, \mathcal{F}) \rightarrow \mathcal{F}^+(U)$ that takes classes, we get a map

$$\iota_{\mathfrak{U}} : \mathcal{F}(U) \rightarrow \mathcal{F}^+(U).$$

We show that this is independent of $\mathfrak{U}$. Indeed, if $\mathfrak{V} = (V_\mu)_{\mu \in M}$ is another covering of U , we consider their common refinement $\mathfrak{U} \times \mathfrak{V}$. Now, a section $f \in \mathcal{F}(U)$ maps to $(f|_{U_\lambda})_{\lambda \in \Lambda} \in H^0(\mathfrak{U}, \mathcal{F})$ and to $(f|_{V_\mu})_{\mu \in M} \in H^0(\mathfrak{V}, \mathcal{F})$, and both of these map to $(f|_{U_\lambda \cap V_\mu})_{(\lambda, \mu)} \in H^0(\mathfrak{U} \times \mathfrak{V}, \mathcal{F})$ under the refinement maps $H^0(\mathfrak{U}, \mathcal{F}) \rightarrow H^0(\mathfrak{U} \times \mathfrak{V}, \mathcal{F})$ and $H^0(\mathfrak{V}, \mathcal{F}) \rightarrow H^0(\mathfrak{U} \times \mathfrak{V}, \mathcal{F})$. Hence, the classes of $(f|_{U_\lambda})_{\lambda}$ and $(f|_{V_\mu})_{\mu}$ in $\mathcal{F}^+(U) = \varinjlim_{\mathfrak{U} \in \mathcal{I}_U} H^0(\mathfrak{U}, \mathcal{F})$ is the same i.e. the class of $(f|_{U_\lambda \cap V_\mu})_{(\lambda, \mu)}$. Thus $\iota_{\mathfrak{U}}$ is independent of $\mathfrak{U}$ and so we have the map $\iota_U : \mathcal{F}(U) \rightarrow \mathcal{F}^+(U)$. Using the explicit description of the restriction maps for $\mathcal{F}^+$, we can check that $\iota : \mathcal{F} \rightarrow \mathcal{F}^+$ is compatible with restrictions : one simply fixes a cover $\mathfrak{U}$ of U and considers the cover $\mathfrak{U}|_V$ of V .

If $\mathcal{F}$ is a sheaf, then we first show that $\iota_U : \mathcal{F}(U) \rightarrow \mathcal{F}^+(U)$ is surjective. Any element of $\mathcal{F}^+(U)$ is a class of some $(f_\lambda)_\lambda \in H^0(\mathfrak{U}, \mathcal{F})$ for some covering $\mathfrak{U}$ of U . Since $\mathcal{F}(U) \xrightarrow{\sim} H^0(\mathfrak{U}, \mathcal{F})$, there exists $f \in \mathcal{F}(U)$ such that $f|_{U_\lambda} = f_\lambda$ for every λ . Thus $\iota_U(f) = [(f_\lambda)_\lambda]$. On the other hand, if $\iota_U(f) = \iota_U(g)$, then, for some covering $\mathfrak{U} = (U_\lambda)_{\lambda \in \Lambda}$, we have $[(f|_{U_\lambda})_\lambda] = [(g|_{U_\lambda})_\lambda]$. Thus there is a refinement $\mathfrak{V} = (V_\mu)_{\mu \in M}$ of $\mathfrak{U}$ such that

$$f|_{V_\mu} = f|_{U_{\tau(\mu)}}|_{V_\mu} = g|_{U_{\tau(\mu)}}|_{V_\mu} = g|_{V_\mu}$$

for every $\mu \in M$. Hence $f = g$ as required. □

Lemma 9.9. – *Let X be a topological space and $\mathcal{F}$ be a presheaf on X .*

- (1) *The assignment $\mathcal{F} \rightarrow \mathcal{F}^+$ is compatible with restriction to open subsets $U \subset X$, i.e. there are canonical isomorphisms $(\mathcal{F}^+)|_U \simeq (\mathcal{F}|_U)^+$.*
- (2) *The presheaf $\mathcal{F}^+$ satisfies the condition that for any open set U and any cover $\mathfrak{U} = (U_\lambda)_{\lambda \in \Lambda}$ of U , if $f, g \in \mathcal{F}^+(U)$ satisfy $f|_{U_\lambda} = g|_{U_\lambda}$ for every λ , then $f = g$.*
- (3) *Assume that the above sheaf condition in (2) holds for $\mathcal{F}$. Then $\mathcal{F}^+$ is a sheaf. Furthermore, $\mathcal{F} \rightarrow \mathcal{F}^+$ is a monomorphism in the sense that the canonical maps $\mathcal{F}(U) \rightarrow \mathcal{F}^+(U)$ for $U \subset X$ open are injective.*

Proof. – The statement (1) follows directly from the definition of $\mathcal{F}^+$. We consider the assertion (2). Fix an open set $U \subset X$ and consider elements $f, g \in \mathcal{F}^+(U)$ such that for an open covering $(U_\lambda)_{\lambda \in \Lambda}$ of U we have $f|_{U_\lambda} = g|_{U_\lambda}$ for all λ . Now we can choose an open covering $\mathfrak{V} = (V_\mu)_\mu$ of U such that f, g are represented by i.e. classes of $f' = (f_\mu)_\mu, g' = (g_\mu)_\mu \in H^0(\mathfrak{V}, \mathcal{F})$ respectively. Thus $f|_{U_\lambda} = [(f_\mu|_{V_\mu \cap U_\lambda})_\mu, \mathfrak{V}|_{U_\lambda}]$ and $g|_{U_\lambda} = [(g_\mu|_{V_\mu \cap U_\lambda})_\mu, \mathfrak{V}|_{U_\lambda}]$. Since we have $f|_{U_\lambda} = g|_{U_\lambda}$ we have

$$[(f_\mu|_{V_\mu \cap U_\lambda})_\mu, \mathfrak{V}|_{U_\lambda}] = [(g_\mu|_{V_\mu \cap U_\lambda})_\mu, \mathfrak{V}|_{U_\lambda}]$$

and thus there exists a refinement $\mathfrak{W}_\lambda$ of $\mathfrak{V}|_{U_\lambda}$ such that the images of $(f_\mu|_{V_\mu \cap U_\lambda})_\mu, (g_\mu|_{V_\mu \cap U_\lambda})_\mu \in H^0(\mathfrak{V}|_{U_\lambda}, \mathcal{F})$ in $H^0(\mathfrak{W}_\lambda, \mathcal{F})$ coincide. Now we set up an open cover $\mathfrak{W}$ of U by using the coverings $\mathfrak{W}_\lambda$ of U_λ with λ varying in Λ . Clearly, this $\mathfrak{W}$ is a refinement of $\mathfrak{V}$, and the images of $(f_\mu)_\mu, (g_\mu)_\mu \in H^0(\mathfrak{V}, \mathcal{F})$ in $H^0(\mathfrak{W}, \mathcal{F})$ coincide by construction. Thus $f = g$ as desired.

Finally, we verify assertion (3). First of all, the fact that $\mathcal{F} \rightarrow \mathcal{F}^+$ is a monomorphism is already established in the proof of the previous fact (see the last paragraph of the proof). Consider now an open set $U \subset X$ and an open covering $U = \bigcup_{\lambda \in \Lambda} U_\lambda$, as well as elements $f_\lambda \in \mathcal{F}^+(U_\lambda)$ such that $f_\lambda|_{U_\lambda \cap U_{\lambda'}} = f_{\lambda'}|_{U_\lambda \cap U_{\lambda'}}$ for all

$\lambda, \lambda' \in \Lambda$. Now fix $\lambda, \lambda' \in \Lambda$, and choose an open covering $\mathfrak{V}_\lambda = (V_{\lambda,\mu})_\mu$ of U_λ such that f_λ is represented by some compatible family $f'_\lambda = (f_{\lambda,\mu})_\mu \in H^0(\mathfrak{V}_\lambda, \mathcal{F})$ where $f_{\lambda,\mu} \in \mathcal{F}(V_{\lambda,\mu})$; similarly choose an open covering $\mathfrak{V}_{\lambda'} = (V_{\lambda',\nu})_\nu$ of $U_{\lambda'}$ such that $f_{\lambda'}$ is represented by some compatible family $f'_{\lambda'} = (f_{\lambda',\nu})_\nu \in H^0(\mathfrak{V}_{\lambda'}, \mathcal{F})$ where $f_{\lambda',\nu} \in \mathcal{F}(V_{\lambda',\nu})$. Take the product covering $\mathfrak{V}_\lambda \times \mathfrak{V}_{\lambda'} = \{(V_{\lambda,\mu} \cap V_{\lambda',\nu})_{(\mu,\nu)}\}$ of $U_\lambda \cap U_{\lambda'}$. Now consider the elements

$$f''_\lambda := \left(f_{\lambda,\mu} \mid_{V_{\lambda,\mu} \cap V_{\lambda',\nu}} \right)_{(\mu,\nu)}, \quad f''_{\lambda'} := \left(f_{\lambda',\nu} \mid_{V_{\lambda,\mu} \cap V_{\lambda',\nu}} \right)_{(\mu,\nu)}$$

of $H^0(\mathfrak{V}_\lambda \times \mathfrak{V}_{\lambda'}, \mathcal{F})$.

Claim. The images of f''_λ and $f''_{\lambda'}$ in $\mathcal{F}^+(U_\lambda \cap U_{\lambda'})$ are precisely $f_\lambda \mid_{U_\lambda \cap U_{\lambda'}}$ and $f_{\lambda'} \mid_{U_\lambda \cap U_{\lambda'}}$ respectively.

Proof. First note that the following is a commutative diagram :

$$\begin{array}{ccc} H^0(\mathfrak{V}_\lambda, \mathcal{F}) & \longrightarrow & \mathcal{F}^+(U_\lambda) \\ \downarrow & & \downarrow \text{res} \\ H^0(\mathfrak{V}_\lambda \mid_{U_\lambda \cap U_{\lambda'}}, \mathcal{F}) & \longrightarrow & \mathcal{F}^+(U_\lambda \cap U_{\lambda'}) \\ \text{ref} \downarrow & \nearrow & \\ H^0(\mathfrak{V}_\lambda \times \mathfrak{V}_{\lambda'}, \mathcal{F}) & & \end{array}$$

which is given by the element chase

$$\begin{array}{ccc} (f_{\lambda,\mu})_\mu & \longmapsto & f_\lambda \\ \downarrow & & \downarrow \\ (f_{\lambda,\mu} \mid_{V_{\lambda,\mu} \cap U_{\lambda'}})_\mu & \longmapsto & f_\lambda \mid_{U_\lambda \cap U_{\lambda'}} \\ \downarrow & \nearrow & \\ (f_{\lambda,\mu} \mid_{V_{\lambda,\mu} \cap V_{\lambda',\nu}})_{(\mu,\nu)} & & \end{array}$$

Here, note that $\mathfrak{V}_\lambda \mid_{U_\lambda \cap U_{\lambda'}} = (V_{\lambda,\mu} \cap U_\lambda \cap U_{\lambda'})_\mu = (V_{\lambda,\mu} \cap U_{\lambda'})_\mu$ and thus $\mathfrak{V}_\lambda \times \mathfrak{V}_{\lambda'} = \{(V_{\lambda,\mu} \cap V_{\lambda',\nu})_{(\mu,\nu)}\}$ is a refinement. The map *ref* denotes the refinement map. The claim can be proved by exactly same diagram chase for λ' . $\square$

Hence, since $f_\lambda \mid_{U_\lambda \cap U_{\lambda'}} = f_{\lambda'} \mid_{U_\lambda \cap U_{\lambda'}}$ we can take a refinement $\mathfrak{W}_{\lambda,\lambda'}$ of $\mathfrak{V}_\lambda \times \mathfrak{V}_{\lambda'}$ such that the images of f''_λ and $f''_{\lambda'}$ coincide in $H^0(\mathfrak{W}_{\lambda,\lambda'}, \mathcal{F})$.

Claim. Actually, f''_λ and $f''_{\lambda'}$ coincide in $H^0(\mathfrak{V}_\lambda \times \mathfrak{V}_{\lambda'}, \mathcal{F})$.

Proof. We have to show that the two elements

$$f''_\lambda := \left(f_{\lambda,\mu} \mid_{V_{\lambda,\mu} \cap V_{\lambda',\nu}} \right)_{(\mu,\nu)}, \quad f''_{\lambda'} := \left(f_{\lambda',\nu} \mid_{V_{\lambda,\mu} \cap V_{\lambda',\nu}} \right)_{(\mu,\nu)}$$

of $H^0(\mathfrak{V}_\lambda \times \mathfrak{V}_{\lambda'}, \mathcal{F})$ are equal i.e. for each pair (μ, ν) we must show that

$$f_{\lambda,\mu} \mid_{V_{\lambda,\mu} \cap V_{\lambda',\nu}} = f_{\lambda',\nu} \mid_{V_{\lambda,\mu} \cap V_{\lambda',\nu}}.$$

So fix a pair (μ, ν) and take an element $x \in V_{\lambda,\mu} \cap V_{\lambda',\nu}$. Since $\mathfrak{W}_{\lambda,\lambda'}$ is a covering of $U_\lambda \cap U_{\lambda'}$ there exists W in $\mathfrak{W}_{\lambda,\lambda'}$ such that $x \in W$. Suppose $W \subset V_{\lambda,\mu'} \cap V_{\lambda',\nu'}$ for some pair (μ', ν') . Consider the neighborhood

$$N_x := V_{\lambda,\mu} \cap V_{\lambda,\mu'} \cap V_{\lambda',\nu} \cap V_{\lambda',\nu'} \cap W$$

of x contained in $V_{\lambda,\mu} \cap V_{\lambda',\nu}$. Then, since the equality of the images of f''_λ and $f''_{\lambda'}$ coincide in $H^0(\mathfrak{W}_{\lambda,\lambda'}, \mathcal{F})$ we have $f_{\lambda,\mu'}|_W = f_{\lambda',\nu'}|_W$, it follows that

$$\begin{aligned} f_{\lambda,\mu}|_{N_x} &= \left(f_{\lambda,\mu}|_{V_{\lambda,\mu} \cap V_{\lambda,\mu'}} \right)|_{N_x} = \left(f_{\lambda,\mu'}|_{V_{\lambda,\mu} \cap V_{\lambda,\mu'}} \right)|_{N_x} \\ &= f_{\lambda,\mu'}|_{N_x} = (f_{\lambda,\mu'}|_W)|_{N_x} = (f_{\lambda',\nu'}|_W)|_{N_x} \\ &= f_{\lambda',\nu'}|_{N_x} = \left(f_{\lambda',\nu'}|_{V_{\lambda',\nu} \cap V_{\lambda',\nu'}} \right)|_{N_x} = \left(f_{\lambda',\nu}|_{V_{\lambda',\nu} \cap V_{\lambda',\nu'}} \right)|_{N_x} \\ &= f_{\lambda',\nu}|_{N_x}. \end{aligned}$$

Thus, for x varying over $V_{\lambda,\mu} \cap V_{\lambda',\nu}$ these N_x form a cover of $V_{\lambda,\mu} \cap V_{\lambda',\nu}$, and using the sheaf locality hypothesis on $\mathcal{F}$ we conclude that $f_{\lambda,\mu}|_{V_{\lambda,\mu} \cap V_{\lambda',\nu}} = f_{\lambda',\nu}|_{V_{\lambda,\mu} \cap V_{\lambda',\nu}}$ as required. $\square$

Now we set up an open covering $\mathfrak{V}$ of U from the coverings $\mathfrak{V}_\lambda$ with $\lambda \in \Lambda$; then, the elements f'_λ , $\lambda \in \Lambda$, give rise to an element $f' \in H^0(\mathfrak{V}, \mathcal{F})$ by the previous claim. Let $f \in \mathcal{F}^+(U)$ be the image of $f' \in H^0(\mathfrak{V}, \mathcal{F})$.

Claim. For every $\lambda \in \Lambda$ we have $f|_{U_\lambda} = f_\lambda$.

Proof. Write $\mathfrak{V} = (V_\alpha)_{\alpha \in A}$ as the covering of U obtained by collecting all the $V_{\lambda,\mu}$, and thus α ranges over all pairs (λ, μ) . Also write $f' \in H^0(\mathfrak{V}, \mathcal{F})$ as $(s_\alpha)_{\alpha \in A}$ where $s_\alpha|_{V_\alpha \cap V_{\alpha'}} = s_{\alpha'}|_{V_\alpha \cap V_{\alpha'}}$. Fix λ . Consider the two coverings $\mathfrak{V}|_{U_\lambda} = \{V_\alpha \cap U_\lambda \mid \alpha \in A\}$ and $\mathfrak{V}_\lambda = \{V_{\lambda,\mu} \mid \mu\}$. Let $\mathfrak{R}_\lambda := \mathfrak{V}|_{U_\lambda} \times \mathfrak{V}_\lambda = \{V_\alpha \cap V_{\lambda,\mu} \mid \alpha, \mu\}$. Now there are canonical maps

$$H^0(\mathfrak{V}, \mathcal{F}) \xrightarrow{\text{res}} H^0(\mathfrak{V}|_{U_\lambda}, \mathcal{F}) \xrightarrow{\text{ref}} H^0(\mathfrak{R}_\lambda, \mathcal{F})$$

and

$$H^0(\mathfrak{V}_\lambda, \mathcal{F}) \xrightarrow{\text{ref}} H^0(\mathfrak{R}_\lambda, \mathcal{F}).$$

Then $f' = (s_\alpha)_{\alpha \in A} \mapsto f'|_{U_\lambda} = (s_\alpha|_{U_\lambda \cap V_\alpha})_\alpha \mapsto (s_\alpha|_{V_\alpha \cap V_{\lambda,\mu}})_{(\alpha,\mu)}$ and $f'_\lambda \mapsto (f_{\lambda,\mu}|_{V_\alpha \cap V_{\lambda,\mu}})_{(\alpha,\mu)}$. Fix a pair (α, μ) . Now V_α equals some $V_{\lambda',\nu}$ and

$$s_\alpha|_{V_\alpha \cap V_{\lambda,\mu}} = f_{\lambda',\nu}|_{V_{\lambda',\nu} \cap V_{\lambda,\mu}} = f_{\lambda,\mu}|_{V_\alpha \cap V_{\lambda,\mu}}$$

by the previous claim. We have proved that the images of $f' \in H^0(\mathfrak{V}, \mathcal{F})$ and $f'_\lambda \in H^0(\mathfrak{V}_\lambda, \mathcal{F})$ under the two maps above coincide in $H^0(\mathfrak{R}_\lambda, \mathcal{F})$. Passing to colimits we get $f|_{U_\lambda} = f_\lambda$. $\square$

This completes the proof of the lemma. $\square$

Theorem 9.10. – Let $\mathcal{F}$ be a presheaf on a topological space X . Then there exists a sheaf $\mathcal{F}'$ together with a morphism of presheaves $\mathcal{F} \rightarrow \mathcal{F}'$ such that the following universal property is satisfied :

If $\mathcal{F} \rightarrow \mathcal{G}$ is a morphism of presheaves into a sheaf $\mathcal{G}$, there exists a unique morphism of sheaves $\mathcal{F}' \rightarrow \mathcal{G}$ such that the diagram

$$\begin{array}{ccc} \mathcal{F} & \longrightarrow & \mathcal{F}' \\ \downarrow & \swarrow & \\ \mathcal{G} & & \end{array}$$

is commutative.

We call $\mathcal{F}'$ together with the morphism $\mathcal{F} \rightarrow \mathcal{F}'$ the sheafification of $\mathcal{F}$, or the sheaf associated to $\mathcal{F}$.

Proof. – Step 1 : Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism into a sheaf $\mathcal{G}$. We show that this factors through a unique morphism $\varphi^+ : \mathcal{F}^+ \rightarrow \mathcal{G}$ such that $\varphi^+ \circ \iota = \varphi$ where $\iota : \mathcal{F} \rightarrow \mathcal{F}^+$ is the canonical map.

We first define the map φ^+ . Fix an open set $U \subset X$. An element of $\mathcal{F}^+(U)$ is represented by some $((s_\lambda)_{\lambda \in \Lambda}, \mathfrak{U})$ for some $(s_\lambda)_\lambda \in H^0(\mathfrak{U}, \mathcal{F})$ where $\mathfrak{U} = (U_\lambda)_{\lambda \in \Lambda}$ is an open covering of U . Now $(\varphi(U_\lambda)(s_\lambda))_{\lambda \in \Lambda} \in H^0(\mathfrak{U}, \mathcal{G}) \simeq \mathcal{G}(U)$ since $\mathcal{G}$ is a sheaf. Define

$$S := \text{the unique section such that } S|_{U_\lambda} = \varphi(U_\lambda)(s_\lambda) \text{ for all } \lambda.$$

Now we show that this S does not depend on the choice of the representative $((s_\lambda)_{\lambda \in \Lambda}, \mathfrak{U})$. Indeed, suppose $((s_\lambda)_{\lambda \in \Lambda}, \mathfrak{U})$ and $((t_\mu)_{\mu \in M}, \mathfrak{V}) \in H^0(\mathfrak{V}, \mathcal{F})$ with $\mathfrak{V} = (V_\mu)_{\mu \in M}$ a covering of U and $t_\mu \in \mathcal{F}(V_\mu)$, be two representatives of the same class in $\mathcal{F}^+(U)$. Also suppose that $((t_\mu)_{\mu \in M}, \mathfrak{V})$ corresponds to $T \in \mathcal{G}(U)$. Now, there exists a common refinement $\mathfrak{W}$ of $\mathfrak{U}$ and $\mathfrak{V}$ such that both $(s_\lambda)_{\lambda \in \Lambda}$, $(t_\mu)_{\mu \in M}$ map to the same element in $H^0(\mathfrak{W}, \mathcal{F})$. Writing $\mathfrak{W} = (W_i)_{i \in I}$ we know that for each i there exists $\lambda(i) \in \Lambda$ and

$\mu(i) \in M$ such that $W_i \subset U_{\lambda(i)} \cap V_{\mu(i)}$ and $s_{\lambda(i)}|_{W_i} = t_{\mu(i)}|_{W_i}$ in $\mathcal{F}(W_i)$. Applying $\varphi(W_i)$ we obtain that $\varphi(W_i)(s_{\lambda(i)}|_{W_i}) = \varphi(W_i)(t_{\mu(i)}|_{W_i})$, which by naturality of φ implies $\varphi(U_{\lambda(i)})(s_{\lambda(i)})|_{W_i} = \varphi(V_{\mu(i)})(t_{\mu(i)})|_{W_i}$ i.e. $(S|_{U_{\lambda(i)}})|_{W_i} = (T|_{V_{\mu(i)}})|_{W_i}$. But the W_i cover U and we have $S|_{W_i} = T|_{W_i}$ for all i , whence $S = T$. Thus, we can define $\varphi^+ : \mathcal{F}^+ \rightarrow \mathcal{G}$ by

$$\varphi^+(U) : \mathcal{F}^+(U) \rightarrow \mathcal{G}(U), \quad [((s_\lambda)_{\lambda \in \Lambda}, \mathfrak{U})] \mapsto S.$$

Next, we show that φ^+ is compatible with restrictions. Let $V \subset U$ be open sets. So take a class $[((s_\lambda)_{\lambda \in \Lambda}, \mathfrak{U} = (U_\lambda)_{\lambda \in \Lambda})] \in \mathcal{F}^+(U)$ and consider its image under the composite

$$\mathcal{F}^+(U) \xrightarrow{\varphi^+(U)} \mathcal{G}(U) \xrightarrow{\text{res}} \mathcal{G}(V)$$

and

$$\mathcal{F}^+(U) \xrightarrow{\text{res}} \mathcal{F}^+(V) \xrightarrow{\varphi^+(V)} \mathcal{G}(V).$$

The first one gives $[((s_\lambda)_{\lambda \in \Lambda}, \mathfrak{U} = (U_\lambda)_{\lambda \in \Lambda})] \mapsto S \mapsto S|_V$ where $S \in \mathcal{G}(U)$ satisfies $S|_{U_\lambda} = \varphi(U_\lambda)(s_\lambda)$ for all λ . The second one carries $[((s_\lambda)_{\lambda \in \Lambda}, \mathfrak{U} = (U_\lambda)_{\lambda \in \Lambda})] \mapsto [((s_\lambda|_{V \cap U_\lambda})_{\lambda \in \Lambda}, \mathfrak{U}|_V = (V \cap U_\lambda)_{\lambda \in \Lambda})] \mapsto S'$ where $S' \in \mathcal{G}(V)$ satisfies $S'|_{V \cap U_\lambda} = \varphi(V \cap U_\lambda)(s_\lambda|_{V \cap U_\lambda}) = (S|_{U_\lambda})|_{V \cap U_\lambda} = (S|_V)|_{V \cap U_\lambda}$ for every λ . But as the $V \cap U_\lambda$ cover V , we have $S' = S|_V$ as required. Thus φ^+ is a natural map.

Next, we check that $\varphi^+ \circ \iota = \varphi$ where $\iota : \mathcal{F} \rightarrow \mathcal{F}^+$ is the canonical map. Now recall that by the arguments in the proof of Fact 9.8, we can consider $\iota(U) : \mathcal{F}(U) \rightarrow \mathcal{F}^+(U)$ to be $s \mapsto [(s, \mathfrak{U} = \{U\})]$. Thus $\varphi^+(U)(\iota(U)(s)) = \varphi^+(U)([(s, \mathfrak{U} = \{U\})]) = \varphi(U)(s)$. This verifies $\varphi^+ \circ \iota = \varphi$.

Next, we show the uniqueness of φ^+ subject to the property $\varphi^+ \circ \iota = \varphi$. So suppose $\psi : \mathcal{F}^+ \rightarrow \mathcal{G}$ is another map satisfying $\psi \circ \iota = \varphi$. Fix an open set $U \subset X$. We have to show that $\psi(U) = \varphi^+(U)$. Let $\alpha := [((s_\lambda)_{\lambda \in \Lambda}, \mathfrak{U} = (U_\lambda)_{\lambda \in \Lambda})] \in \mathcal{F}^+(U)$ for some open cover $\mathfrak{U}$ of U and a compatible sequence $(s_\lambda)_{\lambda \in \Lambda} \in H^0(\mathfrak{U}, \mathcal{F})$. Since $\mathcal{G}$ is a sheaf, we show that for each λ we have

$$\psi(U)(\alpha)|_{U_\lambda} = \varphi^+(U)(\alpha)|_{U_\lambda} = \varphi(U_\lambda)(s_\lambda),$$

where the second equality is by definition of $\varphi^+(U)$. Now $\psi(U)(\alpha)|_{U_\lambda} = \psi(U_\lambda)(\alpha|_{U_\lambda})$ and so it suffices to show $\alpha|_{U_\lambda} = \iota(U_\lambda)(s_\lambda)$. For this consider the restriction map

$$\text{res} : H^0(\mathfrak{U}, \mathcal{F}) \rightarrow H^0(\mathfrak{U}|_{U_\lambda}, \mathcal{F})$$

and the canonical refinement map

$$\text{ref} : H^0(\{U_\lambda\}, \mathcal{F}) \rightarrow H^0(\mathfrak{U}|_{U_\lambda}, \mathcal{F}).$$

We then have $\text{res}((s_\mu)_{\mu \in \Lambda}) = (s_\mu|_{U_\lambda \cap U_\mu})_{\mu \in \Lambda}$ and $\text{ref}((s_\lambda)) = (s_\lambda|_{U_\lambda \cap U_\mu})_{\mu \in \Lambda}$. These two images are equal by $(s_\mu)_\mu$ being a compatible sequence. Finally, we get

$$\alpha|_{U_\lambda} = [(\text{res}((s_\mu)_{\mu \in \Lambda}), \mathfrak{U}|_{U_\lambda})] = [(\text{ref}((s_\lambda)), \mathfrak{U}|_{U_\lambda})] = [(s_\lambda, \{U_\lambda\})] = \iota(U_\lambda)(s_\lambda)$$

as required.

Step 2 : Now we set $\mathcal{F}' := \mathcal{F}^{++} := (\mathcal{F}^+)^+$ with the map $\mathcal{F} \rightarrow \mathcal{F}'$ being the composite $\mathcal{F} \xrightarrow{\iota} \mathcal{F}^+ \xrightarrow{\iota^+} \mathcal{F}^{++}$. Now suppose $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ is a morphism into a sheaf $\mathcal{G}$. Then we have the commuting diagram

$$\begin{array}{ccccc} \mathcal{F} & \xrightarrow{\iota} & \mathcal{F}^+ & \xrightarrow{\iota^+} & \mathcal{F}^{++} \\ \varphi \downarrow & \nearrow \varphi^+ & & \nearrow \varphi^{++} & \\ \mathcal{G} & & & & \end{array}$$

where $\varphi^+ : \mathcal{F}^+ \rightarrow \mathcal{G}$ is the unique map such that $\varphi^+ \circ \iota = \varphi$ and $\varphi^{++} : \mathcal{F}^{++} \rightarrow \mathcal{G}$ is the unique map such that $\varphi^{++} \circ \iota^+ = \varphi^+$. Thus, φ^{++} satisfies $\varphi^{++} \circ (\iota^+ \circ \iota) = \varphi$. Conversely, if $\psi : \mathcal{F}^{++} \rightarrow \mathcal{G}$ is a map satisfying $\psi \circ (\iota^+ \circ \iota) = \varphi$, then $\psi \circ \iota^+ = \varphi^+$ and thus $\psi = \varphi^{++}$ as required. $\square$

We will now give an alternative proof of Theorem 9.10. This is a slightly less technical approach than the previous one. We prove this as a series of standard facts.

First, for each open $U \subset X$, we define

$$\mathcal{F}'(U) := \left\{ (s_x)_{x \in U} \in \prod_{x \in U} \mathcal{F}_x \mid \exists \mathfrak{U} = (U_\lambda)_\lambda \in \mathcal{I}_U \text{ and } (t_\lambda)_\lambda \in \prod_\lambda \mathcal{F}(U_\lambda) \text{ s.t. } (t_\lambda)_x = s_x \text{ for } x \in U_\lambda \right\}.$$

For open sets $V \subset U$, the restriction $\mathcal{F}'(U) \rightarrow \mathcal{F}'(V)$ is given by $(s_x)_{x \in U} \mapsto (s_x)_{x \in V}$. Indeed, we check that $(s_x)_{x \in V} \in \mathcal{F}'(V)$. For this choose the open cover $\mathfrak{U} = (U_\lambda)_\lambda$ that locally induces $(s_x)_{x \in U}$ and consider $\mathfrak{U}|_V = (U_\lambda \cap V)_\lambda$. Then $(t_\lambda|_{V \cap U_\lambda})_x = (t_\lambda)_x = s_x$ for all $x \in V \cap U_\lambda \subset U_\lambda$.

Fact 9.11. – $\mathcal{F}'$ is a sheaf.

Proof. – Suppose $U = \bigcup_\lambda U_\lambda$ and $t = (s_x)_{x \in U}$, $t' = (s'_x)_{x \in U} \in \mathcal{F}'(U)$ be such that $t|_{U_\lambda} = t'|_{U_\lambda}$ for all λ . This means $(s_x)_{x \in U_\lambda} = (s'_x)_{x \in U_\lambda}$ for all λ . Now since U_λ cover U , given any $x \in U$ we know $x \in U_\lambda$ for some λ and so $s_x = s'_x$. Thus $t = t'$.

Next, suppose given an open cover $U = \bigcup_{\lambda \in \Lambda} U_\lambda$ and $t_\lambda = (s_x^\lambda)_{x \in U_\lambda} \in \mathcal{F}'(U_\lambda)$ such that $t_\lambda|_{U_\lambda \cap U_\mu} = t_\mu|_{U_\lambda \cap U_\mu}$ for all λ, μ . This means $(s_x^\lambda)_{x \in U_\lambda \cap U_\mu} = (s_x^\mu)_{x \in U_\lambda \cap U_\mu}$ i.e. $s_x^\lambda = s_x^\mu$ for all $x \in U_\lambda \cap U_\mu$. So we set $t := (s_x^\lambda)_{x \in U_\lambda, \lambda \in \Lambda} \in \prod_{x \in U} \mathcal{F}_x$, and this makes sense by the previous line. Then clearly $t|_{U_\lambda} = (s_x^\lambda)_{x \in U_\lambda} = t_\lambda$. Thus it remains to show that $t \in \mathcal{F}'(U)$. Since each t_λ is locally induced there exists an open cover $\mathfrak{V}_\lambda$ of U_λ and sections on the open sets of $\mathfrak{V}_\lambda$. Putting together we get $(\mathfrak{V}_\lambda)_{\lambda \in \Lambda}$ which is an open cover of U such that the sections on the open sets of $(\mathfrak{V}_\lambda)_{\lambda \in \Lambda}$ induce the components of t . This completes the proof that $\mathcal{F}'$ is a sheaf. $\square$

Now we define $\kappa_{\mathcal{F}} : \mathcal{F} \rightarrow \mathcal{F}'$ that sends a section $s \in \mathcal{F}(U)$ to $(s_x)_{x \in U}$. Obviously, $(s_x)_{x \in U} \in \mathcal{F}'(U)$ since it is locally induced by the one-term sequence $(s) \in \mathcal{F}(U)$ corresponding to the trivial cover $\{U\}$ of U . Also, it is clear that $\kappa_{\mathcal{F}}$ is natural in the sense that it is compatible with restrictions.

Fact 9.12. – If $\mathcal{F}$ is a sheaf, then $\kappa_{\mathcal{F}}$ is an isomorphism.

Proof. – Fix an open set U and suppose $\kappa_{\mathcal{F}}(U)(s) = \kappa_{\mathcal{F}}(U)(t)$ for $s, t \in \mathcal{F}(U)$. This means $(s_x)_{x \in U} = (t_x)_{x \in U}$. By Lemma 9.2, we have $s = t$.

Now suppose $t \in \mathcal{F}'(U)$; write $t = (t_x)_{x \in U}$ and suppose $U = \bigcup_\lambda U_\lambda$ with $t_\lambda \in \mathcal{F}(U_\lambda)$ which locally induce t i.e. $(t_\lambda)_x = t_x$ for all $x \in U_\lambda$. Thus for all $x \in U_\lambda \cap U_\mu$ we have $(t_\lambda)_x = t_x = (t_\mu)_x$. Again using Lemma 9.2, we get $t_\lambda|_{U_\lambda \cap U_\mu} = t_\mu|_{U_\lambda \cap U_\mu}$ for all λ, μ . But since $\mathcal{F}$ is a sheaf, these t_λ glue to some $s \in \mathcal{F}(U)$ i.e. $s|_{U_\lambda} = t_\lambda$ for all λ . So, for any $x \in U$ there is some λ such that $x \in U_\lambda$, whence $s_x = (s|_{U_\lambda})_x = (t_\lambda)_x = t_x$ i.e. $\kappa_{\mathcal{F}}(U)(s) = t$. $\square$

Fact 9.13. – $\kappa_{\mathcal{F}}$ induces an isomorphism of stalks : $\mathcal{F}_x \simeq \mathcal{F}'_x$ for all $x \in X$.

Proof. – Let $t_x, s_x \in \mathcal{F}_x$ and suppose $t \in \mathcal{F}(U_1)$, $s \in \mathcal{F}(U_2)$ be sections that induce the germs t_x and s_x respectively, where U_1, U_2 are two open sets containing x . We can replace U_1 and U_2 by $U := U_1 \cap U_2$ and assume $s, t \in \mathcal{F}(U)$. Now $\kappa_{\mathcal{F}}(U) : \mathcal{F}(U) \rightarrow \mathcal{F}'(U)$ takes $t \mapsto (t_y)_{y \in U}$ and $s \mapsto (s_y)_{y \in U}$. If $(\kappa_{\mathcal{F}})_x(s_x) = (\kappa_{\mathcal{F}})_x(t_x)$, i.e. $(\kappa_{\mathcal{F}}(U)(s))_x = (\kappa_{\mathcal{F}}(U)(t))_x$ i.e. $((s_y)_{y \in U})_x = ((t_y)_{y \in U})_x$, then there exists a neighborhood $U_x \subset U$ of x such that $((s_y)_{y \in U})|_{U_x} = ((t_y)_{y \in U})|_{U_x}$ i.e. $(s_y)_{y \in U_x} = (t_y)_{y \in U_x}$. In particular, $s_x = t_x$, whence we have shown that $(\kappa_{\mathcal{F}})_x$ is injective.

Now we show that $(\kappa_{\mathcal{F}})_x : \mathcal{F}_x \rightarrow \mathcal{F}'_x$ is surjective for all $x \in X$. Take some $T_x \in (\mathcal{F}')_x$ and suppose it is a germ of some $T \in \mathcal{F}'(U)$ where U is an open set with $x \in U$. We write $T = (s_y)_{y \in U}$ and obtain an open cover $U = \bigcup_\lambda U_\lambda$ with $s_\lambda \in \mathcal{F}(U_\lambda)$ such that $(s_\lambda)_z = s_z$ for all $z \in U_\lambda$. Since $x \in U$, there is some λ such that $x \in U_\lambda$ and so $(s_\lambda)_x = s_x$. Now for this λ with $x \in U_\lambda$, T_x is a germ of $T|_{U_\lambda} = (s_y)_{y \in U_\lambda} = ((s_\lambda)_y)_{y \in U_\lambda}$ as well. Consider $\kappa_{\mathcal{F}}(U_\lambda) : \mathcal{F}(U_\lambda) \rightarrow \mathcal{F}'(U_\lambda)$ which takes $s_\lambda \in \mathcal{F}(U_\lambda)$ to $((s_\lambda)_y)_{y \in U_\lambda} = T|_{U_\lambda} \in \mathcal{F}'(U_\lambda)$. Therefore, we have

$$(\kappa_{\mathcal{F}})_x((s_\lambda)_x) = (\kappa_{\mathcal{F}}(U_\lambda)(s_\lambda))_x = T_x$$

i.e. $(\kappa_{\mathcal{F}})_x$ is surjective. $\square$

Fact 9.14. – If $\mathcal{G}$ is a sheaf and $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ is a presheaf morphism, then there exists a unique morphism $\varphi' : \mathcal{F}' \rightarrow \mathcal{G}$ such that $\varphi' \circ \kappa_{\mathcal{F}} = \varphi$.

Proof. – First observe that the $\varphi' \circ \kappa_{\mathcal{F}} = \varphi$ if and only if $\kappa_{\mathcal{G}} \circ \varphi = (\kappa_{\mathcal{G}} \circ \varphi') \circ \kappa_{\mathcal{F}}$ that is to say the following diagram

$$\begin{array}{ccc} \mathcal{F} & \xrightarrow{\kappa_{\mathcal{F}}} & \mathcal{F}' \\ \varphi \downarrow & & \downarrow \kappa_{\mathcal{G}} \circ \varphi' \\ \mathcal{G} & \xrightarrow{\kappa_{\mathcal{G}}} & \mathcal{G}' \end{array}$$

is commutative, since $\kappa_{\mathcal{G}}$ is an isomorphism. Thus, it suffices to show that there exists a unique morphism $\varphi^+ : \mathcal{F}' \rightarrow \mathcal{G}'$ such that

$$\begin{array}{ccc} \mathcal{F} & \xrightarrow{\kappa_{\mathcal{F}}} & \mathcal{F}' \\ \varphi \downarrow & & \downarrow \varphi^+ \\ \mathcal{G} & \xrightarrow{\kappa_{\mathcal{G}}} & \mathcal{G}' \end{array}$$

is commutative.

Now define $\varphi^+ : \mathcal{F}' \rightarrow \mathcal{G}'$ on an open set U by $\varphi^+(U)((s_x)_{x \in U}) := (\varphi_x(s_x))_{x \in U}$. First, we note that $(\varphi_x(s_x))_{x \in U}$ is indeed an element of $\mathcal{G}'(U)$, since for a given open cover $U = \bigcup_{\lambda} U_{\lambda}$ with $s_{\lambda} \in \mathcal{F}(U_{\lambda})$ that locally induce $(s_x)_{x \in U}$, we have $\varphi(U_{\lambda})(s_{\lambda}) \in \mathcal{G}(U_{\lambda})$ satisfying $\varphi(U_{\lambda})(s_{\lambda})_x = \varphi_x((s_{\lambda})_x) = \varphi_x(s_x)$ for every $x \in U_{\lambda}$. Now, for a section $t \in \mathcal{F}(U)$, consider the following equalities :

$$((\varphi^+ \circ \kappa_{\mathcal{F}})(U))(t) = \varphi^+(U)((t_x)_{x \in U}) = (\varphi_x(t_x))_{x \in U}.$$

On the other hand, we have

$$(\kappa_{\mathcal{G}} \circ \varphi)(U)(t) = \kappa_{\mathcal{G}}(U)(\varphi(U)(t)) = ((\varphi(U)(t))_x)_{x \in U} = (\varphi_x(t_x))_{x \in U}.$$

We have thus established the existence.

It remains to show the uniqueness of φ^+ subject to the commutativity property. So suppose $\psi : \mathcal{F}' \rightarrow \mathcal{G}'$ is such that $\psi \circ \kappa_{\mathcal{F}} = \kappa_{\mathcal{G}} \circ \varphi$. Since taking stalks commutes with composition of presheaf morphisms (recall that for a given morphism $\eta : \mathcal{F} \rightarrow \mathcal{G}$ of presheaves, the stalk map $\eta_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$ is defined by $\eta_x([s, U]) = [\eta(U)(s), U]$, which directly yields the commutation assertion), we get $\psi_x \circ (\kappa_{\mathcal{F}})_x = (\kappa_{\mathcal{G}})_x \circ \varphi_x$ for all $x \in X$. But $(\kappa_{\mathcal{F}})_x$ is an isomorphism, hence $\psi_x = \varphi_x^+$ for all $x \in X$, and so Lemma 9.4 implies $\psi = \varphi^+$. $\square$

9.1. Kernels, images, and cokernels of morphisms of sheaves.

Definition 9.15. – Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of presheaves of abelian groups on a topological space X . We define the *presheaf kernel* of φ , *presheaf cokernel* of φ , and *presheaf image* of φ to be the presheaves given by $U \rightsquigarrow \text{Ker}(\varphi(U))$, $U \rightsquigarrow \text{Coker}(\varphi(U))$, and $U \rightsquigarrow \text{Im}(\varphi(U))$ respectively.

Notation 9.16. – The presheaf kernel of φ will be denoted by $\text{Ker } \varphi$ and the presheaf image will be denoted by $\text{Im}_{\text{pre}} \varphi$. Similarly, the presheaf cokernel will be denoted by $\text{Coker}_{\text{pre}} \varphi$.

Proposition 9.17. – If $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ is a morphism of sheaves of abelian groups on X , then $\text{Ker } \varphi$ is a sheaf.

Proof. – Let U be open in X and $(U_i)_{i \in I}$ be an open cover of U . The locality axiom for $\text{Ker } \varphi$ follows from the locality axiom of the sheaf $\mathcal{F}$. For the gluing axiom, suppose $s_i \in (\text{Ker } \varphi)(U_i) \subset \mathcal{F}(U_i)$ such that $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j} \in (\text{Ker } \varphi)(U_i \cap U_j)$ for all i, j . Then, we glue together these s_i to get $s \in \mathcal{F}(U)$. We show that $\varphi(U)(s) = 0$. This follows from the fact that $\varphi(U)(s)|_{U_i} = \varphi(U_i)(s|_{U_i}) = \varphi(U_i)(s_i) = 0$ for all i . $\square$

Definition 9.18. – Let $\mathcal{F}$ be a sheaf of abelian groups on a topological space X . A *subsheaf* of $\mathcal{F}$ is a sheaf $\mathcal{G}$ such that $\mathcal{G}(U)$ is a subgroup of $\mathcal{F}(U)$ for every open set $U \subset X$, and the restriction maps of the sheaf $\mathcal{G}$ are induced by those of $\mathcal{F}$.

Remark 9.19. – Clearly, the presheaf kernel of a morphism $\mathcal{F} \rightarrow \mathcal{G}$ of sheaves is a subsheaf of $\mathcal{F}$.

Fact 9.20. – If $\mathcal{G}$ is a subsheaf of $\mathcal{F}$, then for any point $x \in X$, the stalk $\mathcal{G}_x$ is a subgroup of $\mathcal{F}_x$.

Proof. – Fix an $x \in X$. Then for pairs of open sets $U \subset V$ containing x we have commuting squares

$$\begin{array}{ccc} \mathcal{G}(V) & \xrightarrow{\text{res}} & \mathcal{G}(U) \\ \iota_V \downarrow & & \downarrow \iota_U \\ \mathcal{F}(V) & \xrightarrow{\text{res}} & \mathcal{F}(U) \end{array}$$

thus yielding a morphism $\iota_x : \mathcal{G}_x \rightarrow \mathcal{F}_x$. We show that this is injective. So suppose $s_x \in \mathcal{G}_x$ is a germ represented by some section $s \in \mathcal{G}(U) \subset \mathcal{F}(U)$ for some open set $U \ni x$, and such that $\iota_x(s_x) = \iota(U)(s)_x = 0$ i.e. $s_x = 0$ in $\mathcal{F}_x$. but this means there exists an open set $V \ni x$ such that $s|_V = 0$ in $\mathcal{F}(V)$. Since $\mathcal{G}(V)$ is a subgroup of $\mathcal{F}(V)$, it follows that $s|_V = 0$ in $\mathcal{G}(V)$ as well, whence $s_x = 0$ in $\mathcal{G}_x$. $\square$

Definition 9.21. – We say that a morphism of sheaves of abelian groups $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ is a *monomorphism* if $\text{Ker } \varphi = 0$. Thus φ is a monomorphism if and only if $\varphi(U)$ is injective for every open set $U \subset X$.

Definition 9.22. – If $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ is a morphism of sheaves, we define the *image* of φ , denoted $\text{Im } \varphi$, to be the sheafification of the presheaf image $\text{Im}_{\text{pre}} \varphi$ of φ .

Definition 9.23. – For a map of sheaves $\varphi : \mathcal{F} \rightarrow \mathcal{G}$, we define $\text{Im}' \varphi$ by

$$(\text{Im}' \varphi)(U) := \{t \in \mathcal{G}(U) \mid \exists \text{ a cover } (U_\lambda)_{\lambda \in \Lambda} \text{ of } U \text{ and } s_\lambda \in \mathcal{F}(U_\lambda) \text{ such that } t|_{U_\lambda} = \varphi(U_\lambda)(s_\lambda)\}.$$

It is easy to check that $\text{Im}' \varphi$ is actually a sheaf.

Proposition 9.24. – We have $\text{Im } \varphi \simeq \text{Im}' \varphi$.

Proof. – Recall that for any open set $U \subset X$, by definition

$$(\text{Im } \varphi)(U) := \left\{ (g_x)_{x \in U} \in \prod_{x \in U} (\text{Im}_{\text{pre}} \varphi)_x \mid \begin{array}{l} \exists \text{ an open cover } (U_\lambda)_\lambda \text{ of } U \text{ and } t_\lambda \in (\text{Im}_{\text{pre}} \varphi)(U_\lambda) \\ \text{such that } g_x = (t_\lambda)_x \text{ for all } x \in U_\lambda \end{array} \right\}.$$

Now take a sequence $(g_x)_{x \in U} \in (\text{Im } \varphi)(U)$, an open cover $(U_\lambda)_{\lambda \in \Lambda}$ of U , and $t_\lambda = \varphi(U_\lambda)(s_\lambda)$ with $s_\lambda \in \mathcal{F}(U_\lambda)$ such that $g_x = (t_\lambda)_x$ for all $x \in U_\lambda$. The sections $t_\lambda \in \mathcal{G}(U_\lambda)$ agree on overlaps : for all $x \in U_\lambda \cap U_\mu$ we have $(t_\lambda)_x = g_x = (t_\mu)_x$, and so by Lemma 9.2 we get $t_\lambda|_{U_\lambda \cap U_\mu} = t_\mu|_{U_\lambda \cap U_\mu}$. Thus t_λ glue to a unique section $t \in \mathcal{G}(U)$ i.e. $t|_{U_\lambda} = t_\lambda = \varphi(U_\lambda)(s_\lambda)$ for all λ . Hence, $t \in (\text{Im}' \varphi)(U)$. We define a map

$$\Phi(U) : (\text{Im } \varphi)(U) \rightarrow (\text{Im}' \varphi)(U), \quad (g_x)_{x \in U} \mapsto t$$

as constructed above. We now show that this definition is independent of the choice of the cover $(U_\lambda)_\lambda$ of U . Indeed, suppose $(V_\mu)_{\mu \in M}$ is another cover of U with $p_\mu = \varphi(V_\mu)(r_\mu) \in (\text{Im}_{\text{pre}} \varphi)(V_\mu)$ such that $g_x = (p_\mu)_x$ for all $x \in V_\mu$. We wish to show that the section $p \in \mathcal{G}(U)$ obtained by gluing p_μ is the same as $t \in \mathcal{G}(U)$. For this take any pair (λ, μ) and consider $W_{\lambda, \mu} := U_\lambda \cap V_\mu$. Then, we have the equality of restrictions

$$t|_{W_{\lambda, \mu}} = p|_{W_{\lambda, \mu}}$$

since for every $x \in W_{\lambda, \mu}$ we have $t_x = (t_\lambda)_x = g_x = (p_\mu)_x = p_x$. Since $W_{\lambda, \mu}$ cover U , we get $t = p$ as $\mathcal{G}$ is a sheaf.

Next, we show that $\Phi : \text{Im } \varphi \rightarrow \text{Im}' \varphi$ defined above is a natural transformation. Indeed, if $V \subset U$ are open sets in X , then we have to show : $\text{res}_V^U \circ \Phi(U) = \Phi(V) \circ \text{res}_V^U$. So we compute the left hand side on some $(g_x)_{x \in U}$ to obtain $(\text{res}_V^U \circ \Phi(U))(g_x)_{x \in U} = t|_V$. On the other hand, the right hand side gives $\Phi(V)(g_x)_{x \in V}$. We show that this must be $t|_V$. For this consider the cover $(V \cap U_\lambda)_\lambda$ of V corresponding to the cover $(U_\lambda)_\lambda$ that gives t . Consider the sections $(t_\lambda|_{V \cap U_\lambda})_\lambda$ and note that for all $x \in V \cap U_\lambda$ we have $(t_\lambda|_{V \cap U_\lambda})_x = g_x$. Clearly, these glue to give $t|_V$ since $t|_{V \cap V \cap U_\lambda} = t|_{U_\lambda \cap V \cap U_\lambda} = t_\lambda|_{V \cap U_\lambda}$. Thus we have shown that Φ is a well-defined natural transformation.

Now we will construct the inverse $\Psi : \text{Im}' \varphi \rightarrow \text{Im } \varphi$ of Φ . Fix an open set $U \subset X$ and consider $t \in (\text{Im}' \varphi)(U)$; by definition there exists an open cover $(U_\lambda)_\lambda$ of U and sections $s_\lambda \in \mathcal{F}(U_\lambda)$ such that $t|_{U_\lambda} = \varphi(U_\lambda)(s_\lambda)$ for all λ . For each $x \in U$ we define

$$g_x := \text{germ of } t \text{ at } x \text{ i.e. } t_x.$$

Then each $g_x \in (\text{Im}_{\text{pre}} \varphi)_x$ since each $x \in U$ belongs to some $x \in U_\lambda$ so that $g_x = (t|_{U_\lambda})_x = \varphi(U_\lambda)(s_\lambda)_x$. Setting $t_\lambda := t|_{U_\lambda}$, we see that $(g_x)_{x \in U} \in (\text{Im } \varphi)(U)$. Thus, we can define $\Psi(U)(t) := (t_x)_{x \in U}$. It is easy to check that Ψ is in fact a natural transformation.

We now finally show that Φ and Ψ are inverses of each other. We first show $\Phi(U) \circ \Psi(U) = \text{id}$. Indeed, take a section $t \in (\text{Im}' \varphi)(U)$ as in the previous paragraph and set $t_\lambda := t|_{U_\lambda} = \varphi(U_\lambda)(s_\lambda)$ as before. Then, $\Phi(U)(\Psi(U)(t)) = \Phi(U)((t_x)_{x \in U})$ is obtained by gluing the t_λ and it is therefore t itself. On the other hand, $\Psi(U) \circ \Phi(U) = \text{id}$. To see this, take some $(g_x)_{x \in U} \in (\text{Im } \varphi)(U)$, an open cover $(U_\lambda)_{\lambda \in \Lambda}$ of U , and $t_\lambda = \varphi(U_\lambda)(s_\lambda)$ with $s_\lambda \in \mathcal{F}(U_\lambda)$ such that $g_x = (t_\lambda)_x$ for all $x \in U_\lambda$. Then we glue the t_λ to obtain $t \in (\text{Im}' \varphi)(U)$. But $\Psi(U)(t) = (t_x)_{x \in U}$. Now, we have to check that $t_x = g_x$ for all $x \in U$. So take an arbitrary $x \in U$ such that $x \in U_\lambda$ for some λ , then $t_x = (t|_{U_\lambda})_x = (t_\lambda)_x = g_x$. Thus, we have shown that Φ and Ψ are inverses of each other and this completes the proof. $\square$

The above proposition allows us to identify $\text{Im } \varphi$ as a subsheaf $\text{Im}' \varphi$ of $\mathcal{G}$. From now on we will use the single notation $\text{Im } \varphi$ to mean both the formulations.

Lemma 9.25. – Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of sheaves of abelian groups on a topological space X and $x \in X$. Then, we have $(\text{Ker } \varphi)_x = \text{Ker } \varphi_x$ and $(\text{Im } \varphi)_x = \text{Im } \varphi_x$.

Proof. – Take a germ $s_x \in (\text{Ker } \varphi)_x$ which is represented by some section $s \in (\text{Ker } \varphi)(U)$ for some opens set $U \ni x$. Then $\varphi(U)(s) = 0$, and so taking germ at x , we get $\varphi_x(s_x) = 0$ i.e. $s_x \in \text{Ker } \varphi_x$. Conversely, suppose $\varphi_x(s_x) = 0$ for some $s_x \in \mathcal{F}_x$ represented by a section $s \in \mathcal{F}(U)$ on some open set $U \ni x$. This implies that $\varphi(U)(s)_x = 0$ and so there is some open set $V \subset U$ containing x such that $\varphi(U)(s)|_V = \varphi(V)(s|_V) = 0$. Taking germs again we get $\varphi_x((s|_V)_x) = \varphi_x(s_x) = 0$.

Let $t_x \in \text{Im } \varphi_x$ and suppose $t_x = \varphi_x(s_x)$ for some $s_x \in \mathcal{F}_x$. We can always choose sections $s \in \mathcal{F}(V)$ and $t \in \mathcal{G}(V)$ representing s_x and t_x respectively. Since $\varphi(V)(s)_x = t_x$, by shrinking V we may assume $\varphi(V)(s) = t$. Thus $t \in (\text{Im } \varphi)(V)$ and so $t_x \in (\text{Im } \varphi)_x$. Conversely, let $t_x \in (\text{Im } \varphi)_x$. By definition, there exists an open set $U \ni x$ and a section $t \in (\text{Im } \varphi)(U)$ representing t_x . Now there exists an open cover $(U_\lambda)_{\lambda \in \Lambda}$ of U and sections $t_\lambda = \varphi(U_\lambda)(s_\lambda)$ with $s_\lambda \in \mathcal{F}(U_\lambda)$ such that $t|_{U_\lambda} = t_\lambda$ for all λ . But the given x belongs to some U_λ , and thus, taking germs we get $t_x = (t|_{U_\lambda})_x = \varphi(U_\lambda)(s_\lambda)_x = \varphi_x((s_\lambda)_x) \in \text{Im } \varphi_x$. $\square$

Definition 9.26. – We say that a morphism of sheaves of abelian groups $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ is an *epimorphism* if $\text{Im } \varphi = \mathcal{G}$.

Lemma 9.27. – Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of sheaves on abelian groups on a topological space X . Then, φ is a monomorphism (resp. epimorphism) if and only if φ_x is injective (resp. surjective) for every $x \in X$.

Proof. – This is essentially a restatement of Proposition 9.5. Indeed, φ being a monomorphism i.e. $\text{Ker } \varphi = 0$ is equivalent to saying $\varphi(U)$ is injective for each open set U , and φ being an epimorphism i.e. $\text{Im } \varphi = \mathcal{G}$ is equivalent to saying that φ is locally surjective. $\square$

Lemma 9.28. – Two subsheaves $\mathcal{H}, \mathcal{G}$ of a sheaf $\mathcal{F}$ are equal if and only if $\mathcal{H}_x = \mathcal{G}_x$ for every $x \in X$.

Proof. – The “only if” direction is trivial. For the “if” direction, let $U \subset X$ be an open set and $s \in \mathcal{G}(U)$ a section. Then, fixing an arbitrary $x \in X$, we have that $s_x \in \mathcal{G}_x = \mathcal{H}_x$. So for each $x \in U$, there exists an open set $U_x \subset U$ containing x and a section $t^x \in \mathcal{H}(U_x)$ that represents s_x . Now the open sets $\{U_x\}_{x \in U}$ cover U and the any two sections, say t^x and t^y , agree on the overlap $U_x \cap U_y$: to see this, for any $z \in U_x \cap U_y$ the germs $t^x_z = t^y_z = s_z$, and so we can cover $U_x \cap U_y$ with open sets V_z such that $t^x|_{V_z} = t^y|_{V_z}$ for all $z \in U_x \cap U_y$, and so $t^x|_{U_x \cap U_y} = t^y|_{U_x \cap U_y}$. Thus we can glue these t^x to obtain a section $t \in \mathcal{H}(U)$. But then $t_x = (t|_{U_x})_x = t^x_x = s_x$ for all $x \in U$, and hence $s = t$ as sections in $\mathcal{F}(U)$. Thus, we have shown $s \in \mathcal{H}(U)$ i.e. $\mathcal{G}(U) \subset \mathcal{H}(U)$ for every open set $U \subset X$. By a symmetric argument, we also have $\mathcal{H}(U) \subset \mathcal{G}(U)$ for every open set $U \subset X$. This completes the proof. $\square$

Definition 9.29. – Let $\mathcal{G}$ be a subsheaf of a sheaf $\mathcal{F}$ of abelian groups on a topological space X . We define the *quotient sheaf* to be the sheafification of the presheaf $(\mathcal{F}/\mathcal{G})_{\text{pre}} : U \mapsto \mathcal{F}(U)/\mathcal{G}(U)$. It is denoted by $\mathcal{F}/\mathcal{G}$.

Lemma 9.30. – For any point $x \in X$, the stalk $(\mathcal{F}/\mathcal{G})_x$ is isomorphic to the quotient $\mathcal{F}_x/\mathcal{G}_x$.

Proof. – Recall that $(\mathcal{F}/\mathcal{G})_x \simeq ((\mathcal{F}/\mathcal{G})_{\text{pre}})_x$ by Fact 9.13. So it suffices to show that $((\mathcal{F}/\mathcal{G})_{\text{pre}})_x \simeq \mathcal{F}_x/\mathcal{G}_x$. We do this as follows. First, we define, for each open set $U \ni x$, the map

$$\alpha_U : \mathcal{F}(U)/\mathcal{G}(U) \rightarrow \mathcal{F}_x/\mathcal{G}_x, \quad s + \mathcal{G}(U) \mapsto s_x + \mathcal{G}_x.$$

We check that α_U is well-defined. Indeed, if $s' = s + t$ with $t \in \mathcal{G}(U)$, then $s'_x = s_x + t_x$ i.e. $s_x + \mathcal{G}_x = s'_x + \mathcal{G}_x$. Then, note that for $V \subset U$ containing x , we have $\alpha_V \circ \text{res}_V^U = \alpha_U$; this is easy to see. Thus, the α_U form a co-cone below the diagram $\{\mathcal{F}(U)/\mathcal{G}(U)\}_{U \ni x}$, and we have an induced map

$$\alpha : \varinjlim_{x \in U} \mathcal{F}(U)/\mathcal{G}(U) \rightarrow \mathcal{F}_x/\mathcal{G}_x, \quad [s + \mathcal{G}(U)] \mapsto s_x + \mathcal{G}_x.$$

It is then clear that α is surjective: for some $s_x + \mathcal{G}_x \in \mathcal{F}_x/\mathcal{G}_x$ we choose a section $s \in \mathcal{F}(U)$ representing s_x , so that $\alpha([s + \mathcal{G}(U)]) = s_x + \mathcal{G}_x$. To show that α is injective, suppose $\alpha([s + \mathcal{G}(U)]) = s_x + \mathcal{G}_x = 0$ i.e. $s_x \in \mathcal{G}_x$. So there exists an open set $V \subset U$ containing x such that $s|_V \in \mathcal{G}(V)$. But then $[s + \mathcal{G}(U)] = [s|_V + \mathcal{G}(V)] = 0$. This completes the proof that α is an isomorphism, and hence that of the lemma. $\square$

Lemma 9.31. – Let $\mathcal{G}$ be a subsheaf of $\mathcal{F}$ as in the previous lemma. Then, the natural map from $\mathcal{F}$ to $\mathcal{F}/\mathcal{G}$ is an epimorphism having kernel $\mathcal{G}$.

Proof. – First, we note that the natural map from $\mathcal{F}$ to $\mathcal{F}/\mathcal{G}$ is obtained by composing the presheaf morphism $\pi : \mathcal{F} \rightarrow (\mathcal{F}/\mathcal{G})_{\text{pre}}$ with the sheafification map $\kappa_{(\mathcal{F}/\mathcal{G})_{\text{pre}}} : (\mathcal{F}/\mathcal{G})_{\text{pre}} \rightarrow \mathcal{F}/\mathcal{G}$. Thus, we set the natural map $\eta : \mathcal{F} \rightarrow \mathcal{F}/\mathcal{G}$ to be $\eta := \kappa_{(\mathcal{F}/\mathcal{G})_{\text{pre}}} \circ \pi$. Then, for any $x \in X$, we have $\eta_x = (\kappa_{(\mathcal{F}/\mathcal{G})_{\text{pre}}})_x \circ \pi_x$, and since $(\kappa_{(\mathcal{F}/\mathcal{G})_{\text{pre}}})_x$ is an isomorphism, we have $\text{Ker } \eta_x = \text{Ker } \pi_x$. We show that $(\text{Ker } \eta)_x = \text{Ker } \eta_x = \text{Ker } \pi_x = \mathcal{G}_x$ for all $x \in X$, and then by Lemma 9.28 we can finally conclude $\text{Ker } \eta = \mathcal{G}$.

So take a germ $s_x \in \text{Ker } \pi_x$ for some section $s \in \mathcal{F}(U)$ with U an open set containing x . This means $0 = \pi_x(s_x) = [s + \mathcal{G}(U)]$ and so there exists an open set $V \subset U$ containing x such that $s|_V \in \mathcal{G}(V)$. But then $s_x = (s|_V)_x \in \mathcal{G}_x$ i.e. $\text{Ker } \pi_x \subset \mathcal{G}_x$. Conversely, if $s_x \in \mathcal{G}_x$ is represented by some section $s \in \mathcal{G}(U)$ for some open set $U \ni x$, then $\pi_x(s_x) = \pi(U)(s)_x = [s + \mathcal{G}(U)] = 0$ i.e. $\mathcal{G}_x \subset \text{Ker } \pi_x$.

Next, we show that $\eta : \mathcal{F} \rightarrow \mathcal{F}/\mathcal{G}$ is an epimorphism i.e. $\eta_x = (\kappa_{(\mathcal{F}/\mathcal{G})_{\text{pre}}})_x \circ \pi_x$ is a surjection by Lemma 9.27. Again, it suffices to show that π_x is a surjection. But this is clear since any given element of $((\mathcal{F}/\mathcal{G})_{\text{pre}})_x = \varinjlim_{x \in U} \mathcal{F}(U)/\mathcal{G}(U)$ is of the form $[t + \mathcal{G}(U)]$ which is the image $\pi_x(t_x)$ of the germ $t_x \in \mathcal{F}_x$. $\square$

Lemma 9.32. – Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of sheaves of abelian groups on a topological space X . Then, $\text{Im } \varphi \simeq \mathcal{F}/\text{Ker } \varphi$.

Proof. – For each open set $U \subset X$ we consider the maps $\varphi'(U) : \mathcal{F}(U)/\text{Ker } \varphi(U) \rightarrow (\text{Im } \varphi)(U)$ given by $s + \text{Ker } \varphi(U) \mapsto \varphi(U)(s)$, noting that $\varphi(U)(s) \in (\text{Im } \varphi)(U)$ by definition of the image sheaf $\text{Im } \varphi$ identified as a subsheaf of $\mathcal{G}$. Then these maps give a presheaf morphism $(\mathcal{F}/\text{Ker } \varphi)_{\text{pre}} \rightarrow \text{Im } \varphi$. Also, note that this fits in the following diagram

$$\begin{array}{ccc} \mathcal{F} & & \\ \downarrow & \searrow & \\ (\mathcal{F}/\text{Ker } \varphi)_{\text{pre}} & \longrightarrow & \text{Im } \varphi \end{array}$$

which is commutative. Since $\text{Im } \varphi$ is a sheaf we have a factoring of the above morphism through $\mathcal{F}/\text{Ker } \varphi$ yielding the diagram

$$\begin{array}{ccc} \mathcal{F} & & \\ \downarrow & \searrow & \\ (\mathcal{F}/\text{Ker } \varphi)_{\text{pre}} & \xrightarrow{\Psi} & \text{Im } \varphi \\ \downarrow & \nearrow & \\ \mathcal{F}/\text{Ker } \varphi & & \end{array}$$

which is commutative, and also the composition of the vertical maps is the canonical morphism $\eta : \mathcal{F} \rightarrow \mathcal{F}/\text{Ker } \varphi$ which is an epimorphism with kernel $\text{Ker } \varphi$ by the previous lemma. Now taking stalks we get a diagram

$$\begin{array}{ccc} \mathcal{F}_x & & \\ \downarrow & \searrow & \\ ((\mathcal{F}/\text{Ker } \varphi)_{\text{pre}})_x & \xrightarrow{\Psi_x} & \text{Im } \varphi_x \\ \downarrow \wr & \nearrow & \\ (\mathcal{F}/\text{Ker } \varphi)_x & & \end{array}$$

which is commutative. To show $(\mathcal{F}/\text{Ker } \varphi)_x \rightarrow \text{Im } \varphi_x$ is an isomorphism for every x it suffices to show that $\Psi_x : ((\mathcal{F}/\text{Ker } \varphi)_{\text{pre}})_x \rightarrow \text{Im } \varphi_x$ is an isomorphism. Now, a germ in $((\mathcal{F}/\text{Ker } \varphi)_{\text{pre}})_x$ is represented by $[s + (\text{Ker } \varphi)(U)]$ for some $s \in \mathcal{F}(U)$. So, we have $\Psi_x([s + (\text{Ker } \varphi)(U)]) = \varphi(U)(s)_x = \varphi_x(s_x)$. Next, we identify

$$\alpha_x : ((\mathcal{F}/\text{Ker } \varphi)_{\text{pre}})_x \xrightarrow{\sim} \mathcal{F}_x/\text{Ker } \varphi_x, \quad [s + (\text{Ker } \varphi)(U)] \mapsto s_x + \text{Ker } \varphi_x$$

and then we get $\Psi_x \circ \alpha_x^{-1} : \mathcal{F}_x/\text{Ker } \varphi_x \rightarrow \text{Im } \varphi_x$ which is explicitly given by

$$s_x + \text{Ker } \varphi_x \mapsto [s + (\text{Ker } \varphi)(U)] \mapsto \varphi_x(s_x)$$

and thus this composite is just the canonical isomorphism as expected. Thus Ψ_x is also an isomorphism as required. $\square$

Definition 9.33. – For a morphism of sheaves $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ of abelian groups on X , we define its *cokernel sheaf* $\text{Coker } \varphi$ as the sheafification of the presheaf cokernel $\text{Coker}_{\text{pre}} \varphi : U \rightsquigarrow \text{Coker } \varphi(U)$.

Lemma 9.34. – We have $\text{Coker } \varphi \simeq \mathcal{G}/\text{Im } \varphi$.

Proof. – First, note that for every open set U we have a natural map

$$(\text{Coker}_{\text{pre}} \varphi)(U) = \mathcal{G}(U)/\text{Im } \varphi(U) \rightarrow \mathcal{G}(U)/(\text{Im } \varphi)(U)$$

owing to the fact that $\text{Im } \varphi(U) \subset (\text{Im } \varphi)(U)$. These assemble to yield a morphism of presheaves

$$\text{Coker}_{\text{pre}} \varphi \rightarrow (\mathcal{G}/\text{Im } \varphi)_{\text{pre}}$$

and composing with the sheafification map of $(\mathcal{G}/\text{Im } \varphi)_{\text{pre}}$ we get a morphism

$$\text{Coker}_{\text{pre}} \varphi \rightarrow \mathcal{G}/\text{Im } \varphi$$

and then by the universal property of sheafification we get a morphism of sheaves

$$\text{Coker } \varphi \rightarrow \mathcal{G}/\text{Im } \varphi.$$

Now these maps fit into the following diagram

$$\begin{array}{ccccc} \text{Coker}_{\text{pre}} \varphi & \longrightarrow & (\mathcal{G}/\text{Im } \varphi)_{\text{pre}} & \longrightarrow & \mathcal{G}/\text{Im } \varphi \\ \downarrow & & \nearrow & & \\ \text{Coker } \varphi & & & & \end{array}$$

which is commutative. Now, taking stalks at x we obtain

$$\begin{array}{ccccc} \text{Coker}_{\text{pre}} \varphi & \longrightarrow & (\mathcal{G}/\text{Im } \varphi)_{\text{pre}} & \longrightarrow & \mathcal{G}/\text{Im } \varphi \\ \downarrow & & \nearrow & & \\ \text{Coker } \varphi & & & & \end{array}$$

which commutes. Thus, to show that $(\text{Coker } \varphi)_x \rightarrow (\mathcal{G}/\text{Im } \varphi)_x$ is isomorphism for all x , it suffices to show that $(\text{Coker}_{\text{pre}} \varphi)_x \rightarrow ((\mathcal{G}/\text{Im } \varphi)_{\text{pre}})_x$ is an isomorphism. Written explicitly, this map is

$$[s + \text{Im } \varphi(U)] \mapsto [s + (\text{Im } \varphi)(U)], \quad (s \in \mathcal{G}(U)).$$

To show injectivity, suppose $[s + (\text{Im } \varphi)(U)] = 0$ so that there exists an open set $V \subset U$ containing x with $s|_V \in (\text{Im } \varphi)(V)$. Now since $x \in V$, there exists an open set $W \subset V$ containing x such that $s|_W = \varphi(W)(t)$ for some $t \in \mathcal{F}(W)$ i.e. $s|_W \in \text{Im } \varphi(W)$ and so $[s + \text{Im } \varphi(U)] = [s|_W + \text{Im } \varphi(W)] = 0$.

Next, we prove surjectivity. But this is clear, since an arbitrary germ in $((\mathcal{G}/\text{Im } \varphi)_{\text{pre}})_x$ looks like $[s + (\text{Im } \varphi)(U)]$ for some section $s \in \mathcal{G}(U)$ and an open set $U \ni x$, which is the image of $[s + \text{Im } \varphi(U)] \in (\text{Coker}_{\text{pre}} \varphi)_x$. This completes the proof. $\square$

There is also a notion of an exact sequence of sheaves of abelian groups. A sequence of morphisms

$$\mathcal{F}' \xrightarrow{\varphi} \mathcal{F} \xrightarrow{\psi} \mathcal{F}''$$

is called *exact* if $\text{Im } \varphi = \text{Ker } \psi$. We have the following criterion :

Proposition 9.35. – A sequence

$$0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$$

of sheaves of abelian groups on a topological space X is exact if and only if the sequence of stalks

$$0 \rightarrow \mathcal{F}'_x \rightarrow \mathcal{F}_x \rightarrow \mathcal{F}''_x \rightarrow 0$$

is exact for all $x \in X$.

Proof. – Suppose that $0 \rightarrow \mathcal{F}' \xrightarrow{\varphi} \mathcal{F} \xrightarrow{\psi} \mathcal{F}'' \rightarrow 0$ is exact i.e. $\text{Ker } \varphi = 0$, $\text{Im } \varphi = \text{Ker } \psi$, and $\text{Im } \psi = \mathcal{F}''$. Taking stalks at any $x \in X$ and using Lemma 9.25, we get $\text{Ker } \varphi_x = 0$, $\text{Im } \varphi_x = \text{Ker } \psi_x$, and $\text{Im } \psi_x = \mathcal{F}''_x$ i.e. $0 \rightarrow \mathcal{F}'_x \rightarrow \mathcal{F}_x \rightarrow \mathcal{F}''_x \rightarrow 0$ is exact.

Conversely, suppose $0 \rightarrow \mathcal{F}'_x \rightarrow \mathcal{F}_x \rightarrow \mathcal{F}''_x \rightarrow 0$ is exact for all $x \in X$. We use Lemma 9.25 and Lemma 9.28. Thus $\text{Ker } \varphi_x = 0 = (\text{Ker } \varphi)_x$ for all x implies $\text{Ker } \varphi = 0$. Similarly, $(\text{Im } \varphi)_x = \text{Im } \varphi_x = \text{Ker } \psi_x = (\text{Ker } \psi)_x$ for all x implies $\text{Im } \varphi = \text{Ker } \psi$. Finally, $\text{Im } \psi_x = (\text{Im } \psi)_x = \mathcal{F}''_x$ for all x implies $\text{Im } \psi = \mathcal{F}''$. We have prove that $0 \rightarrow \mathcal{F}' \xrightarrow{\varphi} \mathcal{F} \xrightarrow{\psi} \mathcal{F}'' \rightarrow 0$ is exact. $\square$

Lemma 9.36. – Let $\mathcal{F}'$ be a subsheaf of $\mathcal{F}$. Then, there is an exact sequence $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}/\mathcal{F}' \rightarrow 0$. Conversely, if $0 \rightarrow \mathcal{F}' \xrightarrow{\varphi} \mathcal{F} \xrightarrow{\psi} \mathcal{F}'' \rightarrow 0$ is an exact sequence then $\mathcal{F}'$ is isomorphic to a subsheaf of $\mathcal{F}$ and $\mathcal{F}''$ is isomorphic to the quotient of $\mathcal{F}$ by this subsheaf.

Proof. – The first assertion follows from Lemma 9.31. For the second assertion we first show that $\text{Im}_{\text{pre}} \varphi$ which is a subpresheaf of $\mathcal{F}$ is actually a sheaf by showing that it coincides with the image sheaf i.e. $\text{Im } \varphi = \text{Im}_{\text{pre}} \varphi$. Now for each open set U we already know that $(\text{Im}_{\text{pre}} \varphi)(U) = \text{Im } \varphi(U) \subset (\text{Im } \varphi)(U)$. So suppose $t \in \mathcal{F}(U)$ is a section for which there is a cover $(U_i)_{i \in I}$ of U together with sections $s_i \in \mathcal{F}'(U_i)$ satisfying $\varphi(U_i)(s_i) = t|_{U_i}$ for all i . Then we have

$$\varphi(U_i \cap U_j)(s_i|_{U_i \cap U_j}) = t|_{U_i \cap U_j} = \varphi(U_i \cap U_j)(s_j|_{U_i \cap U_j}), \quad \forall i, j \in I.$$

But since $\varphi(U_i \cap U_j) = 0$ for all i, j , we have $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all i, j . We glue these s_i to get a section $s \in \mathcal{F}'(U)$. Now we show that $\varphi(U)(s) = t$; but this is clear since $\varphi(U)(s)|_{U_i} = \varphi(U_i)(s_i) = t|_{U_i}$ for all i and the locality axiom holds for the sheaf $\mathcal{F}$. Thus $(\text{Im}_{\text{pre}} \varphi)(U) = (\text{Im } \varphi)(U)$. We can easily see that $\mathcal{F}' \simeq \text{Im } \varphi$: indeed the restricted maps $\mathcal{F}'(U) \rightarrow \text{Im } \varphi(U)$ given by $s \mapsto \varphi(U)(s)$ are bijections for every open set U . Next, we know $\text{Im } \varphi = \text{Ker } \psi$ and $\text{Im } \psi = \mathcal{F}''$, also by Lemma 9.32 we have $\mathcal{F}'' = \text{Im } \psi \simeq \mathcal{F}/\text{Ker } \psi = \mathcal{F}/\text{Im } \varphi$. This completes the proof. $\square$

Proposition 9.37. – Let U be an open set of X and suppose $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}''$ is an exact sequence of sheaves of abelian groups on X . Then, $0 \rightarrow \mathcal{F}'(U) \rightarrow \mathcal{F}(U) \rightarrow \mathcal{F}''(U)$ is exact.

Proof. – Let $\varphi : \mathcal{F}' \rightarrow \mathcal{F}$ and $\psi : \mathcal{F} \rightarrow \mathcal{F}''$ be the morphisms in the exact sequence. Then it is given that $\text{Ker } \varphi = 0$ and $\text{Im } \varphi = \text{Ker } \psi$. Hence $\varphi(U) : \mathcal{F}'(U) \rightarrow \mathcal{F}(U)$ is injective. Thus it remains to show that $\text{Im } \varphi(U) = \text{Ker } \psi(U)$. Now $\text{Im } \varphi(U) \subset (\text{Im } \varphi)(U) = (\text{Ker } \psi)(U) = \text{Ker } \psi(U)$ is the obvious direction. On the other hand, suppose $t \in \text{Ker } \psi(U)$ i.e. $\psi(U)(t) = 0$. Therefore, $\psi_x(t_x) = 0$ for all $x \in U$ i.e. $t_x \in \text{Ker } \psi_x$ for all $x \in U$. But also $\text{Ker } \psi_x = \text{Im } \varphi_x$ and $0 \rightarrow \mathcal{F}'_x \rightarrow \mathcal{F}_x \rightarrow \mathcal{F}''_x$ is exact for all $x \in U$. Thus $t_x = \varphi_x(s_x)$ for some $s_x \in \mathcal{F}'_x$. Representing s_x be some section $s^{(x)} \in \mathcal{F}'(V_x)$ with $V_x \subset U$ being an open set containing x , we get $t_x = \varphi(V_x)(s^{(x)})_x$. Hence, for each $x \in U$ we obtain an open set $W_x \subset V_x$ such that $t|_{W_x} = \varphi(W_x)(s^{(x)}|_{W_x})$. This yields an open cover $(W_x)_{x \in U}$ of U and sections $s^x := s^{(x)}|_{W_x} \in \mathcal{F}'(W_x)$ such that $t|_{W_x} = \varphi(W_x)(s^x)$. Now we check that the sections s^x agree on overlaps: indeed, we have

$$\varphi(W_x \cap W_{x'})(s^x|_{W_x \cap W_{x'}}) = t|_{W_x \cap W_{x'}} = \varphi(W_x \cap W_{x'})(s^{x'}|_{W_x \cap W_{x'}})$$

and then noting that $\varphi(W_x \cap W_{x'})$ is injective, we get $s^x|_{W_x \cap W_{x'}} = s^{x'}|_{W_x \cap W_{x'}}$ for all $x, x' \in U$. Glue these $(s^x)_{x \in U}$ to obtain a section $s \in \mathcal{F}'(U)$. Finally, note that $\varphi(U)(s) = t$ since

$$\varphi(U)(s)|_{W_x} = \varphi(W_x)(s^x) = t|_{W_x}$$

for all $x \in U$ and by the locality of the sheaf $\mathcal{F}$ we have the required equality. Thus, $t \in \text{Im } \varphi(U) \subset (\text{Im } \varphi)(U)$. This completes the proof of exactness. $\square$

9.2. Direct sums and products of sheaves. Let $(\mathcal{F}_i)_{i \in I}$ be a family of sheaves of abelian groups over a topological space X . We define the *direct sum* of $\mathcal{F}_i$ to be the sheafification of the presheaf

$$\left(\bigoplus_{i \in I} \mathcal{F}_i \right)_{\text{pre}} : U \rightsquigarrow \bigoplus_{i \in I} \mathcal{F}_i(U).$$

We denote this direct sum sheaf by $\bigoplus_{i \in I} \mathcal{F}_i$.

Proposition 9.38. – The direct sum $\bigoplus_{i \in I} \mathcal{F}_i$ has canonical inclusions $\epsilon_i : \mathcal{F}_i \rightarrow \bigoplus_{i \in I} \mathcal{F}_i$, satisfying the following universal property : for any family of maps $\eta_i : \mathcal{F}_i \rightarrow \mathcal{G}$, where $\mathcal{G}$ is a sheaf of abelian groups on X , there exists a unique map $\eta : \bigoplus_{i \in I} \mathcal{F}_i \rightarrow \mathcal{G}$ such that $\eta_i = \eta \circ \epsilon_i$ for all i .

Proof. – For every $i \in I$, there is the obvious presheaf inclusion

$$j_i : \mathcal{F}_i \rightarrow \left(\bigoplus_{i \in I} \mathcal{F}_i \right)_{\text{pre}}$$

which after composition with the sheafification map $\kappa : \left(\bigoplus_{i \in I} \mathcal{F}_i \right)_{\text{pre}} \rightarrow \bigoplus_{i \in I} \mathcal{F}_i$ gives the required ϵ_i – that is to say – we set $\epsilon_i := \kappa \circ j_i$ for each $i \in I$.

Next, given a sheaf $\mathcal{G}$ and morphisms $\eta_i : \mathcal{F}_i \rightarrow \mathcal{G}$ we define first the morphism of presheaves

$$\theta : \left(\bigoplus_{i \in I} \mathcal{F}_i \right)_{\text{pre}} \rightarrow \mathcal{G}$$

by

$$\theta(U)((s_i)_{i \in I}) := \sum_{i \in I} \eta_i(U)(s_i), \quad \text{where } U \text{ is open and } s_i \in \mathcal{F}_i(U).$$

It is easy to see that θ is indeed a morphism of presheaves i.e. it is compatible with restrictions. By universal property of sheafification there exists a unique morphism of sheaves

$$\eta : \bigoplus_{i \in I} \mathcal{F}_i \rightarrow \mathcal{G}$$

such that

$$\theta = \eta \circ \kappa.$$

Now we see that

$$\eta \circ \epsilon_i = \eta \circ \kappa \circ j_i = \theta \circ j_i = \eta_i$$

for every $i \in I$. Finally, we check that η is the unique morphism of sheaves satisfying the above composition law. Indeed, if $\eta' : \bigoplus_i \mathcal{F}_i \rightarrow \mathcal{G}$ is another sheaf morphism satisfying $\eta' \circ \epsilon_i = \eta_i$ for all i , then, it suffices to show that $\eta \circ \kappa = \eta' \circ \kappa$ (and thus $\eta' \circ \kappa = \theta$ so that by uniqueness of η we get $\eta = \eta'$). So fix any open set U and take a *eventually zero* sequence $(s_i)_{i \in I} \in \bigoplus_{i \in I} \mathcal{F}_i(U)$. Then, we have

$$\kappa(U)((s_i)_{i \in I}) = \kappa(U) \left(\sum_{i \in I} j_i(U)(s_i)_{i \in I} \right) = \sum_{i \in I} (\kappa \circ j_i)(U)(s_i)_{i \in I} = \sum_{i \in I} \epsilon_i(U)(s_i)_{i \in I}.$$

Therefore, we get

$$(\eta \circ \kappa)(U)((s_i)_{i \in I}) = \sum_{i \in I} (\eta \circ \epsilon_i)(U)(s_i)_{i \in I} = \sum_{i \in I} (\eta' \circ \epsilon_i)(U)(s_i)_{i \in I} = (\eta' \circ \kappa)(U)((s_i)_{i \in I}).$$

This completes the proof. $\square$

We can also define the *direct product* of a family $(\mathcal{F}_i)_{i \in I}$ of sheaves of abelian groups on X as the presheaf

$$\prod_{i \in I} \mathcal{F}_i : U \rightsquigarrow \prod_{i \in I} \mathcal{F}_i(U).$$

The following is a rather easy and routing check.

Fact 9.39. – The above presheaf is a sheaf.

Proposition 9.40. – The direct product $\prod_i \mathcal{F}_i$ comes equipped with canonical projections $\pi_i : \prod_i \mathcal{F}_i \rightarrow \mathcal{F}_i$ satisfying the following universal property : for a sheaf $\mathcal{G}$ of abelian groups in X and any collection of maps $\epsilon_i : \mathcal{G} \rightarrow \mathcal{F}_i$ there exists a unique morphism of sheaves $\eta : \mathcal{G} \rightarrow \prod_i \mathcal{F}_i$ such that $\pi_i \circ \eta = \epsilon_i$ for all i .

Proof. – Set $\eta(U) : \mathcal{G}(U) \rightarrow \prod_i \mathcal{F}_i(U)$ as $s \mapsto (\epsilon_i(U)(s_i))_{i \in I}$. Then, it is clear that it is a morphism of presheaves as it is obviously compatible with restrictions as ϵ_i are. Also, $\pi_i \circ \eta = \epsilon_i$ for all i is clear as well. For the uniqueness part, suppose $\eta' : \mathcal{G} \rightarrow \prod_i \mathcal{F}_i$ is another morphism of sheaves such that $\pi_i \circ \eta' = \epsilon_i$ for all i . Thus, the i -components of both $\eta(U)$ and $\eta'(U)$ are given by $\epsilon_i(U)$, whence $\eta(U) = \eta'(U)$ for all open sets U i.e. $\eta = \eta'$. $\square$

9.3. **Sheaf $\mathcal{H}om$.** Let $\mathcal{F}, \mathcal{G}$ be sheaves of abelian groups on a topological space X for this subsection.

Proposition 9.41. – *Let $U \subset X$ be an open set. Then, the set $\text{Hom}(\mathcal{F}|_U, \mathcal{G}|_U)$ of morphisms of restricted sheaves has a natural structure of an abelian group.*

Proof. – For $\varphi, \psi \in \text{Hom}(\mathcal{F}|_U, \mathcal{G}|_U)$, and $V \subset U$ open, define $(\varphi + \psi)(V) := \varphi(V) + \psi(V) \in \text{Hom}(\mathcal{F}(V), \mathcal{G}(V))$ which is the usual way to define addition of two homomorphisms of abelian groups. Remaining details are routine checks. $\square$

Proposition 9.42. – *The following assignment*

$$U \rightsquigarrow \text{Hom}(\mathcal{F}|_U, \mathcal{G}|_U)$$

is a presheaf of abelian groups. Furthermore, it is a sheaf and it is denoted by $\mathcal{H}om(\mathcal{F}, \mathcal{G})$.

Proof. – The fact that it is a presheaf is clear : for $V \subset U$, the restriction is simply $\alpha \mapsto \alpha|_V$ for any $\alpha \in \text{Hom}(\mathcal{F}|_U, \mathcal{G}|_U)$. Now fix an open set U and a cover $(U_i)_{i \in I}$ of U .

We first check the locality axiom. Suppose $\alpha|_{U_i} = \beta|_{U_i}$ for every i , where $\alpha, \beta \in \text{Hom}(\mathcal{F}|_U, \mathcal{G}|_U)$. Let $V \subset U$ be any open set, and we have to show that $\alpha(V) = \beta(V)$. Take an arbitrary $s \in \mathcal{F}(V)$. Consider the cover $(V \cap U_i)_{i \in I}$ of V . Since $\mathcal{G}$ is a sheaf, it suffices to check that $\alpha(V)(s)|_{V \cap U_i} = \beta(V)(s)|_{V \cap U_i}$ for every i . But we see that

$$\alpha(V)(s)|_{V \cap U_i} = \alpha|_{U_i}(V \cap U_i)(s|_{V \cap U_i}) = \beta|_{U_i}(V \cap U_i)(s|_{V \cap U_i}) = \beta(V)(s)|_{V \cap U_i}$$

for all $i \in I$ as required.

Next, we check the gluing axiom. So suppose $\alpha_i \in \text{Hom}(\mathcal{F}|_{U_i}, \mathcal{G}|_{U_i})$ are given such that $\alpha_i|_{U_i \cap U_j} = \alpha_j|_{U_i \cap U_j}$ for all i, j . We have to construct $\alpha \in \text{Hom}(\mathcal{F}|_U, \mathcal{G}|_U)$ satisfying $\alpha|_{U_i} = \alpha_i$ for all i . So take an open set $V \subset U$ and cover it with $(V \cap U_i)_{i \in I}$. Take $s \in \mathcal{F}(V)$. Now for each i define

$$t_i := \alpha_i(V \cap U_i)(s|_{V \cap U_i}).$$

Since α_i and α_j agree on $U_i \cap U_j$, it follows that $t_i|_{V \cap U_i \cap U_j} = t_j|_{V \cap U_i \cap U_j}$. Glue these to obtain $t \in \mathcal{G}(V)$ and set $\alpha(V)(s) := t$. We have to show that this $\alpha \in \text{Hom}(\mathcal{F}|_U, \mathcal{G}|_U)$. Let $V \subset U$ be open and suppose $s, s' \in \mathcal{F}(V)$. We show that $\alpha(V)(s + s') = \alpha(V)(s) + \alpha(V)(s')$ i.e. $\alpha(V)$ is a group homomorphism. For each i , we have

$$t_i(s + s') = \alpha_i(V \cap U_i)((s + s')|_{V \cap U_i}) = \alpha_i(V \cap U_i)(s|_{V \cap U_i}) + \alpha_i(V \cap U_i)(s'|_{V \cap U_i}) = t_i(s) + t_i(s').$$

Therefore, we obtain

$$\alpha(V)(s + s')|_{V \cap U_i} = t_i(s + s') = t_i(s) + t_i(s') = \alpha_i(V)(s)|_{V \cap U_i} + \alpha_i(V)(s')|_{V \cap U_i}.$$

By locality we get

$$\alpha(V)(s + s') = \alpha(V)(s) + \alpha(V)(s').$$

Next, suppose $W \subset V \subset U$ are open sets. We need to prove $\alpha(V)(s)|_W = \alpha(W)(s|_W)$. Set $t := \alpha(V)(s)$. By definition of $\alpha(V)(s)$, for each i , we have

$$t|_{V \cap U_i} = \alpha_i(V \cap U_i)(s|_{V \cap U_i}).$$

Restricting to $W \cap U_i \subset V \cap U_i$, we have

$$t|_{W \cap U_i} = \alpha_i(W \cap U_i)(s|_{W \cap U_i}) = t|_W|_{W \cap U_i}.$$

But $\alpha(W)(s|_W)$ was defined as the unique section in $\mathcal{G}(W)$ whose restriction to each $W \cap U_i$ equals $\alpha_i(W \cap U_i)(s|_{W \cap U_i})$. Hence $t|_W = \alpha(W)(s|_W)$ as required. This completes the proof of the gluing part and that of the proposition. $\square$

9.4. Direct images and inverse images of sheaves.

Definition 9.43. – Let $f : X \rightarrow Y$ be a continuous map of topological spaces. For any sheaf $\mathcal{F}$ of sets, abelian groups or rings on X , we define the *direct image* sheaf $f_*\mathcal{F}$ on Y by $(f_*\mathcal{F})(V) := \mathcal{F}(f^{-1}(V))$ for open sets V of Y . For any sheaf $\mathcal{G}$ on Y , we define the *inverse image* sheaf $f^{-1}\mathcal{G}$ on X as the sheafification of the presheaf $U \rightsquigarrow \varinjlim_{V \supset f(U)} \mathcal{G}(V)$, where U is any open set in X . Note that the limit is taken over the family of all open sets in Y that contain $f(U)$; this family is partially ordered with reversed inclusion, and it is a therefore directed family.

Remark 9.44. – First, let us understand the presheaf $U \rightsquigarrow \varinjlim_{V \supset f(U)} \mathcal{G}(V)$. An element in $\varinjlim_{V \supset f(U)} \mathcal{G}(V)$ is actually an equivalence class $[s, V]$ represented by a section $s \in \mathcal{G}(V)$ and an open set $V \supset f(U)$. Two such classes $[s, V]$ and $[s', V']$ are equal if there exists $W \subset V \cap V'$ an open set containing $f(U)$ such that $s|_W = s'|_W$. For open sets $U' \subset U$, then $f(U') \subset f(U)$. Thus, the family $\{\mathcal{G}(V) \mid V \supset f(U)\}$ has $\varinjlim_{V \supset f(U')} \mathcal{G}(V)$ as a co-cone below it. So there exists a unique map, which is what we define to be the restriction, from $\varinjlim_{V \supset f(U)} \mathcal{G}(V) \rightarrow \varinjlim_{V \supset f(U')} \mathcal{G}(V)$. Explicitly, this map is easily seen to be of the form $[s, V] \mapsto [s|_W, W]$, where in the right hand side the equivalence class is taken over a larger family. For the time being let us denote this presheaf by $\mathcal{P}$.

Now, we show that for any $x \in X$, there is an isomorphism of the stalks

$$\mathcal{P}_x \xrightarrow{\sim} \mathcal{G}_{f(x)}.$$

For a germ in $\mathcal{P}_x$ written as $[s, V], U$, where U is an open set containing x and $[s, V] \in \mathcal{P}(U)$ is a class for some open set $V \supset f(U)$ and $s \in \mathcal{G}(V)$, we note that $f(x) \in f(U)$ is contained in such a V . Thus, we define $\Phi : \mathcal{P}_x \rightarrow \mathcal{G}_{f(x)}$ by $[s, V], U \mapsto s_{f(x)}$. To see that this is well defined, suppose $[s, V], U = [s', V'], U'$, so that there is some small open set U'' containing x over which the restrictions of both $[s, V]$ and $[s', V']$ coincide i.e. $[s, V] = [s', V']$ in $\mathcal{P}(U'')$. Note that $f(U'') \subset f(U \cap U') \subset f(U) \cap f(U') \subset V \cap V'$ so the above equality makes sense. Thus, there exists $W \subset V \cap V'$ and $f(U'') \subset W$ such that $s|_W = s'|_W$. But since $f(x) \in W$, we have $s_{f(x)} = s'_{f(x)}$. It can be easily checked that Φ is a group homomorphism. We define another map $\Psi : \mathcal{G}_{f(x)} \rightarrow \mathcal{P}_x$ as follows : any germ in $\mathcal{G}_{f(x)}$ can be written as $s_{f(x)}^V$ where V is an open set containing $f(x)$ and $s^V \in \mathcal{G}(V)$. Thus $f^{-1}(V)$ is an open set containing x and obviously $f(f^{-1}(V)) \subset V$. So we define $\Psi : s_{f(x)}^V \mapsto [s^V, V], f^{-1}(V)$. To see well-definedness, suppose $s_{f(x)}^V = s_{f(x)}^{V'}$; then there exists $V'' \subset V \cap V'$ with $f(x) \in V''$ such that $s^V|_{V''} = s^{V'}|_{V''}$. Now $f^{-1}(V'') \subset f^{-1}(V) \cap f^{-1}(V')$ and $x \in f^{-1}(V'')$. It follows readily that $[s^V, V] = [s^{V'}, V']$ in $\mathcal{P}(f^{-1}(V''))$ since $s^V|_{V''} = s^{V'}|_{V''}$.

It remains to check that $\Psi \circ \Phi$ and $\Phi \circ \Psi$ are both identities. Take $[s, V], U$ so that $\Phi([s, V], U) = s_{f(x)}$. Then, $\Psi(s_{f(x)}) = [s^V, V], f^{-1}(V)$ by construction. But since $f(U) \subset V$ we have $U \subset f^{-1}(V)$ and so $[s, V], f^{-1}(V) = [s, V], U$. Hence $\Psi \circ \Phi = \text{id}$. On the other hand, take an arbitrary germ $s_{f(x)}^V \in \mathcal{G}_{f(x)}$, so that $\Psi(s_{f(x)}^V) = [s^V, V], f^{-1}(V)$ by definition. But again by definition of Φ we obtain $\Phi([s^V, V], f^{-1}(V)) = s_{f(x)}^V$, whence $\Phi \circ \Psi = \text{id}$. Thus we have shown the following :

$$\mathcal{P}_x = \varinjlim_{x \in U} \left(\varinjlim_{V \supset f(U)} \mathcal{G}(V) \right) \simeq \varinjlim_{V \ni f(x)} \mathcal{G}(V) = \mathcal{G}_{f(x)}.$$

Consequently, we have a canonical isomorphism :

$$(f^{-1}\mathcal{G})_x \simeq \mathcal{G}_{f(x)}.$$

Definition 9.45. – If Z is a subset of the topological space X , regarded as a topological subspace with the induced topology, if $i : Z \hookrightarrow X$ is the inclusion map, and if $\mathcal{F}$ is a sheaf of sets, abelian groups, or rings on X , then we call $i^{-1}\mathcal{F}$ the *restriction* of $\mathcal{F}$ to Z , and we often denote it by $\mathcal{F}|_Z$. By the previous remark, the stalk of $\mathcal{F}|_Z$ at any point $z \in Z$ is isomorphic to $\mathcal{F}_z$.

Remark 9.46. – Let $Z = U$ be an open set in X , and $\mathcal{F}$ be a sheaf on X . Denote by $\mathcal{F}_U$ the sheaf on U given by $V \rightsquigarrow \mathcal{F}(V)$ for open sets $V \subset U$. This follows from the fact that the presheaf

$$(i^{-1}\mathcal{F})_{\text{pre}} : V \rightsquigarrow \varinjlim_{W \supset V} \mathcal{F}(W)$$

is actually isomorphic to the sheaf $\mathcal{F}_U$. Indeed, it is easy to check that the canonical map

$$\iota_V : \mathcal{F}(V) \rightarrow \varinjlim_{W \supset V} \mathcal{F}(W), \quad s \mapsto [s, V]$$

is an isomorphism; its inverse is $\pi_V : \varinjlim_{W \supset V} \mathcal{F}(W) \rightarrow \mathcal{F}(V), [s, W] \mapsto s|_V$. Thus $(i^{-1}\mathcal{F})_{\text{pre}} \simeq \mathcal{F}_U \simeq i^{-1}\mathcal{F}$.

Proposition 9.47. – *Let $f : X \rightarrow Y$ be a continuous map of topological spaces. Suppose $\mathcal{F}$ and $\mathcal{F}'$ are two sheaves of sets, abelian groups, or rings on X . Then, given any morphism $\varphi : \mathcal{F} \rightarrow \mathcal{F}'$ we have a canonical map*

$$f_*\varphi : f_*\mathcal{F} \rightarrow f_*\mathcal{F}'.$$

Similarly, if $\mathcal{G}$ and $\mathcal{G}'$ are two sheaves of abelian groups on Y and $\psi : \mathcal{G} \rightarrow \mathcal{G}'$ is a morphism, then we have an induced morphism

$$f^{-1}\psi : f^{-1}\mathcal{G} \rightarrow f^{-1}\mathcal{G}'.$$

Proof. – For the first statement we define $(f_*\varphi)(U) := \varphi(f^{-1}(U)) : \mathcal{F}(f^{-1}(U)) \rightarrow \mathcal{F}'(f^{-1}(U))$ for every open set U of X . It is clear that $f_*\varphi$ is restriction compatible since φ is.

Now, consider the later statement. Fix an open set $U \subset X$. We define a map

$$\widetilde{f^{-1}\psi}(U) : \varinjlim_{V \supset f(U)} \mathcal{G}(V) \rightarrow \varinjlim_{V \supset f(U)} \mathcal{G}'(V)$$

explicitly by

$$[s, V] \mapsto [\psi(V)(s), V].$$

This is well-defined. If $[s, V] = [s', V']$, then there exists an open set $W \subset V \cap V'$ with $f(U) \subset W$ such that $s|_W = s'|_W$ in $\mathcal{G}(W)$. But then we have $\psi(V)(s)|_W = \psi(W)(s|_W) = \psi(W)(s'|_W) = \psi(V)(s')|_W$, and thus $[\psi(V)(s), V] = [\psi(V')(s'), V']$. It is readily checked that the above canonical map is restriction compatible. Thus, we get a morphism of presheaves

$$\widetilde{f^{-1}\psi} : (f^{-1}\mathcal{G})_{\text{pre}} \rightarrow (f^{-1}\mathcal{G}')_{\text{pre}}.$$

Composing this with the sheafification map we get a morphism $(f^{-1}\mathcal{G})_{\text{pre}} \rightarrow f^{-1}\mathcal{G}'$ which then induces our required morphism

$$f^{-1}\psi : f^{-1}\mathcal{G} \rightarrow f^{-1}\mathcal{G}'.$$

□

Keeping the setup of Definition 9.43, we now define a map

$$\Phi : \text{Hom}(\mathcal{G}, f_*\mathcal{F}) \rightarrow \text{Hom}(f^{-1}\mathcal{G}, \mathcal{F})$$

in the following way. Let $\sigma : \mathcal{G} \rightarrow f_*\mathcal{F}$ be a morphism of sheaves on Y . Then, given an open subset $U \subset X$, for every open set $V \subset Y$ with $f(U) \subset V$ (or, $U \subset f^{-1}(V)$), we consider the composite

$$\mathcal{G}(V) \xrightarrow{\sigma(V)} \mathcal{F}(f^{-1}(V)) \xrightarrow{\text{res}} \mathcal{F}(U).$$

Thus, $\mathcal{F}(U)$ becomes a co-cone below the diagram $\{\mathcal{G}(V) \mid V \supset f(U)\}$ and hence we get a map

$$\widetilde{\Phi(\sigma)}(U) : \varinjlim_{V \supset f(U)} \mathcal{G}(V) \rightarrow \mathcal{F}(U), \quad U \subset X \text{ open}.$$

We first see that $\widetilde{\Phi(\sigma)}$ is restriction compatible. This can be easily shown once the above map is explicitly written : $\widetilde{\Phi(\sigma)}(U)([s, V]) = \sigma(V)(s)|_U$. So we have a morphism

$$\widetilde{\Phi(\sigma)} : (f^{-1}\mathcal{G})_{\text{pre}} \rightarrow \mathcal{F}.$$

Since $\mathcal{F}$ is a sheaf, we get an induced morphism

$$\Phi(\sigma) : f^{-1}\mathcal{G} \rightarrow \mathcal{F}$$

due to the universal property of sheafification.

Fact 9.48. – *The resulting map $\Phi : \text{Hom}(\mathcal{G}, f_*\mathcal{F}) \rightarrow \text{Hom}(f^{-1}\mathcal{G}, \mathcal{F})$ is functorial in $\mathcal{F}$.*

Proof. – We have to show that the following diagram commutes for a morphism $\alpha : \mathcal{F} \rightarrow \mathcal{F}'$:

$$\begin{array}{ccc} \mathrm{Hom}(\mathcal{G}, f_*\mathcal{F}) & \xrightarrow{\Phi_{\mathcal{F}}} & \mathrm{Hom}(f^{-1}\mathcal{G}, \mathcal{F}) \\ (f_*\alpha) \circ - \downarrow & & \downarrow \alpha \circ - \\ \mathrm{Hom}(\mathcal{G}, f_*\mathcal{F}') & \xrightarrow{\Phi_{\mathcal{F}'}} & \mathrm{Hom}(f^{-1}\mathcal{G}, \mathcal{F}') \end{array}$$

which means we must show that, for every $\sigma : \mathcal{G} \rightarrow f_*\mathcal{F}$, we have

$$\Phi_{\mathcal{F}'}((f_*\alpha) \circ \sigma) = \alpha \circ \Phi_{\mathcal{F}}(\sigma).$$

However, it suffices to show

$$\Phi_{\mathcal{F}'}(\widetilde{f_*\alpha \circ \sigma}) = \alpha \circ \widetilde{\Phi_{\mathcal{F}}(\sigma)} : (f^{-1}\mathcal{G})_{\mathrm{pre}} \rightarrow \mathcal{F}'.$$

This is because $\widetilde{\Phi_{\mathcal{F}'}(f_*\alpha \circ \sigma)}$ factors through $f^{-1}\mathcal{G}$ to yield $\Phi_{\mathcal{F}'}(f_*\alpha \circ \sigma)$, wherein this morphism is the unique morphism whose composite with the sheafification map $\kappa_{(f^{-1}\mathcal{G})_{\mathrm{pre}}} : (f^{-1}\mathcal{G})_{\mathrm{pre}} \rightarrow f^{-1}\mathcal{G}$ should yield $\Phi_{\mathcal{F}'}(\widetilde{f_*\alpha \circ \sigma}) = \alpha \circ \widetilde{\Phi_{\mathcal{F}}(\sigma)}$. But $\alpha \circ \Phi_{\mathcal{F}}(\sigma) \circ \kappa_{(f^{-1}\mathcal{G})_{\mathrm{pre}}} = \alpha \circ \widetilde{\Phi_{\mathcal{F}}(\sigma)}$ by construction of $\Phi_{\mathcal{F}}(\sigma)$. So we must have $\Phi_{\mathcal{F}'}((f_*\alpha) \circ \sigma) = \alpha \circ \Phi_{\mathcal{F}}(\sigma)$.

Now fix an open set U and an element $[s, V] \in (f^{-1}\mathcal{G})_{\mathrm{pre}}(U)$. Then, applying $\widetilde{\Phi_{\mathcal{F}}(\sigma)}(U)$ we obtain

$$\widetilde{\Phi_{\mathcal{F}}(\sigma)}(U)([s, V]) = \sigma(V)(s) \mid_U.$$

Next applying $\alpha(U)$ we get

$$\alpha(U)(\sigma(V)(s) \mid_U) = (\alpha(f^{-1}(V))(\sigma(V)(s))) \mid_U.$$

Similarly, since $(f_*\alpha)(V) = \alpha(f^{-1}(V))$, we get

$$\Phi_{\mathcal{F}'}(\widetilde{f_*\alpha \circ \sigma})(U)([s, V]) = (\alpha(f^{-1}(V)) \circ \sigma(V))(s) \mid_U.$$

This completes the proof. $\square$

Now we will show that Φ is an isomorphism, we will construct an inverse map

$$\Psi : \mathrm{Hom}(f^{-1}\mathcal{G}, \mathcal{F}) \rightarrow \mathrm{Hom}(\mathcal{G}, f_*\mathcal{F}).$$

Let $\tau : f^{-1}\mathcal{G} \rightarrow \mathcal{F}$ be a morphism of sheaves. For each open set V of Y consider the following composite

$$\mathcal{G}(V) \xrightarrow{\iota_V} \varinjlim_{V' \supset f(f^{-1}(V))} \mathcal{G}(V') = (f^{-1}\mathcal{G})_{\mathrm{pre}}(f^{-1}(V)) \xrightarrow{\kappa_{(f^{-1}\mathcal{G})_{\mathrm{pre}}}} (f^{-1}\mathcal{G})(f^{-1}(V)) \xrightarrow{\tau_{(f^{-1}\mathcal{G})_{\mathrm{pre}}}} \mathcal{F}(f^{-1}(V)) = (f_*\mathcal{F})(V)$$

which assemble to yield

$$\Psi(\tau) : \mathcal{G} \rightarrow f_*\mathcal{F}.$$

Indeed, we have to check that $\Psi(\tau)(V) := \tau(f^{-1}(V)) \circ \kappa_{(f^{-1}\mathcal{G})_{\mathrm{pre}}} \circ \iota_V$ is restriction compatible. So take an open subset $W \subset V$ and we must show that

$$\Psi(\tau)(V)(s) \mid_{f^{-1}(W)} = \Psi(\tau)(W)(s \mid_W), \quad \text{for all } s \in \mathcal{G}(V).$$

The left hand side is precisely

$$(\tau(f^{-1}(V))(\kappa_{(f^{-1}\mathcal{G})_{\mathrm{pre}}}([s, V]))) \mid_{f^{-1}(W)} = (\tau(f^{-1}(W)) \circ \kappa_{(f^{-1}\mathcal{G})_{\mathrm{pre}}})([s, V])$$

where the $[s, V]$ at the right hand side of the previous equation is in $(f^{-1}\mathcal{G})_{\mathrm{pre}}(f^{-1}(W))$. Hence it is equal to $[s \mid_W, W]$. Thus, the restriction

$$\Psi(\tau)(V)(s) \mid_{f^{-1}(W)} = (\tau(f^{-1}(W)) \circ \kappa_{(f^{-1}\mathcal{G})_{\mathrm{pre}}})([s \mid_W, W]).$$

Obviously, we have, by definition, the following

$$\Psi(\tau)(W)(s \mid_W) = (\tau(f^{-1}(W)) \circ \kappa_{(f^{-1}\mathcal{G})_{\mathrm{pre}}})([s \mid_W, W]).$$

Fact 9.49. – *The map $\Psi : \mathrm{Hom}(f^{-1}\mathcal{G}, \mathcal{F}) \rightarrow \mathrm{Hom}(\mathcal{G}, f_*\mathcal{F})$ is functorial in $\mathcal{F}$.*

Proof. – Let $\alpha : \mathcal{F} \rightarrow \mathcal{F}'$ be a morphism. We have to show that the following square commutes :

$$\begin{array}{ccc} \mathrm{Hom}(f^{-1}\mathcal{G}, \mathcal{F}) & \xrightarrow{\Psi_{\mathcal{F}}} & \mathrm{Hom}(\mathcal{G}, f_*\mathcal{F}) \\ \alpha \circ - \downarrow & & \downarrow (f_*\alpha) \circ - \\ \mathrm{Hom}(f^{-1}\mathcal{G}, \mathcal{F}') & \xrightarrow{\Psi_{\mathcal{F}'}} & \mathrm{Hom}(\mathcal{G}, f_*\mathcal{F}') \end{array}$$

That is to say – for every $\tau : f^{-1}\mathcal{G} \rightarrow \mathcal{F}$ we must show that

$$\Psi_{\mathcal{F}'}(\alpha \circ \tau) = (f_*\alpha) \circ (\Psi_{\mathcal{F}}(\tau)).$$

Fix an open set V of Y and a section $s \in \mathcal{G}(V)$. The left hand side yields

$$\Psi_{\mathcal{F}'}(\alpha \circ \tau)(V)(s) = (\alpha \circ \tau)(f^{-1}(V))(\kappa(f^{-1}(V))([s, V])) = \alpha(f^{-1}(V))(\tau(f^{-1}(V))(\kappa(f^{-1}(V))([s, V]))) .$$

The right hand side yields

$$((f_*\alpha) \circ (\Psi_{\mathcal{F}}(\tau)))(V)(s) = \alpha(f^{-1}(V))(\Psi_{\mathcal{F}}(\tau)(V)(s)) = \alpha(f^{-1}(V))(\tau(f^{-1}(V))(\kappa(f^{-1}(V))([s, V]))) .$$

Thus the two sides are equal and we have the desired functoriality. $\square$

Proposition 9.50. – *The Φ and Ψ constructed above are inverses to each other.*

Proof. – We first show that $\Psi(\Phi(\sigma)) = \sigma$ for every $\sigma : \mathcal{G} \rightarrow f_*\mathcal{F}$. Now set $\tau := \Phi(\sigma)$ and note that $\tau \circ \kappa = \widetilde{\Phi(\sigma)}$. Thus, for a fixed open set V and $s \in \mathcal{G}(V)$, we have

$$\begin{aligned} \Psi(\tau)(V)(s) &= (\tau(f^{-1}(V)) \circ \kappa(f^{-1}(V)))([s, V]) \\ &= \widetilde{\Phi(\sigma)}(f^{-1}(V))([s, V]) \\ &= \sigma(V)(s) \mid_{f^{-1}(V)} \\ &= \sigma(V)(s), \end{aligned}$$

where the last equality follows because $\sigma(V)(s)$ is already a section of $\mathcal{F}(f^{-1}(V))$. Thus $\Psi(\Phi(\sigma)) = \sigma$ as asserted.

Next, we prove the more technical part $\Phi(\Psi(\tau)) = \tau$ for every $\tau : f^{-1}\mathcal{G} \rightarrow \mathcal{F}$. Set $\sigma := \Psi(\tau) : \mathcal{G} \rightarrow f_*\mathcal{F}$. We need to show that for every open set U of X we have

$$\Phi(\sigma)(U) = \tau(U) : (f^{-1}\mathcal{G})(U) \rightarrow \mathcal{F}(U).$$

Set $\mathcal{P} := (f^{-1}\mathcal{G})_{\mathrm{pre}}$ so that $f^{-1}\mathcal{G}$ is the sheafification of $\mathcal{P}$. Take a section $S = (s_x)_{x \in U} \in (f^{-1}\mathcal{G})(U)$ where $s_x \in \mathcal{P}_x$. Then there exists an open cover $(U_\lambda)_{\lambda \in \Lambda}$ of U and sections $t^\lambda \in \mathcal{P}(U_\lambda)$ such that $t_x^\lambda = s_x$ for all $x \in U_\lambda$. Write each t^λ as a colimit class : $t^\lambda = [s_\lambda, V_\lambda]$ where V_λ is open, $s_\lambda \in \mathcal{G}(V_\lambda)$, and $f(U_\lambda) \subset V_\lambda$. Now the statement “ $t_x^\lambda = s_x$ for all $x \in U_\lambda$ ” can be rewritten as

$$\kappa(U_\lambda)(t^\lambda) = S \mid_{U_\lambda} .$$

Thus to show that $\Phi(\sigma)(U)(S) = \tau(U)(S)$ it suffices to show that the following restrictions are equal for all λ :

$$\Phi(\sigma)(U)(S) \mid_{U_\lambda} = \tau(U)(S) \mid_{U_\lambda} .$$

Now we observe that

$$\begin{aligned} \Phi(\sigma)(U_\lambda)(\kappa(U_\lambda)(t^\lambda)) &= \widetilde{\Phi(\sigma)}(U_\lambda)(t^\lambda) = \widetilde{\Phi(\sigma)}(U_\lambda)([s_\lambda, V_\lambda]) \\ &= \sigma(V_\lambda)(s_\lambda) \mid_{U_\lambda} = \Psi(\tau)(V_\lambda)(s_\lambda) \mid_{U_\lambda} \\ &= (\tau(f^{-1}(V_\lambda)) \circ \kappa(f^{-1}(V_\lambda)))([s_\lambda, V_\lambda]) \mid_{U_\lambda} \\ &= \tau(U_\lambda)(\kappa(f^{-1}(V_\lambda))([s_\lambda, V_\lambda])) \mid_{U_\lambda} \\ &= \tau(U_\lambda)(\kappa(U_\lambda)([s_\lambda, V_\lambda] \mid_{U_\lambda})) \\ &= \tau(U_\lambda)(\kappa(U_\lambda)([s_\lambda, V_\lambda])) = \tau(U_\lambda)(S \mid_{U_\lambda}) \\ &= \tau(U)(S) \mid_{U_\lambda} . \end{aligned}$$

Thus, by locality, we get $\Phi(\sigma)(U)(S) = \tau(U)(S)$. Since U and S were chosen arbitrarily, we have

$$\Phi(\sigma) = \Phi(\Psi(\tau)) = \tau$$

as required. $\square$

Summarizing all these we have the following important theorem.

Theorem 9.51. – Let $f : X \rightarrow Y$ be a continuous map of topological spaces, and suppose $\mathcal{G}$ be a sheaf of sets, abelian groups or rings on Y . Then there is a sheaf $f^{-1}\mathcal{G}$ of sets, abelian groups or rings respectively on X with a functorial isomorphism

$$\text{Hom}(f^{-1}\mathcal{G}, \mathcal{F}) \xrightarrow{\sim} \text{Hom}(\mathcal{G}, f_*\mathcal{F})$$

of functors on the category of sheaves $\mathcal{F}$ of abelian groups on X .

Proof. – The only thing that remains to be checked is that Φ and Ψ are group and homomorphisms. It suffices to show Ψ is a group homomorphism. But observe that

$$\begin{aligned} \Psi(\tau_1 + \tau_2)(V)(s) &= (\tau_1 + \tau_2)(f^{-1}(V))(\kappa(f^{-1}(V))([s, V])) \\ &= \tau_1(f^{-1}(V))(\kappa(f^{-1}(V))([s, V])) + \tau_2(f^{-1}(V))(\kappa(f^{-1}(V))([s, V])) \\ &= \Psi(\tau_1)(V)(s) + \Psi(\tau_2)(V)(s). \end{aligned}$$

$\square$

10. SCHEME THEORY 2 : MODULE SHEAVES

Let $\mathcal{O}$ be a presheaf of rings on a topological space X and $\mathcal{F}$ be a presheaf of abelian groups on X . Then the cartesian product $\mathcal{O} \times \mathcal{F}$ is defined as a functor that associates to an open subset $U \subset X$ the cartesian product $\mathcal{O}(U) \times \mathcal{F}(U)$ and to an inclusion $U \subset V$ the product of restriction morphisms $\mathcal{O}(V) \rightarrow \mathcal{O}(U)$ and $\mathcal{F}(V) \rightarrow \mathcal{F}(U)$. Thus $\mathcal{O} \times \mathcal{F} : \mathbf{Opn}(X) \rightarrow \mathbf{AbGrp}$ is a functor. A *composition law* $\mathcal{O} \times \mathcal{F} \rightarrow \mathcal{F}$ is a natural transformation.

Definition 10.1. – An $\mathcal{O}$ -module is a presheaf $\mathcal{F}$ of abelian groups together with a composition law $\mu : \mathcal{O} \times \mathcal{F} \rightarrow \mathcal{F}$ such that, for any open $U \subset X$, the map $\mu(U) : \mathcal{O}(U) \times \mathcal{F}(U) \rightarrow \mathcal{F}(U)$ defines an $\mathcal{O}(U)$ -module structure on $\mathcal{F}(U)$. For $a \in \mathcal{O}(U)$ and $s \in \mathcal{F}(U)$, we denote $\mu(U)(a, s)$ by $a \cdot s \in \mathcal{F}(U)$. Thus, the compatibility of μ with restrictions can be written as follows : given open sets $V \subset U$ and $a \in \mathcal{O}(U), s \in \mathcal{F}(U)$ we have $(a \cdot s)|_V = a|_V \cdot s|_V$.

A morphism of $\mathcal{O}$ -modules $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ is a morphism of presheaves of abelian groups that is compatible with $\mathcal{O}$ -module structures – that is to say – for each open set U the map $\varphi(U) : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ is $\mathcal{O}(U)$ -linear. Furthermore, for open sets $V \subset U$ we have

$$\varphi(U)(a \cdot s)|_V = \varphi(V)(a|_V \cdot s|_V) \text{ i.e. } (a \cdot \varphi(U)(s))|_V = a|_V \cdot \varphi(V)(s|_V).$$

From here onwards we fix a topological space X and we let $\mathcal{O}_X$ denote a presheaf of rings on X . We will now carry over some of the constructions we did in the section of sheaves to module sheaves.

Lemma 10.2. – Let $\mathcal{F}$ be a presheaf of $\mathcal{O}_X$ -modules. Then, its sheafification $\mathcal{F}'$ is also naturally an $\mathcal{O}_X$ -module.

Proof. – Recall that for an open set U we have

$$\mathcal{F}'(U) := \left\{ (s_x)_{x \in U} \in \prod_{x \in U} \mathcal{F}_x \mid \exists \text{ a cover } (U_\lambda)_{\lambda \in \Lambda} \text{ of } U \text{ and } t^\lambda \in \mathcal{F}(U_\lambda) \text{ with } t^\lambda_x = s_x \text{ for all } x \in U_\lambda \right\}.$$

Now take $a \in \mathcal{O}_X(U)$ and $S := (s_x)_{x \in U} \in \mathcal{F}'(U)$. We define $a \cdot S \in \mathcal{F}'(U)$ as follows : given $x \in U$ choose λ with $x \in U_\lambda$ and set the x -component of $a \cdot S$ to be $(a|_{U_\lambda} \cdot t^\lambda)_x$. Observe that this is independent of the choice of λ : since if λ' is another index with $x \in U_{\lambda'}$, then $t^{\lambda'}_x = t^\lambda_x$ and so there exists $W \subset U_\lambda \cap U_{\lambda'}$ such that $t^{\lambda'}|_W = t^\lambda|_W$, whence $a|_W \cdot t^{\lambda'}|_W = a|_W \cdot t^\lambda|_W$ i.e. $(a|_{U_{\lambda'}} \cdot t^{\lambda'})_x = (a|_{U_\lambda} \cdot t^\lambda)_x$. Next, we show that $a \cdot S$ is also independent of the choice of the cover $(U_\lambda)_{\lambda \in \Lambda}$. Since, if $(V_\mu)_{\mu \in M}$ was another cover of U with $r^\mu \in \mathcal{F}(V_\mu)$ satisfying $r^\mu_y = s_y$ for all $y \in V_\mu$, and $x \in U$ satisfied $x \in V_\mu$, then the x -components defined as above $(a|_{U_\lambda} \cdot t^\lambda)_x$ and $(a|_{V_\mu} \cdot r^\mu)_x$ should coincide. Indeed, $x \in U_\lambda \cap V_\mu$ and so $t^\lambda_x = r^\mu_x$, therefore there exists some $W \subset U_\lambda \cap V_\mu$ with $t^\lambda|_W = r^\mu|_W$, whence $a|_W \cdot t^\lambda|_W = a|_W \cdot r^\mu|_W$. Hence, $(a|_{U_\lambda} \cdot t^\lambda)_x = (a|_{V_\mu} \cdot r^\mu)_x$ as asserted.

The module axioms are easy to check, so it remains to verify the compatibility with restrictions. Suppose $V \subset U$ be open sets. Write $a \cdot S = ((a|_{U_{\lambda(x)}} \cdot t^{\lambda(x)})_x)_{x \in U}$, where $\lambda : U \rightarrow \Lambda$ is the choice function. We use the cover $(V \cap U_{\lambda})_{\lambda \in \Lambda}$ of V and sections $t^{\lambda}|_{V \cap U_{\lambda}} \in \mathcal{F}(V \cap U_{\lambda})$ for the restriction $S|_V = (s_x)_{x \in V}$, whence the restricted choice function $\lambda|_V : V \rightarrow \Lambda$ can be used to define $b \cdot S|_V$ for any $b \in \mathcal{O}_X(V)$. Then $(a \cdot S)|_V = ((a|_{U_{\lambda(x)}} \cdot t^{\lambda(x)})_x)_{x \in V}$ and $a|_V \cdot S|_V = a|_V \cdot (s_x)_{x \in V} = ((a|_{V \cap U_{\lambda(x)}} \cdot t^{\lambda(x)}|_{V \cap U_{\lambda(x)}})_x)_{x \in V} = ((a|_{V \cap U_{\lambda(x)}} \cdot t^{\lambda(x)}|_{V \cap U_{\lambda(x)}})_x)_{x \in V}$. But it is clear that $(a|_{U_{\lambda(x)}} \cdot t^{\lambda(x)})_x$ and $(a|_{V \cap U_{\lambda(x)}} \cdot t^{\lambda(x)}|_{V \cap U_{\lambda(x)}})_x$ are equal for any $x \in V$. The proof is now complete. $\square$

Lemma 10.3. – *Let $\mathcal{F}$ be a presheaf of $\mathcal{O}_X$ -modules. Then, the sheafification map $\kappa_{\mathcal{F}} : \mathcal{F} \rightarrow \mathcal{F}'$ is an $\mathcal{O}_X$ -module map.*

Proof. – Recall that for any open set U the map $\kappa_{\mathcal{F}}(U) : s \mapsto (s_x)_{x \in U}$ for $s \in \mathcal{F}(U)$. For $a \in \mathcal{O}_X(U)$, we have $\kappa(U)(a \cdot s) = ((a \cdot s)_x)_{x \in U}$, whereas, taking the singleton cover $\{U\}$ for $(s_x)_{x \in U}$ we defined $a \cdot \kappa(U)(s) = a \cdot (s_x)_{x \in U} = ((a \cdot s)_x)_{x \in U}$. Thus $\kappa(U)$ is $\mathcal{O}_X(U)$ -linear. $\square$

Lemma 10.4. – *Let $\mathcal{F}$ be a presheaf of $\mathcal{O}_X$ -modules. Then, for any $x \in X$ the stalk $\mathcal{F}_x$ is an $\mathcal{O}_{X,x}$ -module.*

Proof. – Recall that $\mathcal{O}_{X,x} = \varinjlim_{x \in U} \mathcal{O}_X(U)$ and $\mathcal{F}_x = \varinjlim_{x \in U} \mathcal{F}(U)$. So take germs $a_x \in \mathcal{O}_{X,x}$ and $s_x \in \mathcal{F}_x$ represented by, say, $a \in \mathcal{O}_X(U)$ and $s \in \mathcal{F}(U)$ respectively. We define

$$a_x \cdot s_x := (a|_{U \cap V} \cdot s|_{U \cap V})_x.$$

That the above is well defined can be easily checked after observing that various sections representing the same germ agree on smaller neighborhoods of x , and so we can always shrink neighborhoods. We omit the verification of the module axioms which are routine. $\square$

Remark 10.5. – Note that if $a \in \mathcal{O}_X(U)$ and $s \in \mathcal{F}(U)$ for some open set $U \ni x$, then

$$(a \cdot s)_x = a_x \cdot s_x.$$

This is clear from the definition of $a_x \cdot s_x$ in the proof of the lemma above.

Lemma 10.6. – *Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of $\mathcal{O}_X$ -module presheaves. Then, $\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$ is $\mathcal{O}_{X,x}$ -linear.*

Proof. – For $a_x \in \mathcal{O}_{X,x}$ and $s_x \in \mathcal{F}_x$, we have to show that $\varphi_x(a_x \cdot s_x) = a_x \cdot \varphi_x(s_x)$. So suppose, after shrinking neighborhoods, a_x and s_x are represented by $a \in \mathcal{O}_X(U)$ and $s \in \mathcal{F}(U)$ respectively, for some open set U containing x . Then, $\varphi_x(s_x)$ is represented by $\varphi(U)(s)$. So $a_x \cdot s_x = (a \cdot s)_x$ and $\varphi_x(a_x \cdot s_x) = \varphi(U)(a \cdot s)_x = (a \cdot \varphi(U)(s))_x = a_x \cdot \varphi_x(s_x)$. Remaining matters are easy to check. This completes the proof. $\square$

Lemma 10.7. – *Let $\mathcal{F}$ be a presheaf of $\mathcal{O}_X$ -modules and $\mathcal{G}$ a sheaf of $\mathcal{O}_X$ -modules. If $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ is a morphism of $\mathcal{O}_X$ -modules, then there exists a unique morphism $\varphi' : \mathcal{F}' \rightarrow \mathcal{G}$ of $\mathcal{O}_X$ -modules such that $\varphi' \circ \kappa_{\mathcal{F}} = \varphi$.*

Proof. – We just have to check that the morphism $\varphi' : \mathcal{F}' \rightarrow \mathcal{G}$ constructed in the proof of the Fact 9.14 is an $\mathcal{O}_X$ -module morphism. Recall that this morphism was defined as $\kappa_{\mathcal{G}}^{-1} \circ \varphi^+$ where $\varphi^+ : \mathcal{F}' \rightarrow \mathcal{G}'$ was defined by the formula : $\varphi^+(U)((s_x)_{x \in U}) := (\varphi_x(s_x))_{x \in U}$. So we start with such a $(s_x)_{x \in U}$ with an open cover $(U_{\lambda})_{\lambda \in \Lambda}$ and sections $t^{\lambda} \in \mathcal{F}(U_{\lambda})$ satisfying $t_y^{\lambda} = s_y$ for all $y \in U_{\lambda}$. Take $a \in \mathcal{O}_X(U)$, and $a \cdot (s_x)_{x \in U} = ((a|_{U_{\lambda(x)}} \cdot t^{\lambda(x)})_x)_{x \in U}$ by definition. We have to show $\varphi^+(U)(a \cdot (s_x)_{x \in U}) = a \cdot \varphi^+(U)(s_x)_{x \in U} = a \cdot (\varphi_x(s_x))_{x \in U}$. The left hand side is $(\varphi_x(a|_{U_{\lambda(x)}} \cdot t^{\lambda(x)})_x)_{x \in U}$. Now fix an $x \in U$ and note that $a_x \cdot t_x^{\lambda(x)} = (a|_{U_{\lambda(x)}} \cdot t^{\lambda(x)})_x$ by the proof of Lemma 10.4 and by Lemma 10.6, we have

$$(\varphi_x(a|_{U_{\lambda(x)}} \cdot t^{\lambda(x)})_x)_{x \in U} = (a_x \cdot \varphi_x(s_x))_{x \in U}.$$

Now the right hand side is $a \cdot (\varphi_x(s_x))_{x \in U}$, and using the same cover we obtain the sections $\varphi(U_{\lambda})(t^{\lambda})$ satisfying $\varphi(U_{\lambda})(t^{\lambda})_x = \varphi_x(t_x^{\lambda}) = \varphi_x(s_x)$ for every $x \in U_{\lambda}$. Thus we get

$$a \cdot (\varphi_x(s_x))_{x \in U} = ((a|_{U_{\lambda(x)}} \cdot \varphi(U_{\lambda})(t^{\lambda(x)}))_x)_{x \in U}.$$

For a fixed $x \in U$, we have

$$\left(a|_{U_{\lambda(x)}} \cdot \varphi(U_{\lambda(x)})(t^{\lambda(x)}) \right)_x = \left(\varphi(U_{\lambda(x)})(a|_{U_{\lambda(x)}} \cdot t^{\lambda(x)}) \right)_x = \varphi_x \left((a|_{U_{\lambda(x)}} \cdot t^{\lambda(x)})_x \right) = a_x \cdot \varphi_x(s_x).$$

This completes the proof that φ^+ is an $\mathcal{O}_X$ -module morphism, and therefore φ' must be the same. $\square$

Lemma 10.8. – Let $\mathcal{G}$ be a subsheaf of $\mathcal{O}_X$ -modules of an $\mathcal{O}_X$ -module $\mathcal{F}$. Then, $\mathcal{F}/\mathcal{G}$ is also naturally an $\mathcal{O}_X$ -module.

Proof. – It suffices to show that $(\mathcal{F}/\mathcal{G})_{\text{pre}}$ is a presheaf of $\mathcal{O}_X$ -module owing to Lemma 10.2. For an open set U the $\mathcal{O}_X(U)$ -module structure on $(\mathcal{F}/\mathcal{G})_{\text{pre}}(U) = \mathcal{F}(U)/\mathcal{G}(U)$ is naturally induced from that of $\mathcal{F}(U)$. Compatibility with restrictions follows. $\square$

Lemma 10.9. – Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of sheaves of $\mathcal{O}_X$ -modules. Then, $\text{Ker } \varphi$, $\text{Im } \varphi$, and $\text{Coker } \varphi$ are all $\mathcal{O}_X$ -modules.

Proof. – The statement is clear for $\text{Ker } \varphi$. We first prove that $\text{Im } \varphi$ is an $\mathcal{O}_X$ -module. Fix an open set U and take $a \in \mathcal{O}_X(U)$ and $t \in (\text{Im } \varphi)(U)$. So there is an open cover $(U_\lambda)_{\lambda \in \Lambda}$ such that $t|_{U_\lambda} = \varphi(U_\lambda)(s_\lambda)$ for some sections $s_\lambda \in \mathcal{F}(U_\lambda)$. Then, $(a \cdot t)|_{U_\lambda} = a|_{U_\lambda} \cdot \varphi(U_\lambda)(s_\lambda) = \varphi(U_\lambda)(a|_{U_\lambda} \cdot s_\lambda)$ for every λ i.e. $a \cdot t$ also locally lies in the presheaf image of φ . Lastly, the fact that $\text{Coker } \varphi$ is also an $\mathcal{O}_X$ -module follows from Lemma 10.2 once we show that the presheaf $\text{Coker}_{\text{pre}} \varphi : U \rightsquigarrow \mathcal{G}(U)/\text{Im } \varphi(U)$ is an $\mathcal{O}_X$ -module. But clearly $\mathcal{G}(U)/\text{Im } \varphi(U)$ inherits $\mathcal{O}_X(U)$ -module structure, and compatibility with restrictions is also easy to check. $\square$

Lemma 10.10. – Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of sheaves of $\mathcal{O}_X$ -modules on a topological space X . Then, $\text{Im } \varphi \simeq \mathcal{F}/\text{Ker } \varphi$ as $\mathcal{O}_X$ -modules.

Proof. – Recall that in the proof of Lemma 9.32 we constructed the following diagram

$$\begin{array}{ccc} \mathcal{F} & & \\ \downarrow & \searrow & \\ (\mathcal{F}/\text{Ker } \varphi)_{\text{pre}} & \xrightarrow{\Psi} & \text{Im } \varphi \\ \downarrow & \nearrow & \\ \mathcal{F}/\text{Ker } \varphi & & \end{array}$$

in which the morphisms in the top triangle are all $\mathcal{O}_X$ -module morphisms, and thus by the previous Lemma 10.7, we have the $\mathcal{O}_X$ -module isomorphism $\mathcal{F}/\text{Ker } \varphi \rightarrow \text{Im } \varphi$. $\square$

Lemma 10.11. – Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of sheaves of $\mathcal{O}_X$ -modules on a topological space X . We have $\text{Coker } \varphi \simeq \mathcal{G}/\text{Im } \varphi$.

Proof. – In this case we constructed the following diagram

$$\begin{array}{ccccc} \text{Coker}_{\text{pre}} \varphi & \longrightarrow & (\mathcal{G}/\text{Im } \varphi)_{\text{pre}} & \longrightarrow & \mathcal{G}/\text{Im } \varphi \\ \downarrow & & & \nearrow & \\ \text{Coker } \varphi & & & & \end{array}$$

in the proof of Lemma 9.34 in which it can be easily checked that the top left horizontal morphism is an $\mathcal{O}_X$ -module morphism, and the right horizontal one is also $\mathcal{O}_X$ -module morphism being the sheafification map. Thus the factoring through $\text{Coker } \varphi$ yielding the required isomorphism $\text{Coker } \varphi \rightarrow \mathcal{G}/\text{Im } \varphi$ is an $\mathcal{O}_X$ -module morphism by Lemma 10.7. $\square$

Let $(\mathcal{F}_i)_{i \in I}$ be a family of sheaves of $\mathcal{O}_X$ -modules over a topological space X .

Proposition 10.12. – The direct sum $\bigoplus_{i \in I} \mathcal{F}_i$ of sheaves of abelian groups is naturally an $\mathcal{O}_X$ -module and is equipped with canonical inclusions $\epsilon_i : \mathcal{F}_i \rightarrow \bigoplus_{i \in I} \mathcal{F}_i$ of $\mathcal{O}_X$ -module morphisms satisfying the following universal property : for any family of $\mathcal{O}_X$ -module morphisms $\eta_i : \mathcal{F}_i \rightarrow \mathcal{G}$, where $\mathcal{G}$ is an $\mathcal{O}_X$ -module sheaf on X , there exists a unique $\mathcal{O}_X$ -module morphism $\eta : \bigoplus_{i \in I} \mathcal{F}_i \rightarrow \mathcal{G}$ such that $\eta_i = \eta \circ \epsilon_i$ for all i .

Proof. – Recall that ϵ_i was defined to be the composite of the sheafification $(\bigoplus_{i \in I} \mathcal{F}_i)_{\text{pre}} \rightarrow \bigoplus_{i \in I} \mathcal{F}_i$ and the canonical inclusion $j_i : \mathcal{F}_i \rightarrow (\bigoplus_{i \in I} \mathcal{F}_i)_{\text{pre}}$. Since $(\bigoplus_{i \in I} \mathcal{F}_i)_{\text{pre}}$ can be canonically made into an $\mathcal{O}_X$ -module presheaf, it follows that j_i are morphisms of $\mathcal{O}_X$ -modules, and $\bigoplus_{i \in I} \mathcal{F}_i$ is also naturally an $\mathcal{O}_X$ -module. Thus, ϵ_i are naturally morphisms of $\mathcal{O}_X$ -modules.

Next, recall that in the proof of Proposition 9.38, the morphism $\eta : \bigoplus_{i \in I} \mathcal{F}_i \rightarrow \mathcal{G}$ was obtained by factoring the morphism of presheaves

$$\theta : \left(\bigoplus_{i \in I} \mathcal{F}_i \right)_{\text{pre}} \rightarrow \mathcal{G}$$

by

$$\theta(U)((s_i)_{i \in I}) := \sum_{i \in I} \eta_i(U)(s_i), \quad \text{where } U \text{ is open and } s_i \in \mathcal{F}_i(U).$$

It can be easily checked that θ is actually canonically a morphism of $\mathcal{O}_X$ -modules. The final assertion follows by Lemma 10.7. $\square$

Now we define the *direct product* of a family $(\mathcal{F}_i)_{i \in I}$ of sheaves of $\mathcal{O}_X$ -modules on X as follows : first consider the presheaf of $\mathcal{O}_X$ -modules

$$\prod_{i \in I} \mathcal{F}_i : U \rightsquigarrow \prod_{i \in I} \mathcal{F}_i(U).$$

As in case of sheaves of abelian groups, here also it follows that the above presheaf is actually a sheaf. Thus it is actually a sheaf of $\mathcal{O}_X$ -modules.

Proposition 10.13. – *The direct product $\prod_i \mathcal{F}_i$ comes equipped with canonical projections $\pi_i : \prod_i \mathcal{F}_i \rightarrow \mathcal{F}_i$ of morphisms of $\mathcal{O}_X$ -modules satisfying the following universal property : for a sheaf of $\mathcal{O}_X$ -modules $\mathcal{G}$ on X and any collection of morphisms of $\mathcal{O}_X$ -modules $\epsilon_i : \mathcal{G} \rightarrow \mathcal{F}_i$ there exists a unique morphism of $\mathcal{O}_X$ -modules $\eta : \mathcal{G} \rightarrow \prod_i \mathcal{F}_i$ such that $\pi_i \circ \eta = \epsilon_i$ for all i .*

Proof. – All the maps defined in the proof of Proposition 9.40 are canonical maps and thus they are morphisms of $\mathcal{O}_X$ -modules. $\square$

If $\mathcal{F}, \mathcal{G}$ are two sheaves of $\mathcal{O}_X$ -modules, then the set of morphisms of $\mathcal{O}_X$ -modules from $\mathcal{F}$ to $\mathcal{G}$ is denoted by $\text{Hom}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$. It is a group naturally as in the case of sheaves of abelian groups.

Proposition 10.14. – *The presheaf*

$$U \rightsquigarrow \text{Hom}_{\mathcal{O}_X|_U}(\mathcal{F}|_U, \mathcal{G}|_U)$$

is actually a sheaf, and is denoted by $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$.

Proof. – The only thing to check here is that the α constructed in the proof of Proposition 9.42 by gluing the $\alpha_i \in \text{Hom}_{\mathcal{O}_X|_{U_i}}(\mathcal{F}|_{U_i}, \mathcal{G}|_{U_i})$ is $\mathcal{O}_X$ -linear on every open set $V \subset U$. Recall that $(U_i)_{i \in I}$ is a cover of U and for any open set $V \subset U$ and $s \in \mathcal{F}(V)$, the section $\alpha(V)(s)$ is the unique one satisfying $\alpha(V)(s)|_{V \cap U_i} = \alpha_i(V \cap U_i)(s|_{V \cap U_i})$ for all i . Now suppose $a \in \mathcal{O}_X(V)$, and we have to show that $\alpha(V)(a \cdot s) = a \cdot \alpha(V)(s)$. Consider the cover $(V \cap U_i)_{i \in I}$ of V and the restriction

$$\alpha(V)(a \cdot s)|_{V \cap U_i} = \alpha_i(V \cap U_i)((a \cdot s)|_{V \cap U_i})$$

by definition of $\alpha(V)(a \cdot s)$. Now we see that the right hand side is

$$a|_{V \cap U_i} \cdot \alpha_i(V \cap U_i)(s|_{V \cap U_i}) = a|_{V \cap U_i} \cdot \alpha(V)(s)|_{V \cap U_i} = (a \cdot \alpha(V)(s))|_{V \cap U_i}.$$

Thus $\alpha(V)(a \cdot s)|_{V \cap U_i} = (a \cdot \alpha(V)(s))|_{V \cap U_i}$ for all i . By locality $\alpha(V)(a \cdot s) = a \cdot \alpha(V)(s)$ as required. $\square$

Proposition 10.15. – *$\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$ is naturally an $\mathcal{O}_X$ -module.*

Proof. – Fix an open set $U \subset X$. Let

$$a \in \mathcal{O}_X(U), \quad \varphi \in \mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})(U) := \text{Hom}_{\mathcal{O}_X|_U}(\mathcal{F}|_U, \mathcal{G}|_U).$$

We define $a \cdot \varphi$ on each open set $V \subset U$ by

$$(a \cdot \varphi)(V) : \mathcal{F}(V) \rightarrow \mathcal{G}(V), \quad s \mapsto a|_V \cdot \varphi(V)(s).$$

We first check that $a \cdot \varphi$ is a morphism of sheaves and is $\mathcal{O}_X|_U$ -linear.

To see that $a \cdot \varphi$ is a morphism of sheaves, take open sets $W \subset V \subset U$ and $s \in \mathcal{F}(V)$. Using the definition of $a \cdot \varphi$ and the restriction compatibility of φ we get

$$\begin{aligned} (a \cdot \varphi)(V)(s)|_W &= (a|_V \cdot \varphi(V)(s))|_W \\ &= a|_W \cdot \varphi(W)(s|_W) \\ &= (a \cdot \varphi)(W)(s|_W) \end{aligned}$$

which means $a \cdot \varphi$ is restriction compatible.

Next, we show that $a \cdot \varphi$ is $\mathcal{O}_X|_U$ -linear. So fix an open set $V \subset U$ and suppose $b \in \mathcal{O}_X(V)$, $s \in \mathcal{F}(V)$. Then

$$\begin{aligned} (a \cdot \varphi)(V)(b \cdot s) &= a|_V \cdot \varphi(V)(b \cdot s) \\ &= a|_V \cdot (b \cdot \varphi(V)(s)) \\ &= (a|_V b) \cdot \varphi(V)(s) \\ &= b \cdot (a|_V \cdot \varphi(V)(s)) \\ &= b \cdot (a \cdot \varphi)(V)(s). \end{aligned}$$

Thus $a \cdot \varphi$ is $\mathcal{O}_X|_U$ -linear and hence $a \cdot \varphi \in \text{Hom}_{\mathcal{O}_X|_U}(\mathcal{F}|_U, \mathcal{G}|_U)$.

Finally, for the verification of the module axioms, namely, for U open and $a, a' \in \mathcal{O}_X(U)$, we must check :

$$(a + a') \cdot \varphi = a \cdot \varphi + a' \cdot \varphi, \quad a \cdot (\varphi + \psi) = a \cdot \varphi + a \cdot \psi, \quad (aa') \cdot \varphi = a \cdot (a' \cdot \varphi), \quad 1 \cdot \varphi = \varphi$$

are verified pointwise on every open set $V \subset U$. But these are routine so we omit them.

Thus it remains to check restriction compatibility i.e. for $a \in \mathcal{O}_X(U)$, $s \in \mathcal{F}(U)$ and $V \subset U$ open, we have to show

$$(a \cdot \varphi)|_V = a|_V \cdot \varphi|_V.$$

Both sides are morphisms $\mathcal{F}|_V \rightarrow \mathcal{G}|_V$, so it suffices to check equality on each open set $W \subset V$. Fix $s \in \mathcal{F}(W)$. Then

$$(a \cdot \varphi)|_V(W)(s) = (a \cdot \varphi)(W)(s) = a|_W \cdot \varphi(W)(s)$$

whereas

$$(a|_V \cdot \varphi|_V)(W)(s) = a|_{V|_W} \cdot (\varphi|_V)(W)(s) = a|_W \cdot \varphi(W)(s).$$

This completes the verification of restriction compatibility and the proof. $\square$

We will now prove an analogous adjunction theorem like Theorem 9.51 for module sheaves. But first, we introduce the notion of a ringed space.

Definition 10.16. – A *ringed space* is a pair $(X, \mathcal{O}_X)$ where X is a topological space and $\mathcal{O}_X$ a sheaf of rings on X . We call $\mathcal{O}_X$ the *structure sheaf* of the ringed space.

A *morphism of ringed spaces* $(X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ is a pair $(f, f^\#)$ where $f : X \rightarrow Y$ is a continuous map and $f^\# : \mathcal{O}_Y \rightarrow f_*(\mathcal{O}_X)$ a morphism of sheaves of rings on Y .

Remark 10.17. – If $(X, \mathcal{O}_X)$ is a ringed space and $U \subset X$ is open, then we can consider the restriction $(U, \mathcal{O}_X|_U)$. The inclusion $i : U \hookrightarrow X$ is canonically a morphisms of ringed spaces. It is canonical in the sense that for each open $W \subset X$ the canonical restriction maps $\mathcal{O}_X(W) \rightarrow i_*(\mathcal{O}_X)(W) = \mathcal{O}_X(i^{-1}(W)) = \mathcal{O}_X(W \cap U)$ assemble to give $i^\# : \mathcal{O}_X \rightarrow i_*(\mathcal{O}_X)$. The pair $(i, i^\#)$ is called *open immersion* of ringed spaces.

Lemma 10.18. – Any morphism of ringed spaces $(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ canonically induces a homomorphism of rings $f_x^\# : \mathcal{O}_{Y, f(x)} \rightarrow \mathcal{O}_{X, x}$ for all $x \in X$.

Proof. – Let $x \in X$ and $V \subset U$ be open sets containing $f(x)$, so that $f^{-1}(V) \subset f^{-1}(U)$ are open sets containing x . Then we have the diagram

$$\begin{array}{ccc} \mathcal{O}_Y(U) & \xrightarrow{\quad} & \mathcal{O}_Y(V) \\ \downarrow f^\#(U) & & \downarrow f^\#(V) \\ \mathcal{O}_X(f^{-1}(U)) & \xrightarrow{\quad} & \mathcal{O}_X(f^{-1}(V)) \\ & \searrow & \swarrow \\ & \mathcal{O}_{X,x} & \end{array}$$

where the square and the triangle commute. Thus $\mathcal{O}_{X,x}$ is a co-cone below the diagram $\{\mathcal{O}_Y(U) \mid f(x) \in U, U \text{ open in } Y\}$. Hence, there exists a unique map $f_x^\# : \mathcal{O}_{Y,f(x)} \rightarrow \mathcal{O}_{X,x}$. $\square$

Let $(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a morphism of ringed spaces. The following is evident.

Fact 10.19. – $f^{-1}(\mathcal{O}_Y)$ is a sheaf of rings on X and $f_*\mathcal{O}_X$ is a sheaf of rings on Y .

We now prove two general lemmas.

Lemma 10.20. – If $\mathcal{G}$ is a presheaf of abelian groups, and $\mathcal{O}$ a presheaf of rings on X . Suppose further that $\mathcal{G}$ is an $\mathcal{O}$ -module. Then $\mathcal{G}'$ is naturally an $\mathcal{O}'$ -module, where $(\cdot)'$ indicates sheafification.

Proof. – First recall that by Lemma 10.4 we know $\mathcal{G}_x$ is an $\mathcal{O}_x$ -module for every $x \in X$. Now, fix an open set U in X and take $a := (a_x)_{x \in U} \in \mathcal{O}'(U)$ and $s := (s_x)_{x \in U} \in \mathcal{G}'(U)$. We define

$$a \cdot s := (a_x \cdot s_x)_{x \in U}.$$

We have to show that $a \cdot s \in \mathcal{G}'(U)$. Suppose a is locally represented by sections $A_i \in \mathcal{O}(U_i)$ for some open cover $(U_i)_{i \in I}$ of U , and suppose s is locally represented by sections $S_j \in \mathcal{G}(V_j)$ for another open cover $(V_j)_{j \in J}$ of U . Consider the common refinement $(W_{ij} := U_i \cap V_j)_{(i,j) \in I \times J}$. Set $T_{ij} := A_i|_{W_{ij}} \cdot S_j|_{W_{ij}} \in \mathcal{G}(W_{ij})$. Then, for any $x \in W_{ij}$, we have

$$(T_{ij})_x = (A_i|_{W_{ij}} \cdot S_j|_{W_{ij}})_x = (A_i|_{W_{ij}})_x \cdot (S_j|_{W_{ij}})_x = a_x \cdot s_x.$$

Thus $a \cdot s$ is also locally represented by $T_{ij} \in \mathcal{G}(W_{ij})$ for the cover $(W_{ij})_{i,j}$ of U . The module axioms can be checked easily.

We check restriction compatibility. But this is also routine : indeed, if $V \subset U$ are open sets, then

$$(a \cdot s)|_V = (a_x \cdot s_x)_{x \in V} = a|_V \cdot s|_V.$$

$\square$

Lemma 10.21. – Let $\mathcal{O}$ be a presheaf of rings on X , and $\mathcal{O}'$ be its sheafification, with $\kappa : \mathcal{O} \rightarrow \mathcal{O}'$ the sheafification morphism. Suppose $\mathcal{F}$ is a presheaf of $\mathcal{O}$ -modules and $\mathcal{G}$ is a sheaf of $\mathcal{O}'$ -modules. Then the following hold :

- (1) $\mathcal{G}$ is also naturally an $\mathcal{O}$ -module.
- (2) If $\tilde{\varphi} : \mathcal{F} \rightarrow \mathcal{G}$ is $\mathcal{O}$ -linear, then the factored map $\varphi : \mathcal{F}' \rightarrow \mathcal{G}$ is $\mathcal{O}'$ -linear.

Proof. – Item (1) is clear. For (2), it suffices to show that the map constructed in the proof of Fact 9.14

$$\varphi^+(U) : \mathcal{F}'(U) \rightarrow \mathcal{G}'(U), \quad (s_x)_{x \in U} \mapsto (\tilde{\varphi}_x(s_x))_{x \in U}$$

is $\mathcal{O}'(U)$ -linear for any open set U . So fix an open set U and consider $a := (a_x)_{x \in U} \in \mathcal{O}'(U)$, $s := (s_x)_{x \in U} \in \mathcal{F}'(U)$. Then, $a \cdot s = (a_x \cdot s_x)_{x \in U}$ and since $\tilde{\varphi}_x$ is $\mathcal{O}_x$ -linear by Lemma 10.6, the $\mathcal{O}'(U)$ -linearity of φ^+ follows. Thus, $\varphi = \kappa_{\mathcal{G}}^{-1} \circ \varphi^+$ is also $\mathcal{O}'$ -linear. $\square$

Now suppose $\mathcal{G}$ is a sheaf of $\mathcal{O}_Y$ -modules.

Fact 10.22. – $f^{-1}(\mathcal{G})$ is naturally equipped with a structure of $f^{-1}(\mathcal{O}_Y)$ -modules.

Proof. – Owing to the first lemma above, it suffices to show that $(f^{-1}\mathcal{G})_{\text{pre}}$ is a presheaf of $(f^{-1}\mathcal{O}_Y)_{\text{pre}}$ -modules. Let U be an open set X and take

$$[a, V] \in (f^{-1}\mathcal{O}_Y)_{\text{pre}}(U) \quad \text{and} \quad [s, W] \in (f^{-1}\mathcal{G})_{\text{pre}}(U)$$

where $a \in \mathcal{O}_Y(V)$, $s \in \mathcal{G}(W)$, and $f(U) \subset V \cap W$. Now we define

$$[a, V] \cdot [s, W] := [a|_{V \cap W} \cdot s|_{V \cap W}, V \cap W].$$

We show that this definition is independent of the choice of representatives. So suppose $[a, V] = [a', V']$ and $[s, W] = [s', W']$. We must show that

$$[a|_{V \cap W} \cdot s|_{V \cap W}, V \cap W] = [a|_{V' \cap W'} \cdot s|_{V' \cap W'}, V' \cap W'].$$

But the given conditions imply the existence of an open set $Q \subset V \cap V' \cap W \cap W'$ such that

$$a|_Q = a'|_Q \quad \text{and} \quad s|_Q = s'|_Q$$

whence $a|_Q \cdot s|_Q = a'|_Q \cdot s'|_Q$, consequently, $a|_{V \cap W} \cdot s|_{V \cap W}$ and $a|_{V' \cap W'} \cdot s|_{V' \cap W'}$ both restrict to the same section on Q .

Next we check restriction compatibility. If $U' \subset U$ are open sets. We must show that

$$([a, V] \cdot [s, W])|_{U'} = [a, V]|_{U'} \cdot [s, W]|_{U'} = [a, V] \cdot [s, W] \in (f^{-1}\mathcal{G})_{\text{pre}}(U').$$

The left hand side is equal to $[a|_{V \cap W} \cdot s|_{V \cap W}, V \cap W]|_{U'} = [a|_{V \cap W} \cdot s|_{V \cap W}, V \cap W] \in (f^{-1}\mathcal{G})_{\text{pre}}(U')$. The module axioms are easy to check by restriction to a common open set. This completes the proof. $\square$

Next suppose $\mathcal{F}$ is an $\mathcal{O}_X$ -module. Since, for every open set V of Y , we know $\mathcal{F}(f^{-1}(V))$ is an $\mathcal{O}_X(f^{-1}(V))$ -module, we have :

Fact 10.23. – $f_*\mathcal{F}$ is naturally a $f_*\mathcal{O}_X$ -module.

Since $f^\# : \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$ is a morphism of sheaves of rings, we have

Fact 10.24. – $f_*\mathcal{F}$ is naturally an $\mathcal{O}_Y$ -module.

Proof. – For an open set V of Y and sections $b \in \mathcal{O}_Y(V)$, $s \in (f_*\mathcal{F})(V)$, we define

$$b * s := f^\#(V)(b) \cdot s$$

since $f^\#(V)(b) \in (f_*\mathcal{O}_X)(V) = \mathcal{O}_X(f^{-1}(V))$ which has an action on $(f_*\mathcal{F})(V) = \mathcal{F}(f^{-1}(V))$. $\square$

Fact 10.25. – By the previously proven adjoint property, $\mathcal{O}_X$ is a $f^{-1}\mathcal{O}_Y$ -module.

Proof. – The adjoint property gives a bijection $\text{Hom}(f^{-1}\mathcal{O}_Y, \mathcal{O}_X) \simeq \text{Hom}(\mathcal{O}_Y, f_*\mathcal{O}_X)$. Corresponding to $f^\# \in \text{Hom}(\mathcal{O}_Y, f_*\mathcal{O}_X)$ we get a morphism in $\text{Hom}(f^{-1}\mathcal{O}_Y, \mathcal{O}_X)$. Call this morphism $f^\flat : f^{-1}\mathcal{O}_Y \rightarrow \mathcal{O}_X$. Note that explicitly, $f^\flat = \Phi(f^\#)$ in our notation, and also by construction we have $\widetilde{\Phi(f^\#)} = \Phi(f^\#) \circ \kappa_{(f^{-1}\mathcal{O}_Y)_{\text{pre}}} = f^\flat \circ \kappa_{(f^{-1}\mathcal{O}_Y)_{\text{pre}}}$. For any open set U of X and sections $a \in (f^{-1}\mathcal{O}_Y)(U)$, $s \in \mathcal{O}_X(U)$ we define $a * s := f^\flat(U)(a) \times s$, where $'\times'$ indicates the ring multiplication in $\mathcal{O}_X(U)$. The restriction compatibility follows from the fact that restrictions are ring homomorphisms and $f^\flat$ is restriction compatible. $\square$

In general, given a sheaf of rings $\mathcal{O}$ on X and two $\mathcal{O}$ -modules $\mathcal{F}, \mathcal{G}$ on X , we define the *tensor product* $\mathcal{F} \otimes_{\mathcal{O}} \mathcal{G}$ to be the sheafification of the presheaf $U \rightsquigarrow \mathcal{F}(U) \otimes_{\mathcal{O}(U)} \mathcal{G}(U)$. Since this presheaf is naturally a $\mathcal{O}$ -module, we have $\mathcal{F} \otimes_{\mathcal{O}} \mathcal{G}$ is also a $\mathcal{O}$ -module.

Returning to our set-up, we define $f^*\mathcal{G}$ to be the tensor product

$$f^{-1}\mathcal{G} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X.$$

Thus by construction $f^*\mathcal{G}$ is a $f^{-1}\mathcal{O}_Y$ -module.

Remark 10.26. – Note that the presheaf $(f^*\mathcal{G})_{\text{pre}}$ defined by $U \rightsquigarrow (f^{-1}\mathcal{G})(U) \otimes_{(f^{-1}\mathcal{O}_Y)(U)} \mathcal{O}_X(U)$ is naturally also an $\mathcal{O}_X$ -module. Here, $b \in \mathcal{O}_X(U)$ acts on $s \otimes c \in (f^{-1}\mathcal{G})(U) \otimes_{(f^{-1}\mathcal{O}_Y)(U)} \mathcal{O}_X(U)$ by

$$b \cdot_1 (s \otimes c) = s \otimes (bc).$$

Whereas, a section $a \in (f^{-1}\mathcal{O}_Y)(U)$ acts by

$$a \cdot_2 (s \otimes c) = (a \cdot s) \otimes c.$$

But by $f^{-1}(\mathcal{O}_Y)(U)$ -balancedness, we get

$$a \cdot_2 (s \otimes c) = s \otimes (a \cdot c) = s \otimes (f^b(U)(a)c) = f^b(U)(a) \cdot_1 (s \otimes c).$$

Now denote the $\mathcal{O}_X$ -module structure on the sheafification $f^*\mathcal{G}$ by $\cdot_1$ and the $f^{-1}\mathcal{O}_Y$ -module structure by $\cdot_2$. We wish to show that given any $S \in (f^*\mathcal{G})(U)$, we have a similar relation i.e.

$$a \cdot_2 S = f^b(U)(a) \cdot_1 S$$

for $a \in (f^{-1}\mathcal{O}_Y)(U)$. Choose a cover $(U_\lambda)_{\lambda \in \Lambda}$ of U and sections $s^\lambda \in (f^*\mathcal{G})_{\text{pre}}(U)$ such that

$$\kappa_{(f^*\mathcal{G})_{\text{pre}}}(U_\lambda)(s^\lambda) = S|_{U_\lambda}.$$

We know that $a|_{U_\lambda} \cdot_2 s^\lambda = f^b(U_\lambda)(a|_{U_\lambda}) \cdot_1 s^\lambda$ for every λ . Applying $\kappa_{(f^*\mathcal{G})_{\text{pre}}}(U_\lambda)$ and noting that it is both $\mathcal{O}_X$ -linear and $f^{-1}\mathcal{O}_Y$ -linear, we obtain

$$a|_{U_\lambda} \cdot_2 \kappa(U_\lambda)(s^\lambda) = f^b(U_\lambda)(a|_{U_\lambda}) \cdot_1 \kappa(U_\lambda)(s^\lambda)$$

that is same as

$$a|_{U_\lambda} \cdot_2 (S|_{U_\lambda}) = f^b(U_\lambda)(a|_{U_\lambda}) \cdot_1 (S|_{U_\lambda}), \quad \forall \lambda.$$

By locality in $f^*\mathcal{G}$ we have $a \cdot_2 S = f^b(U)(a) \cdot_1 S$ as required.

Proposition 10.27. – *We have an isomorphism*

$$\text{Hom}_{f^{-1}(\mathcal{O}_Y)}(f^{-1}\mathcal{G}, \mathcal{F}) \xrightarrow{\sim} \text{Hom}_{\mathcal{O}_Y}(\mathcal{G}, f_*\mathcal{F})$$

which is functorial in $\mathcal{F}$.

Proof. – First, note that since $\mathcal{F}$ is an $\mathcal{O}_X$ -module and $\mathcal{O}_X$ naturally has a $f^{-1}\mathcal{O}_Y$ -module structure coming from $f^\#$, we have a $f^{-1}\mathcal{O}_Y$ -module structure on $\mathcal{F}$. More precisely, we consider $f^b : f^{-1}\mathcal{O}_Y \rightarrow \mathcal{O}_X$, and for any open set U and sections $a \in (f^{-1}\mathcal{O}_Y)(U)$, $s \in \mathcal{F}(U)$ we set $a * s := f^b(U)(a) \cdot s$. Consequently, we have a $(f^{-1}\mathcal{O}_Y)_{\text{pre}}$ -module structure on $\mathcal{F}$ as well; explicitly, this is given by :

$$\alpha * s := \kappa_{(f^{-1}\mathcal{O}_Y)_{\text{pre}}}(U)(\alpha) * s = \left(f^b(U) \left(\kappa_{(f^{-1}\mathcal{O}_Y)_{\text{pre}}}(U) \right) (\alpha) \right) \cdot s = \widetilde{\Phi(f^\#)}(U)(\alpha) \cdot s$$

where U is any opens set in X and arbitrary sections $\alpha \in (f^{-1}\mathcal{O}_Y)_{\text{pre}}(U)$, $s \in \mathcal{F}(U)$.

As in the case of sheaves of abelian groups, we first define a map

$$\Phi : \text{Hom}_{\mathcal{O}_Y}(\mathcal{G}, f_*\mathcal{F}) \rightarrow \text{Hom}_{f^{-1}(\mathcal{O}_Y)}(f^{-1}\mathcal{G}, \mathcal{F})$$

in the following way. Let $\sigma : \mathcal{G} \rightarrow f_*\mathcal{F}$ be a morphism of $\mathcal{O}_Y$ -modules. Then, given an open subset $U \subset X$, for every open set $V \subset Y$ with $f(U) \subset V$ (or, $U \subset f^{-1}(V)$), we consider the composite

$$\mathcal{G}(V) \xrightarrow{\sigma(V)} \mathcal{F}(f^{-1}(V)) \xrightarrow{\text{res}} \mathcal{F}(U).$$

Thus, $\mathcal{F}(U)$ becomes a co-cone below the diagram $\{\mathcal{G}(V) \mid V \supset f(U)\}$ and hence we get a map

$$\widetilde{\Phi(\sigma)}(U) : \varinjlim_{V \supset f(U)} \mathcal{G}(V) \rightarrow \mathcal{F}(U), \quad U \subset X \text{ open.}$$

We first see that $\widetilde{\Phi(\sigma)}$ is restriction compatible. This can be easily shown once the above map is explicitly written : $\widetilde{\Phi(\sigma)}(U)([s, V]) = \sigma(V)(s)|_U$. So we have a morphism

$$\widetilde{\Phi(\sigma)} : (f^{-1}\mathcal{G})_{\text{pre}} \rightarrow \mathcal{F}.$$

We show that this is $f^{-1}\mathcal{O}_Y$ -linear. Owing to Lemma 10.21, it suffices to show that $\widetilde{\Phi(\sigma)}$ is $(f^{-1}\mathcal{O}_Y)_{\text{pre}}$ -linear.

So fix any open set U of X and sections $[b, V] \in (f^{-1}\mathcal{O}_Y)_{\text{pre}}(U)$, $[s, W] \in (f^{-1}\mathcal{G})_{\text{pre}}$. We have to show that

$$\widetilde{\Phi(\sigma)}(U)([b, V] \cdot [s, W]) = [b, V] \star \widetilde{\Phi(\sigma)}(U)([s, W]).$$

Now, we know that $f(U) \subset V \cap W$. Then, we know that $[b, V] \cdot [s, W] = [b|_{V \cap W} \cdot s|_{V \cap W}, V \cap W]$. Thus, for the left and side, we have

$$\begin{aligned}
\widetilde{\Phi(\sigma)}(U)([b, V] \cdot [s, W]) &= \sigma(V \cap W)(b|_{V \cap W} \cdot s|_{V \cap W})|_U \\
&= (f^\#(V \cap W)(b|_{V \cap W}) \cdot \sigma(V \cap W)(s|_{V \cap W}))|_U \\
&= (f^\#(V \cap W)(b|_{V \cap W}))|_U \cdot (\sigma(V \cap W)(s|_{V \cap W}))|_U \\
&= (f^\#(V)(b)|_{f^{-1}(V \cap W)})|_U \cdot (\sigma(W)(s)|_{f^{-1}(V \cap W)})|_U \\
&= f^\#(V)(b)|_U \cdot \sigma(W)(s)|_U
\end{aligned}$$

where we have used the $\mathcal{O}_Y(V \cap W)$ -linearity of $\sigma(V \cap W)$ and obviously the restriction compatibility of $f^\#$ and σ . Now for the right hand side, we get

$$\begin{aligned}
[b, V] \star \widetilde{\Phi(\sigma)}(U)([s, W]) &= [b, V] \star (\sigma(W)(s)|_U) \\
&= \widetilde{\Phi(f^\#)}(U)([b, V]) \cdot (\sigma(W)(s)|_U) \\
&= f^\#(V)(b)|_U \cdot \sigma(W)(s)|_U.
\end{aligned}$$

Next we define the map

$$\Psi : \text{Hom}_{f^{-1}(\mathcal{O}_Y)}(f^{-1}\mathcal{G}, \mathcal{F}) \rightarrow \text{Hom}_{\mathcal{O}_Y}(\mathcal{G}, f_*\mathcal{F})$$

as in the case of the adjunction for sheaves of abelian groups by the following way :

$$\Psi(\tau)(V)(s) = (\tau(f^{-1}(V)) \circ \kappa_{f^{-1}(\mathcal{G})_{\text{pre}}}(f^{-1}(V)))([s, V])$$

for $\tau : f^{-1}\mathcal{G} \rightarrow \mathcal{F}$ a $f^{-1}\mathcal{O}_Y$ -linear morphism, open set V of Y and section $s \in \mathcal{G}(V)$. We have to show that $\Psi(\tau)(V)$ is $\mathcal{O}_Y(V)$ -linear. So take a section $b \in \mathcal{O}_Y(V)$, and we must show that

$$\Psi(\tau)(V)(b \cdot s) = b * (\Psi(\tau)(V)(s)) = f^\#(V)(b) \cdot (\Psi(\tau)(V)(s)).$$

Now the left hand side, by definition of Ψ , evaluates to

$$\Psi(\tau)(V)(b \cdot s) = \tau(f^{-1}(V))(\kappa_{f^{-1}(\mathcal{G})_{\text{pre}}}(f^{-1}(V))([b \cdot s, V])).$$

We write $[b \cdot s, V] = [b, V] \cdot [s, V]$ indicating the action of $[b, V] \in f^{-1}(\mathcal{O}_Y)_{\text{pre}}(f^{-1}(V))$ on $[s, V] \in f^{-1}(\mathcal{G})_{\text{pre}}(f^{-1}(V))$. Now $f^{-1}\mathcal{G}$ is a $f^{-1}\mathcal{O}_Y$ -module and so a $f^{-1}(\mathcal{O}_Y)_{\text{pre}}$ -module; explicitly, for the open set $f^{-1}(V)$ of X the action of $[b, V]$ on a section $\alpha \in f^{-1}\mathcal{G}$ is given by

$$[b, V] \bullet \alpha = \kappa_{f^{-1}(\mathcal{O}_Y)_{\text{pre}}}(f^{-1}(V))([b, V]) \cdot \alpha.$$

Also, the morphism $\kappa_{f^{-1}(\mathcal{G})_{\text{pre}}} : f^{-1}(\mathcal{G})_{\text{pre}} \rightarrow f^{-1}(\mathcal{G})$ is $f^{-1}(\mathcal{O}_Y)_{\text{pre}}$ -linear. Therefore, we have

$$\begin{aligned}
\kappa_{f^{-1}(\mathcal{G})_{\text{pre}}}(f^{-1}(V))([b \cdot s, V]) &= \kappa_{f^{-1}(\mathcal{G})_{\text{pre}}}(f^{-1}(V))([b, V] \cdot [s, V]) \\
&= \kappa_{f^{-1}(\mathcal{O}_Y)_{\text{pre}}}(f^{-1}(V))([b, V]) \cdot \kappa_{f^{-1}(\mathcal{G})_{\text{pre}}}(f^{-1}(V))([s, V]).
\end{aligned}$$

Consequently, using $f^{-1}\mathcal{O}_Y$ -linearity of τ , we get

$$\begin{aligned}
\Psi(\tau)(V)(b \cdot s) &= \tau(f^{-1}(V))(\kappa_{f^{-1}(\mathcal{G})_{\text{pre}}}(f^{-1}(V))([b \cdot s, V])) \\
&= \tau(f^{-1}(V))(\kappa_{f^{-1}(\mathcal{O}_Y)_{\text{pre}}}(f^{-1}(V))([b, V]) \cdot \kappa_{f^{-1}(\mathcal{G})_{\text{pre}}}(f^{-1}(V))([s, V])) \\
&= \kappa_{f^{-1}(\mathcal{O}_Y)_{\text{pre}}}(f^{-1}(V))([b, V]) \cdot \tau(f^{-1}(V))(\kappa_{f^{-1}(\mathcal{G})_{\text{pre}}}(f^{-1}(V))([s, V])) \\
&= \kappa_{f^{-1}(\mathcal{O}_Y)_{\text{pre}}}(f^{-1}(V))([b, V]) \cdot \Psi(\tau)(V)(s)
\end{aligned}$$

Now note that $\Psi(\tau)(V)(s) \in (f_*\mathcal{F})(V) = \mathcal{F}(f^{-1}(V))$ on which $\kappa_{f^{-1}(\mathcal{O}_Y)_{\text{pre}}}(f^{-1}(V))([b, V]) \in (f^{-1}\mathcal{O}_Y)(f^{-1}(V))$ is acting. This action was defined via $f^b : f^{-1}\mathcal{O}_Y \rightarrow \mathcal{O}_X$, that is to say :

$$\kappa_{f^{-1}(\mathcal{O}_Y)_{\text{pre}}}(f^{-1}(V))([b, V]) \cdot \Psi(\tau)(V)(s) = f^b(f^{-1}(V))(\kappa_{f^{-1}(\mathcal{O}_Y)_{\text{pre}}}(f^{-1}(V))([b, V])) \cdot \Psi(\tau)(V)(s).$$

But we also know that

$$\begin{aligned} f^b(f^{-1}(V))(\kappa_{f^{-1}(\mathcal{O}_Y)_{\text{pre}}}(f^{-1}(V))([b, V])) &= \widetilde{\Phi(f^\#)}(f^{-1}(V))([b, V]) \\ &= f^\#(V)(b) \mid_{f^{-1}(V)} \\ &= f^\#(V)(b) \end{aligned}$$

since $f^\#(V)(b)$ is already a section on $f^{-1}(V)$. Finally, putting all these together we obtain

$$\begin{aligned} \Psi(\tau)(V)(b \cdot s) &= \kappa_{f^{-1}(\mathcal{O}_Y)_{\text{pre}}}(f^{-1}(V))([b, V]) \cdot \Psi(\tau)(V)(s) \\ &= f^b(f^{-1}(V))(\kappa_{f^{-1}(\mathcal{O}_Y)_{\text{pre}}}(f^{-1}(V))([b, V])) \cdot \Psi(\tau)(V)(s) \\ &= f^\#(V)(b) \cdot \Psi(\tau)(V)(s) \end{aligned}$$

as asserted in the beginning.

Finally, note that functoriality of $\Phi_{\mathcal{F}}$ or $\Psi_{\mathcal{F}}$ in $\mathcal{F}$ is proved in the exact same way as in the case of sheaves of abelian groups, since the module structure does not come in to verifications involved in the previous proof. Similarly, Φ and Ψ are inverses of each other, which can be proved by the exact same arguments. $\square$

We will now construct a canonical isomorphism

$$\Theta : \text{Hom}_{f^{-1}(\mathcal{O}_Y)}(f^{-1}\mathcal{G}, \mathcal{F}) \xrightarrow{\sim} \text{Hom}_{\mathcal{O}_X}(f^{-1}(\mathcal{G}) \otimes_{f^{-1}(\mathcal{O}_Y)} \mathcal{O}_X, \mathcal{F}).$$

Recall that we defined $f^*\mathcal{G} := f^{-1}(\mathcal{G}) \otimes_{f^{-1}(\mathcal{O}_Y)} \mathcal{O}_X$. We denote the presheaf $U \rightsquigarrow f^{-1}(\mathcal{G})(U) \otimes_{f^{-1}(\mathcal{O}_Y)(U)} \mathcal{O}_X(U)$ by $(f^*\mathcal{G})_{\text{pre}}$. Now take a $f^{-1}(\mathcal{O}_Y)$ -linear morphism $\tau : f^{-1}\mathcal{G} \rightarrow \mathcal{F}$. We construct a $\mathcal{O}_X$ -linear map

$$\widetilde{\Theta(\tau)}(U) : f^{-1}(\mathcal{G})(U) \otimes_{f^{-1}(\mathcal{O}_Y)(U)} \mathcal{O}_X(U) \rightarrow \mathcal{F}(U).$$

First consider the bilinear map

$$B_\tau(U) : (f^{-1}\mathcal{G})(U) \times \mathcal{O}_X(U) \rightarrow \mathcal{F}(U)$$

defined by

$$(s, a) \mapsto a \cdot \tau(U)(s), \quad s \in (f^{-1}\mathcal{G})(U), a \in \mathcal{O}_X(U).$$

It suffices to check that this is $f^{-1}(\mathcal{O}_Y)(U)$ -balanced, whence it will induce the required $\widetilde{\Theta(\tau)}(U)$. Thus, we have to show that, for $r \in f^{-1}(\mathcal{O}_Y)(U)$ and $s \in (f^{-1}\mathcal{G})(U), a \in \mathcal{O}_X(U)$, we have

$$B_\tau(r \cdot s, a) = B_\tau(s, f^b(U)(r) \times a)$$

where we note that $f^{-1}(\mathcal{O}_Y)$ acts on $\mathcal{O}_X$ via f^b . Thus, we get

$$\begin{aligned} B_\tau(r \cdot s, a) &= a \cdot \tau(U)(r \cdot s) \\ &= a \cdot (f^b(U)(r) \cdot \tau(U)(s)) = (a \times f^b(U)(r)) \cdot \tau(U)(s) \\ &= (f^b(U)(r) \times a) \cdot \tau(U)(s) \\ &= B_\tau(s, f^b(U)(r) \times a). \end{aligned}$$

Therefore, we obtain

$$\widetilde{\Theta(\tau)}(U) : f^{-1}(\mathcal{G})(U) \otimes_{f^{-1}(\mathcal{O}_Y)(U)} \mathcal{O}_X(U) \rightarrow \mathcal{F}(U), \quad s \otimes a \mapsto a \cdot \tau(U)(s).$$

It is easy to see that $\widetilde{\Theta(\tau)}$ is restriction compatible. Also, it is easy to show that $\widetilde{\Theta(\tau)}(U)$ is $\mathcal{O}_X(U)$ -linear for every open set U since $\mathcal{O}_X(U)$ acts on $f^{-1}(\mathcal{G})(U) \otimes_{f^{-1}(\mathcal{O}_Y)(U)} \mathcal{O}_X(U)$ via $b \cdot (s \otimes a) := s \otimes (ba)$.

Consequently, by Lemma 10.7, we get a unique $\mathcal{O}_X$ -linear morphism

$$\Theta(\tau) : f^{-1}(\mathcal{G}) \otimes_{f^{-1}(\mathcal{O}_Y)} \mathcal{O}_X \rightarrow \mathcal{F}$$

such that

$$\Theta(\tau) \circ \kappa_{(f^*\mathcal{G})_{\text{pre}}} = \widetilde{\Theta(\tau)}.$$

This completes our construction of the map

$$\Theta : \text{Hom}_{f^{-1}(\mathcal{O}_Y)}(f^{-1}\mathcal{G}, \mathcal{F}) \rightarrow \text{Hom}_{\mathcal{O}_X}(f^{-1}(\mathcal{G}) \otimes_{f^{-1}(\mathcal{O}_Y)} \mathcal{O}_X, \mathcal{F}).$$

We will next construct an inverse of the above map, denoted by

$$\Delta : \text{Hom}_{\mathcal{O}_X}(f^{-1}(\mathcal{G}) \otimes_{f^{-1}(\mathcal{O}_Y)} \mathcal{O}_X, \mathcal{F}) \rightarrow \text{Hom}_{f^{-1}(\mathcal{O}_Y)}(f^{-1}\mathcal{G}, \mathcal{F}).$$

First consider the canonical morphism

$$j : f^{-1}\mathcal{G} \rightarrow f^*\mathcal{G}$$

which for each open set U is given by

$$j(U) : (f^{-1}\mathcal{G})(U) \rightarrow (f^*\mathcal{G})(U), \quad s \mapsto \kappa_{(f^*\mathcal{G})_{\text{pre}}}(U)(s \otimes 1_{\mathcal{O}_X(U)}).$$

Then, note that $j(U)$ is $(f^{-1}\mathcal{O}_Y)(U)$ -linear. To see this, take $r \in (f^{-1}\mathcal{O}_Y)(U)$ and $s \in (f^{-1}\mathcal{G})(U)$, then

$$\begin{aligned} j(U)(r \cdot s) &= \kappa(U)((r \cdot s) \otimes 1) \\ &= \kappa(U)(s \otimes f^b(U)(r) \times 1) \\ &= f^b(U)(r) \cdot_1 \kappa(U)(s \otimes 1) \\ &= r \cdot_2 \kappa(U)(s \otimes 1) \end{aligned}$$

as observed in Remark 10.26. Thus $j \in \text{Hom}_{f^{-1}\mathcal{O}_Y}(f^{-1}\mathcal{G}, f^*\mathcal{G})$.

Now, to define Δ , take a morphism $\eta \in \text{Hom}_{\mathcal{O}_X}(f^*\mathcal{G}, \mathcal{F})$. Since η is $\mathcal{O}_X$ -linear, by restricting along the morphism f^b , it is also $f^{-1}\mathcal{O}_Y$ -linear. Define

$$\Delta(\eta) := \eta \circ j : f^{-1}\mathcal{G} \rightarrow f^*\mathcal{G}.$$

This is clearly $f^{-1}\mathcal{O}_Y$ -linear. We show that Δ is functorial in $\mathcal{F}$. More precisely, if $\alpha : \mathcal{F} \rightarrow \mathcal{F}'$ is a morphism of $\mathcal{O}_X$ -modules, then for every $\eta \in \text{Hom}_{\mathcal{O}_X}(f^*\mathcal{G}, \mathcal{F})$,

$$\Delta_{\mathcal{F}'}(\alpha \circ \eta) = \alpha \circ \Delta_{\mathcal{F}}(\eta).$$

But by definition of Δ the left side is $(\alpha \circ \eta) \circ j$, whereas the right side is $\alpha \circ (\eta \circ j)$. This completes the proof of functoriality. So it remains to show that Δ and Θ are inverses of each other (whence Θ will also be functorial in $\mathcal{F}$).

We first show that $\Theta \circ \Delta = \text{id}$. Take $\eta \in \text{Hom}_{\mathcal{O}_X}(f^*\mathcal{G}, \mathcal{F})$, and we must show that $\Theta(\Delta(\eta)) = \eta$. Set $\tau := \Delta(\eta)$, so that we have to show $\Theta(\tau) = \eta$. By definition, $\Theta(\tau)$ is the unique morphism satisfying $\Theta(\tau) \circ \kappa_{(f^*\mathcal{G})_{\text{pre}}} = \widetilde{\Theta(\tau)}$. So it suffices to show

$$\eta \circ \kappa_{(f^*\mathcal{G})_{\text{pre}}} = \widetilde{\Theta(\tau)}.$$

So fix any open set U of X and we must show that $\eta(U) \circ \kappa(U) = \widetilde{\Theta(\tau)}(U)$. Take an elementary tensor $s \otimes a$ and evaluate the left side to obtain : $\eta(U)(\kappa(U)(s \otimes a))$. Evaluating the right hand side, we get

$$\widetilde{\Theta(\tau)}(U)(s \otimes a) = a \cdot \tau(U)(s) = a \cdot (\eta(U)(\kappa(U)(s \otimes 1))).$$

Now using $\mathcal{O}_X$ -linearity of η and κ , we get

$$\widetilde{\Theta(\tau)}(U)(s \otimes a) = \eta(U)(\kappa(U)(s \otimes a))$$

as required.

Next, we show that $\Delta \circ \Theta = \text{id}$. Fix a $\tau \in \text{Hom}_{f^{-1}\mathcal{O}_Y}(f^{-1}\mathcal{G}, \mathcal{F})$, and we must show that

$$\Delta(\Theta(\tau))(U)(s) = \tau(U)(s)$$

for every open set U and every section $s \in (f^{-1}\mathcal{G})(U)$. Starting from the left side we compute :

$$\begin{aligned} \Delta(\Theta(\tau))(U)(s) &= \Theta(\tau)(U)(\kappa(U)(s \otimes 1)) \\ &= \widetilde{\Theta(\tau)}(U)(s \otimes 1) \\ &= 1 \cdot \tau(U)(s) \\ &= \tau(U)(s). \end{aligned}$$

Thus we have shown that Θ and Δ are inverses of each other and functorial in $\mathcal{F}$.

Summarizing, we have :

Theorem 10.28. – Let $(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a morphism of ringed spaces and $\mathcal{G}$ be a sheaf of $\mathcal{O}_Y$ -modules on Y . Then there is an isomorphism

$$\mathrm{Hom}_{\mathcal{O}_X}(f^*\mathcal{G}, \mathcal{F}) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{G}, f_*\mathcal{F})$$

which is functorial in $\mathcal{F}$.

In the next section we will look at a concrete ringed space of the form $(\mathrm{Spec} A, \mathcal{O}_{\mathrm{Spec} A})$, where A denotes any commutative unital ring, and $\mathrm{Spec} A$ its set of prime ideals. We will first put a topology on $\mathrm{Spec} A$ and then :

- (1) Construct a structure sheaf of rings $\mathcal{O}_{\mathrm{Spec} A}$.
- (2) Given an A -module M , construct a sheaf $\tilde{M}$ of $\mathcal{O}_{\mathrm{Spec} A}$ -modules.

11. SCHEME THEORY 3 : AFFINE SCHEMES

Let A be any commutative unital ring. We denote by $\mathrm{Spec} A$ the set of prime ideals in A . The set of nilpotent elements of A is denoted by $\mathrm{rad}(A)$. One can verify that $\mathrm{rad}(A)$ is an ideal of A .

Proposition 11.1. – For any ring A we have

$$\mathrm{rad}(A) = \bigcap_{\mathfrak{p} \in \mathrm{Spec} A} \mathfrak{p}.$$

Proof. – Let $a \in \mathrm{rad}(A)$ and $a^s = 0$ for some $s > 0$. Then $a^s \in \mathfrak{p}$ for every $\mathfrak{p} \in \mathrm{Spec} A$. Thus $a \in \mathfrak{p}$ for every $\mathfrak{p} \in \mathrm{Spec} A$. Conversely, suppose $a \in A - \mathrm{rad}(A)$. We show that there exists a prime ideal $\mathfrak{p}$ such that $a \notin \mathfrak{p}$. Consider the multiplicative system

$$S := \{a^n \mid n \in \mathbf{N}\} \subset A.$$

Since a is not nilpotent, $0 \notin S$ and hence the localization $S^{-1}A \neq 0$. Let $\mathfrak{q} \subset S^{-1}A$ be a maximal ideal and let $\mathfrak{p} := \tau^{-1}(\mathfrak{q}) \subset A$ be its preimage under the canonical map $\tau : A \rightarrow S^{-1}A$. Then $a \notin \mathfrak{p}$ since otherwise $\tau(a)$ which is a unit in $S^{-1}A$ is in $\mathfrak{q}$, which is not possible. Thus we have shown the existence of a prime ideal $\mathfrak{p}$ not containing a . $\square$

Given an ideal $\mathfrak{a}$ of A we set $\mathrm{rad}(\mathfrak{a}) := \{a \in A \mid a^n \in \mathfrak{a} \text{ for some } n \in \mathbf{N}\}$. This is the *nilradical* of $\mathfrak{a}$. If $\mathrm{rad}(\mathfrak{a}) = \mathfrak{a}$, then $\mathfrak{a}$ is called *reduced*.

Corollary 11.2. – $\mathrm{rad}(\mathfrak{a}) = \bigcap_{\mathfrak{a} \subset \mathfrak{p} \in \mathrm{Spec} A} \mathfrak{p}$.

Proof. – Note that $\mathrm{rad}(\mathfrak{a}) = \pi^{-1}(\mathrm{rad}(A/\mathfrak{a}))$ since

$$a \in \mathrm{rad}(\mathfrak{a}) \iff a^n \in \mathfrak{a} \text{ for some } n \iff \pi(a)^n = 0 \text{ in } A/\mathfrak{a} \iff \pi(a) \in \mathrm{rad}(A/\mathfrak{a}).$$

The assertion follows from the fact that prime ideals of $A/\mathfrak{a}$ are precisely of the form $\mathfrak{p}/\mathfrak{a}$ where $\mathfrak{p}$ is a prime ideal in A containing $\mathfrak{a}$. $\square$

Let A be a ring. We define the *Zariski topology* on $\mathrm{Spec} A$ by declaring the sets of the form $V(E) := \{\mathfrak{p} \in \mathrm{Spec} A \mid E \subset \mathfrak{p}\}$ for subsets E of A as closed sets of this topology. Clearly, $V(E)$ only depends of the ideal generated by E i.e. $V(E) = V(\langle E \rangle)$. So we consider closed sets of the Zariski topology to be sets of the form $V(\mathfrak{a})$ where $\mathfrak{a} \subset A$ are ideals. The following facts are easy to check.

Fact 11.3. – Consider subsets E, E' , and a family of subsets $(E_\lambda)_{\lambda \in \Lambda}$ of a ring A . Then :

- (1) $V(0) = \mathrm{Spec} A$, $V(1) = \emptyset$.
- (2) $E \subset E' \iff V(E') \subset V(E)$.
- (3) $V(\bigcup_{\lambda \in \Lambda} E_\lambda) = \bigcap_{\lambda \in \Lambda} V(E_\lambda)$.
- (4) $V(EE') = V(E) \cup V(E')$, where $EE' = \{ff' \mid f \in E, f' \in E'\}$.
- (5) $V(E) = V(\mathrm{rad}(E))$, where $\mathrm{rad}(E) := \mathrm{rad}(\langle E \rangle)$.

For an ideal $\mathfrak{a} \subset A$ we set $D(\mathfrak{a}) := \mathrm{Spec} A \setminus V(\mathfrak{a})$. Thus sets of the form $D(\mathfrak{a})$ are open sets of the Zariski topology. Then, we have

$$D(\mathfrak{a}) = \bigcup_{f \in \mathfrak{a}} D(f), \quad \text{where} \quad D(f) := \{\mathfrak{p} \in \mathrm{Spec} A \mid f \notin \mathfrak{p}\}$$

since $\mathfrak{p} \in D(\mathfrak{a}) \iff \mathfrak{a} \not\subset \mathfrak{p} \iff \exists f \in \mathfrak{a} \setminus \mathfrak{p}$. So $\{D(f) \mid f \in A\}$ forms a base of the Zariski topology on $\text{Spec } A$.

Fact 11.4. – $D(f) = \emptyset \iff f$ is nilpotent.

Proof. – Indeed, $D(f) = \emptyset \iff f \in \bigcap_{\mathfrak{p}} \mathfrak{p} = \text{rad}(A)$, by Proposition 11.1 and $\text{rad}(A)$ is precisely the set of nilpotent elements. $\square$

Proposition 11.5. – A ring homomorphism $\varphi : A \rightarrow A'$ induces a map $\text{Spec } \varphi : \text{Spec } A' \rightarrow \text{Spec } A$ given by $\mathfrak{p}' \mapsto \varphi^{-1}(\mathfrak{p}')$. Then $\text{Spec } \varphi$ is continuous.

Proof. – Let $\mathfrak{a} \subset R$ be an ideal. Then

$$\mathfrak{p}' \in (\text{Spec } \varphi)^{-1}(V(\mathfrak{a})) \iff \varphi^{-1}(\mathfrak{p}') \in V(\mathfrak{a}) \iff \mathfrak{a} \subset \varphi^{-1}(\mathfrak{p}') \iff \varphi(\mathfrak{a}) \subset \mathfrak{p}' \iff \mathfrak{p}' \in V(\varphi(\mathfrak{a}))$$

i.e. $\text{Spec } \varphi$ pulls back closed sets to closed sets. $\square$

Proposition 11.6. – For a family of elements $(f_\lambda)_{\lambda \in \Lambda}$ in A , we have

$$\text{Spec } A = \bigcup_{\lambda \in \Lambda} D(f_\lambda) \iff A = \langle f_\lambda \mid \lambda \in \Lambda \rangle.$$

Proof. – $\text{Spec } A = \bigcup_{\lambda} D(f_\lambda) \iff$ for every prime $\mathfrak{p}$ there exists some $f_\lambda \notin \mathfrak{p} \iff A = \langle f_\lambda \mid \lambda \in \Lambda \rangle$. $\square$

Corollary 11.7. – $\text{Spec } A$ is quasi-compact w.r.t. the Zariski topology.

Proof. – Suppose $\text{Spec } A = \bigcup_{\lambda} D(f_\lambda)$. Then $A = \langle \{f_\lambda\} \rangle$ and so $1 = \sum_{i=1}^N a_i f_{\lambda_i}$ for some $a_i \in A$. Thus $A = \langle f_{\lambda_1}, \dots, f_{\lambda_N} \rangle$ and so $\text{Spec } R = \bigcup_{i=1}^N D(f_{\lambda_i})$. $\square$

Lemma 11.8. – For $f, g \in A$ we have :

- (1) $D(f) \cap D(g) = D(fg)$.
- (2) $D(f) \supset D(g) \iff g \in \text{rad}(\langle f \rangle)$.

Proof. – (1) Observe that $\mathfrak{p} \in D(f) \cap D(g) \iff f, g \notin \mathfrak{p} \iff fg \notin \mathfrak{p} \iff \mathfrak{p} \in D(fg)$.

(2) Note that $g \notin \text{rad}(\langle f \rangle) \iff \bigcap_{\mathfrak{p} \ni f} \mathfrak{p} \not\subset \langle g \rangle \iff \exists \mathfrak{p}$ such that $f \in \mathfrak{p}$ and $g \notin \mathfrak{p}$ i.e. $\mathfrak{p} \in D(g) \setminus D(f)$. $\square$

Now we set $X := \text{Spec } A$ and suppose $D(f) \supset D(g)$ be two basic open subsets of X . Now since $g^n = fh$ for some $h \in A$, the canonical map $A \rightarrow A_g$ sends g to a unit $\frac{g}{1}$, and hence $\frac{f}{1}$ which is the image of f is also a unit. Hence we have a factoring, denoted as $\text{res}_{D(f), D(g)} : A_f \rightarrow A_g$, by the universal mapping property of localization.

Fact 11.9. – For $D(f) \supset D(g) \supset D(h)$ we have $\text{res}_{D(g), D(h)} \circ \text{res}_{D(f), D(g)} = \text{res}_{D(f), D(h)}$.

Proof. – Consider the following diagram

$$\begin{array}{ccc}
 & A_h & \\
 \text{res}_{D(f), D(h)} \swarrow & \uparrow i_h & \searrow \text{res}_{D(g), D(h)} \\
 & A & \\
 i_f \swarrow & & \searrow i_g \\
 A_f & \xrightarrow{\text{res}_{D(f), D(g)}} & A_g
 \end{array}$$

in which we have to show that the outer triangle commutes. Note that the three inner triangle commute. However, we also have

$$\text{res}_{D(g), D(h)} \circ \text{res}_{D(f), D(g)} \circ i_f = \text{res}_{D(g), D(h)} \circ i_g = i_h.$$

But by the universal mapping property of A_f , the map $\text{res}_{D(f), D(h)}$ is the unique map such that $\text{res}_{D(f), D(h)} \circ i_f = i_h$. Thus $\text{res}_{D(f), D(h)} = \text{res}_{D(g), D(h)} \circ \text{res}_{D(f), D(g)}$. $\square$

Remark 11.10. – We must emphasize that we interpret the elements of A as functions on $X = \operatorname{Spec} A$. To be more precise, for $f \in A$ we define $F : X \rightarrow \bigsqcup_{\mathfrak{p} \in \operatorname{Spec} A} A/\mathfrak{p}$ by $F(\mathfrak{p}) := f + \mathfrak{p}$. The elements of A_f are then interpreted as functions on $X_f = \operatorname{Spec} A_f \simeq D(f)$ (see Corollary 11.17), and hence on $D(f)$. Thus the maps $\tau : A \rightarrow A_f$ is supposed to be interpreted as a restriction of functions on $D(f)$, and similarly the maps $A_f \rightarrow A_g$ are to be thought of as restriction of functions on $D(f)$ to $D(g) \subset D(f)$.

We fix $X = \operatorname{Spec} A$ for the remaining part of this section. Now we will construct the structure sheaf of rings $\mathcal{O}_X$ in two ways.

11.1. Construction of $\mathcal{O}_{\operatorname{Spec} A}$: first approach. We define $\mathcal{O}_X$ as follows : for any open set U of X define

$$\mathcal{O}_X(U) := \left\{ (s_{\mathfrak{p}})_{\mathfrak{p} \in U} \in \prod_{\mathfrak{p} \in U} A_{\mathfrak{p}} \mid \begin{array}{l} \exists \text{ a cover } (U_{\lambda})_{\lambda \in \Lambda} \text{ of } U \text{ and elements } a_{\lambda}, f_{\lambda} \in A \\ \text{such that } \forall \mathfrak{q} \in U_{\lambda} \text{ we have } f_{\lambda} \notin \mathfrak{q} \text{ and } s_{\mathfrak{q}} = a_{\lambda}/f_{\lambda} \in A_{\mathfrak{q}} \end{array} \right\}.$$

Fact 11.11. – $\mathcal{O}_X(U)$ defined above is a ring for every open set U .

Proof. – We only check that $\mathcal{O}_X(U)$ is closed under addition and multiplication (defined pointwise). So let $(s_{\mathfrak{p}})_{\mathfrak{p} \in U}, (t_{\mathfrak{p}})_{\mathfrak{p} \in U} \in \mathcal{O}_X(U)$, with associated covers $(U_{\lambda})_{\lambda \in \Lambda}$ and $(V_{\mu})_{\mu \in M}$, and elements $a_{\lambda}, f_{\lambda} \in A, b_{\mu}, g_{\mu} \in A$ respectively satisfying the defining conditions. We consider the refined cover $(W_{\lambda, \mu} := U_{\lambda} \cap V_{\mu})_{(\lambda, \mu) \in \Lambda \times M}$ of U . Then, for any $\mathfrak{q} \in W_{\lambda, \mu}$ we know $f_{\lambda}, g_{\mu} \notin \mathfrak{q}$ and $s_{\mathfrak{q}} = \frac{a_{\lambda}}{f_{\lambda}}, t_{\mathfrak{q}} = \frac{b_{\mu}}{g_{\mu}}$. Thus, we get

$$s_{\mathfrak{q}} + t_{\mathfrak{q}} = \frac{a_{\lambda}g_{\mu} + f_{\lambda}b_{\mu}}{f_{\lambda}g_{\mu}} \in A_{\mathfrak{q}}.$$

Thus on $W_{\lambda, \mu}$ the sum of the sections is represented by the pair $a_{\lambda}g_{\mu} + f_{\lambda}b_{\mu}, f_{\lambda}g_{\mu}$. The same refinement works for multiplication, with the pair corresponding to $W_{\lambda, \mu}$ being $a_{\lambda}b_{\mu}, f_{\lambda}g_{\mu}$. $\square$

For open sets $V \subset U$ we define the restriction map $\mathcal{O}_X(U) \rightarrow \mathcal{O}_X(V)$ by $(s_{\mathfrak{p}})_{\mathfrak{p} \in U} \mapsto (s_{\mathfrak{p}})_{\mathfrak{p} \in V}$. This makes $\mathcal{O}_X$ a presheaf.

Fact 11.12. – $\mathcal{O}_X$ is a sheaf.

Proof. – Let U be open and $(U_{\lambda})_{\lambda \in \Lambda}$ be an open cover of U , and $s, t \in \mathcal{O}_X(U)$ with $s|_{U_{\lambda}} = t|_{U_{\lambda}}$ for all λ . Write $s = (s_{\mathfrak{p}})_{\mathfrak{p} \in U}$ and $t = (t_{\mathfrak{p}})_{\mathfrak{p} \in U}$. The given condition means that for every λ we have $s_{\mathfrak{p}} = t_{\mathfrak{p}} \quad \forall \mathfrak{p} \in U_{\lambda}$. Since the U_{λ} cover U , we have $s_{\mathfrak{p}} = t_{\mathfrak{p}}$ for all $\mathfrak{p} \in U$.

Now suppose $s^{\lambda} \in \mathcal{O}_X(U_{\lambda})$ be such that $s^{\lambda}|_{U_{\lambda} \cap U_{\mu}} = s^{\mu}|_{U_{\lambda} \cap U_{\mu}}$ for all λ, μ . Write $s^{\lambda} = (s_{\mathfrak{p}}^{\lambda})_{\mathfrak{p} \in U_{\lambda}}$ for each λ . The given condition then means

$$\forall \mathfrak{p} \in U_{\lambda} \cap U_{\mu} : s_{\mathfrak{p}}^{\lambda} = s_{\mathfrak{p}}^{\mu}.$$

We define $s = (s_{\mathfrak{p}})_{\mathfrak{p} \in U} \in \prod_{\mathfrak{p} \in U} A_{\mathfrak{p}}$ as follows : take $\mathfrak{p} \in U$, then if $\mathfrak{p} \in U_{\lambda}$, set $s_{\mathfrak{p}} = s_{\mathfrak{p}}^{\lambda}$. The above condition ensures well-definedness. Now, each $s^{\lambda} \in \mathcal{O}_X(U_{\lambda})$ yields a cover $(W_{\lambda, i})_{i \in I(\lambda)}$ of U_{λ} and elements $a_{\lambda, i}, f_{\lambda, i} \in A$ such that for every $\mathfrak{q} \in W_{\lambda, i}$ we have $f_{\lambda, i} \notin \mathfrak{q}$ and $s_{\mathfrak{q}}^{\lambda} = s_{\mathfrak{q}} = \frac{a_{\lambda, i}}{f_{\lambda, i}}$. But the open sets $W_{\lambda, i}$ cover U and together with the pairs $a_{\lambda, i}, f_{\lambda, i}$ gives the defining condition for s to be in $\mathcal{O}_X(U)$. Thus we have shown $\mathcal{O}_X$ to be a sheaf. $\square$

Proposition 11.13. – Let A be a ring and $(X, \mathcal{O}_X) = (\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})$ be the ringed space constructed above.

- (1) For any $\mathfrak{p} \in X$, the stalk $\mathcal{O}_{X, \mathfrak{p}}$ of the sheaf $\mathcal{O}_X$ is isomorphic to $A_{\mathfrak{p}}$.
- (2) For any element $f \in A$, the ring $\mathcal{O}_X(D(f))$ is isomorphic to A_f .
- (3) In particular, $\mathcal{O}_X(X) \simeq A$.

Proof. – (1) Recall that

$$\mathcal{O}_{X, \mathfrak{p}} = \varinjlim_{\mathfrak{p} \in U} \mathcal{O}_X(U).$$

We write an element of $\mathcal{O}_{X, \mathfrak{p}}$ as a class $[s, U]$ where $s = (s_{\mathfrak{q}})_{\mathfrak{q} \in U} \in \mathcal{O}_X(U)$. Since $\mathfrak{p} \in U$, we define a homomorphism

$$\Phi : \mathcal{O}_{X, \mathfrak{p}} \rightarrow A_{\mathfrak{p}}, \quad [s, U] \mapsto s_{\mathfrak{p}}.$$

We check that this is independent of the choice of the class representative. So if $[s, U] = [t, V]$ then there exists an open set $W \ni \mathfrak{p}$ with $W \subset U \cap V$ such that $s|_W = t|_W$. In particular, $s_{\mathfrak{p}} = t_{\mathfrak{p}}$.

We will show that Φ is an isomorphism of rings. We first show that it is surjective. Take an element $\frac{a}{f} \in A_{\mathfrak{p}}$ with $f \notin \mathfrak{p}$. Now define a sequence

$$s = (s_{\mathfrak{q}})_{\mathfrak{q} \in D(f)} \in \prod_{\mathfrak{q} \in D(f)} A_{\mathfrak{q}}$$

by

$$s_{\mathfrak{q}} := \frac{a}{f} \in A_{\mathfrak{q}} \quad (\mathfrak{q} \in D(f) \iff f \notin \mathfrak{q}).$$

Then $s \in \mathcal{O}_X(D(f))$: take the one-element cover $\{D(f)\}$ of itself and the elements a, f satisfy the defining conditions. Obviously $[s, D(f)] \mapsto \frac{a}{f}$. Next, we show that Φ is injective. Assume $[s, U], [t, U] \in \mathcal{O}_{X, \mathfrak{p}}$ are such that $s_{\mathfrak{p}} = t_{\mathfrak{p}}$: note that we can always choose two arbitrary germs to be represented by sections of the same open set, since otherwise we can still restrict to the intersection of the open sets and then take germs and the germs would still be the same. We must show that $[s, U] = [t, U]$ i.e. there exists an open set $W \subset U$ containing $\mathfrak{p}$ such that $s|_W = t|_W$. Now since $s \in \mathcal{O}_X(U)$ and $\mathfrak{p} \in U$, there exists an open set $W_s \subset U$ containing $\mathfrak{p}$ and elements $a, f \in A$ satisfying: $\mathfrak{q} \in W_s \implies f \notin \mathfrak{q}$ and $s_{\mathfrak{q}} = \frac{a}{f}$. In particular, $s_{\mathfrak{p}} = \frac{a}{f}$. Similarly, since $t \in \mathcal{O}_X(U)$, there exists an open set $W_t \subset U$ containing $\mathfrak{p}$ and elements $b, g \in A$ such that $t_{\mathfrak{p}} = \frac{b}{g}$. However, since $s_{\mathfrak{p}} = t_{\mathfrak{p}}$ we get

$$\frac{a}{f} = \frac{b}{g}.$$

Thus there exists $h \notin \mathfrak{p}$ such that $h(ag - bf) = 0$. We set $W := W_s \cap W_t \cap D(h)$, so that $\mathfrak{p} \in W$. Clearly, for every $\mathfrak{q} \in W$ we have $h \notin \mathfrak{q}$ and $h(ag - bf) = 0$ i.e. $\frac{a}{f} = \frac{b}{g}$ in $A_{\mathfrak{q}}$. Thus we get

$$s_{\mathfrak{q}} = \frac{a}{f} = \frac{b}{g} = t_{\mathfrak{q}}, \quad \text{for all } \mathfrak{q} \in W \quad \text{i.e.} \quad s|_W = t|_W.$$

This completes the proof of (1).

(2) We will now prove the second assertion. Note that this already implies (3) by setting $f = 1$. We define a homomorphism

$$\Psi_f : A_f \rightarrow \mathcal{O}_X(D(f)), \quad \frac{a}{f^n} \mapsto s = (s_{\mathfrak{p}})_{\mathfrak{p} \in D(f)} = \left(\frac{a}{f^n} \in A_{\mathfrak{p}} \right)_{\mathfrak{p} \in D(f)}.$$

We show that Ψ_f is injective. Indeed, suppose $\Psi\left(\frac{a}{f^n}\right) = \Psi\left(\frac{b}{f^m}\right)$. Take any arbitrary $\mathfrak{p} \in D(f)$. Since $\frac{a}{f^n} = \frac{b}{f^m} \in A_{\mathfrak{p}}$, there exists $h \notin \mathfrak{p}$ such that $h(af^m - bf^n) = 0$. Let $\mathfrak{a}$ denote the annihilator of $(af^m - bf^n)$ in A . Then $h \in \mathfrak{a}$. We conclude that $\mathfrak{a} \not\subset \mathfrak{p}$ for all $\mathfrak{p} \in D(f)$ i.e. $V(\mathfrak{a}) \cap D(f) = \emptyset$. Therefore, $f \in \sqrt{\mathfrak{a}}$, since any prime ideal containing $\mathfrak{a}$ will not be in $D(f)$ and thus contains f . Hence some power of f , say f^ℓ , lies in $\mathfrak{a}$. So $f^\ell(af^m - bf^n) = 0$ i.e. $\frac{a}{f^n} = \frac{b}{f^m} \in A_f$.

Next, comes the surjectivity part which is slightly tricky. So let $s = (s_{\mathfrak{p}})_{\mathfrak{p} \in D(f)} \in \mathcal{O}_X(D(f))$, and suppose $D(f)$ can be covered by open sets V_i with elements $a_i, g_i \in A$ such that for all $\mathfrak{p} \in V_i$ we have $g_i \notin \mathfrak{p}$ and $s_{\mathfrak{p}} = \frac{a_i}{g_i} \in A_{\mathfrak{p}}$. Further, we can cover each V_i by basic open sets and the elements $a_i, g_i \in A$ would serve for all the basic open sets contained inside V_i . Thus we may assume $V_i = D(h_i)$ for every i . Since for every $\mathfrak{p} \in V_i$ we have $g_i \notin \mathfrak{p}$, it follows that $V_i = D(h_i) \subset D(g_i)$. Thus $V((h_i)) \supset V((g_i))$, which implies $\sqrt{(h_i)} \subset \sqrt{(g_i)}$, and so, in particular, $h_i^n \in (g_i)$ for some n . Hence there exists $c \in A$ such that $h_i^n = cg_i$ and so $\frac{a_i}{g_i} = \frac{ca_i}{h_i^n} \in A_{\mathfrak{p}}$ for all $\mathfrak{p} \in V_i = D(h_i)$. Thus we may replace a_i by ca_i and h_i by h_i^n (since $D(h_i) = D(h_i^n)$) and assume that $D(f)$ is covered by $D(h_i)$ and that $s_{\mathfrak{p}} = a_i/h_i$ for all $\mathfrak{p} \in D(h_i)$.

Now, observe that $D(f)$ can be covered by finitely many $D(h_i)$. Indeed, $D(f) \subset \bigcup_i D(h_i) \iff V((f)) \supset \bigcap_i V((h_i)) = V(\sum_i (h_i))$. This implies $f \in \sqrt{\sum_i (h_i)}$. So $f^n \in \sum_{i=1}^r (h_i)$. Thus, if $\mathfrak{p} \in D(f)$, it misses f^n and so it will miss one of these $h_1, \dots, h_r$ and consequently $\mathfrak{p} \in \bigcup_{i=1}^r D(h_i)$. Thus we assume from this point that $D(f)$ is covered by finitely many of these $D(h_i)$, say $D(h_1), \dots, D(h_r)$.

Next, note that $D(h_i) \cap D(h_j) = D(h_i h_j)$. Consider the map

$$\Psi_{h_i h_j} : A_{h_i h_j} \rightarrow \mathcal{O}_X(D(h_i h_j))$$

for which

$$\frac{a_i h_j}{h_i h_j} \mapsto \left(\frac{a_i h_j}{h_i h_j} \right)_{\mathfrak{p} \in D(h_i h_j)} \quad \text{and} \quad \frac{a_j h_i}{h_i h_j} \mapsto \left(\frac{a_j h_i}{h_i h_j} \right)_{\mathfrak{p} \in D(h_i h_j)}.$$

So if $\mathfrak{p} \in D(h_i) \cap D(h_j) = D(h_i h_j)$, then $s_{\mathfrak{p}} = \frac{a_i}{h_i} = \frac{a_j}{h_j} \in A_{\mathfrak{p}}$. But $\frac{a_i}{h_i} = \frac{a_i h_j}{h_i h_j} \in A_{\mathfrak{p}}$ and $\frac{a_j}{h_j} = \frac{a_j h_i}{h_i h_j} \in A_{\mathfrak{p}}$ for all $\mathfrak{p} \in D(h_i h_j)$. Thus, by injectivity of $\Psi_{h_i h_j}$ we have $\frac{a_i h_j}{h_i h_j} = \frac{a_j h_i}{h_i h_j} \in A_{h_i h_j}$. Hence, for some n we have

$$(h_i h_j)^n (a_i h_j - a_j h_i) = 0.$$

Since there are only finitely many indices i, j we may choose n so large that it works for all pair i, j . Rewrite the above equation as

$$h_j^{n+1} (h_i^n a_i) - h_i^{n+1} (h_j^n a_j) = 0.$$

Now replace h_i by h_i^{n+1} , and a_i by $h_i^n a_i$. Then, for all $\mathfrak{p} \in D(h_i)$, we have $s_{\mathfrak{p}} = \frac{a_i}{h_i} = \frac{h_i^n a_i}{h_i^{n+1}} \in A_{\mathfrak{p}}$ for all $\mathfrak{p} \in D(h_i) = D(h_i^{n+1})$.

All the above arguments allow us to make the following assumption : $D(f)$ is covered by $\{D(h_i) \mid i = 1, \dots, r\}$ and there exists a_i, h_i such that $s_{\mathfrak{p}} = \frac{a_i}{h_i}$ for all $\mathfrak{p} \in D(h_i)$ and $h_j a_i = h_i a_j$ for all i, j . Now write $f^n = \sum_i b_i h_i$, which is possible for some n since $D(h_i)$ cover $D(f)$. Let $a := \sum_i b_i a_i$. Then for each j we have

$$h_j a = \sum_i b_i a_i h_j = \sum_j b_j h_j a_j = f^n a_j.$$

Thus, for all $\mathfrak{p} \in D(h_j) \subset D(f)$, we have $s_{\mathfrak{p}} = \frac{a_j}{h_j} = \frac{a}{f^n} \in A_{\mathfrak{p}}$. Therefore, $s = (s_{\mathfrak{p}})_{\mathfrak{p} \in D(f)} \in \mathcal{O}_X(D(f))$ is the image of $\frac{a}{f^n}$. $\square$

11.2. Construction of $\mathcal{O}_{\text{Spec } A}$: second approach. Now consider the category $\mathbf{D}(X)$ whose objects are sets of the form $D(f) \subset X$ with f varying in A , and whose morphisms are inclusions $D(f) \subset D(g)$. We define a contravariant functor

$$\mathcal{O}_X : \mathbf{D}(X) \rightarrow \text{Ring}$$

as follows. For each $f \in A$ set

$$S(f) := \{g \in A \mid D(f) \subset D(g)\}.$$

Then the equation $D(g \cdot g') = D(g) \cap D(g')$ for elements $g, g' \in A$ shows that $S(f)$ is a multiplicative system containing all the powers of f . Let $\tau_f : A \rightarrow A_f$ and $\tau_{S(f)} : A \rightarrow A_{S(f)}$ denote the canonical maps. Since $\tau_{S(f)}$ sends the powers of f to units, the universal mapping property of localizations yields a unique homomorphism $\iota_f : A_f \rightarrow A_{S(f)}$ satisfying $\iota_f \circ \tau_f = \tau_{S(f)}$. Similarly, the canonical map $\tau_f : A \rightarrow A_f$ sends f to a unit and hence for every g with $D(f) \subset D(g)$ we have $\tau_f(g) \in (A_f)^\times$. Thus τ_f maps $S(f)$ into units in A_f . Again, the universal mapping property of localizations yield a unique homomorphism $\varphi : A_{S(f)} \rightarrow A_f$ satisfying $\varphi \circ \tau_{S(f)} = \tau_f$. Thus we get

$$\iota_f \circ \tau_f = \iota_f \circ \varphi \circ \tau_{S(f)} = \tau_{S(f)} \quad \text{and} \quad \varphi \circ \tau_{S(f)} = \varphi \circ \iota_f \circ \tau_f = \tau_f.$$

But using the uniqueness of the factored map from the UMP of A_f and $A_{S(f)}$ again, we have

$$\iota_f \circ \varphi = \text{id}_{A_{S(f)}} \quad \text{and} \quad \varphi \circ \iota_f = \text{id}_{A_f}.$$

Also, if $D(f) \subset D(g)$, then we have $S(f) \supset S(g)$. Therefore, we have the canonical map $A_{S(g)} \rightarrow A_{S(f)}$ since $\tau_{S(f)} : A \rightarrow A_{S(f)}$ sends the elements of $S(g) \subset S(f)$ to units. We therefore define the functor

$$\begin{aligned} \mathcal{O}_X : \mathbf{D}(X) &\rightarrow \text{Ring} \\ D(f) &\mapsto A_{S(f)}, \\ D(f) \subset D(g) &\mapsto A_{S(g)} \rightarrow A_{S(f)}. \end{aligned}$$

Since, $A_f \simeq A_{S(f)}$ we will set $\mathcal{O}_X(D(f)) = A_f$ for every $f \in A$. This removes ambiguity, since if $D(f_1) = D(f_2)$ then $S(f_1) = S(f_2)$ and so $A_{S(f_1)} = A_{S(f_2)}$. Also, in place of the map $A_{S(g)} \rightarrow A_{S(f)}$ we will consider $\text{res}_{D(g), D(f)} : A_g \rightarrow A_f$ which is induced by $\tau_f : A \rightarrow A_f$ sending g to units.

Lemma 11.14. – For $f, g \in A$ and $d, e \in \mathbf{N}$ with $d \geq 1$, we have a canonical commuting square

$$\begin{array}{ccc} A & \xrightarrow{\tau_f} & A_f \\ \tau_{fg} \downarrow & & \downarrow T_{f^{-e}g^d} \\ A_{fg} & \xrightarrow{\sim} & (A_f)_{f^{-e}g^d} \end{array}$$

with $A_{fg} \xrightarrow{\sim} (A_f)_{f^{-e}g^d}$ being an isomorphism.

Proof. – Note that f, g go to units in $(A_f)_{f^{-e}g^d}$ by the composite $A \rightarrow A_f \rightarrow (A_f)_{f^{-e}g^d}$. Therefore, $A \rightarrow A_f \rightarrow (A_f)_{f^{-e}g^d}$ factors through A_{fg} to yield $A_{fg} \rightarrow (A_f)_{f^{-e}g^d}$. This map explicitly written looks like : $\frac{a}{(fg)^n} \mapsto \frac{af^{-n(1+e)}g^{n(d-1)}}{(f^{-e}g^d)^n}$. On the other hand, the map $A \rightarrow A_{fg}$ sends f to a unit and factors through A_f to give $A_f \rightarrow A_{fg}$ which sends $f^{-e}g^d$ to a unit. Thus we get $(A_f)_{f^{-e}g^d} \rightarrow A_{fg}$ which, written explicitly, is : $\frac{a/f^m}{f^{-en}g^{dn}} \mapsto \frac{af^{m(dn-1)}g^{dn(m-1)}}{(fg)^{dmn}}$. Now it can be checked that these two maps are inverses of each other by usual universal mapping property arguments. $\square$

Proposition 11.15. – Let $\varphi : A \rightarrow A'$ be a ring homomorphism such that every element $f' \in A'$ is of the form $f' := \varphi(f)h$ where $f \in A$ and $h \in A'^{\times}$. Then, the map ${}^a\varphi : \text{Spec } A' \rightarrow \text{Spec } A$ given by $\mathfrak{p}' \mapsto \varphi^{-1}(\mathfrak{p}')$ is injective, and furnishes a homeomorphism $\text{Spec } A' \xrightarrow{\sim} \text{Im}({}^a\varphi) \subset \text{Spec } A$.

Proof. – First, we show that ${}^a\varphi$ is injective. Indeed, suppose $\varphi^{-1}(\mathfrak{p}'_1) = \varphi^{-1}(\mathfrak{p}'_2)$. Now let $f' \in \mathfrak{p}'_1$. By hypothesis there exists $f \in A$ and $h \in A'^{\times}$ such that $f' = \varphi(f)h$ and so $\varphi(f) = f'h^{-1} \in \mathfrak{p}'_1$. Hence $f \in \varphi^{-1}(\mathfrak{p}'_2)$, and consequently $f'h^{-1} \in \mathfrak{p}'_2$ and so $f' \in \mathfrak{p}'_2$. Thus $\mathfrak{p}'_1 \subset \mathfrak{p}'_2$ and by a symmetric argument $\mathfrak{p}'_2 \subset \mathfrak{p}'_1$.

Now we have a bijection of sets $\text{Spec } A' \xrightarrow{\sim} \text{Im}({}^a\varphi)$. We first show that this is a closed map with respect to the subspace topology on $\text{Im}({}^a\varphi)$. It suffices to show that given $Y' \subset \text{Spec } A'$ closed, there exists a closed $Y \subset \text{Spec } A$ such that $Y' = ({}^a\varphi)^{-1}(Y)$ since this implies ${}^a\varphi(Y') = Y \cap \text{Im}({}^a\varphi)$ which is closed in $\text{Im}({}^a\varphi)$. So write $Y' = V(E')$ for some subset $E' \subset A'$, and note that we can adjust the elements of E' by suitable units in $A'^{\times}$ in such a way that we may assume $E' \subset \varphi(A)$. Thus, since E' can be assumed to be a subset of the image of φ , we may write $E' = \varphi(E)$ for some subset $E \subset A$. We have, as in the proof of Proposition 11.5, we have

$$Y' = V(E') = V(\varphi(E)) = ({}^a\varphi)^{-1}(V(E)) = ({}^a\varphi)^{-1}(Y)$$

with $Y = V(E)$, as desired. $\square$

Corollary 11.16. – Let $\mathfrak{a} \subset A$ be an ideal. Then the map ${}^a\pi : \text{Spec } A/\mathfrak{a} \rightarrow \text{Spec } A$ associated to the canonical projection $\pi : A \rightarrow A/\mathfrak{a}$ defines a homeomorphism $\text{Spec } A/\mathfrak{a} \xrightarrow{\sim} V(\mathfrak{a}) \subset \text{Spec } A$. In this case, ${}^a\pi$ is called a closed immersion of spectra.

Proof. – Use $\varphi = \pi$ and $\text{Im}({}^a\pi) = V(\mathfrak{a})$ in Proposition 11.15. $\square$

Corollary 11.17. – Let $f \in A$ and consider the canonical map $\tau : A \rightarrow A_f$. This induces a homeomorphism ${}^a\tau : \text{Spec } A_f \xrightarrow{\sim} D(f) \subset \text{Spec } A$. In this case, ${}^a\tau$ is called an open immersion of spectra.

Proof. – Note that $\text{Im}({}^a\tau) = \{\tau^{-1}(\mathfrak{p}') \mid \mathfrak{p}' \in \text{Spec } A_f\} = \{\mathfrak{p} \in \text{Spec } A \mid f \notin \mathfrak{p}\} = D(f)$. $\square$

Lemma 11.18. – Let $f \in A$ and set $X_f := \text{Spec } A_f$. We have :

- (1) The map ${}^a\tau : X_f \xrightarrow{\sim} D(f) \subset \text{Spec } A$ induced from the canonical map $\tau : A \rightarrow A_f$ is a homeomorphism.
- (2) For $D(g) \subset D(f)$, we have $({}^a\tau)^{-1}(D(g)) = D_f(\tau(g))$. Furthermore, ${}^a\tau^{-1}$ induces a bijection between sets of type $D(g) \subset D(f)$, $g \in A$, and sets of type $D_f(h)$, $h \in A_f$.
- (3) The restriction of the functor $\mathcal{O}_X : \mathbf{D}(X) \rightarrow \mathbf{Ring}$ to the subcategory induced on $D(f)$ is equivalent to $\mathcal{O}_{X_f} : \mathbf{D}(X_f) \rightarrow \mathbf{Ring}$.

Proof. – The first assertion is just the Corollary 11.17. For the second statement, let $D(g) \subset D(f)$, then

$$\mathfrak{p}' \in ({}^a\tau)^{-1}(D(g)) \Leftrightarrow {}^a\tau(\mathfrak{p}') = \tau^{-1}(\mathfrak{p}') \in D(g) \Leftrightarrow g \notin \tau^{-1}(\mathfrak{p}') \Leftrightarrow \tau(g) \notin \mathfrak{p}' \Leftrightarrow \mathfrak{p}' \notin D_f(\tau(g)).$$

Conversely, suppose $h = \frac{g}{f^m} \in A_f$, then $D_f(h) = D_f\left(\frac{fg}{f^{m+1}}\right) = D_f\left(\frac{fg}{1}\right) = D_f(\tau(fg)) = ({}^a\tau)^{-1}(D(fg))$ noting that $D(fg) = D(f) \cap D(g) \subset D(f)$.

Finally, for the third statement, first note that any $D(g) \subset D(f)$ can be written as $D(fg)$, which is mapped to A_{fg} . On the other hand, $D_f(\tau(g))$ is mapped to $(A_f)_{\tau(g)}$, which by Lemma 11.14 is isomorphic to A_{fg} . $\square$

Proposition 11.19. – *The functor $\mathcal{O}_X : \mathbf{D}(X) \rightarrow \mathbf{Ring}$ satisfies the sheaf properties if we restrict to open coverings of type $D(f) = \bigcup_{\lambda \in \Lambda} D(f_\lambda)$ with $f, f_\lambda \in A$.*

Proof. – Consider a basic open covering $D(f) = \bigcup_{\lambda \in \Lambda} D(f_\lambda)$ as in the statement of the proposition. Then by Lemma 11.18, we can assume without loss of generality $f = 1$ and, hence, $X = D(f)$. Since X is quasi-compact, we have $X = \bigcup_{i=1}^n D(f_i)$ for some $f_1, \dots, f_n \in A$. We first verify the sheaf property for such a finite cover by basic opens.

The first sheaf property we must verify is that the map

$$A \rightarrow \prod_{i=1}^n A_{f_i}, \quad a \mapsto (\tau_{f_i}(a))_{i=1}^n$$

is injective. So suppose for some $g \in A$ we have $\tau_{f_i}(g) = 0$ for every i . This means for each i there exists $r_i \in \mathbf{N}$ such that $f_i^{r_i} g = 0$. Now, since $X = \bigcup_{i=1}^n D(f_i)$, we have

$$\emptyset = \bigcap_{i=1}^n V(f_i) = V(f_1, \dots, f_n) = V(f_1^{r_1}, \dots, f_n^{r_n})$$

where the last equality follows because any prime ideal containing $f_i^{r_i}$ must contain f_i and the converse is obviously true. Thus $A = \langle f_1^{r_1}, \dots, f_n^{r_n} \rangle$ and we can write $1 = \sum_{i=1}^n a_i f_i^{r_i}$ for some $a_i \in A$, and we get

$$g = \sum_{i=1}^n a_i f_i^{r_i} g = 0.$$

This establishes the first sheaf property.

The second sheaf property entails verifying that the following diagram

$$A \rightarrow \prod_{i=1}^n A_{f_i} \rightrightarrows \prod_{i,j=1}^n A_{f_i f_j}$$

is exact : meaning, if $(g_i)_i \in \prod_{i=1}^n A_{f_i}$ satisfies $g_i|_{D(f_i f_j)} = g_j|_{D(f_i f_j)}$ for all i, j then there exists a $g \in A$ such that $g|_{D(f_i)} = g_i$ for all i . Note that here $g|_{D(f_i)}$ or $g_i|_{D(f_i f_j)}$ denotes the image of g or g_i under the canonical map $A \rightarrow A_{f_i}$ or $A_{f_i} \rightarrow A_{f_i f_j}$ respectively (see Remark 11.10). So consider such an element $(g_i)_i \in \prod_{i=1}^n A_{f_i}$. Write $g_i = \frac{h_i}{f_i^{n_i}}$ with $n_i \in \mathbf{N}$ for each i . Since there are only finitely many i , we can further assume $g_i = \frac{h_i}{f_i^s}$

for some $s \in \mathbf{N}$ independent of all the i . Now $g_i|_{D(f_i f_j)} = g_j|_{D(f_i f_j)}$ implies $\frac{f_j^s h_i}{(f_i f_j)^s} = \frac{f_i^s h_j}{(f_i f_j)^s}$. So for each pair i, j there exists some $t_{ij} \in \mathbf{N}$ such that $(f_i f_j)^{t_{ij}} (f_j^s h_i - f_i^s h_j) = 0$. We can choose $t > t_{ij}$ for all i, j (there are finitely many such pairs!) so that $(f_i f_j)^t (f_j^s h_i - f_i^s h_j) = 0$ for all i, j i.e. $f_j^{s+t} (f_i^t h_i) = f_i^{s+t} (f_j^t h_j)$ for all i, j . Now $\sum_{i=1}^n b_i f_i^{s+t} = 1$ for some $b_i \in A$ since $X = \bigcup_{i=1}^n D(f_i)$ and so $A = \langle f_1, \dots, f_n \rangle = \langle f_1^{s+t}, \dots, f_n^{s+t} \rangle$. Now, we set $g := \sum_{i=1}^n b_i f_i^t h_i \in A$ and claim that $g|_{D(f_i)} = g_i$ for all i . Indeed,

$$g|_{D(f_i)} = \sum_{j=1}^n \frac{b_j f_j^{s+t} (f_j^t h_j)}{f_i^{s+t}} = \sum_{j=1}^n \frac{b_j f_j^{s+t}}{1} \cdot \frac{f_i^t h_i}{f_i^{s+t}} = \left(\sum_{j=1}^n \frac{b_j f_j^{s+t}}{1} \right) \cdot \frac{h_i}{f_i^s} = g_i.$$

Hence, we have proved that the second sheaf axiom holds for finite covers.

Now let $\mathfrak{U} = (D(f_\lambda))_{\lambda \in \Lambda}$ be a cover of X admitting the finite subcover $\mathfrak{U}' = (D(f_i))_{i=1}^n$. Consider the following commutative diagram :

$$\begin{array}{ccc} A & \xrightarrow{\iota} & H^0(\mathfrak{U}, \mathcal{O}_X) \\ \parallel & & \downarrow \rho \\ A & \xrightarrow{\iota'} & H^0(\mathfrak{U}', \mathcal{O}_X) \end{array}$$

where

$$H^0(\mathfrak{U}, \mathcal{O}_X) := \text{Ker} \left(\prod_{\lambda} A_{f_{\lambda}} \rightrightarrows \prod_{\lambda, \lambda'} A_{f_{\lambda} f_{\lambda'}} \right) \quad \text{and} \quad H^0(\mathfrak{U}', \mathcal{O}_X) := \text{Ker} \left(\prod_i A_{f_i} \rightrightarrows \prod_{i, j} A_{f_i f_j} \right)$$

and $\rho : H^0(\mathfrak{U}, \mathcal{O}_X) \rightarrow H^0(\mathfrak{U}', \mathcal{O}_X)$ is the restriction of the canonical map $\prod_{\lambda} A_{f_{\lambda}} \rightarrow \prod_i A_{f_i}$. Here, note that $H^0(\mathfrak{U}, \mathcal{O}_X)$ consists of sequences $(g_{\lambda})_{\lambda} \in \prod_{\lambda} A_{f_{\lambda}}$ such that g_{λ} and $g_{\lambda'}$ agree on $D(f_{\lambda}) \cap D(f_{\lambda'})$. Hence, an image of such a sequence in $H^0(\mathfrak{U}, \mathcal{O}_X)$ is a sequence $(g_i)_{i=1}^n \in \prod_{i=1}^n D(f_i)$ such that g_i and g_j agree on $D(f_i) \cap D(f_j)$ so that $(g_i)_i \in H^0(\mathfrak{U}', \mathcal{O}_X)$. By the previous two paragraphs that prove the sheaf axioms for finite covers, we know that ι' is a bijection, and since the square commutes ρ is therefore a surjection. Thus, it remains to show that ρ is injective as well, and this will prove that ι is a bijection as required.

Finally, we show that ρ is injective. Suppose $(g_{\lambda})_{\lambda \in \Lambda} \in H^0(\mathfrak{U}, \mathcal{O}_X)$ be a sequence such that its projection $(g_i)_{i=1}^n \in H^0(\mathfrak{U}', \mathcal{O}_X)$ is zero i.e. each $g_i = 0$. Now fix any $\lambda \in \Lambda$ and consider the cover $D(f_{\lambda}) = \bigcup_{i=1}^n D(f_{\lambda} f_i)$ obtained by intersecting $D(f_{\lambda})$ with the cover $X = \bigcup_i D(f_i)$. So, we have

$$g_{\lambda} \mid_{D(f_{\lambda} f_i)} = g_{\lambda} \mid_{D(f_{\lambda}) \cap D(f_i)} = g_i \mid_{D(f_{\lambda}) \cap D(f_i)} = 0.$$

Therefore, by the first sheaf property applied to the bijection $A_{f_{\lambda}} \rightarrow \text{Ker} \left(\prod_i A_{f_{\lambda} f_i} \rightrightarrows \prod_{i, j} A_{f_{\lambda} f_i f_j} \right)$ we have $g_{\lambda} = 0$. Since λ as chosen arbitrarily, we have $(g_{\lambda})_{\lambda \in \Lambda} = 0$ i.e. ρ is injective. $\square$

Now we extend $\mathcal{O}_X : \mathbf{D}(X) \rightarrow \text{Ring}$ to a presheaf $\overline{\mathcal{O}}_X : \mathbf{Opn}(X) \rightarrow \text{Ring}$ by defining

$$\overline{\mathcal{O}}_X(U) := \varprojlim_{D(f) \subset U} A_f$$

for open subsets U of X . Now if we have a sequence $(a_f)_{D(f) \subset U} \in \varprojlim_{D(f) \subset U} A_f$, then for two terms, say

$a_f, a_{f'}$, the term $a_{ff'}$ corresponding to $D(f) \cap D(f') = D(ff')$ is the image of both a_f and $a_{f'}$, and thus $a_f \mid_{D(f) \cap D(f')} = a_{f'} \mid_{D(f) \cap D(f')}$. We will therefore often call elements of $\varprojlim_{D(f) \subset U} A_f$ as *glueable* sequences. Then

if $U \subset V$ are open sets in X , there is a canonical map

$$\overline{\mathcal{O}}_X(V) = \varprojlim_{D(f) \subset V} A_f \rightarrow \varprojlim_{D(f) \subset U} A_f = \overline{\mathcal{O}}_X(U)$$

as $\varprojlim_{D(f) \subset U} A_f$ is a cone above the diagram $(A_f)_{D(f) \subset U}$. Furthermore, if $U = D(g)$ for some $g \in A$, then

$$\overline{\mathcal{O}}_X(D(g)) = \varprojlim_{D(f) \subset D(g)} A_f \simeq A_g \text{ since } A_g \text{ is clearly a cone itself. In this way } \overline{\mathcal{O}}_X \text{ extends } \mathcal{O}_X.$$

Theorem 11.20. – *The functor $\overline{\mathcal{O}}_X$ is a sheaf of rings on X .*

Proof. – We first prove the sheaf properties for open coverings of an arbitrary open set by basic open sets. So let U be an open set in $X = \text{Spec } A$ and suppose we have an open cover $U = \bigcup_{\lambda \in \Lambda} D(f_{\lambda})$. Let $D(g) \subset U$ and consider the following diagram :

$$\begin{array}{ccccc} \varprojlim_{D(f) \subset U} A_f & \longrightarrow & \prod_{\lambda} A_{f_{\lambda}} & \rightrightarrows & \prod_{\lambda, \lambda'} A_{f_{\lambda} f_{\lambda'}} \\ \downarrow & & \downarrow & & \downarrow \\ A_g & \longrightarrow & \prod_{\lambda} A_{g f_{\lambda}} & \rightrightarrows & \prod_{\lambda, \lambda'} A_{g f_{\lambda} f_{\lambda'}} \end{array}$$

Note that both the squares commute. Also, for the top row we have :

$$\text{Im} \left(\varprojlim_{D(f) \subset U} A_f \rightarrow \prod_{\lambda} A_{f_{\lambda}} \right) \subset \text{Ker} \left(\prod_{\lambda} A_{f_{\lambda}} \rightrightarrows \prod_{\lambda, \lambda'} A_{f_{\lambda} f_{\lambda'}} \right)$$

since we remarked earlier that $\varprojlim_{D(f) \subset U} A_f$ consisted of glueable sequences. Finally, by Proposition 11.19, the bottom row is exact.

We show that the top row is exact as well. Indeed, the injectivity of the maps $A_g \rightarrow \prod_{\lambda} A_{gf_{\lambda}}$ for each $g \in A$ implies that $\varprojlim_{D(f) \subset U} A_f \rightarrow \prod_{\lambda} A_{f_{\lambda}}$ is injective. To see this, take a glueable sequence $(a_f)_{D(f) \subset U} \in \varprojlim_{D(f) \subset U} A_f$ and suppose its image $(a_{f_{\lambda}})_{\lambda} = 0$ i.e. $a_{f_{\lambda}} = 0$ for all λ . We need to show that $a_g = 0$ for all $D(g) \subset U$. So fix an arbitrary $D(g) \subset U$ and consider the image a_g under the left vertical map, and note that the bottom left horizontal map sends this to $(a_g|_{D(gf_{\lambda})})_{\lambda} = (a_{f_{\lambda}}|_{D(gf_{\lambda})})_{\lambda} = 0$, where the first equality follows from $(a_f)_{D(f) \subset U}$ being a glueable sequence. Thus, by the injectivity of the map $A_g \rightarrow \prod_{\lambda} A_{gf_{\lambda}}$, we conclude that $a_g = 0$ as asserted.

Next, we show that any $(h_{\lambda})_{\lambda} \in \text{Ker} \left(\prod_{\lambda} A_{f_{\lambda}} \rightrightarrows \prod_{\lambda, \lambda'} A_{f_{\lambda} f_{\lambda'}} \right)$ has a preimage in $\varprojlim_{D(f) \subset U} A_f$. Now $(h_{\lambda})_{\lambda}$ is a compatible sequence. So, for all $D(g) \subset U$, its image under the middle vertical map is $(h_{\lambda}|_{D(gf_{\lambda})})_{\lambda} \in \text{Ker} \left(\prod_{\lambda} A_{gf_{\lambda}} \rightrightarrows \prod_{\lambda, \lambda'} A_{gf_{\lambda} f_{\lambda'}} \right)$. Hence, for all $g \in A$ with $D(g) \subset U$, we know that there is a unique $h'_g \in A_g$ such that $(h'_g|_{D(gf_{\lambda})})_{\lambda} = (h_{\lambda}|_{D(gf_{\lambda})})_{\lambda}$. We put all these together to obtain $(h'_f)_{D(f) \subset U} \in \prod_{D(f) \subset U} A_f$. We wish to show that its glueable. So suppose $D(f) \subset D(g) \subset U$, and we want to show that $h'_g|_{D(f)} = h'_f$. Consider the image of $h'_g|_{D(f)} \in A_f$ under the map $A_f \rightarrow \prod_{\lambda} A_{ff_{\lambda}}$ which is

$$\begin{aligned} (h'_g|_{D(f)}|_{D(ff_{\lambda})})_{\lambda} &= (h'_g|_{D(f) \cap D(ff_{\lambda})})_{\lambda} = (h'_g|_{D(gff_{\lambda})})_{\lambda} = (h'_g|_{D(gf_{\lambda})}|_{D(gff_{\lambda})})_{\lambda} \\ &= (h_{\lambda}|_{D(gf_{\lambda})}|_{D(gff_{\lambda})})_{\lambda} = (h_{\lambda}|_{D(gff_{\lambda})})_{\lambda} = (h_{\lambda}|_{D(ff_{\lambda})})_{\lambda}, \end{aligned}$$

and $(h_{\lambda}|_{D(ff_{\lambda})})_{\lambda}$ is the image of $h'_f \in A_f$. Thus $h'_g|_{D(f)} = h'_f$ as required. So we have proved that $(h'_f)_{D(f) \subset U} \in \varprojlim_{D(f) \subset U} A_f$. Now it remains to show that $(h'_f)_{D(f) \subset U}$ maps to the compatible sequence $(h_{\lambda})_{\lambda}$. But

$(h'_f)_{D(f) \subset U}$ actually maps to $(h'_{f_{\lambda}})_{\lambda}$, so we must show that $h_{\lambda} = h'_{f_{\lambda}}$ for all λ . Indeed, take some arbitrary $\mu \in \Lambda$ and consider $D(f_{\mu}) = \bigcup_{\lambda \in \Lambda} D(f_{\mu} f_{\lambda})$. Now we have $(h_{\mu}|_{D(f_{\mu} f_{\lambda})})_{\lambda} = (h_{\lambda}|_{D(f_{\mu} f_{\lambda})})_{\lambda}$ since $(h_{\lambda})_{\lambda}$ is compatible. Thus, under the map $A_{f_{\mu}} \rightarrow \prod_{\lambda} A_{f_{\mu} f_{\lambda}}$ the image of h_{μ} , which is $(h_{\mu}|_{D(f_{\mu} f_{\lambda})})_{\lambda}$, coincides with the image of $h'_{f_{\mu}}$, which is $(h_{\lambda}|_{D(f_{\mu} f_{\lambda})})_{\lambda}$. By injectivity, we have $h_{\mu} = h'_{f_{\mu}}$, and μ was chosen arbitrarily. Thus, we have shown that the top row is exact as well. Hence, $\overline{\mathcal{O}}_X$ satisfies the sheaf axioms for the covering $(D(f_{\lambda}))_{\lambda}$ of U .

Finally, we consider an arbitrary open covering $\mathfrak{U} = (U_{\lambda})_{\lambda \in \Lambda}$ of U . We cover each U_{λ} by basic opens to obtain a refinement of $\mathfrak{U}$, which we denote by $\mathfrak{V} = (D(f_{\mu}))_{\mu \in M}$. Thus, there is a map $\tau : M \rightarrow \Lambda$ such that $D(f_{\mu}) \subset U_{\tau(\mu)}$ for every $\mu \in M$. Then, we consider the following square

$$\begin{array}{ccc} \overline{\mathcal{O}}_X(U) & \xrightarrow{\iota} & H^0(\mathfrak{U}, \overline{\mathcal{O}}_X) \\ \parallel & & \downarrow \text{ref} \\ \overline{\mathcal{O}}_X(U) & \xrightarrow{\iota'} & H^0(\mathfrak{V}, \overline{\mathcal{O}}_X) \end{array}$$

which is commutative. Since ι' is a bijection, res is surjective. So, to show that ι is also a bijection (note that commutativity already implies its an injection), it suffices to show that res is an injection. So take a compatible sequence $(s_{\lambda})_{\lambda} \in H^0(\mathfrak{U}, \overline{\mathcal{O}}_X)$ and suppose its image $\text{res}((s_{\lambda})_{\lambda}) = (s_{\tau(\mu)}|_{D(f_{\mu})})_{\mu} = 0$. Fix a $\lambda \in \Lambda$ and consider all the $D(f_{\mu})$ contained in U_{λ} , and thus covering U_{λ} ; it suffices to show, by the sheaf property of $\overline{\mathcal{O}}_X(U_{\lambda})$ by basic open coverings, that $s_{\lambda}|_{D(f_{\mu})} = 0$ for all such $D(f_{\mu}) \subset U_{\lambda}$. So take such a $D(f_{\mu}) \subset U_{\lambda}$, and note that $D(f_{\mu}) \subset U_{\lambda} \cap U_{\tau(\mu)}$. Thus, we have

$$s_{\lambda}|_{D(f_{\mu})} = s_{\lambda}|_{U_{\lambda} \cap U_{\tau(\mu)}}|_{D(f_{\mu})} = s_{\tau(\mu)}|_{U_{\lambda} \cap U_{\tau(\mu)}}|_{D(f_{\mu})} = s_{\tau(\mu)}|_{D(f_{\mu})} = 0.$$

This completes the proof that $\overline{\mathcal{O}}_X$ is a sheaf. $\square$

From now on we will use the notation $\mathcal{O}_X$ in place of $\overline{\mathcal{O}}_X$ to denote the structure sheaf on X . Note that the ring of global sections is $\mathcal{O}_X(X) = \mathcal{O}_X(D(1)) = A$.

Definition 11.21. – A *locally ringed space* is a ringed space $(X, \mathcal{O}_X)$ such that for every $x \in X$ the stalks $\mathcal{O}_{X,x}$ are local rings.

If $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ are locally ringed spaces, a morphism of ringed spaces $(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ is called a *morphism of locally ringed spaces* if all the maps $f_x^\# : \mathcal{O}_{Y, f(x)} \rightarrow \mathcal{O}_{X, x}$ for $x \in X$ are local, i.e. the unique maximal ideal $\mathfrak{m}_{f(x)} \subset \mathcal{O}_{Y, f(x)}$ is mapped inside $\mathfrak{m}_x \subset \mathcal{O}_{X, x} : f_x^\#(\mathfrak{m}_{f(x)}) \subset \mathfrak{m}_x$.

Proposition 11.22. – Let A be a ring and $X := \text{Spec } A$. Let $\mathcal{O}_X$ be the structure sheaf on X . Then, $(X, \mathcal{O}_X)$ is a locally ringed space with $\mathcal{O}_{X, x} \simeq A_x$ for all $x \in X$.

Proof. – Note at first that $\mathcal{O}_{X, x} = \varinjlim_{x \in U} \mathcal{O}_X(U) \simeq \varinjlim_{x \in D(g)} \mathcal{O}_X(D(g)) \simeq \varinjlim_{x \in D(g)} A_g$. We will show that $\varinjlim_{x \in D(g)} A_g$ is canonically isomorphic to A_x . Indeed, for $g, g' \notin x$ and $D(g') \subset D(g)$ we have the following diagram

$$\begin{array}{ccc} A_g & \xrightarrow{\text{res}} & A_{g'} \\ & \searrow \gamma_g \quad \swarrow \gamma_{g'} & \\ & A_x & \end{array}$$

which commutes, and $\gamma_g : A_g \rightarrow A_x$ is the canonical map $a/g^n \mapsto a/g^n$. Now consider any ring R equipped with compatible maps $\sigma_g : A_g \rightarrow R$. Consider the following diagram

$$\begin{array}{ccccc} & & \tau_{g'} & & \\ & \nearrow & & \searrow & \\ A = A_1 & \xrightarrow{\tau_g} & A_g & \xrightarrow{\text{res}} & A_{g'} \\ & \searrow \tau_x & \searrow \gamma_g & \swarrow \gamma_{g'} & \\ & & A_x & & \\ & \searrow \sigma_g & \downarrow \varphi & \swarrow \sigma_{g'} & \\ & & R & & \end{array}$$

σ_1 (curved arrow from $A = A_1$ to R)

in which σ_1 carries every $g \notin x$ to a unit in R , since $\sigma_1 = \sigma_g \circ \tau_g$ where $\tau_g : A \rightarrow A_g$ is the canonical map that carries g to a unit. Hence, by the universal mapping property of the localization A_x , there exists a unique map $\varphi : A_x \rightarrow R$ such that $\varphi \circ \tau_x = \sigma_1$, where $\tau_x : A \rightarrow A_x$ is the canonical map of localization. We show that $\varphi \circ \gamma_g = \sigma_g$ for all $g \notin x$ proving that A_x satisfies the universal property of colimits. Indeed, $\varphi \circ \gamma_g \circ \tau_g = \varphi \circ \tau_x = \sigma_1 = \sigma_g \circ \tau_g$ and so the universal property of A_g implies $\varphi \circ \gamma_g = \sigma_g$ as required. $\square$

Lemma 11.23. – Let A, B be rings, $X = \text{Spec } A$, $Y = \text{Spec } B$, and $(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a morphism of locally ringed spaces. Then the morphism $(f, f^\#)$ is uniquely determined by the ring homomorphism $f^\#(Y) : \mathcal{O}_Y(Y) = B \rightarrow A = \mathcal{O}_X(X)$.

Proof. – The statement means if $(f_1, f_1^\#)$ and $(f_2, f_2^\#)$ are such that $f_1^\#(Y) = f_2^\#(Y)$, then $f_1 = f_2$ and $f_1^\# = f_2^\#$. We set $\varphi = f^\#(Y)$, and we first show that $\varphi = f$. For $\mathfrak{p} \in X$ consider the following commutative diagram

$$\begin{array}{ccc} B & \xrightarrow{\varphi = f^\#(Y)} & A \\ \tau_{f(\mathfrak{p})} \downarrow & & \downarrow \tau_{\mathfrak{p}} \\ B_{f(\mathfrak{p})} & & A_{\mathfrak{p}} \\ \parallel & & \parallel \\ \mathcal{O}_{Y, f(\mathfrak{p})} & \xrightarrow{f_{\mathfrak{p}}^\#} & \mathcal{O}_{X, \mathfrak{p}} \end{array}$$

where $f_{\mathfrak{p}}^\#$ is local by hypothesis. Hence, $f_{\mathfrak{p}}^\#(f(\mathfrak{p})B_{f(\mathfrak{p})}) \subset \mathfrak{p}A_{\mathfrak{p}}$ and so $f(\mathfrak{p})B_{f(\mathfrak{p})} \subset (f_{\mathfrak{p}}^\#)^{-1}(\mathfrak{p}A_{\mathfrak{p}})$. But since $f(\mathfrak{p})B_{f(\mathfrak{p})}$ is a maximal ideal, this implies $f(\mathfrak{p})B_{f(\mathfrak{p})} = (f_{\mathfrak{p}}^\#)^{-1}(\mathfrak{p}A_{\mathfrak{p}})$, since the later is a prime ideal being a preimage of a prime ideal under a ring map. Also, $\tau_{f(\mathfrak{p})}^{-1}(f(\mathfrak{p})B_{f(\mathfrak{p})}) \supset f(\mathfrak{p})$. The reversed inclusion is also true. Indeed, take $b \in \tau_{f(\mathfrak{p})}^{-1}(f(\mathfrak{p})B_{f(\mathfrak{p})})$ so that $b/1 = u/v$ for some $u \in f(\mathfrak{p})$ and $v \notin f(\mathfrak{p})$. Thus there exists $s \in B \setminus f(\mathfrak{p})$ with $s(vb - u) = 0$, whence $svb \in f(\mathfrak{p})$ and so $b \in f(\mathfrak{p})$ as asserted i.e. $\tau_{f(\mathfrak{p})}^{-1}(f(\mathfrak{p})B_{f(\mathfrak{p})}) = f(\mathfrak{p})$.

Similarly, $\mathfrak{p} = \tau_{\mathfrak{p}}^{-1}(\mathfrak{p}A_{\mathfrak{p}})$. Consequently, we get

$$\varphi^{-1}(\mathfrak{p}) = \varphi^{-1}(\tau_{\mathfrak{p}}^{-1}(\mathfrak{p}A_{\mathfrak{p}})) = \tau_{f(\mathfrak{p})}^{-1}\left(\left(f_{\mathfrak{p}}^{\#}\right)^{-1}(\mathfrak{p}A_{\mathfrak{p}})\right) = f(\mathfrak{p})$$

i.e. ${}^a\varphi(\mathfrak{p}) = f(\mathfrak{p})$. Since $\mathfrak{p}$ was chosen arbitrarily, ${}^a\varphi = f$ as required.

Next, we show that for basic open sets $D(g)$, the map $f^{\#}(D(g))$ is uniquely determined by $f^{\#}(Y)$. This is because the following square

$$\begin{array}{ccc} B & \xrightarrow{\varphi=f^{\#}(Y)} & A \\ \text{res} \downarrow & & \downarrow \text{res} \\ B_g & \xrightarrow{f^{\#}(D(g))} & A_{\varphi(g)} \end{array}$$

commutes and by the universal property of B_g the bottom map $f^{\#}(D(g))$ is uniquely determined by $\varphi := f^{\#}(Y)$ and is precisely the localization map $\varphi_g : B_g \rightarrow A_{\varphi(g)}$, $b/g^n \mapsto \varphi(b)/\varphi(g)^n$. Thus, we have shown that given $f^{\#}$ a natural transformation, once $f^{\#}(Y)$ is prescribed, then for each basic open $D(g)$, the map $f^{\#}(D(g))$ is forced to be the map $(f^{\#}(Y))_g$.

In the final step, we show that $f^{\#}(Y)$ forces the description of $f^{\#}(V)$ for every open set $V \subset Y$. Indeed, take any open set V and cover it by basic opens : $V = \bigcup_{\lambda} D(g_{\lambda})$ with $g_{\lambda} \in B$. Then, given a section $s \in \mathcal{O}_Y(V)$ we know that $f^{\#}(V)(s) \in \mathcal{O}_X(f^{-1}(V)) = \mathcal{O}_X(\bigcup_{\lambda} D(\varphi(g_{\lambda})))$ (note that here we are using $f = {}^a\varphi$ and so $f^{-1}(D(g_{\lambda})) = D(\varphi(g_{\lambda}))$). Hence, the restrictions

$$f^{\#}(V)(s) |_{D(\varphi(g_{\lambda}))} = f^{\#}(D(g_{\lambda}))(s |_{D(g_{\lambda})})$$

are uniquely determined. Now as $\mathcal{O}_Y$ is a sheaf, $f^{\#}(V)(s)$ is therefore uniquely determined. But V and s were chosen arbitrarily, thus $f^{\#}$ is uniquely determined by $\varphi := f^{\#}(Y)$. $\square$

Proposition 11.24. – *The following statements hold :*

- (1) *Let A and B be rings with $X = \text{Spec } A$ and $Y = \text{Spec } B$. Then the canonical map*

$$\Phi : \text{Hom}((X, \mathcal{O}_X), (Y, \mathcal{O}_Y)) \rightarrow \text{Hom}(B, A), \quad (f, f^{\#}) \mapsto f^{\#}(Y)$$

is bijective.

- (2) *For any morphism of locally ringed spaces $(f, f^{\#}) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ and a point $x \in X$, the associated map of stalks $f_x^{\#} : \mathcal{O}_{Y, f(x)} \rightarrow \mathcal{O}_{X, x}$ coincides canonically with the map $B_{f(x)} \rightarrow A_x$ obtained via localization from $f^{\#}(Y) : B \rightarrow A$.*

Proof. – We define a map $\Psi : \text{Hom}(B, A) \rightarrow \text{Hom}((X, \mathcal{O}_X), (Y, \mathcal{O}_Y))$ inverse to Φ . Take a ring map $\varphi : B \rightarrow A$ and consider the associated map of spectra ${}^a\varphi : \text{Spec } A \rightarrow \text{Spec } B$, $\mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$. We define

$$\Psi(\varphi) := (f, f^{\#}) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$$

as follows. We take f to be ${}^a\varphi$, so that for every $g \in B$ we should have $f^{\#}(D(g)) : \mathcal{O}_Y(D(g)) = B_g \rightarrow \mathcal{O}_X(f^{-1}(D(g))) = \mathcal{O}_X(D(\varphi(g))) = A_{\varphi(g)}$. So we set $f^{\#}(D(g))$ to be the canonical map $B_g \rightarrow A_{\varphi(g)}$ derived from $\varphi : B \rightarrow A$. Next, for a general open set V we take the open cover by basic opens $V = \bigcup_{D(g) \subset V} D(g)$ and define $f^{\#}(V) : \mathcal{O}_Y(V) \simeq \varprojlim_{D(g) \subset V} B_g \rightarrow \varprojlim_{D(\varphi(g)) \subset f^{-1}(V)} A_{\varphi(g)} \simeq \mathcal{O}_X(f^{-1}(V))$ obtained by putting together all the $f^{\#}(D(g))$. Here, we note that $\varprojlim_{D(\varphi(g)) \subset f^{-1}(V)} A_{\varphi(g)} \simeq \mathcal{O}_X(f^{-1}(V))$ follows from the fact that $\mathcal{O}_X$ is a sheaf, $f^{-1}(V) = \bigcup_{D(g) \subset V} D(\varphi(g))$, and $\varprojlim_{D(\varphi(g)) \subset f^{-1}(V)} A_{\varphi(g)}$ is isomorphic to the set of compatible sequences $H^0(\{D(\varphi(g)) \mid D(g) \subset V\}, \mathcal{O}_X)$. Also, note that the image of an element in $\varprojlim_{D(g) \subset V} B_g$ will indeed lie in $\varprojlim_{D(\varphi(g)) \subset f^{-1}(V)} A_{\varphi(g)}$ because for $D(g') \subset D(g) \subset V$ the following diagrams

$$\begin{array}{ccc} B_g & \xrightarrow{\text{res}} & B_{g'} \\ f^{\#}(D(g)) \downarrow & & \downarrow f^{\#}(D(g')) \\ A_{\varphi(g)} & \xrightarrow{\text{res}} & A_{\varphi(g')} \end{array}$$

are commutative, and thus glueable sequences go to glueable sequences. It remains to check that $f^\#$ is a natural transformation. For $U \subset V$ opens, we must show that

$$\begin{array}{ccc} \mathcal{O}_Y(V) & \xrightarrow{f^\#(V)} & \mathcal{O}_X(f^{-1}(V)) \\ \text{res} \downarrow & & \downarrow \text{res} \\ \mathcal{O}_Y(U) & \xrightarrow{f^\#(U)} & \mathcal{O}_X(f^{-1}(U)) \end{array} \quad \text{which is} \quad \begin{array}{ccc} \varprojlim_{D(g) \subset V} B_g & \longrightarrow & \varprojlim_{D(\varphi(g)) \subset f^{-1}(V)} A_{\varphi(g)} \\ \downarrow & & \downarrow \\ \varprojlim_{D(g) \subset U} B_g & \longrightarrow & \varprojlim_{D(\varphi(g)) \subset f^{-1}(V)} A_{\varphi(g)} \end{array}$$

is commutative. This is clear since the vertical arrows (i.e. restrictions) simply truncates glueable sequences.

Next, we show that this map $(f, f^\#)$ constructed in the previous paragraph is actually a map of locally ringed spaces by showing that the map on stalks $f_x^\# : \mathcal{O}_{Y, f(x)} \rightarrow \mathcal{O}_{X, x}$ coincides with the canonical map $B_{f(x)} \rightarrow A_x$, $\frac{b}{g} \mapsto \frac{\varphi(b)}{\varphi(g)}$ for $g \notin f(x) = \varphi^{-1}(x)$. It is clear that $B_{f(x)} \rightarrow A_x$ is local : the maximal ideal of $B_{f(x)}$ is $\left\{ \frac{b}{g} \mid b \in f(x), g \notin f(x) \right\}$ and it gets mapped to $\left\{ \frac{\varphi(b)}{\varphi(g)} \mid \varphi(b) \in x, \varphi(g) \notin x \right\}$ which clearly lies inside the maximal ideal of A_x . Now to show that $f_x^\# : \mathcal{O}_{Y, f(x)} \rightarrow \mathcal{O}_{X, x}$ coincides with the canonical map $B_{f(x)} \rightarrow A_x$, we first recall that $f_x^\# : \mathcal{O}_{Y, f(x)} \rightarrow \mathcal{O}_{X, x}$ is the unique map that makes the following diagrams

$$\begin{array}{ccc} \mathcal{O}_Y(U) & \xrightarrow{f^\#(U)} & \mathcal{O}_X(f^{-1}(U)) \\ \text{can}_U \downarrow & & \downarrow \text{can}_{f^{-1}(U)} \\ \mathcal{O}_{Y, f(x)} & \xrightarrow{f_x^\#} & \mathcal{O}_{X, x} \end{array}$$

commute for all open sets U containing $f(x)$. Now $\mathcal{O}_{Y, f(x)} \simeq \varinjlim_{f(x) \in D(g)} B_g$ is canonically isomorphic to $B_{f(x)}$ via $\frac{b}{g} \leftrightarrow \left[\frac{b}{g} \right] = \text{can}_{D(g)} \left(\frac{b}{g} \right)$. Similarly, $\mathcal{O}_{X, x} \simeq \varinjlim_{x \in D(h)} A_h \simeq A_x$ via $\frac{a}{h} \leftrightarrow \text{can}_{D(h)} \left(\frac{a}{h} \right)$. Thus, we have

$$f_x^\# \left(\left[\frac{b}{g} \right] \right) = f_x^\# \left(\text{can}_{D(g)} \left(\frac{b}{g} \right) \right) = \text{can}_{D(\varphi(g))} \left(f^\#(D(g)) \left(\frac{b}{g} \right) \right) = \text{can}_{D(\varphi(g))} \left(\frac{\varphi(b)}{\varphi(g)} \right) = \left[\frac{\varphi(b)}{\varphi(g)} \right].$$

It follows that $f_x^\#$ coincides with $\varphi_x : B_{f(x)} \rightarrow A_x$, that is to say, the following diagram

$$\begin{array}{ccc} \mathcal{O}_{Y, f(x)} & \xrightarrow{f_x^\#} & \mathcal{O}_{X, x} \\ \downarrow \wr & & \downarrow \wr \\ B_{f(x)} & \xrightarrow{\varphi_x} & A_x \end{array}$$

is commutative. Consequently, $(f, f^\#)$ is a map of locally ringed spaces. Note that (3) is already proved.

Finally, we show that Ψ is inverse to Φ . Indeed, for $\varphi : B \rightarrow A$ we have $\Psi(\varphi) = ({}^a\varphi, ({}^a\varphi)^\#)$, and so $\Psi({}^a\varphi, ({}^a\varphi)^\#) = ({}^a\varphi)^\#(Y) = ({}^a\varphi)^\#(D(1)) = \varphi_1 = \varphi$. Hence $\Phi \circ \Psi = \text{id}$. On the other hand, given a morphism $(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$, we defined $\Phi(f, f^\#) = f^\#(Y)$. Set $\varphi := f^\#(Y) : \mathcal{O}_Y(Y) = B \rightarrow A = \mathcal{O}_X(X)$. Now $\Psi(\varphi) = ({}^a\varphi, ({}^a\varphi)^\#)$, and so $({}^a\varphi)^\#(Y) = \varphi = f^\#(Y)$. Thus, by the previous lemma, ${}^a\varphi = f$ and $({}^a\varphi)^\# = f^\#$. Hence, $\Psi(\Phi(f, f^\#)) = \Psi(f^\#(Y)) = \Psi(\varphi) = ({}^a\varphi, ({}^a\varphi)^\#) = (f, f^\#)$ as required. $\square$

To recap what we have done till this point, we started with a ring A and then we constructed the associated sheaf $\mathcal{O}_{\text{Spec } A}$ on $X := \text{Spec } A$ and obtained a locally ringed space $(\text{Spec } A, \mathcal{O}_{\text{Spec } A})$. We are now ready to give the definition of a scheme.

Definition 11.25. – An *affine scheme* is a locally ringed space $(X, \mathcal{O}_X)$ such that there is an isomorphism of locally ringed spaces $(X, \mathcal{O}_X) \xrightarrow{\sim} (\text{Spec } A, \mathcal{O}_{\text{Spec } A})$ for some ring A .

A *scheme* is a locally ringed space $(X, \mathcal{O}_X)$ such that there exists an open covering $(X_i)_{i \in I}$ of X where $(X_i, \mathcal{O}_X|_{X_i})$ is an affine scheme for every $i \in I$.

Finally, a *morphism of schemes* is meant as a morphism of the underlying locally ringed spaces.

For emphasis we restate the following as a fact, whose proof was already given in Lemma 11.23 and Proposition 11.24.

Fact 11.26. – *The category of affine schemes is equivalent to the opposite category of rings.*

Remark 11.27. – Let $(X, \mathcal{O}_X)$ be a scheme, and $U \subset X$ an open subset. If $(X_i)_{i \in I}$ is an open cover of X by affines, then $(X_i \cap U)_{i \in I}$ is an open cover of U . Now $X_i \cap U$ is open in $X_i \simeq \text{Spec } A_i$ and so is a union by basic opens which are also specs. Hence, we can in fact cover U by opens which are isomorphic to specs. Hence $(U, \mathcal{O}_X|_U)$ is also a scheme, called an *open subscheme* of X .

12. SCHEME THEORY 3 : GLUINGS

Lemma 12.1. – Let X be a topological space and $(U_i)_{i \in I}$ be an open covering of X .

- (1) Let $\mathcal{F}$ and $\mathcal{G}$ be presheaves of rings on X , and suppose that $\mathcal{G}$ is a sheaf as well. For every $i \in I$, let $\varphi_i : \mathcal{F}|_{U_i} \rightarrow \mathcal{G}|_{U_i}$ be a morphism of presheaves. Assume that for every $i, j \in I$, the morphisms φ_i and φ_j coincide on $U_i \cap U_j$. Then there is a unique morphism of presheaves $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ such that $\varphi|_{U_i} = \varphi_i$ for every $i \in I$. Furthermore, if both $\mathcal{F}$ and $\mathcal{G}$ are sheaves and φ_i is an isomorphism for every $i \in I$, then φ is also an isomorphism.
- (2) For every $i \in I$, let $\mathcal{F}_i$ be a sheaf on U_i , and for every pair $(i, j) \in I \times I$, assume that $\varphi_{ij} : \mathcal{F}_i|_{U_i \cap U_j} \xrightarrow{\sim} \mathcal{F}_j|_{U_i \cap U_j}$ is an isomorphism of sheaves. Also, suppose that the following hold:
 - (a) For $i \in I$, we have $\varphi_{ii} = \text{id}_{\mathcal{F}_i}$;
 - (b) For $i, j \in I$, we have $\varphi_{ij} = \varphi_{ji}^{-1}$;
 - (c) For $i, j, k \in I$, we have the cocycle condition $\varphi_{jk} \circ \varphi_{ij}|_{U_i \cap U_j \cap U_k} = \varphi_{ik}|_{U_i \cap U_j \cap U_k}$.

Then there exists a sheaf $\mathcal{F}$ on X , and for every $i \in I$, an isomorphism $\varphi_i : \mathcal{F}|_{U_i} \xrightarrow{\sim} \mathcal{F}_i$ such that $\varphi_{ij} \circ \varphi_i|_{U_i \cap U_j} = \varphi_j|_{U_i \cap U_j}$ for all i, j . Moreover, if $\mathcal{G}$ is a sheaf on X and $(\psi_i)_{i \in I}$ a family of isomorphisms from $\mathcal{G}|_{U_i}$ to $\mathcal{F}_i$ such that $\varphi_{ij} \circ \psi_i|_{U_i \cap U_j} = \psi_j|_{U_i \cap U_j}$ for all i, j , then there exists a unique isomorphism of sheaves $\psi : \mathcal{F} \rightarrow \mathcal{G}$ such that $\psi|_{U_i} = \psi_i^{-1} \circ \varphi_i$ for all $i \in I$.

Proof. – (1) Let $U \subset X$ be open and $s \in \mathcal{F}(U)$. For any $i \in I$ we have $s|_{U \cap U_i} \in \mathcal{F}(U \cap U_i)$ and so $\varphi_i(U \cap U_i)(s|_{U \cap U_i}) \in \mathcal{G}(U \cap U_i)$. For any $i, j \in I$ we have

$$\begin{aligned} \varphi_i(U \cap U_i)(s|_{U \cap U_i})|_{U \cap U_i \cap U_j} &= \varphi_i(U \cap U_i \cap U_j)(s|_{U \cap U_i \cap U_j}) \\ &= \varphi_j(U \cap U_i \cap U_j)(s|_{U \cap U_i \cap U_j}) \\ &= \varphi_j(U \cap U_j)(s|_{U \cap U_j})|_{U \cap U_i \cap U_j}. \end{aligned}$$

Since $\mathcal{G}$ is a sheaf, there is a unique section, say $\varphi(U)(s) \in \mathcal{G}(U)$ such that $\varphi(U)(s)|_{U \cap U_i} = \varphi_i(U \cap U_i)(s|_{U \cap U_i})$ for every $i \in I$. This yields a map $\varphi(U) : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ for each $U \subset X$ open.

We check that if $U \subset V$ are open in X then

$$\begin{array}{ccc} \mathcal{F}(V) & \xrightarrow{\varphi(V)} & \mathcal{G}(V) \\ \downarrow & & \downarrow \\ \mathcal{F}(U) & \xrightarrow{\varphi(U)} & \mathcal{G}(U) \end{array}$$

commutes. Take a section $s \in \mathcal{F}(V)$; we need to show that

$$\varphi(V)(s)|_U = \varphi(U)(s|_U).$$

Since $\mathcal{G}$ is a sheaf and $U = \bigcup_{i \in I} (U \cap U_i)$, we check that :

$$\varphi(V)(s)|_{U \cap U_i} = \varphi(V)(s)|_{V \cap U_i}|_{U \cap U_i} = \varphi_i(V \cap U_i)(s|_{V \cap U_i})|_{U \cap U_i} = \varphi_i(U \cap U_i)(s|_{U \cap U_i}) = \varphi(U)(s|_U)|_{U \cap U_i}$$

for every $i \in I$. Hence, $\varphi(V)(s)|_U = \varphi(U)(s|_U)$ as required.

Next we show that $\varphi|_{U_i} = \varphi_i$ for all $i \in I$. So fix any $i \in I$ and some $W \subset U_i$. We show that $\varphi(W) = \varphi_i(W)$. Indeed, take some $s \in \mathcal{F}(W)$ and we show that $\varphi(W)(s) = \varphi_i(W)(s) \in \mathcal{G}(W)$ by checking that their restriction on each $W \cap U_j$, $j \in I$ coincide. Indeed, we have

$$\varphi(W)(s)|_{W \cap U_j} = \varphi_j(W \cap U_j)(s|_{W \cap U_j}) = \varphi_i(W \cap U_j)(s|_{W \cap U_j})$$

since $\varphi_i|_{U_i \cap U_j} = \varphi_j|_{U_i \cap U_j}$ and $W \cap U_j \subset U_i \cap U_j$.

Furthermore, if $\tilde{\varphi} : \mathcal{F} \rightarrow \mathcal{G}$ is another morphism such that $\tilde{\varphi}|_{U_i} = \varphi_i$ for all i , then we show that for any open $V \subset X$ and $s \in \mathcal{F}(V)$ the sections $\tilde{\varphi}(V)(s), \varphi(V)(s) \in \mathcal{G}(V)$ are equal by showing

$$\tilde{\varphi}(V)(s)|_{V \cap U_i} = \varphi(V)(s)|_{V \cap U_i}.$$

Indeed, the above equality is true since both are equal to $\varphi_i(V \cap U_i)(s|_{V \cap U_i})$.

Finally, suppose that both $\mathcal{F}$ and $\mathcal{G}$ are sheaves and φ_i is an isomorphism for every i . Then there is a unique morphism $\varphi' : \mathcal{G} \rightarrow \mathcal{F}$ such that $\varphi'|_{U_i} = \varphi_i^{-1}$ for all i . Then $\varphi' \circ \varphi = \text{id}_{\mathcal{F}}$, since it is the unique morphism of sheaves from $\mathcal{F}$ to itself whose restriction to U_i is identity for every i . Similarly, we have $\varphi \circ \varphi' = \text{id}_{\mathcal{G}}$. This completes the proof of (1).

(2) Let $U \subset X$ be open. Now define

$$\mathcal{F}(U) := \left\{ (s_i)_{i \in I} \in \prod_{i \in I} \mathcal{F}_i(U \cap U_i) \mid s_j|_{U \cap U_i \cap U_j} = \varphi_{ij}(U \cap U_i \cap U_j)(s_i|_{U \cap U_i \cap U_j}) \text{ for all } i, j \in I \right\}$$

If $U \subset V$ are open, we define $\rho_U^V : \mathcal{F}(V) \rightarrow \mathcal{F}(U)$ by $\rho_U^V((s_i)_{i \in I}) := (s_i|_{U \cap U_i})_{i \in I}$. Then it is easy to see that $\mathcal{F}$ is a presheaf on X .

We check that $\mathcal{F}$ is actually a sheaf. Let $V = \bigcup_{\lambda} V_{\lambda}$ be an open cover. Suppose $s_{\lambda} \in \mathcal{F}(V_{\lambda})$ are such that $s_{\lambda}|_{V_{\lambda} \cap V_{\mu}} = s_{\mu}|_{V_{\lambda} \cap V_{\mu}}$ for all λ, μ . Write $s_{\lambda} = (s_{\lambda,i})_{i \in I}$. Now fix an $i \in I$ and note that $s_{\lambda,i} \in \mathcal{F}_i(V_{\lambda} \cap U_i)$. Also the condition $s_{\lambda}|_{V_{\lambda} \cap V_{\mu}} = s_{\mu}|_{V_{\lambda} \cap V_{\mu}}$ means $(s_{\lambda,i}|_{V_{\lambda} \cap V_{\mu} \cap U_i})_{i \in I} = (s_{\mu,i}|_{V_{\lambda} \cap V_{\mu} \cap U_i})_{i \in I}$. Hence, we have

$$s_{\lambda,i}|_{V_{\lambda} \cap V_{\mu} \cap U_i} = s_{\mu,i}|_{V_{\lambda} \cap V_{\mu} \cap U_i} \quad \forall \lambda, \mu.$$

Since $\mathcal{F}_i$ is a sheaf on U_i we have a unique $s_i \in \mathcal{F}_i(V \cap U_i) = \mathcal{F}_i(\bigcup_{\lambda} (V_{\lambda} \cap U_i))$ such that $s_i|_{V_{\lambda} \cap U_i} = s_{\lambda,i}$ for all λ . Consider this $(s_i)_{i \in I} \in \prod_{i \in I} \mathcal{F}_i(V \cap U_i)$. We show that

$$s_j|_{V \cap U_i \cap U_j} = \varphi_{ij}(V \cap U_i \cap U_j)(s_i|_{V \cap U_i \cap U_j}) \in \mathcal{F}_j(V \cap U_i \cap U_j)$$

whence $(s_i)_{i \in I} \in \mathcal{F}(V)$. We do this by showing they agree on each set in the open cover $(V_{\lambda} \cap U_i \cap U_j)_{\lambda}$ of $V \cap U_i \cap U_j$. So consider

$$\begin{aligned} s_j|_{V_{\lambda} \cap U_i \cap U_j} &= s_j|_{V_{\lambda} \cap U_j}|_{V_{\lambda} \cap U_i \cap U_j} = s_{\lambda,j}|_{V_{\lambda} \cap U_i \cap U_j} \\ &= \varphi_{ij}(V_{\lambda} \cap U_i \cap U_j)(s_{\lambda,i}|_{V_{\lambda} \cap U_i \cap U_j}) \\ &= \varphi_{ij}(V_{\lambda} \cap U_i \cap U_j)(s_i|_{V_{\lambda} \cap U_i \cap U_j}) \\ &= \varphi_{ij}(V \cap U_i \cap U_j)(s_i|_{V \cap U_i \cap U_j})|_{V_{\lambda} \cap U_i \cap U_j} \end{aligned}$$

and we have thus shown the required.

Next, we show that $s = (s_i)_{i \in I} \in \mathcal{F}(V)$ has the property : $s|_{V_{\lambda}} = s_{\lambda}$ for all λ . Indeed, note that $s|_{V_{\lambda}} = (s_i|_{V_{\lambda} \cap U_i})_{i \in I} = (s_{\lambda,i})_{i \in I} = s_{\lambda}$. Also, if $s = (s_i)_{i \in I}$ and $t = (t_i)_{i \in I}$ are two sections in $\mathcal{F}(V)$ such that $s|_{V_{\lambda}} = t|_{V_{\lambda}}$ for all λ , then we show that $s_i = t_i \in \mathcal{F}_i(V \cap U_i)$ for all i . But, fixing one such i , the condition implies that $s_i|_{V_{\lambda} \cap U_i} = t_i|_{V_{\lambda} \cap U_i}$ for all λ , and since $\mathcal{F}_i$ is a sheaf on U_i and $\{V_{\lambda} \cap U_i\}_{\lambda}$ is an open cover of $V \cap U_i$, the conclusion follows. Hence, we have shown that $\mathcal{F}$ is a sheaf.

Next, we construct the morphisms $\varphi_k : \mathcal{F}|_{U_k} \rightarrow \mathcal{F}_k$ for each $k \in I$. Fix a $k \in I$, and for any open set $U \subset U_k$ define $\varphi_k(U) : \mathcal{F}(U) \rightarrow \mathcal{F}_k(U)$ by $(s_i)_{i \in I} \mapsto s_k$. We check that for $V \subset U \subset U_k$ open sets and $(s_i)_{i \in I} \in \mathcal{F}(U)$ we have $\varphi_k(U)((s_i)_{i \in I})|_V = \varphi_k(V)((s_i|_{V \cap U_i})_{i \in I})$. Indeed, the left hand side is $s_k|_V$ and the right hand side is $s_k|_{V \cap U_k} = s_k|_V$. Thus, $\varphi_k : \mathcal{F}|_{U_k} \rightarrow \mathcal{F}_k$ for $k \in I$ are natural transformation.

Now we check that φ_k are isomorphisms for all $k \in I$. Take $U \subset U_k$ to be an open set. We first show that $\varphi_k(U)$ is injective. So suppose $(s_i)_{i \in I}, (t_i)_{i \in I} \in \mathcal{F}(U)$ are such that $s_k = t_k$. But by definition of $\mathcal{F}(U)$ we have

$$s_i = s_i|_{U \cap U_i} = s_i|_{U \cap U_i \cap U_k} = \varphi_{ki}(U \cap U_i \cap U_k)(s_k|_{U \cap U_i \cap U_k}) = \varphi_{ki}(U \cap U_i)(t_k|_{U \cap U_k}) = t_i$$

for every $i \in I$. Hence, $\varphi_k(U)$ is injective.

Next, if $s \in \mathcal{F}_k(U)$ is a section, we set $s_i := \varphi_{ki}(U \cap U_i)(s|_{U \cap U_i})$ so that $(s_i)_{i \in I} \in \prod_{i \in I} \mathcal{F}_i(U \cap U_i)$ and $s_k = s$ as $\varphi_{kk} = \text{id}$. Moreover, for $i, j \in I$, we have

$$\begin{aligned} \varphi_{ij}(U \cap U_i \cap U_j)(s_i|_{U \cap U_i \cap U_j}) &= \varphi_{ij}(U \cap U_i \cap U_j)(\varphi_{ki}(U \cap U_i)(s|_{U \cap U_i})|_{U \cap U_i \cap U_j}) \\ &= \varphi_{ij}(U \cap U_i \cap U_j)(\varphi_{ki}(U \cap U_i \cap U_j)(s|_{U \cap U_i \cap U_j})) \\ &= \varphi_{kj}(U \cap U_i \cap U_j)(s|_{U \cap U_i \cap U_j}) \\ &= s_j|_{U \cap U_i \cap U_j} \end{aligned}$$

and hence $(s_i)_{i \in I} \in \mathcal{F}(U)$; note that in the second last equality we have used the cocycle condition given in the hypothesis. We have thus proved that $\varphi_k(U)$ is surjective. This completes the proof that φ_k is an isomorphism.

Now we show that $\varphi_{kj} \circ \varphi_k|_{U_j \cap U_k} = \varphi_j|_{U_j \cap U_k}$ for all j, k . Indeed, let $U \subset U_j \cap U_k$ be open and let $(s_i)_{i \in I} \in \mathcal{F}(U)$. Then, we have

$$(\varphi_{kj}(U) \circ \varphi_k(U))(s_i)_{i \in I} = \varphi_{kj}(U)(s_k) = s_j = \varphi_j(U)((s_i)_{i \in I})$$

as asserted. This completes the proof of the first part of item (2).

Finally, suppose $\mathcal{G}$ is another sheaf on X and $(\psi_i)_{i \in I}$ be family with $\psi_i : \mathcal{G}|_{U_i} \xrightarrow{\sim} \mathcal{F}_i$ such that $\varphi_{ij} \circ \psi_i|_{U_i \cap U_j} = \psi_j|_{U_i \cap U_j}$ for all i, j . Then $\theta_i := \varphi_i^{-1} \circ \psi_i : \mathcal{G}|_{U_i} \xrightarrow{\sim} \mathcal{F}|_{U_i}$ are isomorphisms for every $i \in I$. Also, we have

$$\begin{aligned} \theta_i|_{U_i \cap U_j} &= \varphi_i^{-1} \circ \psi_i|_{U_i \cap U_j} = \varphi_j^{-1}|_{U_i \cap U_j} \circ \varphi_{ji}^{-1} \circ \psi_j|_{U_i \cap U_j} \\ &= \varphi_j^{-1}|_{U_i \cap U_j} \circ \psi_j|_{U_i \cap U_j} = \theta_j|_{U_i \cap U_j} \end{aligned}$$

for all i, j . Thus, by part (1) there exists a unique isomorphism $\theta : \mathcal{G} \rightarrow \mathcal{F}$ such that $\theta|_{U_i} = \theta_i$ for every i . Setting $\psi := \theta^{-1}$ we have $\psi|_{U_i} = \psi_i^{-1} \circ \varphi_i$ for all i and the proof is complete. $\square$

Lemma 12.2. – Let $(X_i)_{i \in I}$ be a family of locally ringed spaces. For every pair $i, j \in I$, let X_{ij} be an open subset of X_i and let $\varphi_{ij} : X_{ij} \rightarrow X_{ji}$ be an isomorphism of locally ringed spaces. Assume the following hold :

- (1) For every i , we have $X_{ii} = X_i$ and $\varphi_{ii} = \text{id}_{X_i}$.
- (2) For every i and j , we have $\varphi_{ij} = \varphi_{ji}^{-1}$.
- (3) For every i, j, k one has $\varphi_{ij}(X_{ij} \cap X_{ik}) = X_{ji} \cap X_{jk}$ and additionally we have

$$\varphi_{ik}|_{X_{ij} \cap X_{ik}} = \varphi_{jk} \circ \varphi_{ij}|_{X_{ij} \cap X_{ik}}.$$

Then there is a locally ringed space X , a family $(U_i)_{i \in I}$ of open sets in X , and a family $(\varphi_i)_{i \in I}$ such that for all i, j the following hold :

- (a) The morphism $\varphi_i : U_i \rightarrow X_i$ is an isomorphism of locally ringed spaces.
- (b) $X_{ij} = \varphi_i(U_i \cap U_j)$.
- (c) $\varphi_j|_{U_i \cap U_j} = \varphi_{ij} \circ \varphi_i|_{U_i \cap U_j}$.

Proof. – We set $X_* := \bigsqcup_{i \in I} X_i \times \{i\}$ where the topology on X_* is defined as : $U \subset X_*$ is open $\iff U \cap (X_i \times \{i\})$ is open in each X_i . Define a relation $\sim$ on X_* as follows : $(x, i) \sim (y, j) \iff x \in X_{ij}, y \in X_{ji}$ and $y = \varphi_{ij}(x)$. This is an equivalence relation which can be easily checked. Now set $X := X_*/\sim$ and put the quotient topology on X i.e. $\Omega \subset X$ is open $\iff \pi^{-1}(\Omega) \cap X_i$ is open in X_i for all i , where $\pi : X_* \rightarrow X$ is the canonical quotient map. We have thus constructed our topological space for the required locally ringed space.

First, we show that π is an open map. Since any open set in X_* is a union of opens in $X_i \times \{i\}$, it suffices to show that for any i and any open $U \subset X_i$, $\pi(U \times \{i\})$ is open. But note that

$$\pi^{-1}(\pi(U \times \{i\})) = \bigcup_{j \in I} (\varphi_{ij}(X_{ij} \cap U) \times \{j\})$$

is open since the φ_{ij} are homeomorphisms. Hence we have shown that π is an open map.

Now, for each i , let $U_i := \pi(X_i \times \{i\})$. By the previous paragraph each U_i is open in X and the family $(U_i)_{i \in I}$ obviously covers X . Moreover, $\pi|_{X_i} : X_i \xrightarrow{\sim} U_i$ is a bijection : indeed, if $\pi(x) = \pi(x')$, then it means $x \sim x'$ i.e. $\varphi_{ii}(x) = x' = x$ since $\varphi_{ii} = \text{id}_{X_i}$. Now we set $\pi_i := \pi|_{X_i}$; it is a homeomorphism satisfying the property $\pi_i(X_{ij}) = U_i \cap U_j = \pi_j(X_{ji})$ i.e. $\pi_i \circ \varphi_{ji}|_{X_{ji}} = \pi_j|_{X_{ji}}$ for all i, j .

Now we define the sheaf $\mathcal{O}_{U_i} := (\pi_i)_* \mathcal{O}_{X_i}$. Then, for $i, j \in I$, the isomorphism $\varphi_{ij} : X_{ij} \rightarrow X_{ji}$ of locally ringed spaces induces an isomorphism of sheaves :

$$\begin{aligned} \theta_{ij} : \mathcal{O}_{U_i}|_{U_i \cap U_j} &= ((\pi_i)_* \mathcal{O}_{X_i})|_{U_i \cap U_j} = (\pi_i|_{X_{ij}})_* (\mathcal{O}_{X_i}|_{X_{ij}}) \\ &\xrightarrow[\text{coming from } \varphi_{ji}^\#]{\sim} (\pi_i|_{X_{ij}} \circ \varphi_{ji}|_{X_{ji}})_* \mathcal{O}_{X_j}|_{X_{ji}} = (\pi_j|_{X_{ji}})_* (\mathcal{O}_{X_j}|_{X_{ji}}) = \mathcal{O}_{U_j}|_{U_i \cap U_j}. \end{aligned}$$

Now assumptions (1), (2) and (3) imply that θ_{ij} satisfy the conditions (a), (b) and (c) of Lemma 12.1(2). Consequently, there exist a sheaf $\mathcal{O}_X$ on X and isomorphisms $\theta_i : \mathcal{O}_X|_{U_i} \xrightarrow{\sim} \mathcal{O}_{U_i}$ such that $\theta_{ij} \circ \theta_i|_{U_i \cap U_j} = \theta_j|_{U_i \cap U_j}$.

Finally, we check that $(X, \mathcal{O}_X)$ is a locally ringed space. Indeed, if $y \in X$, then $y \in U_i$ for some $i \in I$ and so there exists a unique $x \in X_i$ such that $y = \pi_i(x)$. Hence, we have

$$\mathcal{O}_{X,y} \simeq \mathcal{O}_{U_i,y} \simeq \mathcal{O}_{X_i,x}$$

which is a local ring. This completes the proof. $\square$

Lemma 12.3. – Let $(X_i)_{i \in I}$ be a family of schemes. Let X_{ij} be an open subscheme of X_i and $\varphi_{ij} : X_{ij} \xrightarrow{\sim} X_{ji}$ be an isomorphism of schemes for all $i, j \in I$. Assume that they satisfy the conditions (1), (2) and (3) of the previous Lemma 12.2. Then the locally ringed space X obtained by gluing the schemes X_i along the open subschemes X_{ij} by means of the isomorphisms φ_{ij} is also a scheme.

Proof. – Each U_i being isomorphic to X_i is a scheme, and hence a union of affines. Thus, X being union of the U_i is again a union of affines. $\square$

Lemma 12.4. – Let X be a scheme with a covering $(X_i)_{i \in I}$ be open subschemes, and let Y be another scheme. If $f_i : X_i \rightarrow Y$ are morphisms such that $f_i|_{X_i \cap X_j} = f_j|_{X_i \cap X_j}$ for all $i, j \in I$, then there exists a unique morphism $f : X \rightarrow Y$ such that $f|_{X_i} = f_i$ for all $i \in I$. In other words, the following diagram

$$\mathrm{Hom}(X, Y) \rightarrow \prod_{i \in I} \mathrm{Hom}(X_i, Y) \rightrightarrows \prod_{i, j \in I} \mathrm{Hom}(X_i \cap X_j, Y)$$

is exact.

Proof. – We simply glue together the maps f_i to get $f : X \rightarrow Y$, in which case $f|_{X_i} = f_i$ for all i automatically. Also f is continuous : indeed, for $V \subset Y$ open, we have $f^{-1}(V) = \bigcup_i (f^{-1}(V) \cap X_i) = \bigcup_i f_i^{-1}(V)$ and $f_i^{-1}(V)$ is open in X_i , so open in X .

Next, we define $f^\# : \mathcal{O}_Y \rightarrow f_* \mathcal{O}_X$. Consider the following diagram :

$$\begin{array}{ccc} & & \mathcal{O}_X(f^{-1}(V)) \\ & \nearrow f^\#(V) & \downarrow \\ \mathcal{O}_Y(V) & \xrightarrow{(f_i^\#(V))_{i \in I}} & \prod_{i \in I} \mathcal{O}_X(f^{-1}(V) \cap X_i) \\ & \searrow & \downarrow \varphi_1 \downarrow \varphi_2 \\ & & \prod_{(i,j) \in I \times I} \mathcal{O}_X(f^{-1}(V) \cap X_i \cap X_j) \end{array}$$

The vertical maps form an exact sequence since $\mathcal{O}_X$ is a sheaf. The horizontal map is derived from $f_i^\# : \mathcal{O}_Y \rightarrow f_{i*} \mathcal{O}_{X_i}$ for $i \in I$: note that $\mathcal{O}_X(f^{-1}(V) \cap X_i) = \mathcal{O}_{X_i}(f_i^{-1}(V))$. Now, for a section $s \in \mathcal{O}_Y(V)$, we have

$$\varphi_1 \left(\left(f_i^\#(V)(s) \right)_{i \in I} \right) = \left(f_j^\#(V)(s) |_{f^{-1}(V) \cap X_i \cap X_j} \right)_{(i,j) \in I \times I}$$

and

$$\varphi_2 \left(\left(\left(f_i^\#(V)(s) \right)_{i \in I} \right) \right) = \left(f_i^\#(V)(s) |_{f^{-1}(V) \cap X_i \cap X_j} \right)_{(i,j) \in I \times I}.$$

Now $f_i|_{X_i \cap X_j} = f_j|_{X_i \cap X_j}$ is equivalent to the commutativity of the following diagram of locally ringed spaces

$$\begin{array}{ccc} (X_i \cap X_j, \mathcal{O}_X|_{X_i \cap X_j}) & \hookrightarrow & (X_i, \mathcal{O}_X|_{X_i}) \\ \downarrow & & \downarrow f_i \\ (X_j, \mathcal{O}_X|_{X_j}) & \xrightarrow{f_j} & (Y, \mathcal{O}_Y) \end{array}$$

which in turn implies the commutativity of the diagram (of the sheaves involved)

$$\begin{array}{ccc} \mathcal{O}_Y(V) & \xrightarrow{f_i^\#(V)} & \mathcal{O}_{X_i}(f_i^{-1}(V)) = \mathcal{O}_X(f^{-1}(V) \cap X_i) \\ f_j^\#(V) \downarrow & & \downarrow \mathrm{res} \\ \mathcal{O}_{X_j}(f_j^{-1}(V)) = \mathcal{O}_X(f^{-1}(V) \cap X_j) & \xrightarrow{\mathrm{res}} & \mathcal{O}_X(f^{-1}(V) \cap X_i \cap X_j) \end{array}$$

for all $V \subset Y$ open. Hence, $\varphi_1 \left(\left(f_i^\#(V)(s) \right)_{i \in I} \right) = \varphi_2 \left(\left(f_i^\#(V)(s) \right)_{i \in I} \right)$, and so we can compose the horizontal map $\mathcal{O}_Y(V) \rightarrow \prod_{i \in I} \mathcal{O}_X(f^{-1}(V) \cap X_i)$ with φ_1 or φ_2 to obtain the map $\mathcal{O}_Y(V) \rightarrow \prod_{i,j} \mathcal{O}_X(f^{-1}(V) \cap X_i \cap X_j)$. Also, in this process we have shown that $\text{Im} \left(\left(f_i^\#(V) \right)_{i \in I} \right) \subset \text{Ker}(\varphi_1, \varphi_2)$. By the injectivity of the top vertical map, we therefore get a unique morphism $\mathcal{O}_Y(V) \rightarrow \mathcal{O}_X(f^{-1}(V))$, making the whole diagram commute.

Now we have to check that the $f^\#$ defined in the previous paragraph is a natural transformation. So let $V \subset U$ be open sets in Y and consider the following diagram :

$$\begin{array}{ccccc}
 \mathcal{O}_X(f^{-1}(U)) & \xrightarrow{\text{res}_{f^{-1}(V)}^{f^{-1}(U)}} & & \mathcal{O}_X(f^{-1}(V)) & \\
 \downarrow (\text{res}_{f^{-1}(U) \cap X_i}^{f^{-1}(U)})_{i \in I} & \swarrow f^\#(U) & \mathcal{O}_Y(U) \xrightarrow{\text{res}_V^U} \mathcal{O}_Y(V) & \searrow f^\#(V) & \downarrow (\text{res}_{f^{-1}(V) \cap X_i}^{f^{-1}(V)})_{i \in I} \\
 & \swarrow (f_i^\#(U))_{i \in I} & & \searrow (f_i^\#(V))_{i \in I} & \\
 \prod_{i \in I} \mathcal{O}_X(f^{-1}(U) \cap X_i) & \xrightarrow{\prod_{i \in I} \text{res}_{f^{-1}(V) \cap X_i}^{f^{-1}(U) \cap X_i}} & & \prod_{i \in I} \mathcal{O}_X(f^{-1}(V) \cap X_i) &
 \end{array}$$

where we have to show that the top square commutes. So take a section $s \in \mathcal{O}_Y(U)$ and consider its restriction $s_V \in \mathcal{O}_Y(V)$. We must show that $f^\#(V)(s_V) = f^\#(U)(s) \mid_{f^{-1}(V)}$. We do this by showing that they have the same image under $(\text{res}_{f^{-1}(V) \cap X_i}^{f^{-1}(V)})_{i \in I}$, which is injective. But the image of $f^\#(V)(s_V)$ under this restriction tuple is same as $\left(f_i^\#(V)(s_V) \right)_{i \in I}$ by commutativity of the triangle on the right. However, since each $f_i^\#$ is a natural transformation, this image is also the image of $\left(f_i^\#(U)(s) \right)_{i \in I}$ under $\prod_{i \in I} \text{res}_{f^{-1}(V) \cap X_i}^{f^{-1}(U) \cap X_i}$: note that the bottom square commutes. But $\left(f_i^\#(U)(s) \right)_{i \in I}$ is the image of $f^\#(U)(s)$ by construction of the map $f^\#(U)$. Thus, using the fact that the outer square commutes (all restrictions!!), we obtain that

$$\begin{aligned}
 (\text{res}_{f^{-1}(V) \cap X_i}^{f^{-1}(V)})_{i \in I} \circ f^\#(V) \circ \text{res}_V^U &= \prod_{i \in I} \text{res}_{f^{-1}(V) \cap X_i}^{f^{-1}(U) \cap X_i} \circ (f_i^\#(U))_{i \in I} \\
 &= \prod_{i \in I} \text{res}_{f^{-1}(V) \cap X_i}^{f^{-1}(U) \cap X_i} \circ (\text{res}_{f^{-1}(U) \cap X_i}^{f^{-1}(U)})_{i \in I} \circ f^\#(U) \\
 &= (\text{res}_{f^{-1}(V) \cap X_i}^{f^{-1}(V)})_{i \in I} \circ \text{res}_{f^{-1}(V)}^{f^{-1}(U)} \circ f^\#(U).
 \end{aligned}$$

But $(\text{res}_{f^{-1}(V) \cap X_i}^{f^{-1}(V)})_{i \in I}$ is injective, and so we have

$$f^\#(V) \circ \text{res}_V^U = \text{res}_{f^{-1}(V)}^{f^{-1}(U)} \circ f^\#(U)$$

which is what we wanted to show.

Finally, we check uniqueness. So suppose $f, g : X \rightarrow Y$ are two morphisms of schemes such that $f|_{X_i} = g|_{X_i} = f_i$ for all $i \in I$. Thus, as maps $f = g$ is obvious. Also the condition written explicitly is equivalent to the commutativity of the following square

$$\begin{array}{ccc}
 (X_i, \mathcal{O}_X|_{X_i}) & \hookrightarrow & (X, \mathcal{O}_X) \\
 \downarrow & & \downarrow (g, g^\#) \\
 (X, \mathcal{O}_X) & \xrightarrow{(f, f^\#)} & (Y, \mathcal{O}_Y)
 \end{array}$$

for all $i \in I$. This in turn implies that the following squares commute

$$\begin{array}{ccc}
 \mathcal{O}_Y(V) & \xrightarrow{f^\#(V)} & \mathcal{O}_X(f^{-1}(V)) \\
 g^\#(V) \downarrow & & \downarrow \text{res} \\
 \mathcal{O}_X(g^{-1}(V)) & \xrightarrow{\text{res}} & \mathcal{O}_X(f^{-1}(V) \cap X_i) = \mathcal{O}_X(g^{-1}(V) \cap X_i)
 \end{array}$$

for all open $V \subset Y$ and for all $i \in I$. Now consider the following diagram

$$\begin{array}{ccccc}
\mathcal{O}_X(g^{-1}(V)) & \xlongequal{\quad} & \mathcal{O}_X(f^{-1}(V)) & & \\
\downarrow & \swarrow g^\#(V) & \searrow f^\#(V) & & \downarrow \\
& \mathcal{O}_Y(V) & & & \\
\downarrow & \swarrow & \searrow & & \downarrow \\
\prod_{i \in I} \mathcal{O}_X(g^{-1}(V) \cap X_i) & \xlongequal{\quad} & \prod_{i \in I} \mathcal{O}_X(f^{-1}(V) \cap X_i) & &
\end{array}$$

where the images of $f^\#(V)$ and $g^\#(V)$ under the vertical maps are same due to commutativity of the previous squares. Since the vertical maps are injective ($\mathcal{O}_X$ being a sheaf), we have that $f^\#(V) = g^\#(V)$ for all $V \subset Y$ open. The proof is complete. $\square$

Theorem 12.5. – *Let A be a ring and Y be a scheme. Then, the canonical map*

$$\Phi : \text{Hom}(Y, \text{Spec } A) \rightarrow \text{Hom}(A, \mathcal{O}_Y(Y)), \quad (f, f^\#) \mapsto f^\#(\text{Spec } A)$$

is bijective.

Proof. – Step 1 : Assume Y to be an open subscheme of some affine scheme $\tilde{Y}$. Consider an open covering $(Y_i)_{i \in I}$ of Y with basic open sets, whence $Y_{ij} := Y_i \cap Y_j$ are also basic opens for all i, j . Set $B := \mathcal{O}_Y(Y)$, $B_i := \mathcal{O}_Y(Y_i)$ and $B_{ij} := \mathcal{O}_Y(Y_{ij})$ for $i, j \in I$. Then, consider the following diagram :

$$\begin{array}{ccccc}
\text{Hom}(Y, \text{Spec } A) & \longrightarrow & \prod_{i \in I} \text{Hom}(Y_i, \text{Spec } A) & \rightrightarrows & \prod_{(i,j) \in I \times I} \text{Hom}(Y_{ij}, \text{Spec } A) \\
\Phi \downarrow & & \downarrow & & \downarrow \\
\text{Hom}(A, B) & \longrightarrow & \prod_{i \in I} \text{Hom}(A, B_i) & \rightrightarrows & \prod_{(i,j) \in I \times I} \text{Hom}(A, B_{ij})
\end{array}$$

where both the squares commute; note that the bottom horizontal maps are obtained by restrictions. For instance, to check that the left square commutes, take any $f \in \text{Hom}(Y, \text{Spec } A)$ and note that its image $f^\#(\text{Spec } A)$ under Φ gets mapped to $(\text{res}_{Y_i}^Y \circ f^\#(\text{Spec } A))_{i \in I}$. On the other hand, dualizing

$$f|_{Y_i} := (Y_i, \mathcal{O}_Y|_{Y_i}) \hookrightarrow (Y, \mathcal{O}_Y) \xrightarrow{f} (\text{Spec } A, \mathcal{O}_{\text{Spec } A})$$

we obtain

$$(f|_{Y_i})^\#(\text{Spec } A) = \text{res}_{Y_i}^Y \circ f^\#(\text{Spec } A)$$

and so $(\text{res}_{Y_i}^Y \circ f^\#(\text{Spec } A))_{i \in I} = ((f|_{Y_i})^\#(\text{Spec } A))_{i \in I}$. Similarly, one can show that the right square commutes. Now the upper row is exact by Lemma 12.4, whereas the lower row is exact by applying the left exact functor $\text{Hom}(A, \cdot)$ to the exact sequence

$$B \rightarrow \prod_{i \in I} B_i \rightrightarrows \prod_{(i,j) \in I \times I} B_{ij}.$$

Note that the above exactness follows from $\mathcal{O}_Y$ being a sheaf. Also, the middle and the right vertical maps are bijections by Proposition 11.24 since Y_i and Y_{ij} are affine schemes being basic opens of $\tilde{Y}$ which is affine. By an elementary diagram chase, it follows that Φ is bijective.

Step 2 : Now let Y be an arbitrary scheme, and $(Y_i)_{i \in I}$ an affine open cover. Then Y_{ij} are open subschemes of the Y_i . Hence, by Step 1, the right vertical maps are bijections, and so is Φ . $\square$